Good moduli spaces of A_r-stable curves

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Abstract

We study obstructions for open substacks of the stack $\mathcal{M}^r_{g,n}$ of pointed A_r-stable curves to admit a good moduli space and we find an open substack $\mathcal{U}^r_{g,n}$ that admits a proper non-projective good moduli space when $r = 5$. Moreover, we show that when $n = 0$ and $g \geq 9$ or $g = 7$, this moduli space is an algebraic space that is not a scheme. When r is small enough, the stack $\mathcal{U}^r_{g,n}$ appears in the Hasset-Keel program. In order to prove that $\mathcal{U}^r_{g,n}$ admits a separated good moduli we prove Θ-completeness and S-completeness, as introduced by Alper, Halpern-Leistner and Heinloth. Along the way, we introduced some new tools to deal with this existence criterion for good moduli spaces. For instance, we show that under reasonable assumptions, an algebraic stack is S-complete if and only if it is generically S-complete.

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"It has always been about the journey..."

1 Introduction

The moduli space of smooth curves is one of the most studied moduli spaces in algebraic geometry, going back to Riemann. In [DM69], Deligne and Mumford constructed the first modular compactification using algebraic methods. In later years, new modular compactifications appeared in the literature. Among the most famous are the Schubert compactification [Sch91] and others arising in the Hassett–Keel program [HH09, HH13], all of which involve curves with A-type singularities.

These first compactifications were constructed using GIT, via an analysis of stability in the Hilbert and Chow schemes. The series of papers [AFSvdW17, AFS17a, AFS17b] introduces a more intrinsic approach, using what is called local VGIT. This approach remarkably involves fewer computations and achieves one further step in the Hassett–Keel program. New compactifications constructed later in [CTV23] use the same technique.

We address the problem of modular compactifications by stepping further toward even more intrinsic techniques, fitting into what is called the "Beyond GIT" program in [HL14]. Namely, we use the criteria introduced in [AHLH23] and verify Θ- and S-completeness. We construct new modular compactifications that admit a proper good moduli space that is not projective (nor even a scheme in some cases). This therefore provides the first examples of compactifications of the moduli space of curves that cannot be constructed using GIT techniques.

1.1 Modularity

While the idea of *modular compactifications* is classical in spirit, we briefly reformulate it within the natural stack-theoretic framework. This section is greatly inspired by [Smy13]; we essentially adapt his definition to make it suitable for our setting.

Through this section, we omit specifying the base scheme S for brevity. Moreover, this framework extends naturally to the case of n-pointed curves, which will play an important role throughout this paper.

The stack of smooth genus g curves $\mathcal{M}_g$ has a coarse moduli space M_g that is quasi-projective but not projective. In [DM69], Deligne and Mumford consider the open immersion $\mathcal{M}_g \subset \overline{\mathcal{M}}_g$, where $\overline{\mathcal{M}}_g$ was constructed as the algebraic stack parametrizing nodal stable curves. The stack $\overline{\mathcal{M}}_g$ is proper and admits a proper (even projective) coarse moduli space. This is the first algebraic compactification where the boundary classifies certain singular curves. More generally, we can consider the algebraic stack of all connected curves $\mathcal{C}_g$ of genus g, whose objects are families of proper, one-dimensional connected schemes of genus g (see [Sta23, Section 0E6H] for the precise definition). This leads us to the following definition.

Definition 1.1. A *modular compactification* of $\mathcal{M}_g$ is an open substack $\mathcal{U}_g \subset \mathcal{C}_g$ such that

- $\mathcal{U}_g$ contains $\mathcal{M}_g$ as a dense open substack;
- $\mathcal{U}_g$ admits a proper moduli space (in a suitable sense).

This is similar to [Smy13, Definition 1.1], with a major distinction that we now explain. Whenever the stack $\mathcal{U}_g$ is proper, then automatically $\mathcal{U}_g$ admits a proper coarse moduli space U_g by the Keel–Mori theorem; this is exactly what happens for $\overline{\mathcal{M}}_g$. Nevertheless, the definition above is more general and also includes the case when $\mathcal{U}_g$ is not separated, allowing stabilizers of points in $\mathcal{U}_g$ to be non-discrete: this is a classical behavior when one allows worse-than-nodal singularities to appear. Many such examples arise in the Hassett–Keel program, as explained in the next section.

In relatively recent times, the theory of *good moduli spaces* developed in [Alp13] has allowed algebraic geometers to expand the study of moduli spaces to the world of algebraic stacks with linearly reductive stabilizers (at closed points), thus not necessarily finite. Moreover, at the time of this writing, future work by David Rydh on non-reductive moduli spaces (or even topological moduli spaces) may allow algebraic geometers to go even further.

This motivates relaxing our notion of modular compactification with respect to what was previously defined in [Smy13, Definition 1.1]. In this setting, a couple of remarks are in order. If $\mathcal{U}_g \to U_g$ is the good moduli space of a modular compactification, then the induced map $M_g \to U_g$ is not necessarily an open immersion; indeed, it may not even be injective. Although this may seem strange, examples of such compactifications appear naturally, see for instance [ADLW25, Thm. 1.1]. Secondly, if we require $\mathcal{U}_g$ to have a proper good moduli space, then $\mathcal{U}_g$ is forced to be quasi-compact (over S), which is a natural choice whichever notion of moduli space one may use. Therefore, we may shrink the algebraic stack $\mathcal{C}_g$ by imposing some boundedness condition. A possible choice (though certainly not the only one) is to study the modular compactifications lying in $\mathcal{G}_g \subset \mathcal{C}_g$ parametrizing reduced Gorenstein curves with ample dualizing bundle. As a byproduct of our discussion, a vague but inspiring question arises:

Question 1.2. Is it possible to classify all modular compactifications of $\mathcal{M}_g$?

Although this question depends on the definition of *modular compactification* and is very broad, some progress has been made (see for instance [Smy13]). Moreover, in future work, the third author, together with Luca Battistella and Andrea Di Lorenzo, will prove a complete classification of modular compactifications of $\mathcal{M}_{1,n}$ inside $\mathcal{G}_{1,n}$, building on the work of [BKN23]. In an analogous direction, future work of the first author will provide a complete classification of modular compactifications of $\mathcal{M}_{g,n}$ inside $\mathcal{M}^3_{g,n}$ (the moduli stack of curves with ample log dualizing sheaf and singularities of type A_i, $i \leq 3$).

1.2 The Hassett-Keel program

Several examples of modular compactifications already exist in the literature. For instance, an early example is the (coarse) moduli space of pseudostable curves, $\overline{M}_g^{ps}$, due to Schubert [Sch91], which removes curves with elliptic tails but it includes cuspidal singularities into the picture. It turns out that Schubert's compactification is obtained by contracting a K-negative extemal ray of the Mori cone, that result in a divisorial contraction of the locus of curves containing an elliptic tail. From the birational perspective, a central role is played by the canonical model. In an effort to study it, [Has05] introduced the Hassett–Keel program, aimed to understand the *log* canonical models of $\overline{M}_g$. More specifically, for any $\alpha \in [0,1] \cap \mathbb{Q}$ such that $K_{\overline{\mathcal{M}}_g} + \alpha\delta$ is big, Hassett defined

$$\overline{M}_g(\alpha) := \operatorname{Proj} \bigoplus_{m \geq 0} \mathrm{H}^0 \left(\overline{\mathcal{M}}_g, m(K_{\overline{\mathcal{M}}_g} + \alpha\delta) \right)$$

and asked whether the spaces $\overline{M}_g(\alpha)$ admit a modular interpretation. The first three steps of the program have been carried out: see [HH09], [HH13], and the series of three papers [AFSvdW17], [AFS17a], and [AFS17b]. Further flips have also been computed for small genera, and a complete answer to Hassett's question is known for $g \leq 4$ (see for instance [ADLW25]). In these works, new projective moduli spaces of curves are constructed using Local Variation of Geometric Invariant Theory (or LVGIT in short). Further interplay between the MMP of $\overline{M}_{g,n}$ and alternative compactifications can be found in [CTV23, CTV21, Gor26].

The modular interpretation of the first steps of the Hassett-Keel program are obtained by allowing singularities of type A_r for some r, that is considering curves with singularities which étale locally are of the form $y^2 = x^{r+1}$ (in further steps other singularities may appear, as it is shown in [ADLW25] and [FS13]). All the curves appearing so far are also still stable in the sense that they are Gorenstein

and they have an ample canonical line bundle. Thus, such modular compactifications are found inside the stack $\mathcal{G}_g$. The latter stack is not Deligne-Mumford, indeed the condition of having ample canonical bundle does not force discrete stabilizers for curves with worse-than-nodal singularities.

1.3 Contribution to the literature

Inspired by all of the above considerations, we decided to investigate possible modular compactifications of $\mathcal{M}_{g,n}$ inside the stack $\mathcal{M}^r_{g,n}$, parametrizing curves in $\mathcal{G}_{g,n}$ with only A-singularities of type at most A_r.

We decided to study the property of *having a separated good moduli space* for open substacks of $\mathcal{M}^r_{g,n}$, which translates in two valuative criteria (see [AHLH23, Theorem A]) called Θ- and S-completeness. To understand these two valuative criteria, it is fundamental to understand isotrivial degenerations, also called very-close degenerations, or filtrations. In the stacky language, these are degenerations realized by a morphism from $\Theta := [\mathbb{A}^1/\mathbb{G}_m]$, see [HL14] and [Hei17], where the terminology of filtrations and very-close degenerations were introduced respectively.

After studying the rigidity of the geometry of isotrivial degenerations for A-singularities, it was soon clear that Θ-completeness could not hold for the whole stack $\mathcal{M}^r_{g,n}$ (for $r \geq 3$). Classifying the possible counterexamples to Θ-completeness and S-completeness, we attempt to construct a *maximal* modular compactification of $\mathcal{M}^r_{g,n}$ for any r, i.e. not contained in any other modular compactification (inside $\mathcal{M}^r_{g,n}$). In the end, this is our main contribution to the literature.

Main Theorem (see Theorem 6.1, Theorem 7.1 and Theorem 8.1)**.** *There exists a modular compactification $\mathcal{U}^r_{g,n}$ of $\mathcal{M}_{g,n}$ in $\mathcal{M}^r_{g,n}$ for $r \leq 5$ which has a proper good moduli space. Moreover, if $r = 5$ and $g \geq 5$, then $U^5_{g,n}$ is not projective. Finally, if $g \geq 9$ or $g = 7$, then the moduli space of $\mathcal{U}^5_g = \mathcal{U}^5_{g,0}$ is not a scheme.*

The result holds if the characteristic is zero or it is large enough: specifically, we want the deformation theory of A_r-singularities to be well-behaved and the automorphism groups of the curves to be linearly reductive. This gives us a lower bound on the characteristic depending on (g, n, r). For the sake of simplicity, we will assume $\operatorname{char} \kappa = 0$ throughout the paper.

This provides the first example of a moduli space of curves that cannot arise via classical GIT techniques. We are confident that the theory developed to reach this result can be used in future solutions for the modular classification problem, specifically when GIT methods do not work. Moreover, we believe it can also be used to understand further steps of the Hasset-Keel program. Indeed, when $r \leq 3$, the stack $\mathcal{U}^r_{g,n}$ appears in the Hasset-Keel program while $\mathcal{U}^4_{g,n}$ properly contains one of the stacks appearing there, but doesn't appear itself (see Remark 4.6 for a more detailed comparison). Before explaining the byproducts of constructing such modular compactifications, we discuss a couple of remarks regarding the Main Theorem.

Firstly, we believe that the statement holds for any $r \geq 5$.

Conjecture 1.3. The moduli stack $\mathcal{U}^r_{g,n}$ (see Definition 4.4) has a proper non-projective good moduli space for $r \geq 5$ and $g \geq 5$.

The conjecture would follow from Conjecture 7.42, together with Theorem 6.1, Lemma 8.6 and Lemma 8.3. Furthermore, while we prove Θ-completeness for any r, we only manage to handle S-completeness for r small. This is due to the fact that in the (2-dimensional) valuative criterion for Θ-completeness, one starts with one isotrivial degeneration glued generically to one induced by a DVR. Therefore, there are some clear geometric constraints for the common generic fiber, thanks to the presence of the isotrivial degeneration. On the contrary, the valuative criterion for S-completeness starts with two (not necessarily isotrivial) degenerations over the same DVR, thus it is impossible to control what happens in the generic fiber.

Secondly, we remark that we do not prove its maximality. Nevertheless, from Theorem 8.9 follows that the open is big enough to lose the projectivity for $r \geq 5$ (intuitively, the bigger the open, the more loci are contracted by the rational map from the moduli of stable curves) and therefore cannot be constructed using the GIT machinery: its construction completely relies on the combinatorics of (Gorenstein) curves and their specializations. Finally, the case of A_5-singularities is particularly interesting, since they should appear in the next step of the Hassett-Keel program. However, we slightly departed from the conditions appearing in the second flip of the HK program, with the precise intent of obtaining a non-projective moduli space.

Finally, here is a list of the results and contributions to the literature that we obtained as a byproduct along the way:

(a) we classify all the modular compactification in the case of $2:1$ covers of $\mathbb{P}^1$, to explain in small scale some of the ideas (see Theorem 2.49);

(b) we completely determined the geometry of isotrivial degenerations for A-singularities, studying curves with positive dimensional stabilizers and their deformation theory (see Section 3);

(c) we prove a useful result for open embeddings of stacks with linearly reductive stabilizers: we give a combinatorial characterization to the open embedding being S-complete and/or Θ-complete (see Proposition A.7);

(d) we discuss the notion of *weak* S-completeness, and show how to combine it with Θ-completeness to obtain S-completeness (see Section 7.2); moreover, we prove that S-completeness can be checked generically (see Section 7.1).

1.4 Outline of the paper

In what follows, we will briefly sum up the structure of the paper.

Section 2 collects the needed preliminaries in the literature regarding good moduli spaces and A_r-stable curves. In Section 2.1, we discuss briefly the theory of good moduli spaces. Afterward, we recall some properties of A_r-stable curves and the description of the crimping spaces of A_r-singularities (the space of deformations that preserve the singularity type) from [Per23] and [vdW10] (in Section 2.2). In Section 2.3 we discuss when automorphisms descend through the normalization morphism. In Section 2.4 we focus on hyperelliptic curves (in fact cyclic covers of the projective line) with A-singularities: there is a powerful link between A-singularities and $2:1$ covers of the projective line, given by the stabilization technique. Finally, we characterize open substacks of the stack of reduced cyclic covers of degree 2 of $\mathbb{P}^1$ that admits a separated good moduli space (see Theorem 2.49).

Section 3 is dedicated to classifying the A_r-stable curves with positive dimensional stabilizers as well as studying their (equivariant) deformation theory and determinating the possible isotrivial degenerations. This chapter is at the intersection of every result we obtain for the moduli stack of A_r-stable curves. Firstly, we characterize all the possible curves with positive dimensional automorphism group (see Proposition 3.22 and Corollary 3.25) and their deformation spaces (see for instance Proposition 3.29). This allows us to have control over all the possible isotrivial degenerations in the moduli stack, which is fundamental to try and prove Θ-completeness and S-completeness. The study of isotrivial degenerations can be found in Section 3.4. Their fundamental combinatorial constraint will be discussed in Remark 3.33.

In Section 4, we perform the combinatorial/topological pruning process of considering potential counterexamples to Θ- and S-completeness. We use this to come to the definition of *admissible* A_r-stable curves and their classifying stack $\mathcal{U}^r_{g,n}$ (which we prove to be an open in the moduli stack of A_r-stable curve). The counterexamples are divided into two families, called tails and chains, reflecting the fact that A-singularities are unibranched or bibranched according to the parity of r. In particular, even A-singularities are unibranched and correspond to tails: one cannot attach them one after another

to create arbitrary long chains, thus they are easier to deal with. The same cannot be said for odd A-singularities.

The rest of the paper is dedicated to prove the Main Theorem.

Section 5 is dedicated to showing that the inclusions $\mathcal{U}^r_{g,n} \subset \mathcal{M}^r_{g,n}$ of admissible curves into A_r-stable curves for all g, n, r are Θ- and S-complete (see Theorem 5.1). The main ideas is that proving an open embedding verifies S-completeness and Θ-completeness can be checked locally (on every point of the complement of the open), if the stacks involved are nice enough. This reduction step is carried out in Appendix A, where we also give equivalent conditions to the valuative criteria, with a more combinatorial flavour. In the rest of the section, we set up the combinatorial problem and we dedicate Appendix B to the solution of it.

In Section 6, we show that the stack of n-pointed, admissible A_r-stable curves $\mathcal{U}^r_{g,n}$ is Θ-complete for all g, n, r, see Theorem 6.1. The proof relies on an inductive argument on the genus g: the rigidity of isotrivial degenerations allows us to show that, except for very special cases that are handled separately, any family of curves on the punctured (stacky) surface appearing in the Θ-complete criterion admits an equisingular section (see Lemma 6.4). Thus by partial normalization, we can reduce the genus and extend the family to the whole surface (to be precise, such an extension will be constructed in $\mathcal{M}^r_{g,n}$, not necessarily in $\mathcal{U}^r_{g,n}$; luckily, we can apply Theorem 5.1 to conclude).

Section 7 is dedicated to the proof of S-completeness of $\mathcal{U}^r_{g,n}$ for $r \leq 5$, see Theorem 7.1. As already mentioned, S-completeness is more difficult to handle. We start by studying S-completeness abstractly in a way to make our life easier in the concrete proof. Firstly, in Section 7.1, we introduce a topological notion of S-completeness (which implies the existence part of S-completeness in the affine diagonal case). Moreover, we use this notion to prove that the existence part of the valuative criterion for S-completeness can be checked generically, in the affine diagonal case (see Proposition 7.18). This helps us restricting the study to generically smooth families. Moreover, we develop a weaker notion of S-completeness, which is closely related to almost properness as introduced in [Hei17] and S-n completeness as introduced in the thesis of the second author [Mod25]: this weaker notion paired with Θ-completeness implies S-completeness (see Corollary 7.33).

The proof of S-completeness for $\mathcal{U}^r_{g,n}$ is by induction on r. To start the inductive procedure, in Section 7.3 we develop an abstract stabilization result (see Lemma 7.36), which it is nothing more than a clever application of Langton's algorithm (see [AHLH23, Theorem 6.3]). Still, it is not enough to prove S-completeness for every r and we finish the section with a discussion about what goes wrong for bigger r's and why we still believe that $\mathcal{U}^r_{g,n}$ verifies S-completeness (and thus admit a separated good moduli space) for any r.

Finally, Section 8 is dedicated to proving that $U^r_{g,n}$, the good moduli space of $\mathcal{U}^r_{g,n}$, is proper but not projective for $r = 5$. Firstly, we prove that $\mathcal{U}^r_{g,n}$ is *generically* universally closed for any $r \geq 1$ (see Lemma 8.6), essentially reversing the stabilization technique introduced in Section 7. Here generically means that we are checking the existence part of the valuative criterion for families that are (generically) smooth. This is enough to prove universal closedness if the stack has a separated good moduli space (see Lemma 8.3). Moreover, we compute the rational Picard groups of $\mathcal{U}^r_{g,n}$ and their moduli space $U^r_{g,n}$, assuming they exist. In this case, we show in Theorem 8.9 that $\mathrm{Pic}_{\mathbb{Q}}(U^r_{g,n}) = 0$ when $r \geq 5$ and $g \geq 5$, which gives us the non-projectivity of $U^5_{g,n}$. We finish the section, and the paper, with a proof that for $g \geq 9$ or $g = 7$, the moduli space U^5_g is not a scheme in Corollary 8.15.

1.5 Related works and future directions

In this subsection, we will discuss some ideas that came up while working on this project. We believe that some could have interesting applications in the future and they could deepen our understanding of the moduli stack of curves.

Counterexamples

As we already mentioned in Section 1.3, one of our initial motivations was to understand how to interpret the valuative criteria for S-completeness and Θ-completeness in the case of moduli of worse-than-nodal curves.

We started by comparing the proof of the criteria to the case of moduli of semistable vector bundles, quickly discovering what ingredient was missing for the curve case. To prove that the stack of semistable vector bundles (of fixed degree and rank) is S-complete and Θ-complete, one uses the fact that the whole stack of coherent sheaves is S-complete and Θ-complete, and then one proves that the embedding of the locus of semistable vector bundles into the moduli of coherent sheaves is S-complete and Θ-complete. In the case of curves, it seems that we are missing an analogue of the stack of coherent sheaves, i.e. an ambient algebraic stack which is S-complete and Θ-complete, where to search for our compactifications. This would make the study of modular compactification much easier: instead of needing to work our way up from the nodal case (as we do), we could simply produce open substacks such that the open embedding is S-complete and Θ-complete (for instance using Proposition A.7, which gives the problem a combinatorial flavour).

An instance of this idea is in the first author's PhD thesis [Gor26], which approaches the classification of all modular compactifications inside $\mathcal{M}^3_{g,n}$ (thus up to tacnodal curves), establishing three maximal open substacks that are both S- and Θ-complete using similar techniques. The classification then proceeds within these maximal stacks, where more combinatorics enters the picture, and a dictionary with the combinatorics of cluster algebras is established. Another instance is the future work mentioned at the end of Section 1.1 of the third author with Battistella and Di Lorenzo.

Thus the question remains:

Question 1.4. Is there a good equivalent of the moduli stack of coherent sheaves for the moduli problem of curves?

Although we have no clear answer, one natural candidate would be $\mathcal{G}_{g,n}$, the moduli stack of reduced Gorenstein curves with ample (log)-dualizing line bundle. Nevertheless, the counterexamples we produce may be still valid in $\mathcal{G}_{g,n}$. To understand this, we formulate the following more precise question.

Question 1.5. Is the open embedding $\mathcal{M}^r_{g,n} \subset \mathcal{G}_{g,n}$ S-complete and Θ-complete?

An affirmative answer would imply that our counterexamples to the valuative criteria will remain such in the bigger stack $\mathcal{G}_{g,n}$. At this moment of writing, we see two main obstacle to the proof of an affirmative answer to Question 1.5. First of all, to handle the problem one would like to reduce to the local picture, where deformation theory can be a powerful tool. Nevertheless, the étale local structure theorem is still a mystery for non-linearly reductive stabilizers (even though future work of David Rydh on topological moduli space could shed light on the question), which may appear depending on the triple (g, n, r) (see Corollary 3.25 for instance). Secondly, even assuming the local reduction step, it is not true that, given a Gorenstein singularity with a $\mathbb{G}_m$-action, all the weights in the deformation space have the same sign (e.g. the planar singularity $y^4 - x^5$). Although seemingly unrelated, it is not too hard to convince oneself that such property would imply the answer to Question 1.5 is affirmative. Since such property holds for A-singularities, it implies for free that the open embedding $\mathcal{M}^r_{g,n} \subset \mathcal{M}^s_{g,n}$ is S-complete and Θ-complete for any $r \leq s$: this is the reason why the problem of classifying isotrivial degeneration for A-singularity felt difficult, but not out of reach.

Toric Geometry and Moment Measure Conjecture

Another very important property of A-singularities is that (except in the case $n = 0$ and $r \geq 2g+1$) the connected component of the identity of the stabilizer groups of $\mathcal{M}^r_{g,n}$ is a torus. This is true for every geometric point, not just the closed ones (see Corollary 3.25) and it is needed to apply Proposition A.7.

This gives a toric (or more generally logarithmic) flavour to the local theory of $\mathcal{M}^r_{g,n}$, as we already mentioned. Although we are allowing singularities which are not logarithmic in nature (such as cusps, tacnodes, etc...), there are two main reason why a logarithmic approach of the modular classification problem (at least for A_r-singularities) can lead to some new insights:

- the test stack $\overline{\mathrm{ST}}$ for S-completeness has a natural logarithmic structure;
- the stabilization procedure is logarithmic in nature.

We have used a stabilization result only locally to prove both S-completeness and universally closedness, knowing Θ-completeness. We believe that one can globalize this stabilization result, namely finding (for any r) a modular stack of hyperelliptic contractions $\mathcal{M}^{\mathrm{hc}}_{g,n}$ with a logarithmic modification $\mathcal{M}^{\mathrm{hc}}_{g,n} \to \overline{\mathcal{M}}_{g,n}$ and a universal contraction morphism $c : \mathcal{M}^{\mathrm{hc}}_{g,n} \to \mathcal{M}^r_{g,n}$ which is universally closed and surjective. Since being S-complete, Θ-complete and universally closed descend through universally closed surjective morphisms, to check if an open of $\mathcal{M}^r_{g,n}$ has a proper good moduli space, it is enough to check if its preimage through the morphism c does. The logarithmic structure of $\mathcal{M}^{\mathrm{hc}}_{g,n}$ may allow us to translate the properties we need into some combinatorial problem, which could help greatly for the modular classification. For $g = 1$, there exists a candidate for $\mathcal{M}^{\mathrm{hc}}_{1,n}$, namely the moduli stack of radially aligned elliptic curves, introduced in [RSPW19]. This has already been used as explained above to produce several new compactifications of $\mathcal{M}_{1,n}$ in $\mathcal{G}_{1,n}$ and to classify the Deligne-Mumford ones (see [BKN23]). However, the radially aligned curves cannot be enough to classify non Deligne-Mumford compactifications, since the moduli stack is still Deligne-Mumford; nevertheless, in the future work of the third author with Battistella and Di Lorenzo, they use the universal radius over the moduli stack of radially aligned curves as the stack of "universal" contractions and they complete the classification of modular compactifications of $\mathcal{M}_{1,n}$ in $\mathcal{G}_{1,n}$. The main difficulty in extending the constructions and proofs in the higher genus scenario is the presence of curves with multiple singularities of positive genus (i.e. worse than nodal).

Finally, there is another interesting consequence of the toric stabilizer condition: the moment measure conjecture (more precisely, a possible future generalization) could apply to give us the classification of modular compactifications almost for free. For stacks with a global toric structure, i.e. that come with a quotient presentation $[X/T]$ where X is a proper normal scheme and T is a torus whose action lifts to an ample line bundle, there is an explicit classification of open substacks admitting proper good moduli spaces in terms of combinatorial data called *moment measures*. This description was conjectured by Bilynicki-Birula and Sommese in [BBS87] and proven in full generality in [HZ25]. In a future work of the last two authors with Jochen Heinloth, we plan to generalize this combinatorial description to a class of stacks with toric stabilizers (as connected component of the identity) which contains properly the ones of the form $[X/T]$ as above, but it would also include for instance $\mathcal{M}^r_{g,n}$ (at least when $r \leq 2g$) and $\mathcal{G}_{1,n}$. Indeed, after working through the description of the opens of $\mathcal{G}_{1,n}$ (for small value of n) admitting proper good moduli spaces (future work of the third author with Battistella and Di Lorenzo), applying the moment measure machinery gives the exact same answer as the logarithmically flavoured classification.

Acknowledgments

We believe that this project is the perfect example of how sometimes research is really about the journey. Although Theorem 8.1 provides an interesting example of how complicated good moduli spaces of curves can be where no GIT is involved, the main contributions of this elaborate are the techniques developed to try and understand a problem as complicated as the classification of modular compactifications. It is yet another testament of how "failure" helps expanding the knowledge and may prepare the road for future travelers to take (or avoid). We explored and tried a lot of different paths, learning so much, and we are tremendously indebted to all the travelers we met along the way. To name a few that had contributed by sharing comments, ideas or just listening, we are grateful to Jarod Alper, Luca Battistella, Andrea Di Lorenzo, Andres Fernandez Herrero, Jochen Heinloth,

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2 Moduli spaces and A_r-stable curves

This section is dedicated to the introductory material regarding good moduli spaces and curves with singularities of type A. Specifically, we start by recalling in Section 2.1 the theory of good moduli spaces, as introduced in [Alp13]. We then recall some basic theory regarding curves with A-singularities and their moduli stack in Section 2.2, where we also study the crimping spaces associated to A-singularities. In Section 2.3, we study the equivariant properties of normalizations of A-singularities, with the goal of understanding which automorphisms (in the connected component of the identity) descends through the normalization. Finally, in Section 2.4 we concentrate our effort in the understanding of hyperelliptic curves with A-singularities, as they play a crucial role in the study of isotrivial degeneration of A-singularities. We end the section proving that, in the case of the stack of cyclic covers of the projective line, the full classification of opens which admit a separated good moduli space coincides with the answer provided by Geometric Invariant Theory (see Theorem 2.49).

We fix a base field κ of characteristic zero. Every time we talk about genus, we intend arithmetic genus, unless specified otherwise. If R is a discrete valuation ring (or dvr for short), then we will denote by Q its fraction field and by k its residue field.

2.1 Local structure and good moduli spaces

In this section we recall some key structural results on algebraic stacks that we will use throughout the paper. Let $\mathcal{X}$ be a noetherian algebraic stack over κ.

Good moduli spaces were introduced in [Alp13] as a characterization of the key properties of morphisms $[\operatorname{Spec} A/G] \to \operatorname{Spec} A//G$, where G is linearly reductive group acting on $\operatorname{Spec} A$ and $\operatorname{Spec} A//G$ is the quotient constructed by geometric invariant theory (i.e. the spectrum of the G-invariants of A). The definition that captures these properties is surprisingly sparse.

Definition 2.1. A quasi-compact morphism $\pi : \mathcal{X} \to X$ to an algebraic space X is a good moduli space if the following holds.

1. the canonical morphism $\mathcal{O}_X \to \pi_*\mathcal{O}_{\mathcal{X}}$ is an isomorphism,
2. The functor $\pi_* : \mathrm{QCoh}(\mathcal{X}) \to \mathrm{QCoh}(X)$ is exact.

Other than geometric invariant theory quotients, a key example to keep in mind is that a classifying stack BG admits a good moduli space if and only if G is a linearly reductive group scheme. Furthermore, if $\mathcal{X}$ admits a good moduli space then so does every closed substack of $\mathcal{X}$, which has the consequence that the automorphism group schemes of the closed points of $\mathcal{X}$ are linearly reductive.

When a stack is not constructed as a quotient of a quasi-projective scheme by a reductive action, one cannot use the methods of geometric invariant theory directly to study whether it admits a good moduli space, there are however strong local structure results that allows one to do this étale locally. We state the first version of the theorem that appeared, which is for stacks locally of finite type over an algebraically closed field, there are also versions over general bases with weaker assumptions than here, namely [AHR23, Theorem A] and [AHHLR24, Theorems 1.3,1.4,1.5 and 1.6].

Theorem 2.2 ([AHR20])**.** *Let $\mathcal{X}$ be an algebraic stack, locally of finite type and quasi-separated over an algebraically closed field κ, with affine stabilizers. Let $x \in \mathcal{X}(\kappa)$ be a point and $H \subset G_x$ a subgroup of the stabilizer of x such that H is linearly reductive and G_x/H is étale. Then there exists an affine scheme $\operatorname{Spec} A$ with an action of H, a κ-point $w \in \operatorname{Spec} A$ fixed by H and an étale morphism of pointed stacks*

$$f : ([\operatorname{Spec} A/H], w) \to (\mathcal{X}, x)$$

such that $f^{-1}(\mathcal{B}G_x) = \mathcal{B}H$, in particular f induces the given inclusion $H \subset G_x$ on stabilizer group schemes at w. Moreover, if $\mathcal{X}$ has affine diagonal, then f can be arranged to be affine.

In particular, this implies that every point of a stack with linearly reductive stabilizer has an étale neighborhood that admits a good moduli space, namely $[\operatorname{Spec} A/G_x] \to \operatorname{Spec} A^{G_x}$. With this in hand one would like to glue these local good moduli spaces to a global one, the linearization of an action on a quasi-projective variety plays this gluing role in geometric invariant theory. It turns out that this boils down to two stacky valuative criteria, which also guarantee the obtained good moduli space will be separated.

To introduce these, we need two stacks associated to any dvr R and an associated punctured version (playing the role of the spectrum of the fraction field in classical valuative criteria). To introduce these stacks, let $(m,n) \in \{(1,0),(1,-1)\}$ and let $\mathbb{G}_m$ act on $\mathbb{A}^2$ linearly with weights (m,n). Then the map $[\mathbb{A}^2/\mathbb{G}_m] \to \mathbb{A}^1$ induced respectively by $t \mapsto y$ and $t \mapsto xy$ on the coordinate rings $k[t]$ and $k[x,y]$ of $\mathbb{A}^1$ and $\mathbb{A}^2$ are good moduli spaces. For any dvr R with uniformizer π, we define the stacks Θ_R and $\overline{\mathrm{ST}}_R$ by the cartesian diagrams

$$\begin{tikzcd} \Theta_R \arrow[r] \arrow[d] & {[\mathbb{A}^2/\mathbb{G}_m]} \arrow[d, "y \leftarrow\!\shortmid t"] \\ \operatorname{Spec} R \arrow[r, "\pi \leftarrow\!\shortmid t"'] & \mathbb{A}^1 \end{tikzcd}$$

with $(m,n) = (1,0)$ and

$$\begin{tikzcd} \overline{\mathrm{ST}}_R \arrow[r] \arrow[d] & {[\mathbb{A}^2/\mathbb{G}_m]} \arrow[d, "xy \leftarrow\!\shortmid t"] \\ \operatorname{Spec} R \arrow[r, "\pi \leftarrow\!\shortmid t"'] & \mathbb{A}^1 \end{tikzcd}$$

with $(m,n) = (1,-1)$. We also respectively denote by $\Theta_R \setminus 0$ and ST_R the stacks obtained for the analogous cartesian diagrams with $[\mathbb{A}^2/\mathbb{G}_m]$ replaced with $[(\mathbb{A}^2 \setminus \{0\})/\mathbb{G}_m]$, inducing open immersions with codimension 2 complements $\iota : \Theta_R \setminus 0 \subset \Theta$ and $\iota : \mathrm{ST}_R \subset \overline{\mathrm{ST}}_R$.

Definition 2.3 ([AHLH23])**.** Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of locally Noetherian algebraic stacks. We say that f is Θ-complete if for all dvr's R and all commutative solid diagrams

$$\begin{tikzcd} \Theta_R \setminus 0 \arrow[r] \arrow[d, "\iota"'] & \mathcal{X} \arrow[d, "f"] \\ \Theta_R \arrow[r] \arrow[ur, dotted] & \mathcal{Y} \end{tikzcd}$$

there is a unique dotted arrow making the diagram commute. We say that f is S-complete if for all dvr's R and all commutative solid diagrams

$$\begin{tikzcd} \mathrm{ST}_R \setminus 0 \arrow[r] \arrow[d, "\iota"'] & \mathcal{X} \arrow[d, "f"] \\ \overline{\mathrm{ST}}_R \arrow[r] \arrow[ur, dotted] & \mathcal{Y} \end{tikzcd}$$

there is a unique dotted arrow making the diagram commute. If $\mathcal{X} \to \operatorname{Spec} \kappa$, the structure morphism of $\mathcal{X}$ is Θ- or S-complete, then we say that $\mathcal{X}$ is respectively Θ-complete or S-complete.

When first encountering these valuative criteria, it is a good idea to keep the example of $f : \mathcal{X} \to \mathcal{Y}$ being a morphism of schemes in mind, in which case S-completeness amounts to a reformulation of the valuative criterion of separatedness, which can be seen by noting that $\mathrm{ST}_R \cong \operatorname{Spec} R \cup \operatorname{Spec} R$ glued along the generic point and the good moduli space of $\overline{\mathrm{ST}}_R$ is $\operatorname{Spec} R$.

We can now state the existence theorem for separated good moduli spaces, [AHLH23, Theorem A].

Theorem 2.4 ([AHLH23]). *Let $\mathcal{X}$ be a stack of finite type with affine stabilizers and separated diagonal over* $\operatorname{Spec}\kappa$. *Then $\mathcal{X}$ admits a separated good moduli space X if and only if it is Θ- and S-complete. Moreover, X is proper if and only if $\mathcal{X}$ satisfies the existence part of the valuative criterion for properness.*

We will apply this theorem in two ways, first heuristically in Section 4, to find a candidate open substack of $\mathcal{M}^r_{g,n}$ that is likely to admit a separated good moduli space and second formally to prove that our candidate open in fact does admit a proper good moduli space. The heuristic application builds on the following specialization diagrams describing Θ_R and $\overline{\mathrm{ST}}_R$ respectively:

$$\begin{tikzcd}
 & \operatorname{Spec} k \arrow[dl, dashed] & \\
\mathcal{B}\mathbb{G}_{m,k} & & \operatorname{Spec} Q \arrow[ul] \arrow[dl, dashed] \\
 & \mathcal{B}\mathbb{G}_{m,Q} \arrow[ul] &
\end{tikzcd}$$

and

$$\begin{tikzcd}
 & \operatorname{Spec} k \arrow[dl, dashed] & \\
\mathcal{B}\mathbb{G}_{m,k} & & \operatorname{Spec} Q \arrow[ul] \arrow[dl] \\
 & \operatorname{Spec} k \arrow[ul, dashed] &
\end{tikzcd}$$

where Q is the fraction field and k the residue field of R, and the solid arrows represent an inclusion of $\operatorname{Spec} R$ or $\mathcal{B}\mathbb{G}_{m,R}$ and the dotted arrows and inclusion of Θ_Q or Θ_k. We often call these dotted specialization arrows *isotrivial degenerations*: we can think of them as families of objects over the "0-dimensional DVR" Θ. The punctured versions are described by removing the $\mathcal{B}\mathbb{G}_{m,k}$ from these diagrams. The heuristic approach to finding a well behaved open is to investigate families over $\Theta_R \setminus 0$ and ST_R and study obstructions to filling them. More precisely, in the case of S-completeness, given two families of curves over $\operatorname{Spec} R$, agreeing on the generic point, we ask whether the two curves over the residue field k can be realized as the generic fiber of families of curves over Θ_k, with both of these families having the same special fiber. The case of Θ-completeness is analogous.

A key component for both the heuristic and formal application of the existence theorem is to understand families over Θ_k, see Section 3 which, via local structure theorems, reduces to understanding the $\mathbb{G}_m$-equivariant deformation theory of A_r-stable curves, which in turn amounts to studying certain $\mathbb{G}_m$-representations. When studying these, we will use the following notation.

Definition 2.5. Let V be a $\mathbb{G}_m$-representation. We denote by $V^>$ (respectively $V^\geq$, $V^<$, $V^\leq$) the subrepresentation of V defined as the sum of the irreducible subrepresentation of V with positive (respectively non negative, negative, non positive) weights. We denote by V^0 the subspace of $\mathbb{G}_m$-invariants vectors inside V.

2.2 A_r-stable curves and crimping

Most of the results in this subsection are probably well-known to experts. For instance, the study of crimping spaces of A-singularity has been extensively discussed in van der Wyck's PhD thesis [vdW10]. In this work, we use the notation introduced in the third author's PhD thesis (specifically [Per23]), where some of the results of [vdW10] have been slightly generalized.

Definition 2.6. A curve C over an algebraically closed field k is a connected, reduced, one-dimensional scheme proper over k. A n-pointed curve $(C, p_1, \dots, p_n)$ over a field k is a curve C together with n rational smooth points $p_1, \dots, p_n$.

Definition 2.7. A closed point $p \in C$ over an algebraically closed field k is called an *A_h-singularity* if

$$\widehat{\mathcal{O}_{C,p}} \simeq k[[x,y]]/(y^2 - x^{h+1}).$$

If we do not want to stress the integer h, we will drop the subscript and write *A-singularity*. An A-singularity p of C is *separating* if the partial normalization of C at p is disconnected. Moreover, we say that C is an A-prestable curve if all the singular points of C are A-singularities.

Let g and n be two nonnegative integers. Fix a nonnegative integer r.

We start by fixing some classical notations.

Definition 2.8 (Partial Normalization)**.** Let C be a reduced curve over an algebraically closed field k, and let $S \subset \mathrm{Sing}(C)$ be a subset of its singular points. A finite morphism

$$\nu \colon \widetilde{C} \to C$$

is called the *partial normalization of C along S* if ν is an isomorphism over $C \setminus S$ and the induced map $\widetilde{C} \setminus \nu(S^c) \to C \setminus S^c$ is the normalization, where $S^c = \mathrm{Sing}(C) \setminus S$.

Definition 2.9 (Pointed partial normalization)**.** Let $(C, \{p_i\}_{i \in I})$ be a pointed curve, and let $S \subset \mathrm{Sing}(C)$ be a finite subset. The *pointed partial normalization of $(C, \{p_i\})$ along S* is the triple $(\widetilde{C}, \{q_j\}_{j \in J}, \nu)$ where

$$\nu \colon \widetilde{C} \to C$$

is partial normalization of C along S an $\{q_j\}_{j \in J} \coloneqq \nu^{-1}(S \cup \{p_i \mid i \in I\})$. If $S = \mathrm{Sing}(C)$, then ν is called the *pointed normalization*.

Notice that the normalization morphism is not a morphism between pointed curvesç indeed it sends markings to special points, which are either markings or singularities.

Definition 2.10 (A_r-stable curve)**.** Let k be an algebraically closed field and $(C, p_1, \dots, p_n)/k$ be a n-pointed A-prestable curve. We say that C is A_r-prestable if every singular point is an A_h-singularity for $h \leq r$. We will say that $(C, p_1, \dots, p_n)$ is *A_r-stable* if it is A_r-prestable and the line bundle $\omega_C(p_1 + \cdots + p_n)$ is ample.

Remark 2.11. Notice that an A-prestable curve C is l.c.i by definition, therefore the dualizing complex ω_C is in fact a line bundle.

Remark 2.12. Let C be a curve over an algebraically closed field and let p be an A_r-singularity. We denote by $b : \widetilde{C} \to C$ the partial normalization at the point p and by J_b the conductor ideal of b. Then a classical computation shows that

1. if $r = 2h$, then $g(C) = g(\widetilde{C}) + h$;
2. if $r = 2h + 1$ and $\widetilde{C}$ is connected, then $g(C) = g(\widetilde{C}) + h + 1$,
3. if $r = 2h + 1$ and $\widetilde{C}$ is not connected, that is p is a separating singularity, then $g(C) = g(\widetilde{C}) + h$.

Furthermore, Noether' conductor formula gives us that $b^*\omega_C \simeq \omega_{\widetilde{C}}(J_b^\vee)$. See for instance of [Cat82, Proposition 1.2].

We can define $\mathcal{M}^r_{g,n}$ as the fibered category over κ-schemes whose objects are the datum of A_r-stable curves over S with n distinct smooth sections $p_1, \dots, p_n$ such that every geometric fiber over S is a n-pointed A_r-stable curve. These families are called *n-pointed A_r-stable curves* over S. Morphisms are just isomorphisms of n-pointed curves.

We recall the following description of $\mathcal{M}^r_{g,n}$. See [Per23, Theorem 2.2] for the proof.

Theorem 2.13. *$\mathcal{M}^r_{g,n}$ is a smooth connected algebraic stack of finite type over κ. Furthermore, it is a quotient stack: that is, there exists a smooth quasi-projective scheme X with an action of GL_N for some positive N, such that $\mathcal{M}^r_{g,n} \simeq [X/\mathrm{GL}_N]$.*

Remark 2.14. Recall that we have an open embedding $\mathcal{M}^r_{g,n} \subset \mathcal{M}^s_{g,n}$ for every $r \leq s$. Notice that $\mathcal{M}^r_{g,n} = \mathcal{M}^{2g+1}_{g,n}$ for every $r \geq 2g+1$, because of Remark 2.12. Thus, we can consider $\mathcal{M}^{2g+1}_{g,n}$ as the moduli stack of all A-stable curves (with no restriction on the singularities).

Crimping morphisms of A_r-singularities

In this subsection, we describe the crimping datum needed to construct an A_r-stable curve starting from its normalization. Contrary to the nodal and cuspidal case, it is not enough to remember only the closed points mapping to a given singularity if it is of type A_l with $l \geq 3$. We want to stress again that the crimping morphism described below is not new: it can be found, for instance, in [vdW10]. We slightly generalize the analysis of *loc. cit.* for A-type singularities by extending the crimping morphism also to the case where the combinatorics of the curve is not fixed. In particular, when we restrict the normalization map in Proposition 2.18 to the locally closed substack given by specified combinatorics of the curve, we recover the description given in [vdW10, Thm. 1.105] for A-type singularities.

The main players in the description is the stack $\mathcal{A}_l$ for $0 \leq l \leq r$, parametrizing families of A_r-stable curves (with n marked points) together with a section that is an A_l-singularity in each fiber. This stack is naturally a substack of the universal curve over $\mathcal{M}^r_{g,n}$, and it is in fact a locally closed substack as is shown in [Per23, Section 4] (in the case $n = 0$, but the argument in the pointed case is identical). Depending on if l is even or odd, the geometry of $\mathcal{A}_l$ and its relation to crimping varies so we treat the two cases separately in what follows. Moreover, we describe the stack of curves together with an A_r-equisingular section, $\mathcal{A}_r$, and study it's relation to the crimping morphisms. This was done in [Per23, Sections 3 and 4] in the case $n = 0$, but since the markings are always smooth, nothing changes in the proofs of the following results.

Even A_r-singularities

Suppose $r := 2h$ for some integer $h \geq 1$. We start by discussing the data one needs to construct an A_{2h}-singularity *crimping* a smooth point.

Consider the affine scheme $E_{h,2}$ which classifies finite flat extensions of degree 2 of the form

$$\kappa[t]/(t^h) \hookrightarrow \kappa[t]/(t^{2h});$$

these are determined by the choice of an element $\phi(t) \in \kappa[t]/(t^{2h-3})$ with $\phi(0) \neq 0$, using the association $t \mapsto t^2\phi(t)$. Therefore $E_{h,2}$ is isomorphic to $(\mathbb{A}^1 \setminus \{0\}) \times \mathbb{A}^{2h-3}$. If we denote by G_h the automorphism group of $\kappa[t]/(t^h)$, we have an algebraic stack $\mathcal{E}_{h,2} := [E_{h,2}/G_h \times G_{2h}]$, which parametrizes finite flat extensions of degree 2 as above, up to isomorphisms of both the source and target. As we are working over a field of characteristic 0, G_h is isomorphic to $\mathbb{G}_m \ltimes U_h$, where U_h is the unipotent radical, and an element of G_h can be identified with the application $t \mapsto t\varphi(t)$ where $\varphi(t) \in \kappa[t]/(t^{h-1})$ such that $\varphi(0) \neq 0$. See [Per23, Sec. 3] for a more detailed discussion. The stack $\mathcal{B}G_h$ classifies finite flat algebras of length h which étale locally are of the form $\kappa[t]/(t^h)$ (cf. [Per23, Cor. 3.5]). The natural morphism

$$\mathcal{E}_{h,2} \longrightarrow \mathcal{B}G_{2h}$$

classify A_{2h} singularities locally as we now explain. The idea is that to construct a A_{2h}-singularity starting from the smooth point in a curve, it is enough to give a finite flat (sub)extension of degree 2 as the one described above, and then consider the pushout of the curve with the finite flat (sub)algebra of the jet space of length $2h$ at the smooth point. The rest of this section will elaborate on this idea, we start with some remarks on the spaces introduced above.

Remark 2.15 ([Per23, Lemma 3.8])**.** The quotient

$$V_{h,2} := [E_{h,2}/G_h]$$

is a scheme and can be described as an affine space $\mathbb{A}^{h-1}$: a point in such a space corresponds to a finite flat extension determined by $\phi(t) \in \kappa[t]/(t^{2h-2})$ such that $\phi(t) = 1 + a_1 t + a_2 t^3 + \cdots + a_{h-1} t^{2h-3}$.

Before describing $\mathcal{A}_{2h}$, we introduce the stack which classifies the partial normalization along the equisingular section without the crimping datum needed to recover the curve.

Definition 2.16. Given a positive integer l, we denote by $\mathcal{M}^r_{g,n,[l]}$ the stack parametrizing objects $(\widetilde{C}, p_1, \ldots, p_n, q)$ where $\widetilde{C}$ is a A_r-prestable curve, $p_1, \ldots, p_n, q$ are smooth distinct sections and $\omega_C(p_1 + \cdots + p_n + lq)$ is ample. By definition, we have that $\mathcal{M}^r_{g,n,[1]} \simeq \mathcal{M}^r_{g,n+1}$.

One can compare the stack of partial normalizations with the moduli stacks of A_r-stable curves using the stabilization morphism $\mathcal{M}_{g,n,[l]} \to \mathcal{M}_{g,n+1}$, defined by contracting unstable irreducible components (see [Per23, Rem. 1.6, Cor. 1.9 and Prop. 1.17]). The proofs of all the results that follow are given in [Per23, Section 3], where everything was proven in the case $n = 0$. The same proofs works for general n, as smooth markings do not play a role in the constructions.

Proposition 2.17 ([Per23, Prop. 4.9])**.** *Given g, n nonnegative integers and $l \geq 2$, we have that*

$$\mathcal{M}^r_{g,n,[l]} \simeq \mathcal{M}^r_{g,n+1} \times [\mathbb{A}^1/\mathbb{G}_m]$$

if $2g - 2 + n \geq 0$. If $g = 0$ and $n = 1$, we have

$$\mathcal{M}^r_{0,1,[l]} \simeq \mathcal{B}\mathbb{G}_m$$

for any $l \geq 2$. If $g = n = 0$, then $\mathcal{M}^r_{g,n,[l]}$ is isomorphic to $\mathcal{B}(\mathbb{G}_m \ltimes \mathbb{G}_a)$ if $l \geq 3$, whereas for $l = 2$ it is empty.

Let $\widetilde{C} \to T$ be a family of curves over an algebraic stack T and $q : T \to \widetilde{C}$ a smooth section. This data induces a map $j_h : T \to \mathcal{B}G_h$ for any $h \in \mathbb{Z}_{\geq 1}$ defined by the $(h-1)$-jet space of the section q. Whenever $\widetilde{C}$ is a family of curves in $\mathcal{M}^r_{g-h,n,[2h]}$ and q is the section corresponding to the last marking, the map is given by the composition

$$j_{2h} \colon T \to \mathcal{M}^r_{g-h,n,[2h]} \to \mathcal{B}G_{2h}.$$

Over a field κ, we can consider a curve (C, s) and an extension of algebras as above $\mathrm{cr}\colon \operatorname{Spec} \kappa[t]/(t^{2h}) \to \operatorname{Spec} \kappa[t]/(t^h)$ and performing a Ferrand pushout as in the following diagram

$$\begin{tikzcd} \operatorname{Spec} \kappa[t]/(t^{2h}) \arrow[r, "\mathrm{jet}_q"'] \arrow[d, "\mathrm{cr}"'] & \widetilde{C} \arrow[d] \\ \operatorname{Spec} \kappa[t]/(t^h) \arrow[r] & C \end{tikzcd}$$

where $\mathrm{jet}_q : \operatorname{Spec}\colon \kappa[t]/t^{2h} \to \widetilde{C}$ denotes the closed inclusion of the length $2h$ infinitesimal neighborhood of $q \in C$. The pushout defines a curve C isomorphic to $\widetilde{C}$ outside the image of the horizontal arrows, with the image of q being an A_{2h}-singularity with $\widetilde{C}$ the partial normalization of C in q, see Proposition 2.18 below.

Conversely, if $(\widetilde{C}, q)$ is the partial normalization of a pointed curve (C, p) over an algebraically closed field $\operatorname{Spec} \kappa$ where p is a A_{2h}-singularity, then there exist an extension of algebras (non necessarily unique) $\mathrm{cr}\colon \operatorname{Spec} \kappa[t]/(t^{2h}) \to \operatorname{Spec} \kappa[t]/(t^h)$ and a Ferrand pushout diagram as above.

Denote by $\mathcal{A}_{2h}$ the locally closed substack of the universal curve $\mathcal{C}^r_{g,n}$ of $\mathcal{M}^r_{g,n}$ classifying objects $(C, p_1, \ldots, p_n, p)$ where p is a A_{2h}-singularity. Then the discussion above can be carried out in families, this is the content of the following [Per23, Corollary 4.12].

Proposition 2.18. *The pointed normalization of a point $(C, p_1, \dots, p_n, p) \in \mathcal{A}_{2h}(T)$ along the equisingular section p define a morphism*

$$\mathcal{A}_{2h} \longrightarrow \mathcal{M}^r_{g-h,n,[2h]}$$

that we call crimping morphism*. This is obtained via the pullback*

$$\begin{tikzcd} \mathcal{A}_{2h} \arrow[r] \arrow[d] & {[V_{h,2}/G_{2h}]} \arrow[d] \\ \mathcal{M}^r_{g-h,n,[2h]} \arrow[r] & \mathcal{B}G_{2h} \end{tikzcd}$$

where the bottom morphism can be constructed by sending a curve $(\widetilde{C}, p_1, \dots, p_n, q)$ to the jet space of length $2h$ of q, which is étale locally of the form $\kappa[t]/(t^{2n})$. In particular, $\mathcal{A}_{2h}$ is an affine bundle of dimension $h-1$ over the stack $\mathcal{M}^r_{g-h,n,[2h]}$.

This motivates the following definition.

Definition 2.19. Given a family $\widetilde{C}$ of curves over an algebraic stack T and a section q over T, a *crimping datum* of type A_{2h} for the pair is a lift $\mathrm{cr} : T \to \mathcal{E}_{2h}$ of $j_{2h} : T \to \mathcal{B}G_{2h}$, i.e. a dotted arrow making the following diagram

$$\begin{tikzcd} & \mathcal{E}_{2h} \arrow[d] \\ T \arrow[ur, dashed, "\mathrm{cr}"] \arrow[r, "j_{2h}"'] & \mathcal{B}G_{2h} \end{tikzcd}$$

commute.

Odd A_r-singularities

We will now deal with the odd case, similar results obtained in the previous section holds and we will adopt the same notation. Suppose that $r = 2h-1$.

Remark 2.20. In [Per23, Section 3], the author proved that the diagonal morphism

$$\mathcal{B}G_h \longrightarrow \mathcal{B}G_h \times \mathcal{B}G_h$$

is the crimping morphism of the A_{2h-1}-singularity. Indeed, given a pair of smooth points (in either one or two distinct curves), it is enough to give an identification of the two jet spaces of length h at the two points.

As in the even case, denote by $\mathcal{A}_{2h-1}$ the locally closed substack of the universal curve $\mathcal{C}^r_{g,n}$ of $\mathcal{M}^r_{g,n}$ classifying objects $(C, p_1, \dots, p_n, q)$ where q is a A_{2h-1}-singularity. Before describing the relation of $\mathcal{A}_{2h-1}$ to crimping morphisms we introduce the stack which classifies the partial normalization along the equisingular section, in the case where the partial normalization is connected.

Definition 2.21. Let l be a positive integer. We denote by $\mathcal{M}^r_{g,n,2[l]}$ the stack parametrizing tuples $(\widetilde{C}, p_1, \dots, p_n, q_1, q_2)$ where $\widetilde{C}$ is a A_r-prestable curve and $p_1, \dots, p_n, q_1, q_2$ are smooth distinct sections such that the line bundle $\omega_C(p_1 + \cdots + p_n + l(q_1 + q_2))$ is ample. By definition we have $\mathcal{M}^r_{g,n,2[1]} \simeq \mathcal{M}^r_{g,n+2}$.

Similarly to the even case, we can compare the stack of partial normalizations with the moduli stacks of A_r-stable curves using the stabilization procedure.

Proposition 2.22. *Given g, n nonnegative integers and $l \geq 2$, we have that*

$$\mathcal{M}^r_{g,n,2[l]} \simeq \mathcal{M}^r_{g,n+2} \times [\mathbb{A}^1/\mathbb{G}_m] \times [\mathbb{A}^1/\mathbb{G}_m]$$

if $2g-1+n \geq 0$. If $g = n = 0$, then $\mathcal{M}^r_{0,0,2[l]}$ is isomorphic to $[\mathbb{A}^1/\mathbb{G}^2_m]$ for $l \geq 2$.

Remark 2.23. Proposition 2.17 and Proposition 2.22 warn us that the normalization of a A_r-stable curve in a A-singularity is not A_r-stable, in contrast with the nodal case. For instance, we can have an irreducible rational curve (Γ, p), attached to a smooth curve in a node p, with an even A-singularity q on Γ. When we normalize, the component $(\widetilde{\Gamma}, p, q)$ is not stable anymore, only prestable. This is taken into account adding the $[\mathbb{A}^1/\mathbb{G}_m]$ in Proposition 2.17. The generic point of $[\mathbb{A}^1/\mathbb{G}_m]$ corresponds to the case when there is no unstable component, thus the point p will be pinched to create the singularity directly in the smooth curve. The closed point corresponds to the case when we pinch an unstable component, which has in fact a $\mathbb{G}_m$ as automorphism group, making it stable. The odd case is similar, with the only difference that we can sprout an unstable component from both the points where are gluing in the normalization.

Contrary to the even case, $\mathcal{A}_{2h-1}$ is the disjoint union of several connected components, namely $\mathcal{A}^{\mathrm{ns}}_{2h-1}$ and $\mathcal{A}^{i,S}_{2h-1}$ for $0 \leq i \leq (g-m+1)/2$ and $S \subset \{1, 2, \ldots, n\}$. The stack $\mathcal{A}^{\mathrm{ns}}_{2h-1}$ parametrizes $(C, p_1, \ldots, p_n, q)$ such that the partial normalization of C at q is connected. Furthermore, $\mathcal{A}^{i,S}_{2h-1}$ parametrizes pairs $(C, p_1, \ldots, p_n, q)$ such that the partial normalization of C at q is the disjoint union of two components, respectively of genus i and $g-h-i+1$, such that p_j belongs to the component of genus i if and only if $j \in S$.

We have the following descriptions in the odd case.

Proposition 2.24. *There exists a finite étale cover of degree* 2

$$F^{\mathrm{ns}} : \mathcal{I}^{\mathrm{ns}}_{2h-1} \longrightarrow \mathcal{A}^{\mathrm{ns}}_{2h-1}$$

where $\mathcal{I}^{\mathrm{ns}}_{2h-1}$ *is affine over* $\mathcal{M}^r_{g-h,n,2[h]}$.

A similar statement is true for the other connected components.

Proposition 2.25. *Let* $S \subset \{1, 2, \ldots, n\}$ *and denote by* s *its cardinality. We have a morphism*

$$F^{i,S} : \mathcal{I}^{i,S}_{2h-1} \longrightarrow \mathcal{A}^{i,S}_{2h-1}$$

where $\mathcal{I}^{i,S}_{2h-1}$ *is affine over the product*

$$\mathcal{M}^r_{i,s,[h]} \times \mathcal{M}^r_{g-h-i+1,n-s,[h]}.$$

If $2i = g-h+1$ *and* $n = 0$*, then* F^i *is finite étale of degree* 2*; otherwise it is an isomorphism.*

Remark 2.26. The algebraic stack $\mathcal{I}^{\mathrm{ns}}_{2h-1}$ is the moduli stack parametrizing the partial normalization of a curve in $\mathcal{A}^{\mathrm{ns}}_{2h-1}$ at the equisingular section plus a crimping datum. The same is true for $\mathcal{I}^{i,S}_{2h-1}$. Exactly as in the even case, one can construct it by considering the following cartesian diagram:

$$\begin{tikzcd}
\mathcal{I}^{i,S}_{2h-1} \arrow[r] \arrow[d] & \mathcal{B}G_h \arrow[d] \\
\mathcal{M}_{g-i-h+1,n-s,[h]} \times \mathcal{M}_{i,s,[h]} \arrow[r] & \mathcal{B}G_h \times \mathcal{B}G_h;
\end{tikzcd}$$

The same is true for the non-separating case. More explicitly, given a crimping datum over $\operatorname{Spec} \kappa$, the Ferrand pushout diagram

$$\begin{tikzcd}
\operatorname{Spec} \kappa[t]/(t^h) \sqcup \operatorname{Spec} \kappa[t]/(t^h) \arrow[r, "\mathrm{jet}_{q_1} \sqcup \mathrm{jet}_{q_2}"'] \arrow[d, "\mathrm{cr}"'] & \widetilde{C} \arrow[d] \\
\operatorname{Spec} \kappa[t]/(t^h) \arrow[r] & C
\end{tikzcd}$$

where by abuse of notation $\mathrm{cr} : \operatorname{Spec} \kappa[t]/(t^h) \sqcup \operatorname{Spec} \kappa[t]/(t^h) \to \operatorname{Spec} \kappa[t]/t^h$ comes from the crimping datum $\mathrm{cr} : \operatorname{Spec} \kappa \to \mathcal{B}G_h$ and $\mathrm{jet}_q : \operatorname{Spec} \kappa[t]/t^h \to \widetilde{C}$ denotes the closed inclusion of the infinitesimal neighborhood as in the even case. This define a curve C isomorphic to $\widetilde{C}$ outside the image of the horizontal arrows, with the image q of q_1 (or equivalently q_2) being an A_{2h}-singularity with $\widetilde{C}$ the partial normalization of C in q.

Remark 2.27. If $2i = g - h + 1$ and $n = 0$, we have a $\mathbb{Z}/2$ action on the target of the crimping morphism. This exchanges the marked points arising via the normalization of the equisingular section. In that case, we have a map from $\mathcal{A}_{2m-1}$ to the quotient via this action and we can recover the crimping via pullback. In the non-separating case, we have

$$\begin{tikzcd} \mathcal{I}^{\mathrm{ns}}_{2m-1} \arrow[r] \arrow[d] & \mathcal{A}^{\mathrm{ns}}_{2m-1} \arrow[d] \\ \mathcal{M}_{g-m,n,2[m]} \arrow[r] & {[\mathcal{M}_{g-m,n,2[m]}/\mathbb{Z}_2]}. \end{tikzcd}$$

The other cases are similar.

The following is the analogue of Definition 2.19 in the odd case.

Definition 2.28. Given a family $(\widetilde{C}, q_1, q_2)$ of curves over a stack T and two smooth distinct sections, a *crimping datum* of type A_{2h-1} for the triplet is a lift $\mathrm{cr} : T \to \mathcal{B}G_h$ of $(j_{h,1}, j_{h,2}) : T \to \mathcal{B}G_h \times \mathcal{B}G_h$, i.e. a dotted arrow making the following diagram

$$\begin{tikzcd} & \mathcal{B}G_h \arrow[d] \\ T \arrow[ur, dashed, "\mathrm{cr}"] \arrow[r, "(j_{h,1}, j_{h,2})"'] & \mathcal{B}G_h \times \mathcal{B}G_h \end{tikzcd}$$

commute, where $j_{h,i}$ is the morphism induced by the family $(\widetilde{C}, q_i) \to T$ for $i = 1, 2$.

2.3 Equivariant properties of normalization

In this subsection we collect some results on descending automorphisms along the partial normalization map of an A_r-(pre)stable curve.

Because A-singularities are either unibranched (in the even case) or double branched (in the odd case), there are strong restrictions on which curves have positive dimensional stabilizers. Moreover, given that the normalization is smooth local (cfr. [Sta23, Lemma 07TD]), we have the following strong restriction.

Remark 2.29. Let $(C, p_1, \ldots, p_n)$ be a n-pointed A_r-stable curve over an algebraically closed field k with $\operatorname{char} k = 0$, and denote $\nu \colon \widetilde{C} \to C$ the partial normalization along some singularities $S \subset \operatorname{Sing} C$. The connected component of the identity of the automorphism group lifts to the partial normalization, and we have a natural inclusion

$$\iota \colon \operatorname{Aut}^\circ(C, p_1, \ldots, p_n) \hookrightarrow \operatorname{Aut}^\circ(\widetilde{C}, \nu^{-1}(p_1), \ldots, \nu^{-1}(p_n)).$$

The image of ι acts trivially on any connected component $\Gamma \subset \widetilde{C}$ not isomorphic to $\mathbb{P}^1$.
Consider a cocharacter $\mathbb{G}_m \to \operatorname{Aut}^\circ(C, p_1, \ldots, p_n)$ and a singularity $p \in C$, we have two possibilities:

- p is an A_{2h}-singularity. Then $\nu^{-1}(p)$ contains a unique topological point q and we have an induced action of $\mathbb{G}_m$ on $T_q\widetilde{C}$.
- p is an A_{2h+1}-singularity. Then $\nu^{-1}(p)$ contains two topological points q_1, q_2 and we have an induced action of $\mathbb{G}_m$ on $T_{q_i}\widetilde{C}$ for $i = 1, 2$.

In these cases, the action of $\mathbb{G}_m$ on the tangent space of p in C is non-trivial only if the connected component (respectively components) of $\widetilde{C}$ containing q (resp. q_1 and q_2) is isomorphic to $\mathbb{P}^1$.

Remark 2.30. With the notation of Remark 2.29, whenever $S \subset C(k)$ are all nodal singularities, then the map ι is an isomorphism.

In what follows, we are going to relate the automorphism group of an A_r-prestable curve with the one of its partial pointed normalization at a singularity. As always, we treat the even and odd case separately. This will be fundamental in the classification of curves with positive dimensional stabilizers.

Even case

Firstly, we describe the moduli stack of curves with a prescribed partial normalization at an even A-singularity. This is a direct consequence of Proposition 2.18.

Lemma 2.31. *Let $(\widetilde{C}, q)$ be a 1-pointed curve over an algebraically closed field. The fiber product*

$$\begin{tikzcd} {[V_{h,2}/\operatorname{Aut}(\widetilde{C},q)]} \arrow[r] \arrow[d] & \mathcal{E}_{h,2} = [V_{h,2}/G_{2h}] \arrow[d] \\ \mathcal{B}\operatorname{Aut}(\widetilde{C},q) \arrow[r] & \mathcal{B}G_{2h} \end{tikzcd}$$

parametrizes curves (C, p) where p is an A_{2h}-singularity such that the (pointed) partial normalization at p is isomorphic to $(\widetilde{C}, q)$.

Definition 2.32. Let $(\widetilde{C}, q)$ be a 1-pointed curve over an algebraically closed field and $G \subset \operatorname{Aut}(\widetilde{C}, q)$ be a subgroup. A G-equivariant crimping datum is a crimping datum for the family $[(\widetilde{C}, q)/G] \to \mathcal{B}G$.

Remark 2.33. A G-equivariant crimping datum

$$\mathrm{cr}\colon \mathcal{B}G \to \mathcal{E}_{h,2}$$

exists if and only if one can crimp an A_{2h}-singularity on $(\widetilde{C}, q)$ in such a way that the subgroup $G \subset \operatorname{Aut}(\widetilde{C}, q)$ descends to a subgroup of the automorphisms of C along the normalization $\widetilde{C} \to C$. Notice that different G-equivariant crimping data may give rise to the same curve. Indeed, two G-equivariant crimping data

$$\mathrm{cr}_1, \mathrm{cr}_2 : \mathcal{B}G \to \mathcal{E}_{h,2}$$

will give rise to the same curve if and only if the two induced morphisms $\mathcal{B}G \to \mathcal{E}_{h,2} \times_{\mathcal{B}G_{2h}} \mathcal{B}\operatorname{Aut}(C, s)$ are isomorphic. This follows directly from the fact that the quotient stack $\mathcal{E}_{h,2} \times_{\mathcal{B}G_{2h}} \mathcal{B}\operatorname{Aut}(C, s)$ parametrizes curves (C, p) where p is a A_{2h}-singularity and the partial normalization at p is isomorphic to $(\widetilde{C}, q)$ (see Lemma 2.31).

The case $G = \mathbb{G}_m$ will play a central role in our study of isotrivial degenerations of A_r-stable curves.

Lemma 2.34. *Let h be a positive integer. Let $\widetilde{C}/k$ be a (possibly pointed) curve over an algebraically closed field k with a rational smooth point $q \in \widetilde{C}(k)$. Let $\mathbb{G}_m \to \operatorname{Aut}^\circ(\widetilde{C}, q)$ be a cocharacter such that the action on $T_q\widetilde{C}$ is non-trivial. Then there exists a unique $\mathbb{G}_m$-equivariant crimping datum, i.e. there exists a unique dotted arrow making the following diagram*

$$\begin{tikzcd} & \mathcal{E}_{h,2} \arrow[d] \\ \mathcal{B}\mathbb{G}_m \arrow[r, "j_{2h}"'] \arrow[ur, dashed, "\exists!"] & \mathcal{B}G_{2h} \end{tikzcd}$$

commute.

Proof. This follows by a straightforward computation. Indeed, we have seen in Section 2.2 that $\mathcal{E}_{h,2}$ can be described as the quotient stack $[V_{h,2}/G_{2h}]$ where $V_{h,2}$ is an affine space. Thanks to Remark 2.29, it is clear that q lies in an irreducible component of $\widetilde{C}$ whose normalization is isomorphic to $\mathbb{P}^1$, and the cocharacter given induces a cocharacter $\mathbb{G}_m \to \mathrm{Aut}^\circ(\mathbb{P}^1, \infty) \simeq \mathbb{G}_m \ltimes \mathbb{G}_\mathrm{a}$. Since the jet morphism j_{2h} is local on q, we can assume $(C, p) \simeq (\mathbb{P}^1, \infty)$ and compute j_{2h} explicitly. Using the explicit description of the G_{2h}-action on $V_{h,2}$, we get that the fiber product $\mathcal{B}\mathbb{G}_m \times_{\mathcal{B}G_{2h}} \mathcal{E}_{h,2}$ is isomorphic to the quotient stack $[V_{h,2}/\mathbb{G}_m]$ where $\mathbb{G}_m$ acts on the affine space $V_{h,2}$ with strictly positive weights. Therefore, we have that there exists a unique lifting of j_{2h}, corresponding to the 0-section of $V_{h,2}$. □

Remark 2.35. Lemma 2.34 implies that there exists a unique curve (C, p) (constructed using the unique crimping datum given by Lemma 2.34) such that the $\mathbb{G}_m$-action on $(\widetilde{C}, q)$ descends to (C, p).

Odd case

In analogy with the even case, we start by describing the moduli stack of curve with a prescribed normalization at an odd A-singularity. This is a direct consequence of Proposition 2.25 and Proposition 2.24.

Lemma 2.36. *Let $(\widetilde{C}, q_1, q_2)$ be a 2-pointed not necessarily connected curve over an algebraically closed field. The fiber product*

$$\begin{tikzcd} {[G_h/\operatorname{Aut}(\widetilde{C}, q_1, q_2)]} \arrow[r] \arrow[d] & \mathcal{B}G_h \arrow[d] \\ \mathcal{B}\operatorname{Aut}(\widetilde{C}, q_1, q_2) \arrow[r] & \mathcal{B}G_h \times \mathcal{B}G_h \end{tikzcd}$$

parametrizes curves (C, p) where p is an A_{2h-1}-singularity such that the (pointed) partial normalization at p is isomorphic to $(\widetilde{C}, q_1, q_2)$.

Definition 2.37. Let $(\widetilde{C}, q_1, q_2)$ be a 2-pointed not necessarily connected curve over an algebraically closed field and $G \subset \operatorname{Aut}(\widetilde{C}, q_1, q_2)$ be a subgroup. A G-equivariant crimping datum is a crimping datum for the family $[(\widetilde{C}, q_1, q_2)/G] \to \mathcal{B}G$.

Remark 2.38. A G-equivariant crimping datum

$$\mathrm{cr}\colon \mathcal{B}G \to \mathcal{B}G_h$$

exists if and only if one can construct an A_{2h-1}-singularity gluing q_1 and q_2 in such a way that the subgroup $G \subset \operatorname{Aut}(\widetilde{C}, q_1, q_2)$ descends to a subgroup of the automorphisms of C along the normalization $\widetilde{C} \to C$. As in the even case, different G-equivariant crimping data may give rise to the same curve. Indeed, two G-equivariant crimping data

$$\mathrm{cr}_1, \mathrm{cr}_2 : \mathcal{B}G \to \mathcal{B}G_h$$

will give rise to the same curve if and only if the two induced morphisms $\mathcal{B}G \to \mathcal{B}G_h \times_{\mathcal{B}G_h \times \mathcal{B}G_h} \mathcal{B}\operatorname{Aut}(C, s)$ are isomorphic, as follows directly from Lemma 2.36.

A very simple consequence of Lemma 2.36 is the following result. It will help us understand how the automorphism group changes when attaching rational curves using odd A-singularities.

Lemma 2.39. *Let h be a positive integer with $h \geq 2$. Let $\widetilde{C}/k$ be a (possibly pointed) curve over an algebraically closed field k with one rational smooth point $q \in \widetilde{C}(k)$. Let $\mathcal{A}_{2h-1}(\widetilde{C} \cup \mathbb{P}^1)$ be the moduli stack parametrizing (possibly pointed) A_r-stable curves C with a separating A_{2h-1}-singularity such that the pointed normalization of C at p is isomorphic to $(\widetilde{C} \sqcup \mathbb{P}^1, p, \infty)$. Then $\mathcal{A}_{2h-1}(\widetilde{C} \cup \mathbb{P}^1)$ is isomorphic to the quotient stack $[(U_h/\mathbb{G}_a)/\operatorname{Aut}(\widetilde{C}, q)]$ where U_h is an $\operatorname{Aut}(C, p)$-representation. If $\mathbb{P}^1$ has another marking (say the point 0), then the quotient stack is of the form $[U_h/\operatorname{Aut}(\widetilde{C}, q)]$.*

Proof. We leave the 2-pointed case to the reader, since the same proof works. By Lemma 2.36, $\mathcal{A}_{2h-1}(\widetilde{C} \cup \mathbb{P}^1)$ is isomorphic to $[G_h/\operatorname{Aut}(\widetilde{C}, q) \times \operatorname{Aut}(\mathbb{P}^1, \infty)]$, where $h \geq 3$ because of the curve is A_r-stable. To prove the statement, we need to study the explicit description of the morphism

$$\eta : \operatorname{Aut}(\mathbb{P}^1, \infty) \simeq \mathcal{B}(\mathbb{G}_m \ltimes \mathbb{G}_a) \longrightarrow \mathcal{B}G_h.$$

We identify an element $(\lambda, t) \in \mathbb{G}_m \ltimes \mathbb{G}_a$ with the automorphism $z \mapsto \lambda^{-1}(z - t)$ of $\mathbb{P}^1$. This automorphism will act on the h-jet space of $\infty \in \mathbb{P}^1$ by the formula

$$\xi \mapsto \frac{\lambda\xi}{1 - \xi t} = \lambda\xi \sum_{i=0}^{m-2} (\xi t)^i = \lambda\xi + \lambda t\xi^2 + \lambda t^2\xi^3 + \dots$$

where ξ is the local parameter of ∞ in $\mathbb{P}^1$. Therefore, the morphism η is induced by an explicit embedding $\mathbb{G}_m \ltimes \mathbb{G}_a \subset G_h \simeq \mathbb{G}_m \ltimes U_h$ given by the formula

$$(\lambda, t) \mapsto (\lambda, \lambda t, \lambda t^2, \dots, \lambda t^{h-2})$$

and the quotient is an affine space, namely $U_h/\mathbb{G}_a$. Thus, we have the statement as claimed. □

Remark 2.40. In the previous corollary, the A_r-stable condition on C ensures that if $h = 2$, then $\mathbb{P}^1$ has to be 2-pointed. Indeed, if that was not the case, there would be a further copy of $\mathbb{G}_{\mathrm{a}}$ in the group we want to quotient with.

We write here the following remark, which will be useful later in showing Θ-completeness of the substack defined in Definition 4.4. It follows directly from Lemma 2.39, Proposition 2.25 and Remark 2.26.

Remark 2.41. Suppose we are in the situation of Proposition 2.25, where $S = \emptyset$, $i = 0$ and $h \geq 2$. We have the affine crimping morphism

$$\mathcal{I} := \mathcal{I}^{0,\emptyset}_{2h-1} \to \mathcal{M}_{g-h+1,n,[h]} \times \mathcal{M}_{0,0,[h]}$$

where $\mathcal{M}_{0,0,[h]} \simeq \mathcal{B}(\mathbb{G}_m \ltimes \mathbb{G}_a)$ (see Proposition 2.17). We can consider the diagram

$$\begin{tikzcd}
\mathcal{I} \arrow[r] \arrow[d] & {[(U_h/\mathbb{G}_{\mathrm{a}})/G_h]} \arrow[r] \arrow[d] & \mathcal{B}G_h \arrow[d] \\
\mathcal{M}^r_{g-h+1,n,[h]} \times \mathcal{M}_{0,0,[h]} \arrow[r] \arrow[d] & \mathcal{B}G_h \times \mathcal{B}(\mathbb{G}_m \ltimes \mathbb{G}_{\mathrm{a}}) \arrow[r] \arrow[d] & \mathcal{B}G_h \times \mathcal{B}G_h \\
\mathcal{M}^r_{g-h+1,n,[h]} \arrow[r] & \mathcal{B}G_h &
\end{tikzcd}$$

where all the square are cartesian thanks to Lemma 2.39 and Remark 2.26. Thus, the composition

$$\mathcal{I} \longrightarrow \mathcal{M}_{g-h+1,n,[h]} \times \mathcal{B}(\mathbb{G}_m \ltimes \mathbb{G}_a) \to \mathcal{M}_{g-h+1,n,[h]}$$

is still affine, since it is the pullback of an affine morphism.

In analogy with the even case, we now characterize $\mathbb{G}_m$-equivariant crimping data. Since we are gluing the two jet-spaces together, the idea is that any cocharacter that descends through the normalization morphism should have balanced action on the tangent spaces of the points we are about to glue. This is summed up in the following result.

Lemma 2.42. *Let h be a positive integer with $h \geq 2$. Let $\widetilde{C}/k$ be a not necessarily connected curve over an algebraically closed field k with two rational smooth point $q_1, q_2 \in \widetilde{C}(k)$. Let $\phi : \mathbb{G}_m \to \operatorname{Aut}^\circ(\widetilde{C}, q_1, q_2)$ be a cocharacter and denote by n_i the integer associated to the 1-dimensional $\mathbb{G}_m$-representation $T_{q_i}\widetilde{C}$ for $i = 1, 2$. The following are equivalent:*

(a) $n_1 = n_2$;

(b) *there exists a* $\mathbb{G}_m$*-equivariant crimping datum, i.e. there exists a dotted arrow making the following diagram*

$$\begin{tikzcd} & \mathcal{B}G_h \arrow[d] \\ \mathcal{B}\mathbb{G}_m \arrow[ur, dashed, "\exists"] \arrow[r, "j_{h,1}\times j_{h,2}"'] & \mathcal{B}G_h \times \mathcal{B}G_h \end{tikzcd}$$

commute, where $j_{h,i}$ *is the jet map associated to the point* q_i *for* $i = 1, 2$ *and action of* $\mathbb{G}_m$ *on their respective jet spaces induced by* ϕ.

Moreover, suppose that $n_1 = n_2 \neq 0$ *and the cocharacter* ϕ *factors through a 2-dimensional subtorus* $\mathbb{G}_m^2 \subset \mathrm{Aut}^\circ(\widetilde{C}, q_1, q_2)$. *Then the curves obtained gluing* p_1 *and* p_2 *with two different* $\mathbb{G}_m$*-equivariant crimping data are isomorphic.*

Proof. As in the proof of Lemma 2.34, p_1 and p_2 lie in irreducible components (possibly the same) whose normalizations are copies of $\mathbb{P}^1$. Therefore, because the jet morphism is local, we can factor the morphism $j_{h,1} \times j_{h,2} : \mathcal{B}\mathbb{G}_m \to \mathcal{B}G_h \times \mathcal{B}G_h$ through two copies of $\mathbb{G}_m \subset \mathrm{Aut}(\mathbb{P}^1, \infty)$. More precisely, we have a commutative diagram

$$\begin{tikzcd} \mathcal{B}\mathbb{G}_m \arrow[r, "(n_1,n_2)"'] \arrow[rr, bend left, "j_{h_1}\times j_{h_2}"] & \mathcal{B}\mathbb{G}_m \times \mathcal{B}\mathbb{G}_m \arrow[r] & \mathcal{B}G_h \times \mathcal{B}G_h \end{tikzcd}$$

where the rightmost morphism can be explicitly computed as (two copies of) the jet morphism in ∞ associated to the cocharacter $\mathbb{G}_m \subset \mathrm{Aut}^\circ(\mathbb{P}^1, \infty) \simeq \mathbb{G}_m \ltimes \mathbb{G}_{\mathrm{a}}$. Since an odd crimping datum is a lift of the jet morphisms along the diagonal, any $\mathbb{G}_m$-equivariant crimping datum should factor through $[G_h/\mathbb{G}_m \times \mathbb{G}_m]$. Recall that $G_h \simeq \mathbb{G}_m \ltimes U_h$ where U_h is the unipotent radical, and the two $\mathbb{G}_m$ actions are respectively left and right multiplication for elements of the subgroup $\mathbb{G}_m \subset G_h$. Notice that the right multiplication is done after taking the inverse of the element in $\mathbb{G}_m$. Therefore, after pulling back the diagonal of $\mathcal{B}G_h$ through the morphism $j_{h_1} \times j_{h_2}$, we get the quotient stack $[G_h/\mathbb{G}_m]$ where $\mathbb{G}_m$ acts on G_h as follows:

$$\lambda.(u_1, u_2, \ldots, u_{h-1}) := (\lambda^{n_1-n_2}u_1, \lambda^{2n_1-n_2}u_2, \ldots, \lambda^{(h-1)n_1-n_2}u_{h-1})$$

where $u_1 \in \mathbb{G}_m$ and $(u_2, \ldots, u_{h-1}) \in U_h$. Thus $[G_h/\mathbb{G}_m] \to \mathcal{B}\mathbb{G}_m$ has a section only if $n_1 = n_2$, otherwise $[G_h/\mathbb{G}_m]$ only have finite stabilizers, thus $(b) \implies (a)$. Furthermore, if $n_1 = n_2$, then we have $[G_h/\mathbb{G}_m] \simeq \mathbb{G}_m \times [U_h/\mathbb{G}_m]$ where U_h is a $\mathbb{G}_m$-representation with nonnegative weights, thus proving $(a) \implies (b)$.

Finally, we are going to prove the moreover part. Since the cocharacter ϕ factors through 2-dimensional subtorus $\mathbb{G}_m^2 \subset \mathrm{Aut}^\circ(\widetilde{C}, q)$, Remark 2.38 tells us that two crimping data give rise to isomorphic curves if the induced morphisms

$$c_1, c_2 : \mathcal{B}\mathbb{G}_m \longrightarrow (\mathcal{B}\mathbb{G}_m \times \mathcal{B}\mathbb{G}_m) \times_{(\mathcal{B}G_h \times \mathcal{B}G_h)} \mathcal{B}G_h$$

are isomorphic. Nevertheless, the explicit nature of the morphism, gives that

$$(\mathcal{B}\mathbb{G}_m \times \mathcal{B}\mathbb{G}_m) \times_{(\mathcal{B}G_h \times \mathcal{B}G_h)} \mathcal{B}G_h \simeq [G_h/\mathbb{G}_m \times \mathbb{G}_m] \simeq [U_h/\mathbb{G}_m]$$

where $\mathbb{G}_m$ acts on U_h as described above. Since $n_1 = n_2 \neq 0$, we have that U_h is a $\mathbb{G}_m$-representation with strictly positive weights, therefore there exists a unique point with $\mathbb{G}_m$-stabilizers. The result follows. $\square$

Remark 2.43. Notice that Lemma 2.42 tell us that, if we start with a curve $(\widetilde{C}, q_1, q_2)$ with two copies of $\mathbb{G}_m \subset \mathrm{Aut}(\widetilde{C}, q_1, q_2)$ acting independently on $T_{q_1}\widetilde{C}$ and $T_{q_2}\widetilde{C}$, then at most one copy of $\mathbb{G}_m$ will survive after we glue q_1 and q_2. Moreover, if such copy survives, the induced restriction to the normalization can be identified with the diagonal embedding $\mathbb{G}_m \subset \mathbb{G}_m^2$. Notice that Lemma 2.42 is stated for $h \geq 2$ thus it does not deal with nodes. Indeed, when gluing two points using nodes, there is no condition $n_1 = n_2$ and thus both copies of $\mathbb{G}_m$ (if exist) will survive: see Remark 2.30.

Remark 2.44. It is important to stress that in the moreover part of Lemma 2.42, the conditions on the cocharacter ϕ are necessary. Indeed, consider two disjoint copies of $(\mathbb{P}^1, 0, \infty)$: the automorphism group of the disjoint union is $\mathbb{G}_m^2$. For simplicity, we can assume $h = 2$, thus we are dealing with tacnodal singularities. If we apply Lemma 2.42 to the diagonal cocharacter $\mathbb{G}_m \to \mathbb{G}_m^2$ gluing the two copies in 0, there exists a unique curve $\widetilde{C}$ which has $\mathbb{G}_m$ as (the connected component of the identity of) the group of automorphisms: it is constructed by gluing the two projective lines in 0 and $\mathbb{G}_m$ acts as the scalar multiplication on both the component.

Suppose that we want to glue the two irreducible components in ∞ as well, i.e. we want to apply Lemma 2.42 to $(\widetilde{C}, q_1, q_2)$ where q_1, q_2 are the points ∞ in the two irreducible components. Following the proof of Lemma 2.42, we have that the moduli stack of curves constructed in this way is isomorphic to $\mathbb{G}_m \times [U_h/\mathbb{G}_m]$ (see also Remark 2.38). Therefore there is a family of non-isomorphic curves over $\mathbb{G}_m$ constructed with different $\mathbb{G}_m$-equivariant crimping datum. Lemma 2.42 fails because the curve $(\widetilde{C}, q_1, q_2)$ does not have a copy of $\mathbb{G}_m^2$ as a subtorus.

Lemma 2.42 has the following interesting consequence.

Corollary 2.45. *Let $(C, p_1, \ldots, p_n)$ be a n-pointed curve of genus g. Let Γ be an irreducible component and $\mathbb{G}_m \to \mathrm{Aut}^\circ(C, p_1, \ldots, p_n)$ be a cocharacter such that the induced morphism $\mathbb{G}_m \to \mathrm{Aut}^\circ(\Gamma, P_\Gamma \cup I_\Gamma)$ is not trivial. Then the same is true for every component Γ' meeting Γ in (at least one) worse-than-nodal odd A-singularity.*

2.4 Hyperelliptic A_r-stable curves

As anticipated, we conclude the section focusing our attention to hyperelliptic A_r-stable curves. We start with the definition.

Definition 2.46. An hyperelliptic n-pointed curve is a tuple $(C, p_1, \ldots, p_n, \sigma)$ where $(C, p_1, \ldots, p_n)$ is a n-pointed curve and σ is a hyperelliptic involution, i.e. a two-torsion element in $\mathrm{Aut}(C, p_1, \ldots, p_n)$ such that

- the scheme of fixed points of σ is finite;
- the geometric quotient C/σ is a nodal curve of genus 0.

We say that a point $q \in C$ is a Weierstrass (possibly singular) point if it is fixed by the involution.

We define $\mathcal{H}^r_{g,n}$ as the following fibered category: its objects are the data of a pair $(C/S, \sigma, p_1, \ldots, p_n)$ where $(C, p_1 \ldots, p_n)$ is an A_r-stable curve over S and σ is an involution of C over S such that (C_s, σ_s) is a A_r-stable hyperelliptic curve of genus g for every geometric point $s \in S$. These are called *hyperelliptic n-pointed A_r-stable curves over S*. A morphism is a morphism of n-pointed A_r-stable curves that commutes with the involutions. We clearly have a morphism of fibered categories

$$\eta : \mathcal{H}^r_{g,n} \longrightarrow \mathcal{M}^r_{g,n}$$

over κ defined by forgetting the involution. This morphism is known to be a closed embedding for $r \leq 1$. Recall that by convention $\mathcal{M}^0_{g,n}$ coincides with the moduli stack of n-pointed smooth curves of genus g.

Theorem 2.47. *The morphism η is a closed embedding for every r. In fact, we can identify $\mathcal{H}^r_{g,n}$ with the closure of the locally closed substack $\mathcal{H}^0_{g,n} \subset \mathcal{M}^0_{g,n} \subset \mathcal{M}^r_{g,n}$ parametrizing smooth n-pointed hyperelliptic curves of genus g.*

Proof. Indeed, we have that $\mathcal{H}^r_{g,n}$ coincide with fiber product $\mathcal{H}^r_g \times_{\mathcal{M}^r_g} \mathcal{M}^r_{g,n}$ where the morphism

$$\mathcal{M}^r_{g,n} \longrightarrow \mathcal{M}^r_g$$

can be constructed using [Per23, Theorem 2.5], which states that $\mathcal{M}^r_{g,n}$ is an open substack of the universal $n-1$-pointed A_r-stable curve. Thus the statement follows from [Per24, Section 3], which proves that $\mathcal{H}^r_g \to \mathcal{M}^r_g$ is a closed immersion. □

To every n-pointed hyperelliptic curve, we can associate the geometric quotient $C \to Z := C/\sigma$, where Z is a nodal curve of genus 0.

Definition 2.48. If Z is irreducible, then we say that C is a n-pointed *honestly hyperelliptic* A_r-stable curve, namely a cyclic cover of $\mathbb{P}^1$ of degree 2.

The terminology is borrowed from [Cat82], specifically Definition 3.18. The substack of $\mathcal{H}^r_{g,n}$ classifying honestly hyperelliptic curves $\mathcal{H}^{r,\circ}_{g,n}$ is an open inside $\mathcal{H}^r_{g,n}$. The rest of this section is dedicated to give a description of $\mathcal{H}^{r,\circ}_{g,n}$ as a quotient stack, in some specific cases we will need. The reason is that these cases are the building blocks for the points in $\mathcal{M}^r_{g,n}$ that have non-trivial positive dimensional stabilizers, and understanding how they behave with respect to isotrivial degenerations is fundamental for the understanding of the local picture of the moduli stack we are interested in.

We start with the case $n = 0$. We have the following identification:

$$\mathcal{H}^{r,\circ}_g \simeq [U_{r+1}/(\mathrm{GL}_2/\mu_{g+1})]$$

where U_{r+1} is an open subset of the representation $\mathrm{Sym}^{2g+2} E$, where E is the standard representation of GL_2. This directly follows from [AV04]. The open U_{r+1} parametrizes homogeneous binary forms $f \in \mathrm{Sym}^{2g+2} E$ of degree $2g+2$ such that they do not have roots of multiplicity strictly bigger than $r+1$. We are identifying every honestly hyperelliptic curve with the branching divisor of lenght $2g+2$ in $\mathbb{P}^1$. A point of length h in this divisor corresponds to an A_{h-1}-singularity in the (honestly) hyperelliptic curve. Notice that U_{r+1} never contains the 0-section of the representation, as it corresponds to a non-reduced cover of $\mathbb{P}^1$, and it is equal to $\mathrm{Sym}^{2g+2} E \setminus 0$ if $r = 2g+1$. The question we are interested in is the following: for which r does $\mathcal{H}^{r,\circ}_g$ admit a good moduli space?

One attempt to solve this problem can be done by using GIT. The moduli stack $\mathcal{H}^{2g+1,\circ}_g$ is strongly related to the quotient $[\mathbb{P}(\mathrm{Sym}^{2g+2} E)/\mathrm{PGL}_2]$ where the action of PGL_2 on E is the standard one: there is a morphism

$$\mathcal{H}^{2g+1,\circ}_g \to [\mathbb{P}(\mathrm{Sym}^{2g+2} E)/\mathrm{PGL}_2]$$

which is a μ_2-gerbe, whose triviality depends on the parity of g. Therefore, it is enough to study the open subsets of the right-hand side of the morphism which admits a separated good moduli space.

If we apply the GIT machinery, more specifically the Hilbert-Mumford criterion, we find that the open of the semistable points corresponds to U_{g+1}, namely the binary forms without roots of multiplicity greater than $g+1$. Although classical, the authors learned about it in [Hos15], specifically Proposition 7.12. Moreover, U_r consists of only stable points if $r \leq g$. We claim that the GIT answer is somehow "maximal". By abuse of notation, we denote by U_r the projectivization of the open subset of $\mathrm{Sym}^{2g+2} E \setminus 0$ and by $\mathcal{U}_r$ the induced open substack of $\mathcal{U}_{2g+2} := [\mathbb{P}(\mathrm{Sym}^{2g+2} E)/\mathrm{PGL}_2]$.

Theorem 2.49. *In the situation above, let $\mathcal{U} \subset \mathcal{U}_{2g+2}$ be any open substack. Then $\mathcal{U}$ admits a separated good moduli space if and only if one of the following holds:*

(i) $\mathcal{U} \subset \mathcal{U}_g$;

(ii) $\mathcal{U} \subset \mathcal{U}_{g+1}$ *and* $\mathcal{U}_{g+1} \setminus \mathcal{U}_g \subset \mathcal{U}$.

Proof. First of all, notice that the if either (i) or (ii) are satisfied, then the result comes from the GIT machinery. In fact, $\mathcal{U}_{g+1}$ (respectively $\mathcal{U}_g$) coincides with the locus of semistable (respectively stable) points. Because $\mathcal{U}_g$ is a separated DM stack, then $\mathcal{U} \subset \mathcal{U}_g$ implies that $\mathcal{U}$ is also a separated DM stack. Suppose then $\mathcal{U}_{g+1} \setminus \mathcal{U} \subset \mathcal{U}_g$. Therefore Proposition A.7 gives us that the open embedding $\mathcal{U} \subset \mathcal{U}_{g+1}$ is S-complete and Θ-complete, because the complement $\mathcal{Z} := \mathcal{U}_{g+1} \setminus \mathcal{U}$ does not have positive dimensional stabilizers. Because $\mathcal{U}_{g+1}$ admits a separated good moduli space, so does $\mathcal{U}$.

Remain to show that whenever $\mathcal{U}$ is Θ-complete, S-complete and $\mathcal{U} \not\subset \mathcal{U}_g$, then (ii) holds. Firstly, we want to prove that for such an $\mathcal{U}$ we have $\mathcal{U} \subset \mathcal{U}_{g+1}$. First of all, we notice that $\mathcal{U}_{2g+2} \setminus \mathcal{U}_{2g+1}$ consists of one (closed) point with non-reductive stabilizer. This can be done by considering the morphism

$$\mathbb{P}(E) \longrightarrow \mathbb{P}(\mathrm{Sym}^{2g+2} E)$$

given by the association $f(x_0, x_1) \mapsto f^{2g+2}$, where f is a binary linear form in E. The morphism is a closed immersion and identifies $\mathcal{U}_{2g+2} \setminus \mathcal{U}_{2g+1}$ with the quotient $[\mathbb{P}^1 / \mathrm{PGL}_2] \simeq \mathcal{B}(\mathbb{G}_m \ltimes \mathbb{G}_a)$. Therefore, $\mathcal{U} \subset \mathcal{U}_{2g+1}$. Now, the only points that have positive-dimensional stabilizers in $\mathcal{U}_{2g+1}$ are the binary forms f_h having exactly two (set-theoretic) roots, one of multiplicity h and the other of multiplicity $2g+2-h$. These points are closed in $\mathcal{U}_{2g+1}$, as there are no further degenerations in $\mathcal{U}_{2g+1}$. Their stabilizer groups are isomorphic to $\mathbb{G}_m$, as it is the automorphism group of $\mathbb{P}^1$ with two distinct points. We claim that $f_h \in \mathcal{U} \implies h = g+1$. Without loss of generality, suppose $h < g+1$. We can consider the following morphism

$$[\mathbb{P}(E)/\mathbb{G}_m] \longrightarrow \mathcal{U} \subset [\mathbb{P}(\mathrm{Sym}^{2g+2} E)/\mathrm{PGL}_2]$$

defined by the association $f(x_0, x_1) \mapsto f^{2(g+1-h)}(x_0 x_1)^h$, where $\mathbb{G}_m \subset \mathrm{PGL}_2$ is the diagonal torus. A standard computation shows that the morphism is finite: indeed, it is a μ_2-torsor where μ_2 acts on E exchanging x_0 and x_1, followed by a closed embedding. Since the map factors through the open $\mathcal{U} \subset [\mathbb{P}(\mathrm{Sym}^{2g+2} E)/\mathrm{PGL}_2]$, we get a a contradiction because $[\mathbb{P}(E)/\mathbb{G}_m]$ does not have a good moduli space (see [Alp13, Lemma 4.14]).

Therefore $f_h \notin \mathcal{U}$ if $h \neq g+1$. We claim that this implies that the complement $\mathcal{Z}$ contains $\mathcal{Z}_h$, namely the locally closed subspace classifying binary forms with one root of multiplicity exactly h, when $h > g+1$, i.e. $\mathcal{U} \subset \mathcal{U}_{g+1}$.

Because $\mathcal{U}$ is S-complete and Θ-complete, so is the open embedding $\mathcal{U} \subset \mathcal{U}_{2g+1}$, because $\mathcal{U}_{2g+1}$ has affine diagonal. This implies that the complement $\mathcal{Z}$ of $\mathcal{U}$ in $\mathcal{U}_{2g+1}$ satisfies the condition of Proposition A.7. Let us apply it in the case where x is the geometric point associated to f_h. First of all, notice that we know that the space of deformations of (the curve associated to) f_h can be described as the direct sum

$$\mathrm{Def}_{f_h} = V_h \oplus V_{2g+2-h}$$

where V_h (respectively V_{2g+2-h}) is the space that preserves the root of multiplicity h (respectively $2g+2-h$). Moreover, such decomposition is not only $\mathbb{G}_m$-equivariant, but sign-preserving, meaning that the irreducible representations inside V_h have positive weights while the ones inside V_{2g+2-h} have negative weights, or vice versa. This follows from the fact that $\mathbb{G}_m$ acts with weights with opposite signs at the tangent space of 0 and ∞. Therefore, Proposition A.7 allows us to conclude that $\mathcal{Z}$ either contains V_h or V_{2g+2-h}, locally around f_h. This actually holds globally because of the following observation: every binary form which has a root of multiplicity h admits an isotrivial degeneration to f_h, thus the local structure (considered in the proof of Proposition A.7) of $\mathcal{U}_{2g+1}$ in f_h is surjective on the locally closed $\mathcal{Z}_h \subset \mathcal{Z}$. Thus we have that $\mathcal{Z}$ contains either $\mathcal{Z}_h$ or $\mathcal{Z}_{2g+2-h}$ globally. Finally, we notice that $\mathcal{Z}_h$ is contained in the closure of $\mathcal{Z}_{2g+2-h}$ when $h > g+1$. Because $\mathcal{U}$ is an open, in both cases $\mathcal{U}$ will not intersect $\mathcal{Z}_h$ for every $h > g+1$. Therefore we have that $\mathcal{U} \subset \mathcal{U}_{g+1}$.

Suppose now that $f_{g+1} \notin \mathcal{U}$. Then applying Proposition A.7 locally on f_{g+1} as before, we can prove that $\mathcal{Z}_{g+1}$ is contained in the complement of $\mathcal{U}$, thus giving $\mathcal{U} \subset \mathcal{U}_g$. Finally, it remains to prove that

$$f_{g+1} \in \mathcal{U} \iff \mathcal{Z}_{g+1} \subset \mathcal{U}.$$

This follows again from the fact that $\mathcal{U}$ is open and every binary form in $\mathcal{Z}_{g+1}$ admits an isotrivial degeneration to f_{g+1}. □

Remark 2.50. The previous proof relies heavily on the fact that the positive dimensional stabilizers, after removing the non-reductive ones, are one-dimensional tori. This allows us to extend the results from the local picture to the global one, giving us a way to classify the opens which admit a separated good moduli space.

With the same spirit, we can answer the open question in [CTV21, Rem. 3.11]. Indeed, in the notation of [CTV21, Fig. 1], we construct two isotrivial degenerations from C_{nnn} to C_{nt} which are not isomorphic, giving a finite map from $[\mathbb{P}^1/\mathbb{G}_m]$.

Proposition 2.51. *Using the notation of* loc.cit.*, the stack* $\overline{\mathcal{M}}_2^{\mathrm{irr}} \subset \mathcal{M}_2^3$ *does not admit a good moduli space.*

Proof. We can argue in the same way of Theorem 2.49. Firstly, notice that we have a finite μ_2-gerbe φ:

$$\overline{\mathcal{M}}_2^{\mathrm{irr}} = \mathcal{H}_2^{3,\circ} \simeq [U_4/(\mathrm{GL}_2/\mu_3)] \xrightarrow{\varphi} \mathcal{U}_4 \subset [\mathbb{P}(\mathrm{Sym}^6 E)/\mathrm{PGL}_2].$$

We can therefore consider the finite map constructed in the proof of Theorem 2.49:

$$\begin{aligned} \phi\colon [\mathbb{P}(E)/\mathbb{G}_m] &\to \mathcal{U}_4 \subset [\mathbb{P}(\mathrm{Sym}^6 E)/\mathrm{PGL}_2] \\ f &\mapsto (x_0x_1)^2 f^2 \end{aligned}$$

where E is the dual of the standard representation of GL_2. Therefore $\overline{\mathcal{M}}_2^{\mathrm{irr}}$ does not have a good moduli space because it has finite map from a stack that does not admit a good moduli space. □

We move to the case $n = 1$. To be precise, we introduce a stratification of $\mathcal{H}_{g,1}^{r,\circ}$. We denote by $\mathcal{H}_{g,w}^{r,\circ} \subset \mathcal{H}_{g,1}^{r,\circ}$ the closed substack parametrizing pairs (C, p, σ) such that p is a (smooth) Weierstrass point, i.e. a point fixed by the involution. We denote by $\mathcal{H}_{g,g_1^2}^{r,\circ}$ its open complement. Notice that $\mathcal{H}_{g,g_1^2}^{r,\circ}$ can be identified with the moduli stack parametrizing A_r-stable honestly hyperelliptic curves of genus g with two (smooth and distinct) markings which are exchanged by the involution. We end the section with the description of these substacks. This is Proposition 3.9 of [ASvdW10].

We start with the Weierstrass case.

Proposition 2.52. *If $r \geq 2g$, the moduli stack $\mathcal{H}_{g,w}^{r,\circ}$ is isomorphic to the quotient $[\mathbb{A}^{2g-1}/\mathbb{G}_m]$, where $\mathbb{G}_m$ acts on $\mathbb{A}^{2g-1}$ with positive weights.*

Corollary 2.53. *The moduli stack $\mathcal{H}_{g,w}^{r,\circ}$ has a separated good moduli space for every r.*

We end this subsection with the remaining case.

Proposition 2.54. *If $r \geq 2g+1$, the moduli stack $\mathcal{H}_{g,g_1^2}^{r,\circ}$ is isomorphic to the quotient $[\mathbb{A}^{2g+1}/\mathbb{G}_m]$, where $\mathbb{G}_m$ acts on $\mathbb{A}^{2g+1}$ with positive weights.*

Corollary 2.55. *The map $\mathcal{H}_{g,g_1^2}^{r,\circ} \to \operatorname{Spec} \kappa$ is a good moduli space.*

Remark 2.56. The stack $\mathcal{H}_{g,1}^{2g+1,\circ}$ does not admit a separated good moduli space. Namely, it fails to be Θ-complete. Indeed, we have the stratification $\mathcal{H}_{g,w}^{2g+1,\circ} \subset \mathcal{H}_{g,1}^{2g+1,\circ}$ where the open stratum $\mathcal{H}_{g,g_1^2}^{2g+1,\circ}$ has a unique closed point, and every other point of the open stratum admits an isotrivial degeneration to the closed point. The same is true for the closed stratum. Because the closed stratum is contained inside the closure of the open stratum, Θ-completeness would imply that the closed point of the closed stratum should be contained inside the closure of the closed point of the open stratum. In Example 4.20, we show that this is not possible, and thus giving a contradiction.

We conclude with the following corollary, which explains why we are interested mainly in the cases above.

Corollary 2.57. *Let $(C, p_1, \ldots, p_n, \sigma) \in \mathcal{H}_{g,n}^{r,\circ}$ and suppose $n \geq 1$ and $g \geq 1$. If $\mathrm{Aut}(C, p_1, \ldots, p_n)$ has positive dimension, then either*

(1) $n = 2$, $r \geq 2g+1$, the two points are exchanged by the involution and C is the curve corresponding to the 0-section of $[\mathbb{A}^{2g}/\mathbb{G}_m] \simeq \mathcal{H}_{g,g_1^2}^{r,\circ}$;

(2) $n = 1$, $r \geq 2g+1$, p_1 is not a Weierstrass point and C is the curve corresponding to the 0-section of $[\mathbb{A}^{2g}/\mathbb{G}_m] \simeq \mathcal{H}_{g,g_1^2}^{r,\circ}$, where we are identifying $\mathcal{H}_{g,g_1^2}^{r,\circ}$ with the open subset of $\mathcal{H}_{g,1}^{r,\circ}$ where the marking is not Weierstrass;

(3) $n = 1$, $r \geq 2g$, p_1 is a Weierstrass point and C is the curve corresponding to the 0-section of $[\mathbb{A}^{2g-1}/\mathbb{G}_m] \simeq \mathcal{H}_{g,w}^{r,\circ}$;

Proof. Notice that if $n \geq 3$, then a $2:1$-cover of $\mathbb{P}^1$ with at least 3 fixed points has finite automorphism group. Indeed, they will correspond to at least two points on the $\mathbb{P}^1$, which means the automorphism group is a subgroup of $\mathbb{G}_m$. Moreover, since the automorphism would have to fix also the ramification of the $2:1$ cover, clearly it will be finite. Thus we can assume $n \leq 2$.

If $n = 2$, and the two points are not exchanged by the hyperelliptic involution, we are in the same situation as above. Thus we end up in case (1) thanks to Proposition 2.54.

If $n = 1$, we can use the stratification of $\mathcal{H}_{g,1}^{r,\circ}$ by the closed substack $\mathcal{H}_{g,w}^{r,\circ}$ and its open complement $\mathcal{H}_{g,g_1^2}^{r,\circ}$. Then again, we can use Proposition 2.52 and Proposition 2.54 to conclude. □

Remark 2.58. Case (1) and (3) of Corollary 2.57 are the building blocks for everything that will appear from now on. We will refer to these curves as *atoms*, c.f. Definitions 3.3 and 3.7.

3 Families of curves over $[\mathbb{A}^1/\mathbb{G}_m]$

In this section, we describe all possible n-pointed A_r-stable curves with positive-dimensional stabilizers, extending the classification carried out in [HH13] and [CTV23] for $r \geq 3$. We start by understanding the *atomic* case, either even or odd. These are explained in Section 3.1. Subsequently in Section 3.2, we characterize the subcurves which contribute non-trivially to the dimension of the automorphism group of the whole curve and we describe the connected component of the identity of the automorphism group of A_r-stable curves. Furthermore, we study how the automorphism group acts on the deformation space of a curve, specifically on the deformation space of A-singularities and crimping space, which is essential for understanding isotrivial degenerations. This is done in Section 3.3. Finally, we prove that the local description of the isotrivial degenerations given by deformation theory can be globalized. This is explained in detailed in Section 3.4.

Definition 3.1. An A_m-singularity p of an n-pointed A_r-stable curve $(C, p_1, \ldots, p_n)$ with $m \geq 2$ is called *hyperelliptic* if there exists a cocharacter $\mathbb{G}_m \to \mathrm{Aut}^\circ(C, p_1, \ldots, p_n)$ such that $\mathbb{G}_m$ acts non-trivially on T_pC.

The reason why we call the singularity *hyperelliptic* will be explained in Lemma 3.5 and Lemma 3.8. Before starting our analysis of isotrivial degenerations of A_r-stable curves, we encourage the reader to read Section 2.3, where we discuss when a cocharacter of the automorphism group of a curve is preserved in creating an A-singularity. Moreover, take the time to recall the "propagation of the action" phenomenon: if Γ is an irreducible component with a non-trivial $\mathbb{G}_m$-action induced by a cocharacter of the automorphism group of the whole curve C, then any other component Γ' intersecting Γ in a worse-than-nodal singularity will have a non-trivial $\mathbb{G}_m$-action (induced by the same cocharacter). This will turn out to be fundamental in understanding which curves have positive dimensional stabilizers.

3.1 Atoms

We start by finding examples of curves with positive dimensional stabilizers. These curves are crucial to our analysis of the valuative criteria for the existence of a separated good moduli space since every non-trivial Θ-degeneration of curves has a special fiber with positive dimensional stabilizers. In this subsection, we assume that every curve is defined over an algebraically closed base field.

We study the even case first.

Remark 3.2. Consider $(\mathbb{P}^1, \infty)$ and suppose we want to crimp ∞ into an A_{2h}-singularity, with $h \geq 1$. By Lemma 2.31, we know that the moduli stack parametrizing curve (C, p) whose (pointed) partial normalization at p is $(\mathbb{P}^1, \infty)$ can be described as $[V_{h,2}/\mathbb{G}_m \ltimes \mathbb{G}_\mathrm{a}]$ where the $\mathbb{G}_m \ltimes \mathbb{G}_\mathrm{a}$ can be described explicitly using the jet morphism $\mathrm{Aut}(\mathbb{P}^1, \infty) \to G_{2h}$. A straightforward computation shows that $\mathbb{G}_\mathrm{a}$ acts freely on $V_{h,2}$, thus the moduli stack is isomorphic to the quotient $[(V_{h,2}/\mathbb{G}_\mathrm{a})/\mathbb{G}_m]$ where $V_{h,2}/\mathbb{G}_\mathrm{a}$ is a $\mathbb{G}_m$-representations with strictly positive weights (see for instance the proof of Lemma 2.34). Therefore there exists a unique curve obtained crimping $\mathbb{P}^1$ in ∞ whose A_{2h}-singularity is hyperelliptic. It is easy to see that the curve C obtained is A_r-stable if and only if $h \geq 2$ (see Remark 2.12). The same construction can be done starting with $(\mathbb{P}^1, 0, \infty)$, crimping in ∞ and remembering the marking 0: it will give rise to a 1-pointed A_r-stable curve (for $h \geq 1$). The automorphism group of both these curves is a one-dimensional torus.

Definition 3.3 (Even atom)**.** The curve constructed in Remark 3.2 will be called the *even atom* of genus h. If it contains a marking, we will denote it as the 1-pointed even atom.

Figure 1: Even atom

Remark 3.4. The term *atom* is first used in [AFSvdW17]. These are the two building blocks (even and odd) of the A_r-stable curves with positive dimensional stabilizers. As we remarked at the beginning of the section, it is clear that if a curve C has positive dimensional automorphisms, it has to contain some components of geometric genus 0 which are not stable in the classical sense, i.e. there are less than 3 special points on these components. This is the reason why we will study how to construct A-singular curves using projective lines, as they are the only possible components that contribute to the positive dimensional automorphisms.

First of all, we explain why we used the term *hyperelliptic* for the singularities. The even atom of genus g is actually the only A_r-stable curve with an A_{2g}-singularity which is hyperelliptic, and it is in fact honestly hyperelliptic.

Lemma 3.5. *Suppose $n \leq 1$ and $r \geq 2g$. The intersection $\mathcal{D}^{2g} \cap \mathcal{H}^r_{g,n} \subset \mathcal{M}^r_{g,n}$, parametrizing hyperelliptic curves with at least one singularity of type $2g$, coincides with the n-pointed even atom of genus g. Moreover, if $n = 1$ the marking has to be a Weierstrass point.*

Proof. We start by studying the case $n = 0$. Suppose that $C \in \mathcal{H}^r_g \cap \mathcal{D}^{2g}$. First of all, because of the absence of markings, the curve C has to be irreducible (see Remark 2.12), thus in $\mathcal{H}^{r,\circ}_g$, namely a cyclic cover of $\mathbb{P}^1$ of degree 2. The ramification divisor of the morphism $C \to \mathbb{P}^1$ has length $2g+2$ and [Per24, Corollary 1.4] gives us that it has length $2g+1$ on the A_{2g}-singularity. This implies that the branching divisor is (set-theoretically) supported in two distinct points of $\mathbb{P}^1$. Using the description of $\mathcal{H}^{r,\circ}_g$ as cyclic covers of degree 2 of $\mathbb{P}^1$ (see [AV04] or Section 2.4), one can prove that $\mathrm{Aut}(C) \simeq \mathbb{G}_m$ when the branching divisor is supported in two distinct points, namely because $\mathrm{Aut}(\mathbb{P}^1, 0, \infty) \simeq \mathbb{G}_m$. Notice that Remark 3.2 implies that there exists a unique curve such that

- it has an A_{2g}-singularity,
- the partial normalization at the singularity is $\mathbb{P}^1$,
- it has positive dimensional automorphism group;

therefore the statement follows. We leave it to the interested reader to prove the case $n = 1$. □

We deal now with the odd case. From now on, for convenience's sake, we will consider A_{2h+1}-singularity (instead of A_{2h-1}), thus the careful reader will encounter a shift in numerical conditions on h.

Remark 3.6. Let $h \geq 1$. Consider the curve $(\widetilde{C}, q_1, q_2)$ defined as the disjoint union of two copies of $(\mathbb{P}^1, \infty)$. By Lemma 2.36 or Lemma 2.39, we know that the moduli stack parametrizing curves (C, p) such that

- p is an A_{2h+1}-singularity,
- the (pointed) partial normalization of C in p is $(\widetilde{C}, q_1, q_2)$

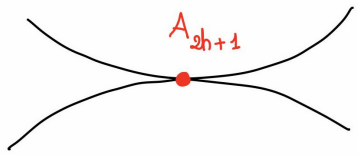
is isomorphic to the quotient $[(U_{h+1}/\mathbb{G}_\mathrm{a})/\mathbb{G}_m \ltimes \mathbb{G}_\mathrm{a}]$, where $U_{h+1}/\mathbb{G}_\mathrm{a}$ is a $\mathbb{G}_m$-representations with strictly positive weights. Therefore there exists a unique curve whose A_{2h+1}-singularity is hyperelliptic

(see also Lemma 2.42). It is easy to see that the curve C is A_r-stable if and only if $h \geq 2$ (see Remark 2.12). Moreover, we can prove that $\mathrm{Aut}^\circ(C) \simeq \mathbb{G}_m \ltimes \mathbb{G}_\mathrm{a}$.

The same construction can be carried out considering 0 as a marking for one (or both) projective line: it will give rise to a 1-pointed (or 2-pointed) A_r-stable curve for $h \geq 2$ (for $h \geq 1$). There is a (non-unique choice of) $\mathbb{G}_m$-equivariant crimping datum, thanks to Lemma 2.42. Moreover, any choice of such a $\mathbb{G}_m$-equivariant crimping datum give rise to isomorphic curves. The (connected component of the identity of the) automorphism group of the pointed curves is a one-dimensional torus.

Definition 3.7 (Odd atom)**.** The curve constructed in Remark 3.6 will be called the *odd atom* of genus h. If it contains n-markings, we will denote it as the n-pointed odd atom.

Figure 2:



As for the even case, the odd atom of genus g is actually the only A_r-stable curve with an A_{2g+1}-singularity which is hyperelliptic, and it is in fact honestly hyperelliptic.

Lemma 3.8. *Suppose $n \leq 2$ and $r \geq 2g+1$. The intersection $\mathcal{D}^{2g+1} \cap \mathcal{H}^r_{g,n} \subset \mathcal{M}^r_{g,n}$ is the n-pointed odd atom. Moreover, if $n = 1$ the marking is not a Weierstrass point, and if $n = 2$ the two markings are exchanged by the hyperelliptic involution.*

Proof. First, notice that $\mathcal{D}^{2g+1} \subset \mathcal{M}^r_{g,n}$ parametrizes curves with *separating* A_{2g+1}-singularities, because of Remark 2.12.

Let us first deal with the case $n = 0$. Suppose that $C \in \mathcal{H}^r_g \cap \mathcal{D}^{2g+1}$. The quotient C/σ has to be irreducible. Indeed, C is the union of two irreducible rational smooth curves attached with an A_{2g+1}-singularity (see Remark 2.12). Because of [Per23, Proposition 1.3], the involution has to exchange the two irreducible components, thus giving $C \in \mathcal{H}^{r,\circ}_g$.

The ramification divisor of the morphism $C \to \mathbb{P}^1$ has length $2g+2$ and [Per24, Corollary 1.4] gives us that it has to be concentrated on the A_{2g+1}-singularity. This implies that the branching divisor is (set-theoretically) supported in a single point of $\mathbb{P}^1$. Using the description of $\mathcal{H}^{r,\circ}_g$ as cyclic covers of degree 2 of $\mathbb{P}^1$ (see [AV04] or Section 2.4), one can prove that $\mathrm{Aut}^\circ(C) \simeq \mathbb{G}_m \ltimes \mathbb{G}_a$, namely because $\mathrm{Aut}(\mathbb{P}^1, 0) \simeq \mathbb{G}_m \ltimes \mathbb{G}_a$, therefore by Remark 3.6, we get that it coincides with the unique point having $\mathbb{G}_m \ltimes \mathbb{G}_a$ stabilizers in $\mathcal{D}^{2g+1}$.

The same proof works with pointed curves if $n \geq 1$. □

Remark 3.9. Notice that having markings on the atoms allows us to attach them to other curves using nodes, with the goal of constructing non-atomic curves with positive dimensional stabilizers.

3.2 Curves with positive-dimensional stabilizers

In this subsection, we are going to classify all the subcurves which contributes to the dimension of the automorphism group, over an algebraically closed field k. Before doing that, we start by introducing the following definitions/notations, which are (almost all) very classical.

Definition 3.10 (Subcurve)**.** Let C be a curve over k. A *subcurve* Γ of C is a proper, reduced, one-dimensional subscheme of C, i.e. the union of some irreducible components of C with the reduced scheme structure.

Let $(C, p_1, \ldots, p_n)$ be a n-pointed A-prestable curve over an algebraically closed field k and let $\Gamma \subset C$ be a subcurve. We denote by $C - \Gamma$ the subcurve of C induced by the closure of the $C \setminus \Gamma$ in C. Moreover, we denote by I_Γ the set of points $\Gamma \cap (C - \Gamma)$ of Γ and by P_Γ the subset of the markings $\{p_1 \ldots, p_n\}$ that lie in the subcurve Γ. Notice that every point in I_Γ is smooth in Γ because there are only A-singularities.

Definition 3.11 (Tails and bridges)**.** Let $(C, p_1, \ldots, p_n)$ be a n-pointed A-prestable curve over an algebraically closed field k and let $\Gamma \subset C$ be a connected subcurve. We say that $(\Gamma, I_\Gamma \cup P_\Gamma)$ is a *tail* if $|I_\Gamma \cup P_\Gamma| = 1$ whereas we say that it is a *bridge* if $|I_\Gamma \cup P_\Gamma| = 2$. We say that a bridge Γ is separating if the curve $C - \Gamma$ is disconnected, otherwise we say it is non-separating.

Notice that in the previous definition of subcurve we do not require Γ to be properly contained in C. Therefore given a 1-pointed curve (C, p) we have that (C, p) itself is a tail. Same is true for bridges and 2-pointed curves.

Definition 3.12. We say that an A-singularity p of C is:

(a) *outer* if it lies in the intersection of (exactly) two irreducible components of C;

(b) *lonely* if it is outer and it is the unique (set-theoretic) intersection of the two irreducible components;

(c) *separating* if the partial normalization at p is disconnected;

(d) *inner* if it is not outer.

Clearly, $(c) \implies (b) \implies (a)$ and p has to be an odd A-singularity, i.e. an A_{2i+1}-singularity for some $i \geq 0$.

It will be useful to study degenerations of A-prestable curves, specifically when they are generically outer. The following proposition will play a fundamental role in the paper, thus we report it here.

Proposition 3.13 ([AFSvdW17, Prop. 2.10])**.** *Let $C \to \operatorname{Spec} R$ be a family of A-prestable curves over a dvr R and let p be a section of the family. Suppose that the (geometric) generic fiber p_Q is an outer A_{2k_1+1}-singularity. Then*

- *the (geometric) special fiber p_k is an outer A_{2m+1}-singularity;*
- *each singularity of the (geometric) generic curve C_Q that approaches p_k must be outer as well and must lie on the same two irreducible components of C_Q as p_Q;*
- *the collection of such (generic) singularities approaching p_k is necessarily of the form $\{A_{2k_1+1}, \ldots, A_{2k_s+1}\}$ where $m = s - 1 + \sum_{i=1}^{s} k_i$;*

Remark 3.14. With the notation of Proposition 3.13, if p_Q is lonely (respectively separating), then p_k will also be lonely (respectively separating).

Definition 3.15 (Gluing morphism)**.** Let $(C, \{p_i\}_{i \in I})$ and $(D, \{q_j\}_{j \in J})$ be (not necessarily connected) pointed curves and $\gamma : C \to D$ a finite morphism. It is called a *gluing morphism* if γ restricts to an open immersion $C \setminus \{p_i | i \in I\} \to D \setminus \{q_j | j \in J\}$ and the image of p_i for $i \in I$ is either a marked point or a singularity.

We are ready to introduce the "molecules" we can construct using our atoms and A-singularities. These will be called *rosaries*, and they are a straighforward generalization of [HH13, Def. 6.1].

Definition 3.16 ((Attached) Rosaries)**.** Let k be an algebraically closed field$/\kappa$, we say that (R, r_1, r_2) is a *2-pointed rosary of length* ℓ if there exists a surjective gluing morphism

$$\gamma\colon \coprod_{i=1}^{\ell}(R_i, p_{2i-1}, p_{2i}) \longrightarrow (R, r_1, r_2)$$

and a sequence of positive integers $k_1, \ldots, k_{\ell-1}$ satisfying:

1. (R_i, p_{2i-1}, p_{2i}) is a 2-pointed smooth rational curve for $i = 1, \ldots, \ell$;
2. $\gamma(p_{2i}) = \gamma(p_{2i+1})$ is an A_{2k_i+1}-type singularity of R for $i = 1, \ldots, \ell - 1$;
3. $\gamma(p_1) = r_1$ and $\gamma(p_{2\ell}) = r_2$.

Moreover, we say that (R, r_1) (resp R) is a 1-*pointed rosary* (resp. *0-pointed rosary*, or simply rosary).

We say that $(C, \{p_i\}_{i=1}^n)$ has an A_{d_1}/A_{d_2}*-attached rosary of length* ℓ if there exists a gluing morphism

$$\gamma\colon (R, r_1, r_2) \longrightarrow (C, \{p_i\}_{i=1}^n)$$

such that

1. (R, r_1, r_2) is a rosary of length ℓ;
2. for $j = 1, 2$, $\gamma(r_j)$ is an A_{d_j}-singularity of C, or if $d_j = 0$ we allow $\gamma(r_j)$ to be a marked point of $(C, \{p_i\}_{i=1}^n)$.

Similarly, we say that $(C, \{p_i\}_{i=1}^n)$ has an A_d*-attached rosary of length* ℓ if there exists a gluing morphism

$$\gamma\colon (R, r_1) \longrightarrow (C, \{p_i\}_{i=1}^n)$$

such that

1. (R, r_1) is a rosary of length ℓ;
2. $\gamma(r_1)$ is an A_d-singularity of C, or if $d = 0$ we allow $\gamma(r_1)$ to be a marked point of $(C, \{p_i\}_{i=1}^n)$.

We say that C is a *closed rosary of length* ℓ if C has an A_{2d+1}/A_{2d+1}-attached rosary

$$\gamma\colon (R, r_1, r_2) \longrightarrow C$$

of length ℓ such that $\gamma(r_1) = \gamma(r_2)$ is an A_{2d+1}-singularity of C.

Remark 3.17. A 2-pointed rosary is always A_r-stable. With the notation of Definition 3.16, the 1-pointed rosary (R, r_1) is stable if and only if $k_{\ell-1} \geq 2$ and the 0-pointed rosary is stable if and only if $k_1 \geq 2$ and $k_{\ell-1} \geq 2$. Finally, notice that crimping an even singularity to a marked point of a rosary create a curve that have an attached rosary whose gluing morphism is surjective.

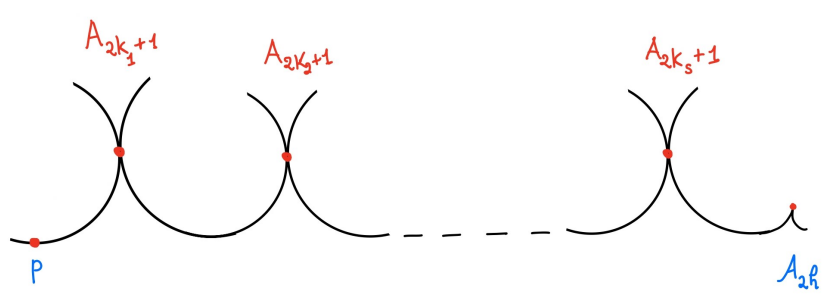


Figure 3: Example of a curve that is not a rosary but it has an attached 2-pointed rosary.

Remark 3.18. An odd atom is a 0-pointed rosary of length 2, whereas a 0 or 1-pointed even atom has respectively a 0 or 1-pointed attached rosary of length 1.

We aim now to compute automorphism of rosaries, we will then prove that attached rosaries are the part of curves that cause Aut° not to be trivial. An explicit description of rosaries with A_3-singularities is carried out in [CTV23, Remark 3.4], in this case 2-pointed rosaries always have a non trivial action of $\mathbb{G}_m$ on the curve, this is not always the case for higher A-type singularities as we will see.

Proposition 3.19. *Let $n \leq 2$ and $\ell \geq 1$. Suppose that (R, r_i) is a n-pointed rosary of length ℓ such that $\mathrm{Aut}^\circ(R, r_i)$ is not trivial. If (R, r_i) is A_r-stable, then we have the following:*

- *if $n + \ell \geq 3$, then $\mathrm{Aut}^\circ(R, r_i) \simeq \mathbb{G}_m$;*
- *if $n = 0$ and $\ell = 2$, then $\mathrm{Aut}^\circ(R)$ is either $\mathbb{G}_\mathrm{a}$ or $\mathbb{G}_m \ltimes \mathbb{G}_\mathrm{a}$.*

Furthermore, a closed rosary R such that $\mathrm{Aut}^\circ(R)$ is not trivial necessarily has even length.

If (R, r_i) is only A_r-prestable, we get that the other possible groups are $\mathbb{G}_m \ltimes \mathbb{G}_\mathrm{a}^2$ and PGL_2. Finally, all the possibilities for Aut° listed above can occur.

Proof. Assume first that the rosary is 2-pointed. Thanks to Remark 2.29, the action of $\mathrm{Aut}^\circ(R, r_i)$ lifts to the pointed normalization

$$\coprod_{i=1}^{\ell} (R_i, p_{2i-1}, p_{2i}) \longrightarrow (R, r_1, r_2).$$

Let $N_i \lhd \mathrm{Aut}^\circ(R, r_1, r_2)$ be the kernel of the map $\mathrm{Aut}^\circ(R, r_1, r_2) \to \mathrm{Aut}^\circ(R_i, p_{2i-1}, p_{2i})$. Thanks to Lemma 2.42 (see also Remark 2.43), we have $N_i = N_{i+1}$ since by definition $(R_i, p_{2i-1}, p_{2i}) \simeq (\mathbb{P}^1, 0, \infty)$ and $\mathrm{Aut}(\mathbb{P}^1, 0, \infty) \simeq \mathbb{G}_m$. Therefore the injectivity of the map

$$\mathrm{Aut}^\circ(R, r_1, r_2) \to \mathrm{Aut}^\circ(\coprod_{i=1}^{\ell} (R_i, p_{2i-1}, p_{2i}))$$

forces N_i to be trivial for any $i = 1, \ldots, \ell$ and therefore $\mathrm{Aut}^\circ(R, r_1, r_2) \simeq \mathbb{G}_m$. Since Lemma 2.42 also allows us to construct a $\mathbb{G}_m$-equivariant crimping datum, we get the statement for the case $n = 2$.

Notice that the 2-pointed case automatically lands us in the first case since $\ell \geq 1$. If (R, r) is 1-pointed and $\ell \geq 2$, then we know that there is one irreducible smooth rational component Γ such that $P_\Gamma = \emptyset$ and $\mathrm{I}_\Gamma = \{p\}$ where p is an odd A_h-singularity, with $h \geq 5$ (since R is A_r-stable). Therefore, $(R - \Gamma, r, p)$ is a 2-pointed A_h/A_0-attached rosary of length $\ell - 1 \geq 1$. Since $\mathrm{Aut}^\circ(R - \Gamma, r, p) \simeq \mathbb{G}_m$ because it is 2-pointed, Lemma 2.39 gives us (R, r) can be identified with an object of $[(U_h/\mathbb{G}_\mathrm{a})/\mathbb{G}_m]$, where $(U_h/\mathbb{G}_\mathrm{a})$ is a $\mathbb{G}_m$-representation with strictly positive weights. Thus $\mathrm{Aut}^\circ(R, r)$ being non trivial implies $\mathrm{Aut}^\circ(R, r) \simeq \mathbb{G}_m$.

Notice that $n = 1$ and $\ell = 1$ is not stable (it is a 1-pointed smooth rational curve), thus it remains to study the case $\ell = 2$ and $n = 0$. This follows again by Lemma 2.39. We avoid writing down the classification of the non-stable cases, as they are not relevant for the current paper, even though it can be easily deduced using the same ideas.

In the case of a closed rosary R, we have $\mathrm{Aut}^\circ(R) \subset \mathbb{G}_m$ with the same strategy. We can then observe that, on any connected component $(\mathbb{P}^1, 0, \infty)$ of the pointed normalization, the action of $\mathbb{G}_m \simeq \mathrm{Aut}^\circ(\mathbb{P}^1, 0, \infty)$ on $T_0\mathbb{P}^1$ and $T_\infty\mathbb{P}^1$ has opposite weight: the claim then follows from Lemma 2.42. $\square$

The following is a consequence of Lemma 2.42 and Lemma 2.39.

Remark 3.20. Given a string of positive integers $k_1, \dots, k_{\ell-1}$ with at least one $k_j > 1$, then it is possible to construct a 0-pointed rosary R as in Definition 3.16 such that $\mathrm{Aut}^\circ(R)$ is trivial. In particular, for any choice of d, we can construct a closed rosary R' such that $\mathrm{Aut}^\circ(R')$ is trivial.

Moreover, the following is a consequence of Lemma 2.42.

Remark 3.21. Let (R, r_1, r_2) be a 2-pointed rosary of even length such that $\mathrm{Aut}^\circ(R, r_1, r_2) \simeq \mathbb{G}_m$, then for any $d \geq 1$ we can construct a closed rosary R' crimping on r_1, r_2 an A_{2d+1} singularity in such a way $\mathrm{Aut}^\circ(R') \simeq \mathbb{G}_m$. This procedure is not unique, it is therefore possible to construct a family of such closed rosaries over $\mathbb{G}_m$ whose geometric fibers are pairwise not isomorphic (see also Remark 2.43).

Finally, we prove the fundamental proposition, which explains why these are the building blocks for A_r-stable curves with positive dimensional stabilizers.

Proposition 3.22. *Let $(C, p_1, \dots, p_n)$ be a n-pointed A_r-stable curve of genus g with* ***no outer nodes*** *and $r \leq 2g$. Then $\mathrm{Aut}^\circ(C, p_1, \dots, p_n)$ is either trivial or $\mathbb{G}_m$. Moreover, if $\mathrm{Aut}^\circ(C, p_1, \dots, p_n) \simeq \mathbb{G}_m$, then $n \leq 2$ and either*

i) $(C, p_1, \dots, p_n)$ has an attached rosary and the gluing morphism is surjective, in this case $n \leq 2$;

ii) $n = 0$ and C is a closed rosary of even lenght;

iii) $n = 0$ and C is irreducible of geometric genus 0 with either one or two hyperelliptic even A-singularities.

Notice that, the case iii) is actually a special case of i), for lenght 1 rosaries.

Proof. We proceed by induction on the genus g. The case $g = 0$ is trivial, because of Remark 2.12. Suppose that $(C, p_1, \dots, p_n)$ is an integral n-pointed A_r-stable curve of genus g such that the group $\mathrm{Aut}^\circ(C, p_1, \dots, p_n)$ is not trivial. It follows easily that $n \leq 1$ and C has geometric genus 0. If $n = 1$, looking at the normalization we can deduce that C is a 1-pointed even atom (see for instance Remark 3.2). If $n = 0$, C cannot have inner odd A-singularities, otherwise $\mathrm{Aut}^\circ(C)$ would be trivial thanks to Lemma 2.34. Therefore C is irreducible of geometric genus 0 with either one or two even A-singularities.

Suppose now that C is not irreducible. Because there are no outer nodes, this implies that there exists at least one outer A_{2k+1}-singularity with $k \geq 1$. We denote by $q \in C$ such A_{2k+1}-singularity, by $\widetilde{C} \to C$ the partial normalization in q and by q_1, q_2 the two closed points of $\widetilde{C}$ which lie over q. Moreover, by abuse of notation we denote by p_i the preimage of p_i in $\widetilde{C}$. Clearly $\mathrm{Aut}^\circ(\widetilde{C}, p_1, \dots, p_n, q_1, q_2)$ is not trivial, otherwise the same would be true for $\mathrm{Aut}^\circ(C, p_1, \dots, p_n)$.

Suppose that q is a non-separating singularity, i.e. $\widetilde{C}$ is connected. Notice that the arithmetic genus of $\widetilde{C}$ cannot be zero, otherwise C is either irreducible or there would be outer nodes in C. This implies that $k \leq g - 1$ and $(\widetilde{C}, p_1, p_2, \dots, p_n, q_1, q_2)$ is an object of $\mathcal{M}^r_{g-h,n,2[k]}$. Because there are no outer nodes, Remark 2.23 gives us that $(\widetilde{C}, p_1, p_2, \dots, p_n, q_1, q_2)$ lives inside the open $\mathcal{M}^r_{g-h,n+2} \subset \mathcal{M}^r_{g-h,n+2} \times [\mathbb{A}^1/\mathbb{G}_m] \times [\mathbb{A}^1/\mathbb{G}_m]$. By the induction procedure, this implies that $n = 0$ and $(\widetilde{C}, q_1, q_2)$ is a 2-pointed rosary of hyperelliptic A-singularities and $\mathrm{Aut}^\circ(\widetilde{C}, q_1, q_2) \simeq \mathbb{G}_m$. By construction, C is a closed rosary, and Proposition 3.19 gives that C has even length and $\mathrm{Aut}^\circ(C) \simeq \mathbb{G}_m$.

Finally, suppose that q is a separating (odd) A-singularity and let C_1 and C_2 be the two connected components of $\widetilde{C}$. We denote by P_i the subset of $(p_1, \dots, p_n)$ that lies in C_i and by n_i the cardinality of P_i for $i = 1, 2$. Suppose that C_1 and C_2 have positive arithmetic genus, then

$$(C_i, P_i, q_i) \in \mathcal{M}_{g_i,n_i,[k]} \simeq \mathcal{M}_{g_i,n_i+1} \times [\mathbb{A}^1/\mathbb{G}_m]$$

where g_i is the arithmetic genus of C_i for $i = 1, 2$. This follows from Proposition 2.25 while the isomorphism is described in Proposition 2.17. Because C does not have outer nodes, again we have

$$(C_i, P_i, q_i) \in \mathcal{M}_{g_i, n_i+1} \subset \mathcal{M}_{g_i, n_i+1} \times [\mathbb{A}^1/\mathbb{G}_m]$$

for $i = 1, 2$. By induction, we get that $n_i \leq 1$, and both (C_1, P_1, q_1) and (C_2, P_2, q_2) are rosaries of hyperelliptic A-singularities. Therefore $n = n_1 + n_2 \leq 2$ and C is the gluing of the two rosaries of hyperelliptic A-singularities along a A_{2k+1}-singularity which is also hyperelliptic, otherwise $\mathrm{Aut}^\circ(C, p_1, \ldots, p_n)$ is trivial. Thus we get case (i).

Notice that if both C_1 and C_2 have arithmetic genus zero, then $k = g$ and therefore $r = 2g + 1$. Thus, it remains only the case where only one curve, say C_1, has genus 0. This is straightforward and it consists of adding an rational component to one of the extremes of the rosary of hyperelliptic A-singularities C_2. By Lemma 2.39, we have the statement. □

Proposition 3.22 motivate the following definition

Definition 3.23 ((Attached) $\mathbb{G}_m$-Rosaries)**.** Let k be an algebraically closed field/κ. A (eventually marked) curve R over k is a *$\mathbb{G}_m$-rosary* if R is a (eventually marked) rosary and there exists an injective map $\mathbb{G}_m \to \mathrm{Aut}(R)$.

We say that an (eventually marked) curve C over k has an *attached $\mathbb{G}_m$-rosary* if there exists an (eventually marked) $\mathbb{G}_m$-rosary R and a gluing morphism $\gamma\colon R \to C$.

We say that C has an *attached rosary of hyperelliptic A-singularities* if it has an attached rosary $\gamma\colon R \to C$, that contain in the image an hyperellyptic A-singularity of odd type.

Similarly, we say that C is a *closed $\mathbb{G}_m$-rosary* if it is a closed rosary and there exists an injective map $\mathbb{G}_m \to \mathrm{Aut}(R)$.

Remark 3.24. Any attached rosary of hyperelliptic A-singularities is an attached $\mathbb{G}_m$-rosary. The converse is not always true: automorphisms of rosaries does not necessarily lift to the whole curve.

In Proposition 3.22, we require that the inequality $r \leq 2g$ holds. When $r > 2g$, the only case is $r = 2g + 1$, that is covered in the following.

Corollary 3.25. *Let r, n and g be three nonnegative integers. Let $(C, p_1, \ldots, p_n)$ be a A_r-stable n-pointed curve of genus g over an algebraically closed field, then $\mathrm{Aut}^\circ(C, p_1, \ldots, p_n)$ is exactly one of the following:*

i) the trivial group,

ii) a (split) torus, i.e. a direct product of copies of $\mathbb{G}_m$,

iii) $\mathbb{G}_\mathrm{a}$ or $\mathbb{G}_m \ltimes \mathbb{G}_\mathrm{a}$;

moreover, the non-reductive automorphisms group appear only if $r \geq 2g+1$, $n = 0$ and C is an A-stable curve with an A_{2g+1}-singularity (not necessarely an odd atom).

Proof. If $r = 2g + 1$, $n = 0$ and C is an A-stable curve with an A_{2g+1}-singularity, then the claim follows from Proposition 3.19. If this is not the case, let $(\widetilde{C}, q_j) \to (C, p_i)$ be the pointed normalization on every outer node. Thanks to Remark 2.29 we have an inclusion $\mathrm{Aut}^\circ(C, p_i) \hookrightarrow \mathrm{Aut}^\circ(\widetilde{C}, q_i)$ (it is indeed an isomorphism). The target is then a torus, thanks to Proposition 3.22, hence the claim. □

We end this section giving an estimate of the dimension of the stabilizer group of an A_r-stable curve.

Lemma 3.26. *Let (g, n) two nonnegative integers such that $3g - 3 + n > 0$ and let $(C, p_1, \ldots, p_n)$ be a n-pointed A_r-stable curve of genus g. Then $\dim \mathrm{Aut}(C, p_1, \ldots, p_n) \leq g$.*

Proof. The result follows by a simple inductive argument on the genus g. If the genus is 0, then the moduli stack is a scheme, so there is nothing to prove. Suppose $g > 0$ and thus we can assume $\dim \mathrm{Aut}(C, p_1, \ldots, p_n) > 0$. Suppose we are not in case (iii) of Corollary 3.25. This implies there exists $p \in C$ an hyperelliptic A_h-singularity with $h \geq 2$. By Lemma 2.34 and Remark 2.43, it is clear that the dimension of the automorphism group of the (pointed) stabilization of the partial normalization is $\dim \mathrm{Aut}(C, p_1, \ldots, p_n) - 1$. By induction, we get that $\dim \mathrm{Aut}(C, p_1, \ldots, p_n) - 1 \leq g'$ where g' is the genus of the partial normalization at p. Since $h \geq 2$, then $g' \leq g - 1$, thus we get the statement. If we are in case (iii) of Corollary 3.25, then by stability $g \geq 2 \geq \dim \mathrm{Aut}(C)$ and we won. □

3.3 Deformation theory

In this subsection, we study the deformation theory of the hyperelliptic A-singularities we introduced in the previous subsection. This will play a key role in understanding the possible isotrivial degenerations in $\mathcal{M}^r_{g,n}$.

Remark 3.27. Recall that given a curve C over an algebraically closed field k and a singularity q, we can always consider the surjective morphism of vector spaces

$$\eta : T^1\mathrm{Def}_C \to T^1_q$$

where T^1_q is the deformation space of the singularity q. The kernel of η classifies first order deformations of C which preserve the singularity q. If q is the only singularity in C, the exact sequence

$$\begin{tikzcd}
 & & T^1\mathrm{Def}_C \arrow[r] \arrow[d, equal] & T^1_q \arrow[d, equal] & \\
0 \arrow[r] & \mathrm{H}^1(\mathcal{H}om(\Omega_C, \mathcal{O}_C)) \arrow[r] & \mathrm{Ext}^1(\Omega_{\mathrm{C}}, \mathcal{O}_{\mathrm{C}}) \arrow[r] & \mathrm{H}^0(\mathcal{E}xt^1(\Omega_C, \mathcal{O}_C)) \arrow[r] & 0
\end{tikzcd}$$

gives us the description of $\ker \eta$.

Recall that we have a closed embedding

$$\mathcal{H}^r_g \hookrightarrow \mathcal{M}^r_g$$

which gives us an injective morphism of tangent spaces

$$T_{[C]}\mathcal{H}^r_g \hookrightarrow T_{[C]}\mathcal{M}^r_g$$

where the right-hand side can be identified with $T^1\mathrm{Def}_C$. For the rest of the subsection, we will consider $r \geq 2g + 1$, so that we can work in the stack $\mathcal{M}^r_{g,n} = \mathcal{M}^{2g+1}_{g,n}$ (see Remark 2.14).

Proposition 3.28. *Let C be an atom of genus g. Then there exists a section of*

$$\eta : T^1\mathrm{Def}_C \to T^1_q$$

which identifies T^1_q with $T_{[C]}\mathcal{H}^r_g$, namely the tangent space of the hyperelliptic locus.

Proof. Notice that C is a cyclic cover of degree 2 of $\mathbb{P}^1$, therefore the tangent space $T_{[C]}\mathcal{H}_g$ coincides with the first-order deformations $T^1\mathrm{Def}_f$ of the finite flat morphism $f : C \to \mathbb{P}^1$of degree 2. By [Sta23, tag 0E3V] we have an exact sequence

$$0 \longrightarrow T^1\mathrm{Def}_f \longrightarrow \mathrm{Ext}^1(\Omega_{\mathrm{C}}, \mathcal{O}_{\mathrm{C}}) \xrightarrow{\gamma} \mathrm{Ext}^1(\mathrm{f}^*\Omega_{\mathbb{P}^1}, \mathcal{O}_{\mathrm{C}})$$

where the injectivity on the left follows from [Per24, Section 3]. Moreover, the exact sequence

$$0 \longrightarrow \mathrm{H}^1(\mathcal{H}om(\Omega_C, \mathcal{O}_C)) \longrightarrow \mathrm{Ext}^1(\Omega_{\mathrm{C}}, \mathcal{O}_{\mathrm{C}}) \longrightarrow \mathrm{H}^0(\mathcal{E}xt^1(\Omega_C, \mathcal{O}_C)) \longrightarrow 0$$

comes from the local-to-global spectral sequence, thus it is functorial in the entries. Therefore we can consider the morphism of short exact sequence

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{H}^1(\mathcal{H}om(\Omega_C, \mathcal{O}_C)) \arrow[r] \arrow[d, "\theta"] & \mathrm{Ext}^1(\Omega_\mathrm{C}, \mathcal{O}_\mathrm{C}) \arrow[r] \arrow[d, "\gamma"] & \mathrm{H}^0(\mathcal{E}xt^1(\Omega_C, \mathcal{O}_C)) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathrm{H}^1(\mathcal{H}om(f^*\Omega_{\mathbb{P}^1}, \mathcal{O}_C)) \arrow[r] & \mathrm{Ext}^1(\mathrm{f}^*\Omega_{\mathbb{P}^1}, \mathcal{O}_\mathrm{C}) \arrow[r] & \mathrm{H}^0(\mathcal{E}xt^1(f^*\Omega_{\mathbb{P}^1}, \mathcal{O}_C)) \arrow[r] & 0
\end{tikzcd}$$

induced by the morphism $f^*\Omega_{\mathbb{P}^1} \to \Omega_C$. Because $\Omega_{C/\mathbb{P}^1}$ is supported in dimension 0, we have that $\mathrm{H}^1(\mathcal{H}om(\Omega_{C/\mathbb{P}^1}, \mathcal{O}_C)) = 0$ which implies the injectivity of θ. Thus the induced morphism

$$T^1\mathrm{Def}_f \simeq \ker\gamma \longrightarrow \mathrm{H}^0(\mathcal{E}xt^1(\Omega_C, \mathcal{O}_C)) \simeq T^1_q$$

is injective as well. Finally, a straightforward computation in deformation theory shows that the two vector spaces have the same dimension, which concludes the proof. □

The same ideas works in the case of pointed atoms.

Proposition 3.29. *Let (C, p_1, p_2) be a 2-pointed odd atom of genus g. Then there exists a section of*

$$\alpha : T^1\mathrm{Def}_{(C,p_1,p_2)} \to T^1\mathrm{Def}_C \to T^1_q$$

which identifies T^1_q with $T_{[C,p_1,p_2]}\mathcal{H}^r_{g,g^2_1}$. Similarly, let (C, p) be a 1-pointed even atom of genus g. Then there exists a section of

$$\beta : T^1\mathrm{Def}_{(C,p)} \to T^1\mathrm{Def}_C \to T^1_q$$

which identifies T^1_q with $T_{[C,p]}\mathcal{H}^r_{g,w}$.

Proof. We write down the proof for the first case. The same proof can be adapted easily to the second one.

Consider the exact sequence

$$\begin{tikzcd}
 & T^1\mathrm{Def}_{(C,p_1,p_2)} \arrow[r] \arrow[d, equal] & T^1\mathrm{Def}_C \arrow[d, equal] \\
\mathrm{Hom}(\Omega_C, \mathcal{O}_{p_1} \oplus \mathcal{O}_{p_2}) \arrow[r, "\phi"] & \mathrm{Ext}^1(\Omega_\mathrm{C}, \mathcal{O}_\mathrm{C}(-\mathrm{p}_1 - \mathrm{p}_2)) \arrow[r, "\psi"] & \mathrm{Ext}^1(\Omega_\mathrm{C}, \mathcal{O}_\mathrm{C})
\end{tikzcd}$$

induced by the short exact sequence

$$0 \longrightarrow \mathcal{O}_C(-p_1 - p_2) \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{O}_{p_1} \oplus \mathcal{O}_{p_2} \longrightarrow 0.$$

We have that $\psi^{-1}(T^1_q) \subset T^1\mathrm{Def}_{(C,p_1,p_2)}$ classifies deformations of (C, p_1, p_2) such that C is hyperelliptic, because of Proposition 3.28. Therefore we have that $T_{[C,p_1,p_2]}\mathcal{H}_{g,g^2_1}$ is contained inside $\psi^{-1}(T^1_q)$; if we prove that the induced morphism

$$T_{[C,p_1,p_2]}\mathcal{H}_{g,g^2_1} \longrightarrow T^1_q$$

is bijective, the statement follows. A straightforward computation shows that the two vector spaces have the same dimension, therefore it is enough to prove that the morphism above is injective, or equivalently that

$$\mathrm{im}\,\phi \cap T_{[C,p_1,p_2]}\mathcal{H}_{g,g^2_1} = \{0\}$$

inside $T^1\mathrm{Def}_{(C,p_1,p_2)}$.

The vector subspace $\operatorname{im}\phi \subset T^1\mathrm{Def}_{(C,p_1,p_2)}$ classifies deformations $(C_\epsilon, p_1^\epsilon, p_2^\epsilon)$ such that C_ϵ is the trivial deformation of C, thus

$$\operatorname{im}\phi \cap T_{[C,p_1,p_2]}\mathcal{H}_{g,g_1^2} = \phi\big(\operatorname{Hom}(\Omega_C, \mathcal{O}_{p_1} \oplus \mathcal{O}_{p_2})^{\mathrm{inv}}\big)$$

where $\operatorname{Hom}(\Omega_C, \mathcal{O}_{p_1} \oplus \mathcal{O}_{p_2})^{\mathrm{inv}} \subset \operatorname{Hom}(\Omega_C, \mathcal{O}_{p_1} \oplus \mathcal{O}_{p_2})$ is the vector subspace of elements invariants under the hyperelliptic involution. Deformation theory helps us describe the kernel of ϕ using the exact sequence

$$\operatorname{Hom}(\Omega_C, \mathcal{O}_C) \xrightarrow{\varphi} \operatorname{Hom}(\Omega_C, \mathcal{O}_{p_1} \oplus \mathcal{O}_{p_2}) \xrightarrow{\phi} \operatorname{Ext}^1(\Omega_{\mathrm{C}}, \mathcal{O}_{\mathrm{C}}(-\mathrm{p}_1 - \mathrm{p}_2));$$

we will prove that

$$\operatorname{Hom}(\Omega_C, \mathcal{O}_{p_1} \oplus \mathcal{O}_{p_2})^{\mathrm{inv}} \subset \operatorname{im}\varphi,$$

which implies the statement. Recall that $\operatorname{Hom}(\Omega_C, \mathcal{O}_C)$ classifies the infinitesimal automorphism of C, namely the tangent space of the automorphism group of C, whereas φ restricts the automorphisms to the points. A simple verification using the explicit description of φ shows what we need, since any infinitesimal automorphism of C commutes with the hyperelliptic involution (because of [Per24, Section 3]). □

The following is a fundamental remark for understanding the proof of Θ-completeness in Section 6.

Remark 3.30. Let (C, p_1, p_2) be a 2-pointed atom of genus g and q be the singularity. The kernel of the morphism

$$\alpha : T^1\mathrm{Def}_{(C,p_1,p_2)} \longrightarrow T^1_q$$

considered in Proposition 3.29 has a very explicit description. In fact, $\ker\alpha$ is the subspace of first-order deformations of (C, p_1, p_2) where the A_{2g+1}-singularity q is preserved (possibly changing the crimping data). Therefore, it is the tangent space of the closed substack $\mathcal{D}^{2g+1} \subset \mathcal{M}_{g,2}$ at the point $[C, p_1, p_2]$. Proposition 3.29 implies that we have a decomposition

$$T_{[C,p_1,p_2]}\mathcal{M}_{g,2} = T_{[C,p_1,p_2]}\mathcal{H}_{g,g_1^2} \oplus T_{[C,p_1,p_2]}\mathcal{D}^{2g+1}$$

of tangent spaces, namely that $\mathcal{D}^{2g+1}$ and $\mathcal{H}_{g,g_1^2}$ intersect transversely in $\mathcal{M}_{g,2}$ at the closed point (C, p_1, p_2). To ease the notation, we will denote this Cr_q, and we will call it the *crimping space* of the singularity q.

More generally, if C is a curve and q is a singularity, we have a decomposition

$$\operatorname{Ext}^1(\Omega_{\mathrm{C}}, \mathcal{O}_{\mathrm{C}}) \simeq \mathrm{T}^1_{\mathrm{q}} \oplus \mathrm{Cr}_{\mathrm{q}} \oplus \mathrm{R}$$

where

- $\mathrm{Cr}_q \oplus R \simeq \mathrm{H}^1(\mathcal{H}om(\Omega_C, \mathcal{O}_C))$;
- R can be identified with first-order deformations of the (pointed) partial normalization of C at q.

The sign decomposition of the $\mathbb{G}_m$-action on the deformation spaces of the atoms coincides with the transversal decomposition given by the hyperelliptic locus and the crimping data.

Corollary 3.31. *Let (C, p_1, p_2) be a 2-pointed atom of genus g and fix an isomorphism $\mathbb{G}_m \simeq \operatorname{Aut}(C, p_1, p_2)$. Then the decomposition introduced in Remark 3.30 coincides with the sign decomposition for the $\mathbb{G}_m$-action, i.e.*

$$(T_{[C,p_1,p_2]}\mathcal{M}_{g,2})^{>} = T_{[C,p_1,p_2]}\mathcal{H}_{g,g_1^2}$$

and

$$(T_{[C,p_1,p_2]}\mathcal{M}_{g,2})^{<} = T_{[C,p_1,p_2]}\mathcal{D}^{2g+1}$$

or viceversa. The same is true for the even case.

Proof. After unraveling definitions, this is a direct consequence of Proposition 6.2 and Proposition 6.3 of [ASvdW10]. □

Now that we have dealt with the case of the atoms, we can focus on understanding the deformation theory of the curves with positive dimensional automorphism group, specifically the subcurves that contributes to its dimension (see Proposition 3.22).

Example 3.32. Let C_0 be a n-pointed rosary of hyperelliptic A-singularities of length 2 and denote by q_1, q_2 the two singularities of type A_{2h_1+1} and A_{2h_2+1} respectively. Moreover, suppose there is an isotrivial degeneration $C_\Theta \to \Theta$ whose special fiber is isomorphic to C_0. Then the combinatorial structure of the generic fiber C_1 is controlled by the one of C_0.

In fact, we have a decomposition

$$\mathrm{Def}_C = T^1_{q_1} \oplus T^1_{q_2} \oplus \mathrm{Cr}_{q_1} \oplus \mathrm{Cr}_{q_2}$$

of the deformation space of C thanks to Remark 3.30. Furthermore, the isotrivial degeneration induces a cocharacter $\mathbb{G}_m \to \mathrm{Aut}(C_0) \simeq \mathbb{G}_m$ which also induces a sign decomposition of Def_C. Because of the explicit description of the action of $\mathrm{Aut}^\circ(C_0)$ on C_0, we have that the $\mathbb{G}_m$-action on the tangent space of q_1 has opposite signs compared to the one on the tangent space of q_2. Therefore, Corollary 3.31 implies that

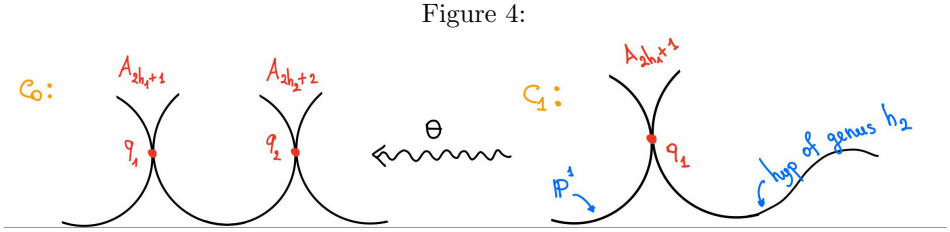
$$(\mathrm{Def}_C)^> = T^1_{q_1} \oplus \mathrm{Cr}_{q_2}$$

and

$$(\mathrm{Def}_C)^< = T^1_{q_2} \oplus \mathrm{Cr}_{q_1}$$

or viceversa. Therefore, Lemma A.2 ensures that if the isotrivial degeneration deforms one singularity, then the other one needs to be preserved (with a possibly different crimping datum) in the generic fiber C_1. Moreover, by Proposition 3.29, the deformation of the odd atom of genus h_2 in the chain will be an (honestly) hyperelliptic curve of genus h_2.

Figure 4:



The same reasoning clearly works if the length of the chain is greater than 2. Suppose that C_0 is a rosary of hyperelliptic A-singularities $q_1, \ldots, q_n$ where q_i and q_{i+1} share one irreducible component for every $i = 1, \ldots, n-1$. Moreover, suppose that the isotrivial degeneration deforms q_{i_0} where i_0 is odd. Then the singularities $q_2, q_4, \ldots, q_{2k}$ for $k = \lfloor n/2 \rfloor$ will also appear in the generic fiber C_1 of the isotrivial degeneration (possibly with a different crimping datum), while the odd atom of genus h_{2i+1} associated to the singularity q_{2i+1} will deform to an (honestly) hyperelliptic curve of genus h_{2i+1} where the two attaching points are exchanged by the involution.

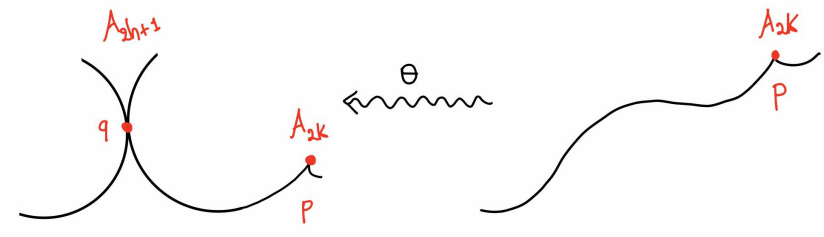
Similar examples can be constructed starting with a 1-pointed odd atom of genus h with singularity q and pinching the marking p into an A_{2k}-singularity using the unique $\mathbb{G}_m$-equivariant crimping datum.

The automorphism group will be a one-dimensional torus, see Proposition 3.22. We have a decomposition

$$\mathrm{Def}_C = T^1_q \oplus T^1_p \oplus \mathrm{Cr_q} \oplus \mathrm{Cr}_p$$

and again, an isotrivial degeneration can either deform the singularity q while preserving p (but not necessarily the crimping datum) or viceversa.

Figure 5:



Before going further, it is crucial to point out that the following remark is a key point in our study of isotrivial degenerations of A-stable curves. Indeed, it gives a significant constraint to the geometry of isotrivial degenerations and it will play a fundamental role in all what follows.

Remark 3.33. Example 3.32 determines a big constraint in the geometry of isotrivial degenerations: whenever we have a rosary of hyperelliptic A-singularities as above, any isotrivial degeneration cannot deform two adiacent A-singularities. If one is deformed, then the ones adiancent to it has to be preserved, and the curve between them is hyperelliptic.

This is true also for nodes, with the extra hypothesis that the $\mathbb{G}_m$-action is trivial on one of the two components meeting in the node. Namely, given a curve C_0 with a subcurve $\Gamma \subset C$ such that (Γ, I_Γ) is an A_1-attached 1-pointed even atom (or equivalently an A_1/A_{2h}-attached rosary of length 1) and suppose we have an isotrivial degeneration to C_0, such that the cocharacter $\mathbb{G}_m \to \mathrm{Aut}(C_0)$ acts trivially on the component $\Gamma' \subset C_0 - \Gamma$ that intersects Γ in the node. Then if the singularity of the atom is deformed, the node has to be preserved and thanks to Proposition 3.29, Γ deforms to an (honestly) hyperelliptic tail attached to the rest of the curve with a node.

The previous example motivates the following definitions.

Definition 3.34 (Tails and bridges)**.** Let $(C, p_1, \dots, p_n)$ be a n-pointed A-prestable curve over an algebraically closed field k and let $\Gamma \subset C$ be a connected subcurve.

- we say that Γ is an A_{2k+1}-attached hyperelliptic tail if Γ is an hyperelliptic curve of positive genus, $P_\Gamma = \emptyset$ and I_Γ consists of a single A_{2k+1}-singularity on C which corresponds to a smooth Weierstrass point on Γ (i.e a point fixed by the hyperelliptic involution of Γ); we say that it is A_0-attached if $I_\Gamma = \emptyset$ (thus $\Gamma = C$) and P_Γ consists of a single point (thus $n = 1$);
- we say that Γ is a A_{2h+1}/A_{2k+1}-attached hyperelliptic bridge if Γ is an hyperelliptic curve of positive genus, $P_\Gamma = \emptyset$ and I_Γ consists of a A_{2h+1} singularity and a A_{2k+1} singularity on C which correspond to two smooth points on Γ exchanged by the involution.

Definition 3.35 ((Attached) Hyperelliptic Chain)**.** Let k be an algebraically closed field/κ, we say that (Γ, r_1, r_2) over k is a 2-*pointed hyperelliptic chain of genus bounded by g of length ℓ* if there exists a surjective gluing morphism

$$\gamma\colon \coprod_{i=1}^{\ell} (\Gamma_i, q_{2i-1}, q_{2i}) \longrightarrow (\Gamma, r_1, r_2)$$

and a sequence of positive integers $k_1, \dots, k_{\ell-1}$ satisfying:

1. Γ_i is an hyperelliptic curve of genus g_i with $1 \leq g_i < g$, for $i = 1, \dots, \ell$;
2. the marked points q_{2i-1}, q_{2i} are exchanged by the hyperelliptic involution of Γ_i, for $i = 1, \dots, \ell$;

3. $\gamma(q_{2i}) = \gamma(q_{2i+1})$ is an A_{2k_i+1}-type singularity of Γ for $i = 1, \dots, \ell - 1$;
4. $\gamma(q_1) = r_1$ and $\gamma(q_{2\ell}) = r_2$.

If the curves Γ_i are honestly hyperelliptic, we say that (Γ, p_1, p_2) is a chain of honestly hyperelliptic curves. We say that $(C, \{p_i\}_{i=1}^n)$ has an A_{d_1}/A_{d_2}*-attached hyperelliptic chain of genus bounded by g of length ℓ* if there exists a gluing morphism

$$\gamma\colon (\Gamma, r_1, r_2) \longrightarrow (C, \{p_i\}_{i=1}^n)$$

such that

1. (Γ, r_1, r_2) is a 2-pointed hyperelliptic chain of genus bounded by g of length ℓ;
2. for $j = 1, 2$, $\gamma(r_j)$ is an A_{d_j}-singularity of C, or if $d_j = 0$ we allow $\gamma(r_j)$ to be a marked point of $(C, \{p_i\}_{i=1}^n)$.

We say that an attached hyperelliptic chain (Γ, r_1, r_2) is separating if the curve $C - \gamma(\Gamma)$ is disconnected, otherwise we say it is non-separating.

We will deal firstly with the specific case of 2-pointed rosaries of hyperelliptic A-singularities, as it is the most frequent case we need to deal with in what follows. We leave it to the interested reader to use the same exact technique to study the other cases.

Proposition 3.36. *Let (C_0, p_1^0, p_2^0) be a 2-pointed rosary of hyperelliptic (odd) A-singularity and (C_1, p_1^1, p_2^1) be a 2-pointed A_r-stable curve. If (C_1, p_1^1, p_2^1) admits an isotrivial degeneration to the curve (C_0, p_1^0, p_2^0) then it is one of the curves described below, depending on the parity of the lenght of the rational chain. Suppose that length of the rational chain is odd, then (C_1, p_1^1, p_2^1) is either*

o1) a 2-pointed chain of (honestly) hyperelliptic curves;

o2) an A_{2h+1}/A_{2k+1}-attached chain of (honestly) hyperelliptic curves, which is attached to two rational 1-pointed smooth curves (see the picture below);

moreover, if the length is even, we have that

e) (C_1, p_1^1, p_2^1) is a 1-pointed chain of (honestly) hyperelliptic curves attached to a 1-pointed rational smooth curve with an odd A-singularity (see the picture below).

Proof. It follows directly from Proposition 3.29 and Example 3.32. Indeed, suppose the length s of the chain is even, i.e. $s = 2t$. Then we can decompose the deformation space of (C_0, p_1^0, p_2^0) as follows:

$$\mathrm{Def}_{(C_0, p_1^0, p_2^0)} \simeq \bigoplus_{i=1}^{s} (T^1_{q_i} \oplus \mathrm{Cr}_{q_i})$$

where q_i is the i-th (odd) A-singularity. The same computations carried out in Example 3.32 gives us that

$$(\mathrm{Def}_{(C_0, p_1^0, p_2^0)})^{>} = \bigoplus_{i=1}^{t} (T^1_{q_{2i-1}} \oplus \mathrm{Cr}_{q_{2i}})$$

and

$$(\mathrm{Def}_{(C_0, p_1^0, p_2^0)})^{<} = \bigoplus_{i=1}^{t} (T^1_{q_{2i}} \oplus \mathrm{Cr}_{q_{2i-1}})$$

or viceversa. Lemma A.2 gives us that any isotrivial degeneration to (C_0, p_1^0, p_2^0) will be contained in one of the two spaces above. The statement follows from Proposition 3.29. The computations for the odd case are identical. □

Finally, we note that the same statement works for rosaries.

Definition 3.37. Let C be an A-prestable curve over an algebraically closed field k. We say that C is a *closed chain of hyperelliptic curves* of genus bounded by g if there exists a A_{2k+1}-singularity with $k \geq 1$ such that the (pointed) normalization $(\widetilde{C}, p_1, p_2)$ is a 2-pointed chain of hyperelliptic curves of genus bounded by g.

We leave the proof of the next proposition to the interested reader. It directly follows from Proposition 3.43.

Proposition 3.38. *Let (C_0, p_1^0, p_2^0) be a closed rosary of hyperelliptic A-singularities (of even length) and (C_1, p_1^1, p_2^1) be an A_r-stable curve which admits an isotrivial degeneration to C_0. Then (C_1, p_1^1, p_2^1) is a closed chain of hyperelliptic curves.*

3.4 Isotrivial specializations

We end the section by globalizing some of the results appearing in Section 3.3. To be more specific, in the previous section we described all the possible isotrivial degenerations to an atom, giving a local description of the Θ-attractor. In this section, we are going to make the description global, completely classifying the possible isotrivial degenerations for A_r-stable curves.

We start with the atomic case. For the sake of simplicity, we deal with the case $n = 0$, but the same proof holds also in the more general setting. Specifically, the next lemma shows us that every curve with an even A-singularity has an isotrivial degeneration to the one glued nodally to the respective even atom.

Lemma 3.39. *Let $(C, q) \in \mathcal{A}_{2h}$ be an A_r-stable curve of genus g with an A_{2h}-singularity q over a field and denote by $(\widetilde{C}, \widetilde{q})$ the stabilization of the (pointed) partial normalization of C at q. Moreover, let C_0 be the A_r-stable curve obtained gluing nodally $(\widetilde{C}, \widetilde{q})$ with the 1-pointed even atom $(\Gamma, \widetilde{q})$ of genus h in $\widetilde{q}$. Then there exists an isotrivial degeneration $C_\Theta \to \Theta$ such that*

- *the generic fiber of the degeneration is isomorphic to C;*
- *the special fiber of the degeneration is isomorphic to C_0.*

Proof. Consider the stabilization morphism (see Definition 2.16)

$$\mathcal{M}^r_{g-h,[h]} \simeq \mathcal{M}^r_{g-h,1} \times [\mathbb{A}^1/\mathbb{G}_m] \longrightarrow \mathcal{M}^r_{g-h,1};$$

the fiber over $[(\widetilde{C}, \widetilde{q})] \in \mathcal{M}^r_{g-h,1}$ is isomorphic to $[\mathbb{A}^1/\mathbb{G}_m]$. More specifically, the generic point of $[\mathbb{A}^1/\mathbb{G}_m]$ corresponds to the case when the (pointed) partial normalization of C at q is already stable, whereas the special point can be interpreted as the curve obtained by gluing a projective line $(\mathbb{P}^1, 0, \infty)$ with $(\widetilde{C}, \widetilde{q})$ using the identification $0 \equiv \widetilde{q}$. The group $\mathbb{G}_m$ corresponds to the automorphisms of $(\mathbb{P}^1, 0, \infty)$.

In the notation of Remark 2.15, we have that the moduli stack $\mathcal{A}_{2h}(\widetilde{C}, \widetilde{q})$ parametrizing curves in $\mathcal{A}_{2h}$ whose stabilization of the partial normalization at the even A-singularity is $(\widetilde{C}, \widetilde{q})$ is isomorphic to $[V_{h,2} \times \mathbb{A}^1/\mathbb{G}_m]$. By definition, (C, q) belongs to $\mathcal{A}_{2h}(\widetilde{C}, \widetilde{q})$. Unraveling the constructions, it is easy to see that $V_{h,2} \times \mathbb{A}^1$ is a $\mathbb{G}_m$-representation with strictly positive (or negative) weights, thus every curve in $\mathcal{A}_{2h}(\widetilde{C}, \widetilde{q})$ admits an isotrivial degeneration to the 0-section of $V_{h,2} \times \mathbb{A}^1$, which is represented by the curve C_0. □

We proceed by doing the same for the other possible attractor of the even atom, namely curves with hyperelliptic tails.

Lemma 3.40. *Suppose r is an even positive integer. Let C be an A_r-stable curve of genus g over some field k and $\Gamma \subset C$ be an A_1-attached honestly hyperelliptic tail of genus $h := r/2$. We denote by $\widetilde{C}$ the subcurve $C - \Gamma$ and by $\widetilde{q}$ the intersection $\Gamma \cap C - \Gamma$. Moreover, let C_0 be the A_r-stable curve obtained gluing nodally $(\widetilde{C}, \widetilde{q})$ with the 1-pointed even atom $(\Gamma, \widetilde{q})$ of genus h in $\widetilde{q}$. Then there exists an isotrivial degeneration $C_\Theta \to \Theta$ such that*

- *the generic fiber of the degeneration is isomorphic to C;*
- *the special fiber of the degeneration is isomorphic to C_0.*

Proof. Notice that there is a morphism

$$\mathcal{H}^r_{h,w} \times \operatorname{Spec} k \to \mathcal{M}^r_{g,n} \times \operatorname{Spec} k$$

that sends every curve $(H, \widetilde{q}) \in \mathcal{H}^r_{h,w} \times \operatorname{Spec} k$ to the curve $H \cup_{\widetilde{q}} \widetilde{C}$. The result follows using Proposition 2.52. □

Similarly to the even case, we describe the basin of attraction for odd A-singularities. As the proof is similar to the one of Lemma 3.39, we avoid write it down here.

Lemma 3.41. *Let $(C, q) \in \mathcal{A}_{2h+1}$ be an A_r-stable curve of genus g with an A_{2h+1}-singularity q over a field and denote by $(\widetilde{C}, \widetilde{q}_1, \widetilde{q}_2)$ the stabilization of the (pointed) partial normalization of C at q. Moreover, let C_0 be the A_r-stable curve obtained gluing nodally $(\widetilde{C}, \widetilde{q}_1, \widetilde{q}_2)$ with the 2-pointed odd atom $(\Gamma, \widetilde{q}_1, \widetilde{q}_2)$ of genus h in $\widetilde{q}_1, \widetilde{q}_2$. Then there exists an isotrivial degeneration $C_\Theta \to \Theta$ such that*

- *the generic fiber of the degeneration is isomorphic to C;*
- *the special fiber of the degeneration is isomorphic to C_0.*

We also describe one of the attractors for a A_1/A_{2h}-attached rosary of hyperelliptic A-singularities. We do it for chains of length less or equal than two, but the idea generalizes trivially.

Lemma 3.42. *Suppose r is an even positive integer. Let C be an A_r-stable curve of genus g over some field k and $\Gamma \subset C$ be an $A_{2h'+1}$-attached (honestly) hyperelliptic tail of genus $h := r/2$ for $h \geq 1$. We denote by $\widetilde{C}$ the subcurve $C - \Gamma$ and by $\widetilde{q}$ the intersection $\Gamma \cap C - \Gamma$. Moreover, let C_0 be the A_r-stable curve obtained gluing nodally $(\widetilde{C}, \widetilde{q})$ with the 1-pointed curve $(D, \widetilde{q})$ of genus $h + h'$ obtained by pinching the 2-pointed odd atom $(\widetilde{D}, \widetilde{q}, q_2)$ of genus h' in q_2 using the hyperelliptic crimping datum (see Figure 5).*

Then there exists an isotrivial degeneration $C_\Theta \to \Theta$ such that

- *the generic fiber of the degeneration is isomorphic to C;*
- *the special fiber of the degeneration is isomorphic to C_0.*

Proof. The case $h' = 0$ was treated in Lemma 3.40. A bit more convoluted is the case $h' \geq 1$. Consider the following diagram

$$\begin{tikzcd}
\mathcal{P} \arrow[r] \arrow[d] & \mathcal{A}^h_{2h'+1} \arrow[r] \arrow[d] & \mathcal{M}^r_{g,n} \\
(\operatorname{Spec} k \times [\mathbb{A}^1/\mathbb{G}_m]) \times \mathcal{H}^{r,\circ}_{h,w} \arrow[r] & (\mathcal{M}^r_{g-k-1,1} \times [\mathbb{A}^1/\mathbb{G}_m]) \times \mathcal{M}^r_{h,[k]} &
\end{tikzcd}$$

where the left-hand square is cartesian and the morphism $\operatorname{Spec} k \to \mathcal{M}^r_{g-k-1,1}$ is induced by $(\widetilde{C}, \widetilde{q})$. Notice that $\mathcal{H}^{r,\circ}_{h,w}$ is a locally closed substack of $\mathcal{M}^r_{h,[k]} \simeq \mathcal{M}^r_{h,1} \times [\mathbb{A}^1/\mathbb{G}_m]$.

By construction, $\mathcal{P}$ parametrizes curves over k constructed attaching an honestly hyperelliptic tail of genus h to $\widetilde{C}$ with an $A_{2h'+1}$-singularity. The factor $[\mathbb{A}^1/\mathbb{G}_m]$ accounts for the possibility of gluing the hyperelliptic tail directly in $\widetilde{q}$ or to an unstable projective line passing through $\widetilde{q}$, and it can be identified with the (stacky) deformation space of the node in $\widetilde{q}$ when the unstable $\mathbb{P}^1$ appears. For a more detailed explanation, see Remark 2.23. Since $r = 2h$, Proposition 2.52 gives that $\mathcal{H}^{r,\circ}_{h,w}$ is the global quotient $[\mathbb{A}^{2h-1}/\mathbb{G}_m]$ and it can be identified with the deformation space of the (hyperelliptic) A_{2h}-singularity, by Proposition 3.29. Thus, Proposition 2.25 implies that $\mathcal{P}$ is the quotient stack $[\mathbb{A}^1 \times \mathbb{A}^{2h-1} \times C_{h'+1}/\mathbb{G}_m]$, where $C_{h'+1} := (U_{h'+1}/\mathbb{G}_{\mathrm{a}})$ is a $\mathbb{G}_m$-representation of dimension h' which accounts for the crimping data of the $A_{2h'+1}$-singularity (see also Remark 3.6). The zero of the representation $\mathbb{A}^1 \times \mathbb{A}^{2h-1} \times C_{h'+1}$ corresponds to the curve $(D, \widetilde{q})$ described in the statement of the lemma. Thus, the statement follows if we show that $\mathbb{G}_m$ acts with strictly positive (or negative) weights on the whole representation. Notice that the three representations $\mathbb{A}^1$, $\mathbb{A}^{2h-1}$ and $C_{h'+1}$ either have strictly positive or strictly negative weights.

By Corollary 3.31, it is clear that the weights of $\mathbb{A}^{2h-1}$ and $C_{h'+1}$ have the same signs: namely, if we set the convention that $\mathbb{G}_m$ acts positively on the A_{2h}-singularity, then they both have strictly positive weights. Since $\mathbb{A}^1$ can be identified with the deformation of the node created by gluing $(\widetilde{C}, \widetilde{q})$ and $(D, \widetilde{q})$, the weight of the $\mathbb{G}_m$-representation $\mathbb{A}^1$ has the same sign of the weights of the other two $\mathbb{G}_m$-representations (see for instance Example 3.32 or Remark 3.33), finally proving the result. □

Finally, we describe the attractor in the case of a 2-pointed rosaries of hyperelliptic A-singularities. This is the inverse of Proposition 3.36. We avoid writing down the proof as it is not particularly enlighting, but it follows by applying the same strategy adopted for Lemma 3.42.

Proposition 3.43. *Let (C_1, p_1^1, p_2^1) be a 2-pointed A_r-stable curve. Then (C_1, p_1^1, p_2^1) admits an isotrivial degeneration to a 2-pointed rosary of hyperelliptic A-singularities if it is one of the curves described below, depending on the parity of the length of the rosary. Suppose that length of the rosary is odd, then (C_1, p_1^1, p_2^1) is either*

- *o1) a 2-pointed chain of (honestly) hyperelliptic curves;*
- *o2) an A_{2h+1}/A_{2k+1}-attached chain of (honestly) hyperelliptic curves, which is attached to two rational 1-pointed smooth curves (see the picture below);*

moreover, if the length is even, we have that

- *e) (C_1, p_1^1, p_2^1) is a 1-pointed chain of (honestly) hyperelliptic curves attached by an odd A-singularity to a 1-pointed rational smooth curve (see the picture below).*

Figure 6:

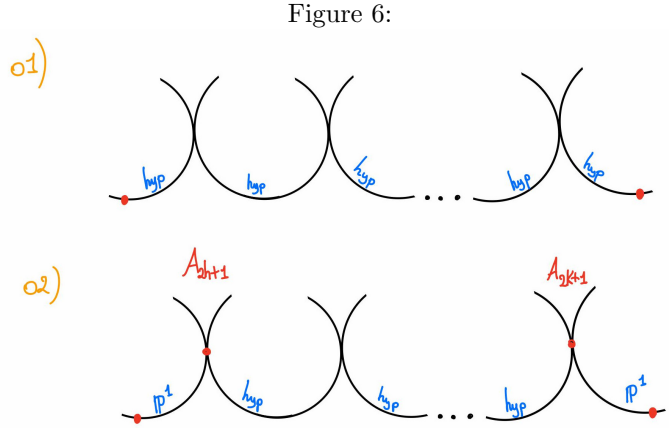
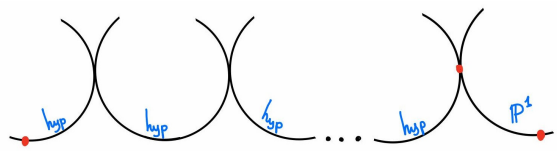


Figure 7:

e)



Remark 3.44. Notice that a rosary of hyperelliptic A-singularities of odd length is a chain of hyperelliptic curves.

4 Counterexamples and the stack of admissible curves

This section is dedicated to studying how Θ-completeness and S-completeness fail to hold for the moduli stack $\mathcal{M}^r_{g,n}$. Moreover, analyzing the possible counterexamples to Θ-completeness and S-completeness, we define an open $\mathcal{U}^r_{g,n}$ of $\mathcal{M}^r_{g,n}$ which we conjecture has a separated good moduli space. In the rest of the paper, we prove that $\mathcal{U}^r_{g,n}$ is Θ-complete for all r and that it is S-complete for $r \leq 5$. Moreover, we prove that such good moduli space (when it exists) does not have ample line bundles if $r \geq 5$, thus giving an example (specifically for $r = 5$) where GIT cannot be applied.

Our modus operandi in the search of $\mathcal{U}^r_{g,n}$ will be the following:

(1) we find counterexamples for Θ-completeness;

(2) we decide what to remove from the stack so that the counterexample cannot appear, preferably avoiding removing curves with positive dimensional automorphisms group;

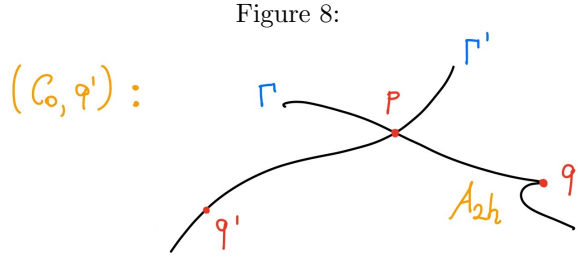
(3) if we remove curves with positive dimensional automorphisms group, we check S-completeness to see if one of the basins of attraction of the curve we removed have to be removed as well;

Remark 4.1. We are interested in finding an open which admits a separated good moduli space but it is as "big" as possible. This implies that the separated good moduli space will be as "small" as possible. This is one of the reason why it does not come from (global) GIT and we prove that it is an algebraic space without non-trivial line bundles (when it exists).

Before going to the counterexamples, we introduce the conditions on A_r-stable curves in Definition 4.4, depending on the integer r. Then we explain why the conditions we give are needed for the existence of the separated good moduli space, referring to the strategy above.

Remark 4.2. One may wonder whether the stack $\mathcal{M}^r_{g,n}$ is S-complete, as the strategy outlined seems to force us to remove closed subset always starting by the failure of Θ-completeness. This is not the case. Let (Γ, p) be a 1-pointed even atom of genus h with singularity q and (Γ', p, q') be a 2-pointed smooth hyperelliptic curve of genus $k \geq 1$ such that p and q' are both Weierstrass points. We denote by (C_0, q') the 1-pointed A_r-stable curve obtained by gluing nodally (Γ, p) with (Γ', p, q') in p.

Figure 8:

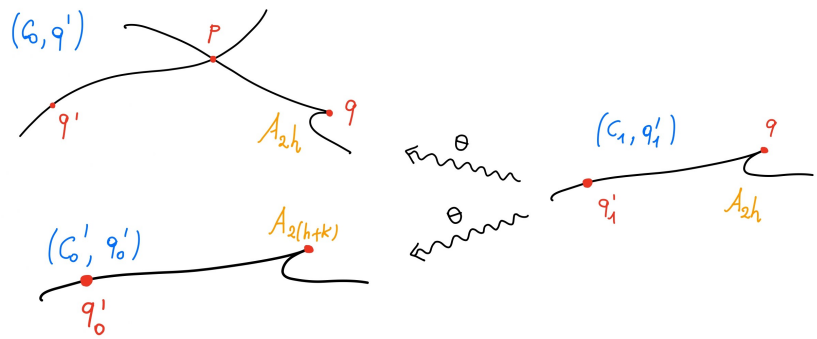


We have $\mathrm{Aut}^\circ(C_0, q') \simeq \mathbb{G}_m$. Moreover, we have a decomposition

$$\mathrm{Def}_{(C,p)} \simeq T^1_q \oplus \mathrm{Cr}_q \oplus T^1_p \oplus \mathrm{Def}_{(\Gamma',p,q')}$$

where the $\mathbb{G}_m$ acts positively on T^1_q, negatively on $\mathrm{Cr}_q \oplus T^1_p$ and trivially on $\mathrm{Def}_{(\Gamma',p,q')}$. Consider the isotrivial degeneration given by the line $(0 \oplus 0 \oplus T^1_p \oplus 0) \subset \mathrm{Def}_{(C,p)}$, which corresponds to the degeneration which deforms the node p, but fixes the singularity q and its crimping datum. The generic point of the degeneration corresponds to an honestly hyperelliptic curve (C_1, q'_1) of genus $h+k$ with an A_{2h}-singularity and a Weierstrass marking q'_1, therefore an object of $\mathcal{H}^{r,\circ}_{h+k,w}$ with $r \geq 2h$. Proposition 2.52 implies that there exists an isotrivial degeneration with generic point (C_1, q'_1) and with special point (C'_0, q'_0), the 1-pointed even atom of genus $h+k$, which is a closed point.

Figure 9:



If we apply Proposition 3.13 to the separating node of (C_0, q'), one can prove that there are no degenerations from (C_0, q') to (C_0, q_0'). This contradicts Θ-completeness and S-completeness: more specifically, it contradicts the filling criterion for morphisms from the open embedding $\Theta^2 \setminus 0 \subset \Theta^2$, which is a weaker condition of both S-complete and Θ-complete. This weaker valuative criterion is the reason why the fibers of the good moduli space morphism always have a unique closed point.

Remark 4.3. In this remark, we will use definitions/notations as they are going to appear in a future work by David Rydh, where the author introduces the notion of *topological moduli space*. The Θ^2-filling condition can be regarded as a valuative criterion of N-separatedness (for locally reductive stacks). We conjecture that this is the only condition in the existence result for topological moduli space that we are missing, i.e. $\mathcal{M}^r_{g,n}$ is M-separated and N-closed, but not N-separated. In Section 7.2, we discuss a valuative criterion for M-separatedness.

Definition 4.4 (The stack $\mathcal{U}^r_{g,n}$ of admissible curves)**.** Given r, g, n non-negative integers, let C be an n-pointed A_r-stable curve of genus g over an algebraically closed field k. We list some conditions on C.

- (t_r) C does not contain A_0-attached or A_{2k+1}-attached hyperelliptic tails of genus $1 \leq h < r/2$ for any $k \geq 0$ (Definition 3.34);
- (c_r) C does not contain non-separating A_1/A_1-attached hyperelliptic chains of genus bounded by $(r-1)/2$ (Definition 3.35);
- $(d_{r,1})$ C does not contain non-separating A_1/A_r-attached hyperelliptic chains of genus bounded by $(r-1)/2$;
- $(d_{r,2})$ C does not contain non-separating A_r/A_r-attached hyperelliptic chains of genus bounded by $(r-1)/2$;
- (e_r) C is not an A_h-self-attached hyperelliptic curve with $h > g$ (see Definition 4.21).

We will often refer to conditions $(d_{r,1}) + (d_{r,2})$ as condition (d_r).

If an A_r-stable curve verifies conditions (t_r), (c_r) and (d_r) and (e_r), we call it an *admissible* curve. We denote by $\mathcal{U}^r_{g,n}$ the moduli stack classifying n-pointed admissible curves of genus g, unless $(g, n) = (1, 1)$ in this case we set $\mathcal{U}^r_{1,1} := \mathcal{M}^r_{1,1} (= \mathcal{M}^2_{1,1})$ for all $r \geq 2$ (see Remark 4.5).

Remark 4.5. We want to remark that condition (e_r) only appears for $n = 0$ and $r > g$. Moreover, notice that if C has a A_0-attached hyperelliptic tail Γ, it means that $\Gamma = C$, $n = 1$ and thus if C does not verify (t_r), then $g < r/2$, which implies $r \geq 2g + 1$. Furthermore, we should mention that if $(g, n) = (1, 1)$, condition (t_r) would imply $\mathcal{U}^3_{1,1} = \emptyset$. This is a pathology which only happens in this case, as Example 4.20 creates a counterexamples as soon as $g \geq 2$.

Remark 4.6. As the reader may notice, Definition 4.4 has strong resemblance with the definition of α_c-stable curves appearing in the Hassett-Keel program (see [AFSvdW17]). Since the last explicit step of the program uses A_h-singularities with $h \leq 4$, we can compare the two definitions for $r \leq 4$.

If $r = 0$ (respectively $r = 1$), the two definitions coincide and they are just the usual definition of smooth (respectively stable) curves. If $r = 2$, then our open corresponds to $\alpha = 9/11$-stable curves [AFSvdW17, Definition 2.5]. If $r = 3$, our open corresponds to $\alpha = 7/10$-stable curves. Indeed conditions (c_r) and (d_r) are empty for $r \leq 3$. Condition (t_r) is empty for $r \leq 2$ and condition (t_3) states that there are no A_0, A_1 and A_3-attached elliptic tails. Notice that if an A_r-stable curve has an A_0-attached elliptic tail, then the curve itself is a 1-pointed genus 1 curve, which is in $\mathcal{M}^r_{1,1}$ (see the discussion in Remark 4.5). Moreover, condition (e_r) is empty if $r \leq 2$, and (e_3) states there C is not an A_3-self-attached hyperelliptic curve of genus $g < 3$. Nevertheless, the definition of α-stable curves is stated in genus ≥ 3 (see [AFSvdW17, Remark 1.1]).

If $r = 4$, then every 2/3-stable curve is actually admissible, but the contrary is not true, as we now show. Firstly, let us recall the definition in the 2/3-stable case: they do not allow

(1) A_1, A_3, A_4-attached elliptic tails;

(2) $A_1/A_1, A_1/A_4, A_4/A_4$-attached elliptic chains;

Condition (t_4) forces us to remove A_1, A_3-attached elliptic tails. Condition (d_4) is trivial, whereas condition (c_r) forces us to remove A_1/A_1-attached *non-separating* elliptic chains. Moreover, condition (e_r) forces us to remove A_4-attached elliptic tails (and nothing else in genus ≥ 3). For a more detailed discussion on our conditions and how they compare to the Hassett-Keel program, see the next three subsections.

Lemma 4.7. *If $r \geq 2g - 1 + n$, then the intersection $\mathcal{H}^r_{g,n} \cap \mathcal{U}^r_{g,n}$ consists of $2:1$-coverings of $\mathbb{P}^1$.*

Proof. First of all, we know that every hyperelliptic curve in $\mathcal{H}^r_{g,n}$ is a curve C with an involution σ and n markings. Recall that the quotient $Z := C/\sigma$ is a genus 0 nodal curve, thus a tree of $\mathbb{P}^1$'s. We want to prove that the quotient Z has only one irreducible component. If Z is not irreducible, we can consider a leaf of our tree of rational curves, namely a component Γ which intersects the rest of the curve $Z - \Gamma$ in one unique point q. Notice that the fiber $\pi^{-1}(q)$ is either a node (which correspond to a Weierstrass point in $\pi^{-1}(\Gamma)$), a tacnode (which correspond to a Weierstrass point in $\pi^{-1}(\Gamma)$) or two nodes (which corresponds to a pair of points in $\pi^{-1}(\Gamma)$ exchanged by the involution). See Proposition 3.2 of [Per24]. Thus $\pi^{-1}(\Gamma)$ is either a non-separating A_1/A_1-attached bridge of genus $h \leq g - 1$ or a A_1 or A_3-attached tail of genus $h \leq g$. Because $r \geq 2g - 1$, condition (t_r) and (c_r) imply that $h = g$, $r \leq 2g$ (thus $n \leq 1$) and Γ is attached to $C - \Gamma$ with a simple node. Finally, the stability condition gives us the contradiction. □

Remark 4.8. Notice that in the previous proof, we have only used condition (t_r) for A_1 and A_3-attached hyperelliptic tails and condition (c_r).

Before discussing the examples that brought us to the conditions of Definition 4.4, it is important to remark that almost none of the conditions have to do anything with the markings, meaning that, except for the A_0-attached tails, no marking lives on the problematic subcurves. Therefore, we will construct almost all examples assuming $n = 0$. They generalize straightforwardly for any number of marked points.

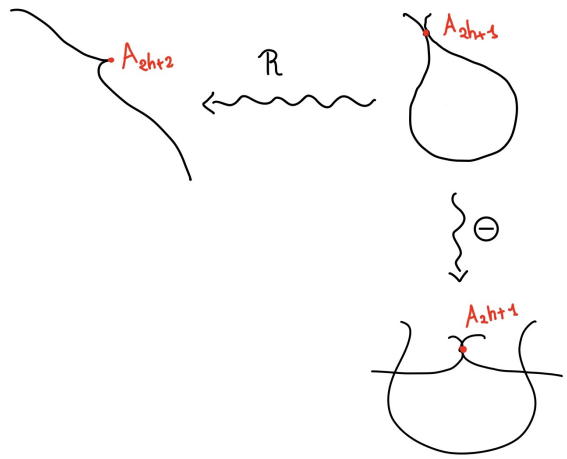
4.1 Attached hyperelliptic chains

Firstly, we explain how conditions (c_r) and (d_r) come up. We start by treating the example of the odd *atom*.

Example 4.9. Let R be a DVR with fraction field Q and $C \to \operatorname{Spec} R$ be any family of A_r-stable curves over R such that the generic fiber C_Q has a *non-separating* A_{2h+1}-singularity which degenerates to an A_k-singularity for $r \geq k > 2h+1$. Such families over $\operatorname{Spec} R$ are "generic", i.e. if we allow non-separating singularities to appear, then such a family exists. Indeed, if the A_{2h+1}-singularity is inner, i.e. it is contained in one irreducible component, and such component has positive geometric genus, then it can always degenerate to an A_k-singularity. It is easy to see that any curve with a non-separating A_{2h+1}-singularity is in the closure of the locus parametrizing curves with an inner A_{2h+1}-singularity.

Thanks to Lemma 3.41, we can construct an isotrivial degeneration $C_\Theta \to \Theta_Q$ such that the generic fiber coincides with C_Q and the A_{2h+1}-singularity of C_Q degenerates to an odd atom of genus h attached to the rest of the curve with two nodes.

Figure 10:



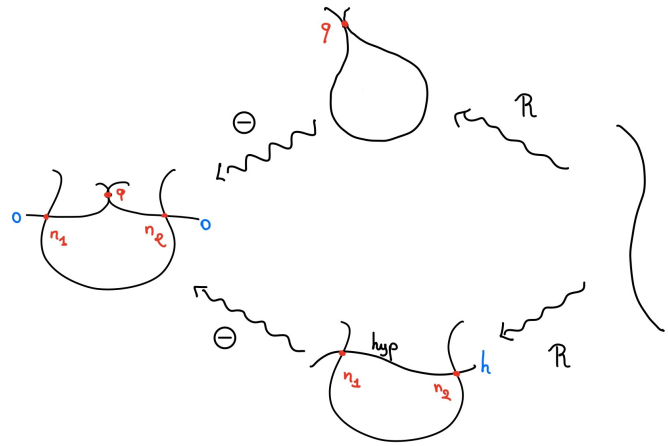
This is equivalent to a morphism $\Theta_R \setminus 0 \to \mathcal{M}_g^r$ which does not extend to a morphism from Θ_R because of Proposition 3.13. Therefore we need to remove either curves with non-separating A_k-singularities with $k > 2h+1$ or with A_1/A_1-attached odd atoms of genus h. We decide to investigate the second case, as in the first one we remove singularities that we feel are "too generic". A priori, a different choice could be made; for the tacnodal case, this is discussed in [Gor26, Sec. 3.8].

Up to quotienting by a finite group, the *stacky* deformation space is of the form

$$\operatorname{Def}_{(C-\Gamma, n_1, n_2)} \oplus [T_q^1 \oplus (\operatorname{Cr}_q \oplus T_{n_1}^1 \oplus T_{n_2}^1)/\mathbb{G}_m]$$

and, because of the previous counterexample to Θ-completeness, we need to remove the zero-section from the deformation space $T_q^1 \oplus (\operatorname{Cr}_q \oplus T_{n_1}^1 \oplus T_{n_2}^1)$. Thus, to get a separated good-moduli space (specifically to have S-completeness), we need to remove one of the two *basin of attractions*, i.e. either $T_q^1 \oplus (0 \oplus 0 \oplus 0)$ or $0 \oplus (\operatorname{Cr}_q \oplus T_{n_1}^1 \oplus T_{n_2}^1)$. As we choose to consider curves with non-separating A_r-singularities in our stack, this forces us to remove $T_q^1 \oplus (0 \oplus 0 \oplus 0)$ which corresponds to the locus where we deform the singularity q but keep the nodes n_1 and n_2 and we preserve the crimping datum. By Corollary 3.31, deforming the singularity gives us an hyperelliptic bridge of genus h attached nodally by two points exchanged by the involution, or equivalently a A_1/A_1-attached hyperelliptic bridge of genus h.

Figure 11:



Therefore the example tells us we are forced to remove non-separating A_1/A_1-attached hyperelliptic bridges of genus $h < (r-1)/2$.

Remark 4.10. We decide to keep the inner singularities for two reasons. Firstly, we believe that removing the inner singularities would give a stack whose geometry is fairly simple, and probably not as interesting as the other choice. Moreover, the strategy of removing the (non-separating) bridges follows closely the three known steps of the Hassett-Keel program and the understanding of future steps of this program isone of the motivations of this project. We believe it could be a nice exercise to understand if the techniques used in this paper applies also to the other choices we do not investigate.

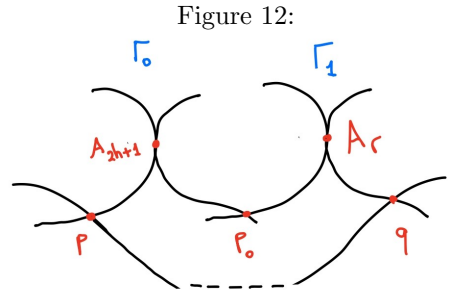
Example 4.9 explains why we need condition (c_r) in the case of chains of length 1, also called bridges. Notice that we can construct the family over $\Theta_R \setminus 0$ only in the non-separating case.

We now explain why condition $(d_{r,1})$ has to hold for hyperelliptic chains of length one, also called bridges.

Example 4.11. Suppose r is odd. Consider the curve C constructed in the following way:

- let (Γ_0, p, p_0) (respectively (Γ_1, q, p_0)) be an odd atom of genus h with $2h+1 < r$ (respectively of genus $(r-1)/2$);
- we denote by (Γ, p, q) the (nodal) gluing of (Γ_0, p, p_0) with (Γ_1, q, p_0) along p_0;
- we denote by C the curve obtained by (nodally) gluing (Γ, p, q) with a 2-pointed smooth curve (C_0, p, q) along p and q.

Figure 12:



We can describe the deformation space of C as

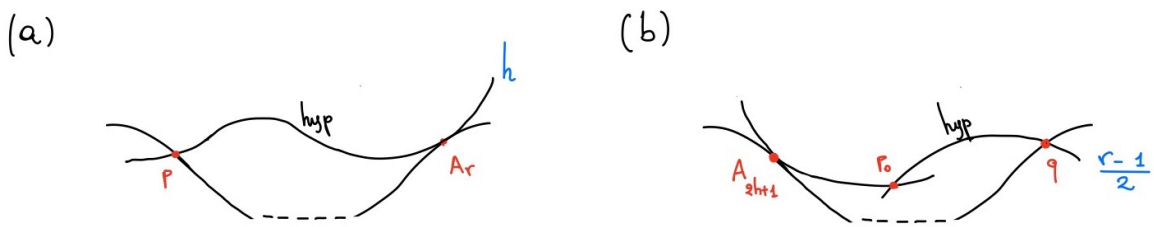
$$\mathrm{Def}_C = \mathrm{Def}_{(C_0,p,q)} \oplus T^1_p \oplus \mathrm{Def}_{(\Gamma_0,p,p_0)} \oplus T^1_{p_0} \oplus \mathrm{Def}_{(\Gamma_1,p_0,q)} \oplus T^1_q$$

and Corollary 3.25 gives us that $\mathrm{Aut}^\circ(C) \simeq \mathbb{G}_m^2$. Using Corollary 3.31, one can find a cocharacter $\mathbb{G}_m \to \mathbb{G}_m^2$ such that

(a) $(\mathrm{Def}_C)^>$ classifies deformations where C_0 is fixed, as well as the node p and the A_r-singularity, while the other singularities deform;

(b) $(\mathrm{Def}_C)^<$ classifies deformations where C_0 is fixed, as well as the node q, the node p_0 and the A_{2h+1}-singularity, while the other singularities deforms.

Figure 13:



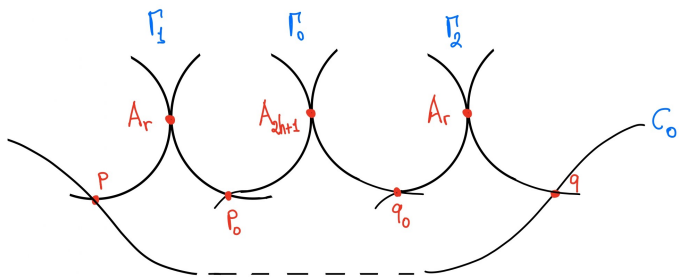
Notice that in case (a), we end up with an A_1/A_r-attached (non-separating) hyperelliptic bridge (chains of length one) of genus $< (r-1)/2$. Because we remove the 0-section of the $\mathbb{G}_m^2$-representation Def_C as it contains an odd atom of genus $h < (r-1)/2$, S-completeness forces us to remove one of the two basins of attraction for the cocharacter we constructed. The choice in this case is arbitrary. Either we remove A_1/A_r-attached hyperelliptic bridges of genus $< (r-1)/2$ or we remove A_1/A_1-attached hyperelliptic chains of genus $(r-1)/2$. We choose to remove the first as it is easier to fit with the previous condition and it does not conflict with the known steps of the Hassett-Keel program.

The next counterexample explains why condition $(d_{r,2})$ is needed.

Example 4.12. Consider the curve C constructed in the following way:

- let (Γ_0, p_0, q_0) be an odd atom of genus h with $5 \leq 2h+1 < r$;
- let (Γ_1, p, p_0) (respectively (Γ_2, q, q_0)) be an odd atom of genus $(r-1)/2$;
- we denote by (Γ, p, q) the (nodal) gluing of (Γ_0, p_0, q_0) with (Γ_1, p, p_0) and (Γ_2, q, q_0) along p_0 and q_0;
- we denote by C the curve obtained by (nodally) gluing (Γ, p, q) with a 2-pointed smooth curve (C_0, p, q) along p and q.

Figure 14:



We can describe the deformation space of C as

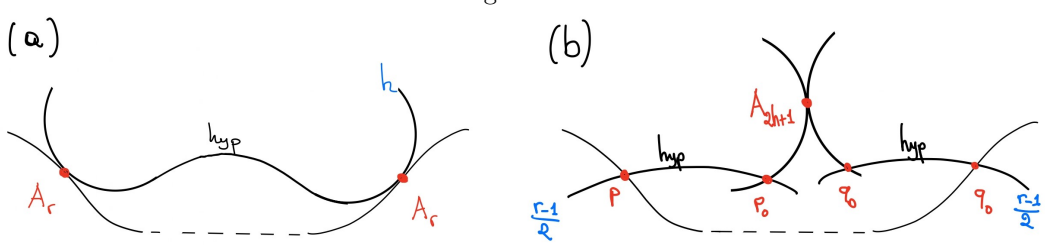
$$\mathrm{Def}_C = \mathrm{Def}_{(C_0,p,q)} \oplus T^1_p \oplus \mathrm{Def}_{(\Gamma_1,p,p_0)} T^1_{p_0} \oplus \mathrm{Def}_{(\Gamma_0,p_0,q_0)} \oplus T^1_{q_0} \oplus \mathrm{Def}_{(\Gamma_2,q,q_0)} \oplus T^1_q$$

and Corollary 3.25 gives us that $\mathrm{Aut}^\circ(C) \simeq \mathbb{G}_m^3$. Using Corollary 3.31, one can find a cocharacter $\mathbb{G}_m \to \mathbb{G}_m^3$ such that

(a) $(\mathrm{Def}_C)^>$ classifies deformations where C_0 is fixed, as well as the two A_r-singularities, while the other singularities deform;

(b) $(\mathrm{Def}_C)^<$ classifies deformations where C_0 is fixed, as well as all the nodes and the A_{2h+1}-singularity, while the other singularities deforms.

Figure 15:



Notice that in case (a), we end up with an A_r/A_r-attached (non-separating) hyperelliptic bridge (chains of length one) of genus $< (r-1)/2$. Because we remove the 0-section of the $\mathbb{G}_m^2$-representation Def_C as it contains an odd atom of genus $h < (r-1)/2$, S-completeness forces us to remove one of the two basins of attraction for the cocharacter we constructed. The choice in this case is arbitrary. Either we remove A_r/A_r-attached hyperelliptic bridges of genus $< (r-1)/2$ or we remove A_1/A_1-attached hyperelliptic chains of genus $(r-1)/2$. Again, we choose to remove the first as it is easier to fit with the previous condition and it does not conflict with the known steps of the Hassett-Keel program.

Remark 4.13. Notice that if $h = 1$, in case (b), the deformation of the subcurve (Γ_0, p_0, q_0) will still be hyperelliptic, because it is constructed as two projective lines attached by a tacnode. Thus it is already removed by condition (c_r) and it does not provide a counterexample. If $h \geq 2$, we can deform the subcurve (Γ_0, p_0, q_0) keeping the singularity but changing its crimping datum, which will not give a hyperelliptic bridge anymore. Indeed, this counterexample makes sense only if $h \geq 2$, thus $r \geq 7$. We decided to avoid dealing with such subtlety by having condition $(d_{r,2})$ for every r.

The next example will explain why the conditions (c_r) and $(d_r) = (d_{r,1}) + (d_{r,2})$ need to hold for chains, not just bridges.

Example 4.14. Let C be a curve in $\mathcal{M}^r_{g,n}$ constructed as follows:

- let (Γ_0, p, p_0) be a 2-pointed odd atom of genus h with $h < (r-1)/2$ and (Γ_1, q, p_0) be a 2-pointed rosary of hyperelliptic A-singularities of length 2, namely one A_{2k_1+1} (the one closer to p_0) and one A_{2k_2+1} singularity;
- let (Γ, p, q) be the curve obtained gluing (nodally) Γ_0 and Γ_1 along p_0;
- let (C_0, p, q) be a 2-pointed smooth curve; we denote by C the curve obtained by gluing (nodally) (C_0, p, q) and (Γ, p_0, p_1) along p and q.

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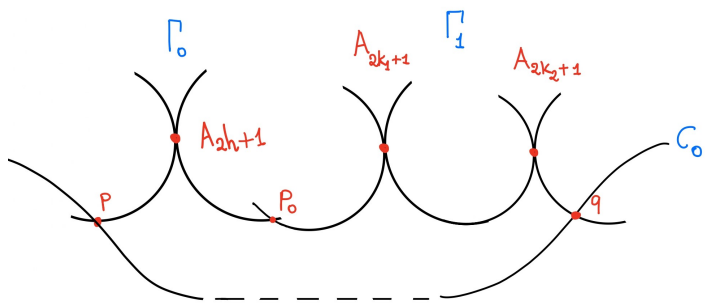
Figure 16:



We can describe the deformation space of C as

$$\mathrm{Def}_C = \mathrm{Def}_{(C_0,p,q)} \oplus T^1_p \oplus \mathrm{Def}_{(\Gamma_0,p,p_0)} \oplus T^1_{p_0} \oplus \mathrm{Def}_{(\Gamma_1,p_0,q)} \oplus T^1_q$$

and Corollary 3.25 gives us that $\mathrm{Aut}^\circ(C) \simeq \mathbb{G}^2_m$. Using the computations carried out in Example 3.32 it is easy to find a cocharacter $\mathbb{G}_m \to \mathbb{G}^2_m$ such that

(a) $(\mathrm{Def})^>$ classifies deformations where C_0 is fixed, as well as the nodes p, q and the A_{2k_1+1}-singularity, while the three other singularities deform;

(b) $(\mathrm{Def}_C)^<$ classifies deformations where C_0 is fixed, as well as the node p_0 and the A_{2h+1} and A_{2k_2+1} singularities, while the A_{2k_1+1}-singularity and the node p deform.

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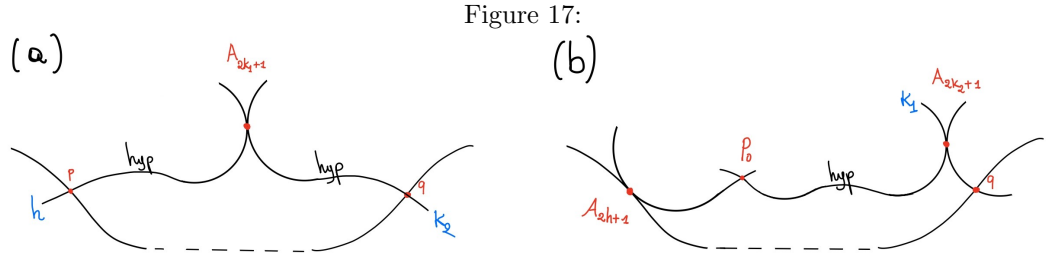
Figure 17:



Notice that in case (a), we end up with an A_1/A_1-attached hyperelliptic chain. Because we remove the 0-section of the $\mathbb{G}^2_m$-representation Def_C, S-completeness forces us to remove one of the two basins of attraction for the cocharacter we constructed. The choice in this case is more arbitrary, due the parity problem. Either we remove A_1/A_1-attached hyperelliptic chains, which are the "generalization"

of A_1/A_1-attached rosaries of hyperelliptic A-singularities of odd length, or we remove A_1/A_{2k_2+1}-attached hyperelliptic chains with a A_1/A_{2h+1}-attached rational curve. We choose to remove the first for two reasons:

(i) it is more in line with the first steps of the Hasset-Keel program (see [AFSvdW17]);

(ii) it is simply easier to write as a condition.

One might argue that the previous example would imply we need to remove A_1/A_1-attached hyperelliptic chains without restrictions on the genus. This is not true though, as if r is odd and $k = (r-1)/2$, then $(\mathrm{Def}_C)^<$ is already removed from the stack because of Example 4.11. Moreover, notice that this example can be generalized easily to enforce conditions (c_r) and (d_r) for chains of arbitrary length.

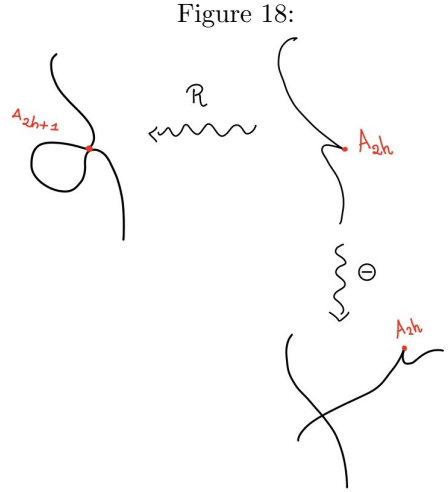
4.2 Attached hyperelliptic tails

We now move to explaining why condition (t_r) appears.

The first example shows us that we need to remove A_1-attached hyperelliptic tails of genus $< r/2$.

Example 4.15. Let R be a DVR with fraction field Q and $C \to \operatorname{Spec} R$ be any family of A_r-stable curves over R such that the generic fiber C_Q has an A_{2h}-singularity which degenerates to an A_k-singularity for $r \geq k > 2h$. Thanks to Lemma 3.39, we can construct a isotrivial degeneration $C_\Theta \to \Theta_Q$ such that the generic fiber coincides with C_Q and the A_{2h}-singularity of C_Q degenerates to an even atom of genus h attached to the rest of the curve by a separating node. This would define a morphism $\Theta_R \setminus 0 \to \mathcal{M}_g^r$ which does not extend to a morphism from Θ_R thanks to [AFSvdW17, Proposition 2.10]. Because our goal is to find an open of $\mathcal{M}_{g,n}^r$ that contains at least one curve with an A_r-singularity, we need to remove the curves that contain an A_1-attached even atom of genus $h < r/2$.

Figure 18:



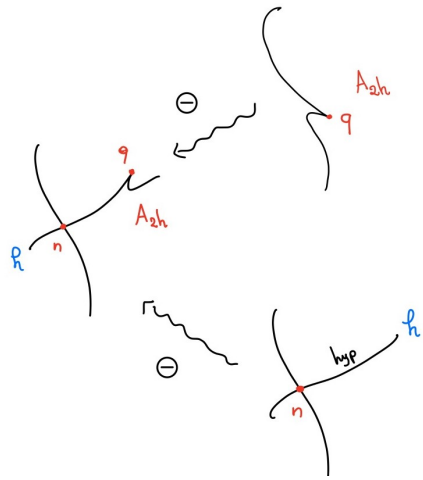
Up to quotienting by a finite group, the *stacky* deformation space is of the form

$$\mathrm{Def}_{(C-\Gamma,n)} \oplus [T_q^1 \oplus (\mathrm{Cr}_q \oplus T_n^1)/\mathbb{G}_m]$$

and, because of the previous counterexample to Θ-completeness, we need to remove the zero-section from the deformation space $T_q^1 \oplus (\mathrm{Cr}_q \oplus T_n^1)/\mathbb{G}_m$. This implies that to get a separated good-moduli space, we need to remove either $T_q^1 \oplus (0 \oplus 0)$ or $0 \oplus (\mathrm{Cr}_q \oplus T_n^1)$. Because we do not want to remove "generic" A_r-singularities, this forces us to remove $T_q^1 \oplus (0 \oplus 0)$ which corresponds to the locus where

we deform the singularity q but keep the node n. The smoothing of q gives rise to an hyperelliptic tail of genus h attached nodally by a Weierstrass point, or equivalently a A_1-attached hyperelliptic tail of genus h (see Corollary 3.31).

Figure 19:

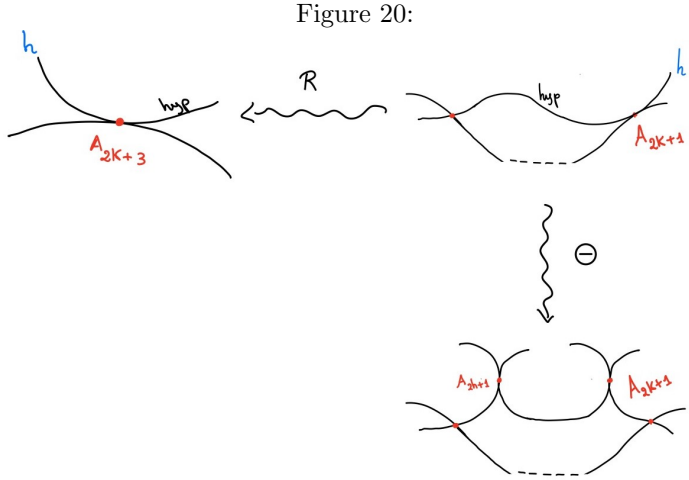


Therefore, we remove A_1-attached hyperelliptic tails of genus $h < r/2$.

The next counterexample do not give a choice as canonical as the previous ones, but it follows the strategy described at the beginning of the section.

Example 4.16. Let (D, p_1, p_2) be a 2-pointed rosary of hyperelliptic A-singularities of length 2 and (C_0, p_1, p_2) be a 2-pointed smooth curve. We denote by C the curve obtained by gluing (nodally) (C_0, p_1, p_2) and (D, p_1, p_2) along p_1 and p_2. Proposition 3.43 allow us to construct an isotrivial degeneration to C which deforms one of the nodes and one of the A-singularities of D. Therefore, the generic fiber would consists of two smooth curves meeting in two outer singularities of type A_{2h+1} and a node. If $2h + 3 \leq r$, we can degenerates these two singularities into an outer (in fact separating) A_{2h+3}-singularity. Putting everything together, we have constructed a morphism $\Theta_R \setminus 0 \to \widetilde{\mathcal{M}}_g^r$ which does not extend to a morphism from Θ_R because of Proposition 2.10 of [AFSvdW17]. A similar counterexample can be constructed if (D, p_1, p_2) has length 3 or length 1.

Figure 20:



We need to decide what to remove between the two choices. Trying again to follow the strategy explained at the beginning of the chapter, we want to avoid removing curves with positive dimensional stabilizers and therefore we end up removing

- A_{2k+1}-attached hyperelliptic tails of genus $1 \leq h < (r-1)/2$ for $2 \leq k < (r-1)/2$ (case of length 2);
- A_3-attached hyperelliptic tails of genus $(r-1)/2$ (case of length 1).

Notice that if $h < (r-1)/2$ then we can degenerate a non-separating A_1/A_1-attached hyperelliptic bridge of genus h to an A_3-attached hyperelliptic tail of genus h. The first condition then becomes

- A_{2k+1}-attached hyperelliptic tails of genus $1 \leq h < (r-1)/2$ for $1 \leq k < (r-1)/2$.

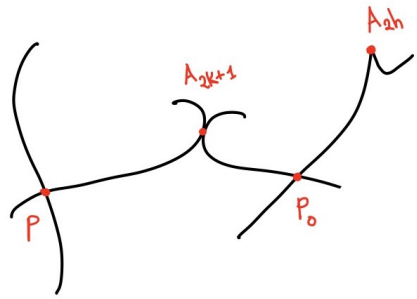
Remark 4.17. Notice that this is not quite condition (t_r) as in Definition 4.4, in fact we have not removed yet the A_{2k+1}-attached hyperelliptic tails of genus $h = (r-1)/2$ when r is odd for $k \geq 2$. This will follow from the next example.

The next example of this subsection is the only step where we decide to not follow the Hassett-Keel program.

Example 4.18. Let C_0 be the curve constructed as follows:

- Let (Γ_0, p_0, p) be a 2-pointed odd atom of genus h with $h < (r-1)/2$ and (Γ_1, p_0) be a 1-pointed even atom of genus h;
- we denote by (Γ, p) the 1-pointed curve obtained by gluing (nodally) (Γ_0, p, p_0) with (Γ_1, p_0) along p_0;
- we denote by C the curve obtained by gluing (nodally) (Γ, p) with the 1-pointed smooth curve (C_0, p) along p;

Figure 21:



We can describe the deformation space of C as

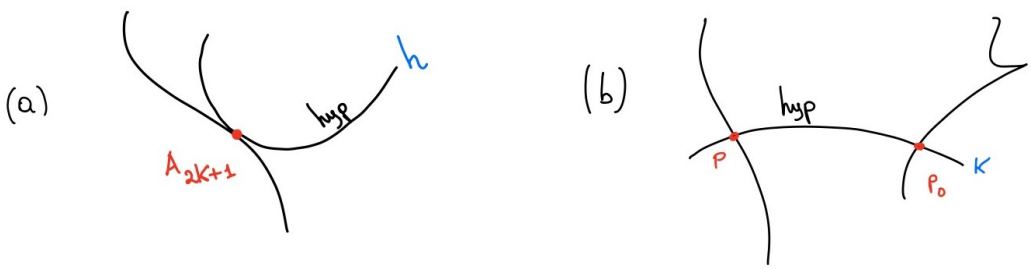
$$\mathrm{Def}_C = \mathrm{Def}_{(C_0,p)} \oplus T^1_p \oplus \mathrm{Def}_{(\Gamma_0,p_0,p)} \oplus T^1_{p_0} \oplus \mathrm{Def}_{(\Gamma_1,p_0)}$$

and Corollary 3.25 gives us that $\mathrm{Aut}^\circ(C) \simeq \mathbb{G}_m^2$. Again, a simple computation using Corollary 3.31 gives a cocharacter $\mathbb{G}_m \to \mathbb{G}_m^2$ such that

(a) $(\mathrm{Def}_C)^>$ parametrizes curves with an A_{2k+1}-attached tail of genus h;

(b) $(\mathrm{Def}_C)^<$ parametrizes curves with an A_1/A_1-attached *separating* bridge of genus k with a rational tail that contains an A_{2h}-singularity;

Figure 22:



The choice is now between removing the separating bridges and removing the A_{2k+1}-attached tails of genus $< r/2$. We opt for the latter, although we believe that our methods can be readily adapted to the alternative scenario.

Compared with the initial steps of Hassett Keel Program, one of the main differences is that in our construction we remove fewer hyperelliptic chains. From this perspective, we choose not to remove bridges that arise as chain instances.

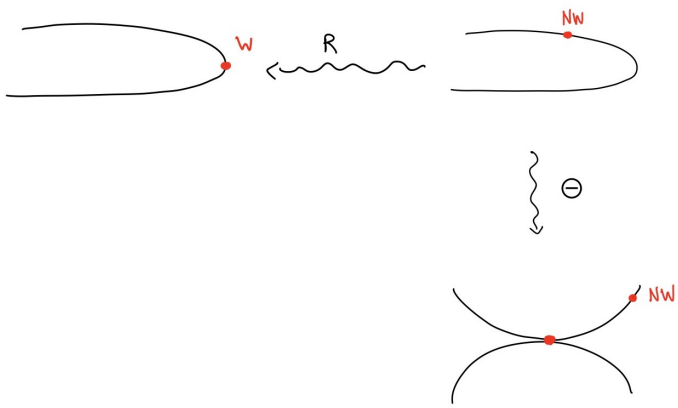
Remark 4.19. Notice that in the previous choice scenario, we need that the A_{2h}-singularity does have non-trivial crimping space, i.e. $r \geq 2h \geq 4$. Nevertheless, it was shown in [Gor26, Theorem F] that for $r = 3$, we still have to remove A_3-attached elliptic tails if we want to construct modular compactifications that contains curves with non-separating tacnodes (see also *Case 1* of the proof of [Gor26, Theorem 3.8.2]).

Finally, we introduce an example which only appears in a very specific "asymmetric" situation, and it is strictly related with the fact that the stack $\mathcal{H}^{2g+1,\circ}_{g,1}$ of 1-pointed double covers of $\mathbb{P}^1$ of genus g does not have a good moduli space, even though it has a stratification $\mathcal{H}^{2g+1,\circ}_{g,w} \subset \mathcal{H}^{2g+1,\circ}_{g,1}$ where both

$\mathcal{H}^{2g+1,\circ}_{g,w}$ and $\mathcal{H}^{r,\circ}_{g,g_1^2} := \mathcal{H}^{2g+1,\circ}_{g,1} \setminus \mathcal{H}^{2g+1,\circ}_{g,w}$ have a good moduli space (which are points in both cases). We believe that this two are exactly the Θ-strata of $\mathcal{H}^{2g+1,\circ}_{g,1}$. See also Remark 2.56.

Example 4.20. Let R be a DVR with fraction field Q and $C \to \operatorname{Spec} R$ be a family of 1-pointed double covers of $\mathbb{P}^1$ over R that such that the marking of the generic fiber C_Q is not Weiestrass, but it degenerates through R to a Weierstrass point. We can construct a isotrivial degeneration $C_\Theta \to \Theta_Q$ such that the generic fiber coincides with C_Q and the special fiber is a 1-pointed rational hyperelliptic A_{2g+1}-singularity.

Figure 23:



This is equivalent to a morphism $\Theta_R \setminus 0 \to \mathcal{M}^r_g$ which does not extend to a morphism from Θ_R because of Proposition 2.10 of [AFSvdW17]. Following the strategy mentioned at the beginning of the section, we remove the closure of the closed substack $\mathcal{H}^{2g+1,\circ}_{g,w}$, which coincides with the substack of $\mathcal{M}^{2g+1}_{g,1}$ parametrizing A_0-attached hyperelliptic tails of genus $< r/2$. Notice that this condition is not empty only if $r \geq 2g+1$, $g \geq 2$ and $n = 1$. Therefore it did not appear in the definition of α_c-stable curves of [AFSvdW17].

4.3 Condition (e_r)

Finally, we have the example that only works for $n = 0$. The problem is related to the fact that if $n = 0$ (and $r \geq 2g+1$), there is a closed point in $\mathcal{H}^r_g$ which has stabilizers of the form $\mathbb{G}_m \ltimes \mathbb{G}_{\mathrm{a}}$ (see Corollary 3.25). We believe that it is not necessary to remove that point if we want to construct a non-reductive good-moduli space, at least for $\mathcal{H}^r_g$.

In Theorem 2.49, we classify the open substacks of $\mathcal{H}^{2g+1,\circ}_g$ which admits a separated good moduli space. Notice that because $\mathcal{H}^r_g$ is closed inside $\mathcal{M}^r_g$, every obstruction for Θ-completeness (or S-completeness) of $\mathcal{H}^r_g$ is preserved in $\mathcal{M}^r_g$. The same is true if we restrict to study the open $\mathcal{H}^{r,\circ}_g \subset \mathcal{H}^r_g$ because the open embedding is affine, as it is the complement of a Cartier divisor, namely $\Delta^{\mathrm{sep}} \cap \mathcal{H}^r_g \subset \mathcal{H}^r_g$, where Δ^{sep} is the closed locus parametrizing curves with a separating node.

Thus, Θ-completeness holds for $\mathcal{M}^r_g$ only if it holds for $\mathcal{H}^{r,\circ}_g$. Therefore, Theorem 2.49 forces us to remove all the curves C in $\mathcal{M}^r_g$ which appear as $2:1$-covers of $\mathbb{P}^1$ with branching divisor with one root of multiplicity $> g+1$.

Definition 4.21 (A_h-self-attached hyperelliptic curves)**.** Let g, h be positive integers with $h > g$. We say that an A_r-stable curve C of genus g is an *A_h-self-attached hyperelliptic curve* if one of the following is true:

- if $h = 2h_0$, then C is obtained from an hyperelliptic A_r-prestable 1-pointed curve (C_0, p), where p is a Weierstrass point, by pinching p creating a A_{2k}-singularity; we also include the case (C_0, p) have genus 0;
- if $h = 2h_0 + 1$, then C is obtained from an hyperelliptic A_r-prestable 2-pointed curve (C_0, p_1, p_2), where p_1 and p_2 are exchanged by the hyperelliptic involution, by gluing p_1 and p_2 creating a A_{2k+1}-singularity; we also include the case (C_0, p_1, p_2) has genus 0;
- if $h = 2g + 1$, then C is obtained by gluing two $\mathbb{P}^1$'s along ∞ to create an A_{2g+1}-singularity.

Remark 4.22. Notice that the third case of condition (e_r) can only appear if $n = 0$ and $r = 2g + 1$. Moreover, we do not require the curve C to be hyperelliptic. We conjecture that condition (e_r) is not needed if one wants to prove that $\mathcal{M}_g^{2g+1}$ has a non-reductive moduli space.

4.4 Openness of the stack of admissible curves

In the last subsection, we prove that the conditions defining admissible curves are open. This is not surprising as it is already proved in [AFSvdW17], but, for the sake of completeness, we sketch a proof here as well. We remark that condition (e_r) was not introduced before, thus the openness is actually new (and indeed more delicate).

Lemma 4.23. *The stack $\mathcal{U}_{g,n}^r$ is an open substack of $\mathcal{M}_{g,n}^r$.*

Proof. The result is essentially a consequence of Subsection 2.2 of [AFSvdW17]. Firstly, we will prove that (t_r) is an open condition. Then we will show that, assuming condition (t_r), (c_r) and (d_r) are open conditions as well and finally we prove the same for (e_r) assuming (t_r) and (c_r).

We start with condition (t_r). Suppose given a family $C \to \operatorname{Spec} R$ of n-pointed A_r-stable curves over a dvr R such that the generic fiber $C_Q \to \operatorname{Spec} Q$ contains an A_{2k+1}-attached hyperelliptic tail Γ_Q of genus $1 \leq h < r/2$. The A_{2k+1}-singularity p_Q (which exists up to passing to a finite extension of dvr's) will be separating by definition, thus it will give rise to an equisingular separating section p_R, because Proposition 3.13 assures us that it will still be a separating A_{2k+1}-singularity. Let $\widetilde{C} \to \operatorname{Spec} R$ be the partial normalization along p_R, which exists by Proposition 3.13, and denote by $\Gamma_R \subset \widetilde{C}$ the connected component which restricts to Γ_Q on the generic fiber. Because $\mathcal{H}_{h,w}^r$ is closed inside $\mathcal{M}_{h,1}^r$, we have that Γ_R is also an hyperelliptic curve and p_R is a Weierstrass point. Notice that the same argument gives us that being an A_0-attached hyperelliptic tail is a closed condition. Thus (t_r) is an open condition.

Suppose now that the generic fiber C_Q has a non-separating A_i/A_j-attached hyperelliptic chain of genus $< (r-1)/2$, where $i, j \in \{1, r\}$. By construction every odd A-singularity connecting the hyperelliptic components (together with the two extremal singularities) is *lonely*, i.e. it is the unique intersection of the two components it belongs to. Because of Proposition 3.13, lonely A-singularities are preserved along degenerations, thus giving an equisingular section over the whole family for every such singularity in the generic fiber. If we consider the partial normalization $\widetilde{C} \to \operatorname{Spec} R$ along these lonely A-singularities, it is enough to prove that the property of being a 2-pointed hyperelliptic curve (with the two markings exchanged by the involution) is closed. This is straightforward. Moreover, if the chain is actually a bridge, i.e. is an A_1/A_1-attached hyperelliptic bridge of genus $1 \leq h < (r-1)/2$, the two nodal singularities p_1 and p_2 can either be preserved or can degenerate through the family $C \to \operatorname{Spec} R$ to the same singularity p, which will be an A_3-singularity. Again, this follows from Proposition 3.13. Nevertheless, the same argument as above will give us an A_3-attached hyperelliptic tail of genus $h < (r-1)/2 < r/2$, which therefore is still in the complement of $\mathcal{U}_{g,n}^r$, proving that (c_r) and (d_r) are both open conditions.

Showing that (e_r) is an open condition needs a bit more care, even assuming condition $(t_r), (c_r)$ and (d_r), which is enough to show that $\mathcal{U}_{g,n}^r$ is an open substack. Suppose $n = 0$ and the generic fiber

C_Q is an A_h-self-attached hyperelliptic curve where $h > g$. The A_h-singular section p_Q extends to a section of C which lands in the special fiber in a A_k-singularity with $k \geq h$. If $k = h$, p_R is an equisingular section thus we can normalize along it and the same proof for the previous cases gives us the result. Thus we can assume $k > h$ and for simplicity we deal with the case $h = 2h_0 + 1$. We leave to the reader to verify that the same proof works in the even case. First of all, consider the iterated blowup procedure as described in Definition 4.3 of [Per23], which will give us a curve $\widetilde{C} \to C$ over $\operatorname{Spec} R$, where we resolved the generic fiber of C along the A_h-singularity. Suppose that the genus of $\widetilde{C}$ is greater or equal than 1. We claim that the (non-pointed) curve $\widetilde{C}$ is an A_r-stable hyperelliptic curve. First of all, the generic fiber $\widetilde{C}_Q$ is an hyperelliptic A_r-prestable curve. Since it is the partial normalization of an A_r-stable curve C_Q, we have that, if it is not stable, the unstable components in the odd case are two A_1-attached $\mathbb{P}^1$s exchanged by the involution (because of the definition of self-attached). Therefore the rest of the curve is an A_1/A_1-attached hyperelliptic curve of genus $g - h_0 - 1 < h_0 = (h-1)/2 \leq (r-1)/2$ who appears also in C_Q (as the morphism $\widetilde{C} \to C$ is an isomorphisms outside the section p). This is impossible since C_Q verifies (c_r). Therefore the curve $\widetilde{C}_Q$ is an hyperelliptic A_r-stable curve with two smooth markings q_1 and q_2 exchanged by the involutions. A priori, the curve $\widetilde{C}_k$ over the closed point may not be A_r-stable, only A_r-prestable. We aim to prove that it is infact A_r-stable. First, consider the stabilization $\widetilde{C} \to C'$. Then C' is an A_r-stable curve over R whose generic fiber is hyperelliptic. Since the stack of hyperelliptic curves is closed (see [Per24]), we get that C'_k is an hyperelliptic A_r-stable curve. If $\widetilde{C}_k$ has unstable components, again this implies that it has an A_1/A_1-attached hyperelliptic curve of genus $g - h_0 - 1 < (r-1)/2$, and the same would be true for C_k. But it is impossible since we are assuming that C_k does verifies (c_r) and (t_r). Thus $\widetilde{C}$ is A_r-stable, and using the closedness of hyperelliptic curves, we get that $\widetilde{C}$ is hyperelliptic and A_r-stable of genus $g - h_0$. Since $g - h_0 - 1 < (r-1)/2$, we can apply Lemma 4.7 to get that $\widetilde{C}$ is indeed honestly hyperelliptic. Therefore (up to take finite extensions) $\widetilde{C}_k$ has a morphism to $\mathbb{P}^1$ (and it is connected), thus it is an hyperelliptic smooth curve. By definition, the curve C_k is a A_k-self-attached curve of genus $g < h < k$, and we win. We leave to genus 0 case, to the interested reader. It is easier to handle since the section p_R is equisingular. □

Finally, we end up with a straightforward, but useful lemma, which will prove to be crucial in the inductive steps for the proof of S-completeness.

Lemma 4.24. *Let C be an n-pointed A_r-stable curve of genus g over an algebraically closed field k. Then*

- *if C satisfies (t_r), then it satisfies (t_{r-1});*
- *if C satisfies (c_r), then it satisfies (c_{r-1});*
- *if C satisfies (e_r), then it satisfies (e_{r-1});*

Moreover, (d_r) is an empty condition if r is even.

Proof. It follows straightforwardly from the definitions. □

5 Good moduli properties of $\mathcal{U}^r_{g,n} \subset \mathcal{M}^r_{g,n}$

This section will focus on the proof of the following theorem.

Theorem 5.1. *The open embedding $\mathcal{U}^r_{g,n} \subset \mathcal{M}^r_{g,n}$ is Θ-complete and S-complete.*

Proof. The idea is to verify the hypothesis of Proposition A.7. We denote by $\mathcal{Z}$ be the closed complement of $\mathcal{U}^r_{g,n}$ in $\mathcal{M}^r_{g,n}$ and suppose given $x : \operatorname{Spec} k \to \mathcal{Z}$ a geometric point. We want to prove that for every cocharacter $\mathbb{G}_m \to G_x$, the linearized closed subset $Z_x \subset T_x\mathcal{X}$ contains either $T_x\mathcal{X}^>$ or $T_x\mathcal{X}^\leq$ (see Remark A.1). By contradiction, suppose it does not. Then

- Lemma 5.4 implies that the curve parametrized by x verifies condition (t_r) of Definition 4.4;
- Lemma 5.6 implies that the curve parametrized by x verifies conditions (c_r) and (d_r) of Definition 4.4;
- Lemma 5.7 implies that the curve parametrized by x verifies condition (e_r) of Definition 4.4.

which implies that x lands in the admissible open $\mathcal{U}^r_{g,n}$, which is a contradiction. □

Throughout this section, $\mathcal{Z}$ denotes the complement of $\mathcal{U}^r_{g,n}$ in $\mathcal{M}^r_{g,n}$.

Remark 5.2. It is important to remark that the statement of Theorem 5.1 is not as surprising as it may seem. Indeed, because $\mathcal{M}^r_{g,n}$ has affine diagonal, if $\mathcal{U}^r_{g,n}$ satisfies Θ-completeness (respectively S-completeness) then the open embedding $\mathcal{U}^r_{g,n} \subset \mathcal{M}^r_{g,n}$ satisfies it as well since extensions of maps $\Theta_R \setminus \{0\} \to \mathcal{X}$ to maps $\Theta_R \to \mathcal{X}$ are unique if $\mathcal{X}$ has affine diagonal.

In the rest of this section, we will prove the lemmas used in the proof of the theorem.

5.1 Condition on tails

Before starting with the proof of Lemma 5.4, we explain how the proof goes in an easy example.

Example 5.3. Suppose x correspond to a curve C/k which is constructed by attaching a 1-pointed even atom (Γ, p) of genus h (for $2h < r$) to a smooth curve C_0 in a node p.

Notice that $\Gamma \subset C$ is an hyperelliptic tail of genus $h < r/2$, and thus C is contained in $\mathcal{Z}$. First of all, $\operatorname{Aut}(C)^\circ \simeq \mathbb{G}_m$ and we can describe the tangent space

$$T_{[C]}\mathcal{M}^r_{g,n} \simeq \operatorname{Def}_{(C_0,p)} \oplus T^1_p \oplus \operatorname{Def}_{(\Gamma,p)}$$

where T^1_p is the deformation space of the node. Recall that we have the decomposition

$$\operatorname{Def}_{(\Gamma,p)} \simeq \operatorname{Def}^h_{(\Gamma,p)} \oplus \operatorname{Def}^{nh}_{(\Gamma,p)}$$

where we denote $\operatorname{Def}^h_{(\Gamma,p)} := T^1_q$ represents the deformation space of the A_{2h}-singularity q, which coincides with the deformations which keep Γ hyperelliptic and p a Weierstrass point (see Proposition 3.29). Furthermore, we identified the complement $\operatorname{Def}^{nh}_{(\Gamma,p)}$ with the crimping space of the singularity. Without loss of generality, we can assume $\mathbb{G}_m \subset \operatorname{Aut}(C)$ acts with weight one on T^1_p, thus negatively on $\operatorname{Def}^h_{(\Gamma,p)}$ and positively on $\operatorname{Def}^{nh}_{(\Gamma,p)}$ because of Corollary 3.31 (see also Example 4.15). Because of the condition (t_r) in Definition 4.4, we have that $\mathcal{Z}_{[C]}$ contains (in this specific example it is actually equal to) the linear subspace

$$\operatorname{Def}_{(C_0,p)} \oplus 0 \oplus \operatorname{Def}^h_{(\Gamma,p)}.$$

Suppose there exists a cocharacter $\mathbb{G}_m \to \mathbb{G}_m \subset \operatorname{Aut}(C)$, i.e. an integer $a \in \mathbb{Z}$, such that $\mathcal{Z}_{[C]}$ does not contain neither $(T_{[C]}\mathcal{M}^r_{g,n})^>$ or $(T_{[C]}\mathcal{M}^r_{g,n})^\leq$. Notice that the action induced by the cocharacter

will have weight a on T^1_p, (strictly) positive multiples of a on $\mathrm{Def}^{nh}_{(\Gamma,p)}$, (stricly) negative multiples of a on $\mathrm{Def}^{h}_{(\Gamma,p)}$ and it will be trivial on $\mathrm{Def}_{(C_0,p)}$. Thus if $a \leq 0$ then

$$(T_{[C]}\mathcal{M}^r_{g,n})^{>} \subset 0 \oplus 0 \oplus \mathrm{Def}^{h}_{(\Gamma,p)} \subset \mathcal{Z}_{[C]};$$

therefore $a > 0$. In the same way, the condition $(T_{[C]}\mathcal{M}^r_{g,n})^{\leq} \not\subseteq \mathcal{Z}_{[C]}$ implies $a \leq 0$. Thus we have a contradiction.

Following the previous example as a prototype, we now prove that if $C \in \mathcal{M}^r_{g,n}$ is a curve that does not verify the hypothesis of Proposition A.7, then C verifies condition (t_r) of Definition 4.4, i.e. there cannot be hyperelliptic tails of genus $h < r/2$.

Lemma 5.4. *If $C : \operatorname{Spec} k \to \mathcal{Z} \subset \mathcal{M}^r_{g,n}$ is a curve that does not verify the hypothesis of Proposition A.7, then C verifies condition (t_r) of Definition 4.4*

Proof. By contradiction, suppose there is an hyperelliptic tail Γ of genus $2h < r$, attached by a A_{2k+1}-singularity p_0. Let us call C_0 the subcurve $C - \Gamma$. Again, we have a decomposition

$$T_{[C]}\mathcal{M}^r_{g,n} \simeq \mathrm{Def}_{(C_0,p_0)} \oplus \mathrm{Def}_{p_0} \oplus \mathrm{Def}_{(\Gamma,p_0)}$$

where Def_{p_0} is the deformation space of the singularity p_0, which can be decomposed as the local deformations $T^1_{p_0}$ and the crimping space Cr_{p_0}. Given a cocharacter $\mathbb{G}_m \to \mathrm{Aut}(C)$, we have an induced $\mathbb{G}_m$-action on the (one-dimensional) tangent space $T_{p_0}C_0$ (respectively $T_{p_0}\Gamma$) of the point p_0 in C_0 (respectively in Γ). Notice that p_0 is indeed a smooth point of C_0. If $k \geq 1$, then the two $\mathbb{G}_m$-actions need to be balanced (i.e. have the same weight) as the k-jets are identified (see Remark 2.43). Suppose this is the case and denote by a the weight. If $a = 0$, we have

$$0 = (T_{[C]}\mathcal{M}^r_{g,n})^{>} \subset \mathrm{Def}_{(C_0,p_0)} \oplus 0 \oplus 0 \subset \mathcal{Z}_{[C]}$$

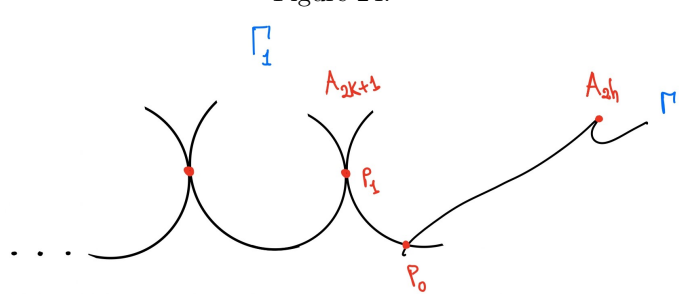
thus we can assume $a \neq 0$. Therefore, we have that (Γ, p_0) is actually a 1-pointed even atom of genus h attached in p_0 to C_0 by Lemma 3.5. Then it is clear that

$$\mathrm{Def}_{(C_0,p_0)} \oplus (0 \oplus \mathrm{Cr}_{p_0}) \oplus \mathrm{Def}^{h}_{(\Gamma,p_0)} \subset \mathcal{Z}_{[C]}$$

We can then repeat a similar argument as in Example 5.3 to get that for every integer a either $(T_{[C]}\mathcal{M}^r_{g,n})^{>}$ or $(T_{[C]}\mathcal{M}^r_{g,n})^{\leq}$ are contained in $\mathrm{Def}_{(C_0,p_0)} \oplus (0 \oplus \mathrm{Cr}_{p_0}) \oplus \mathrm{Def}^{h}_{(\Gamma,p_0)}$. Therefore the same is true for $\mathcal{Z}_{[C]}$.

We will now deal with the case when the two $\mathbb{G}_m$ actions do not need to be balanced, thus p_0 has to be a node. If the action of $\mathbb{G}_m$ on $T_{p_0}C_0$ is trivial, we leave it to the reader to check that one can reduce to the situation analyzed in Example 5.3. Suppose that it is not the case, then the proof of Corollary 3.25 gives us that there exists a subcurve $\Gamma_1 \subset C_0 \subset C$ such that $(\Gamma_1, P_{\Gamma_1} \cup I_{\Gamma_1})$ is of type (i) as in Proposition 3.22, because $p_0 \in I_{\Gamma_1}$. More specifically, Γ_1 will be a rosary of hyperelliptic A-singularities. Firstly, suppose that Γ_1 has at least one odd A-singularity. Let p_1 be the A_{2k+1}-singularity for $k \geq 1$ which lies in the same rational irreducible component (of Γ_1) of p_0.

Figure 24:



Then we get the decomposition

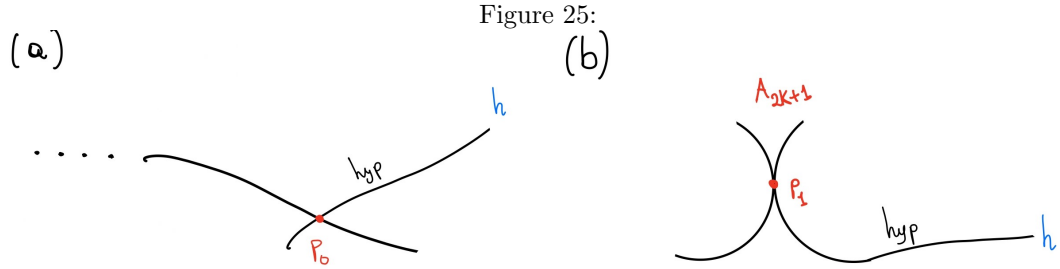
$$T_{[C]}\mathcal{M}^r_{g,n} \simeq R \oplus \mathrm{Def}_{p_1} \oplus T^1_{p_0} \oplus \mathrm{Def}_{(\Gamma,p_0)}$$

where R is just the ($\mathbb{G}_m$-equivariant) complement of the right-most three spaces (i.e. the deformation space of the rest of the curve). As before, recall that we have the decomposition

$$\mathrm{Def}_{p_1} \simeq T^1_{p_1} \oplus \mathrm{Cr}_{p_1}.$$

By construction, we have that $\mathcal{Z}_{[C]}$ contains two linear subspaces, namely $V_1 := R \oplus \mathrm{Def}_{p_1} \oplus 0 \oplus \mathrm{Def}^h_{(\Gamma,p_0)}$ and $V_2 := R \oplus (0 \oplus \mathrm{Cr}_{p_1}) \oplus T^1_{p_0} \oplus \mathrm{Def}^h_{(\Gamma,p_0)}$, which correspond to the following two generic deformations:

Figure 25:



Given a cocharacter $\mathbb{G}_m \to \mathrm{Aut}(C)$, we denote by the integer a the weight induced on the (one dimensional) tangent space $T_{p_0}\Gamma$ and by b the weight induced on the (one-dimensional) tangent space $T_{p_0}\Gamma_1$. If a is zero, then an easy verification shows that $(T_{[C]}\mathcal{M}^r_{g,n})^>$ is contained in either V_1 or V_2, otherwise we will have $b > 0$ and $b \leq 0$, a contradiction. Then it remains to study the case $a \neq 0$, which implies that (Γ, p_0) is the 1-pointed even atom of genus h.

We can now describe the action of $\mathbb{G}_m$ on $\mathrm{Def}_{p_1} \oplus T^1_{p_0} \oplus \mathrm{Def}_{(\Gamma,p_0)}$. Because only the sign will matter to reach a contradiction, we will say that a $\mathbb{G}_m$-equivariant subspace V has sign b, and we will write $s(V) = b$ if all the $\mathbb{G}_m$-weights of V have the same sign of the integer b. We recall that we consider three possible signs, namely 0, 1 and -1. Then we have the following:

- $\mathrm{Def}_{p_1} \simeq T^1_{p_1} \oplus \mathrm{Cr}_{p_1}$ with $s(T^1_{p_1}) = -b$ and $s(\mathrm{Cr}_{p_1}) = b$;
- $s(T^1_{p_0}) = b + a$;
- $\mathrm{Def}_{(\Gamma,p_0)} \simeq \mathrm{Def}^h_{(\Gamma,p_0)} \oplus \mathrm{Def}^{nh}_{(\Gamma,p_0)}$ where we have $s(\mathrm{Def}^h_{(\Gamma,p_0)}) = -a$ while $s(\mathrm{Def}^{nh}_{(\Gamma,p_0)}) = a$.

We will prove that either $(T_{[C]}\mathcal{M}^r_{g,n})^>$ or $(T_{[C]}\mathcal{M}^r_{g,n})^\leq$ is contained inside either V_1 or V_2, for $(a,b) \in \mathbb{Z}^2$. Suppose that it is not the case. Then we have following implications

- $(T_{[C]}\mathcal{M}^r_{g,n})^> \not\subseteq V_1$ implies that $a + b > 0$ or $a > 0$;
- $(T_{[C]}\mathcal{M}^r_{g,n})^\leq \not\subseteq V_1$ implies that $a + b \leq 0$ or $a \leq 0$;
- $(T_{[C]}\mathcal{M}^r_{g,n})^> \not\subseteq V_2$ implies that $-b > 0$ or $a > 0$;
- $(T_{[C]}\mathcal{M}^r_{g,n})^\leq \not\subseteq V_2$ implies that $-b \leq 0$ or $a \leq 0$.

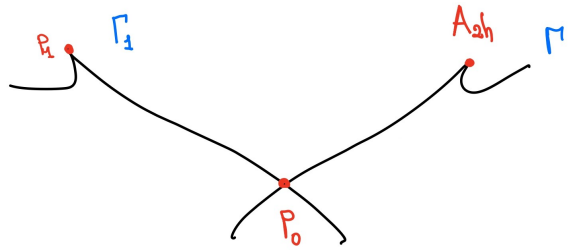
To help the reader, we will explain the first implication. The others follows in the same way. Suppose that $a + b \leq 0$ and $a \leq 0$, then the explicit description of (the signs of) the action gives us

$$(T_{[C]}\mathcal{M}^r_{g,n})^> \subset R \oplus (T^1_{p_1} \oplus 0) \oplus 0 \oplus \mathrm{Def}^h_{(\Gamma,p_0)} \subset V_1$$

which corresponds to the claimed implication. An easy verification shows that the intersection of the four conditions above is empty.

Finally, suppose that Γ_1 does not have any odd A-singularity, which implies that it is actually an even atom as well.

Figure 26:



We denoted by Γ_1 the irreducible component and by p_1 the even A-singularity. In this case, the deformation space is of the form

$$\mathrm{Def}_{(\Gamma_1,p_0)} \oplus T^1_{p_0} \oplus \mathrm{Def}_{(\Gamma,p_0)}$$

where we have the decomposition $\mathrm{Def}_{(\Gamma_1,p_0)} \simeq T^1_{p_1} \oplus \mathrm{Cr}_{p_1}$. As before, $\mathcal{Z}_{[C]}$ contains the representation $V_1 := \mathrm{Def}_{(\Gamma_1,p_0)} \oplus 0 \oplus \mathrm{Def}^h_{(\Gamma,p_0)}$ because of condition (t_r) in Definition 4.4. If $2k < r$, then condition (t_r) gives us that it also contains the representation $V_2 := (T^1_{p_1} \oplus 0) \oplus 0 \oplus \mathrm{Def}_{(\Gamma,p_0)}$; if $r = 2k > 2h$, then we have that $\mathcal{Z}_{[C]}$ contains the representation $V_2 := (0 \oplus \mathrm{Cr}_{p_1}) \oplus T^1_{p_0} \oplus \mathrm{Def}^h_{(\Gamma,p_0)}$ because of condition (e_r) in Definition 4.4. Indeed, the generic deformation in V_2 would give an A_r-self-attached hyperelliptic curve of genus $h+k$ with $r > h+k$. We leave it as an exercise to the interested reader to check that either $(T_{[C]}\mathcal{M}^r_{g,n})^>$ or $(T_{[C]}\mathcal{M}^r_{g,n})^\leq$ is contained inside V_1 or V_2, in both cases.

Finally, we leave it to the reader to check that if $r = 2g+1$, $n = 1$ and C is a A_0-attached hyperelliptic tail of genus g, the same holds. The idea is the same. □

5.2 Condition on chains

It is now time to deal with the chain conditions in Definition 4.4. Although the combinatorics is substantially more complicated, the general idea remains the same. To help the reader, we set up and solve the combinatorial problem in Appendix B. We start by showing it with the following easy example. This is the perfect analog of Example 5.3.

Example 5.5. Suppose x correspond to a curve C/k which is constructed by attaching an hyperelliptic chain Γ of genus bounded by $(r-1)/2$ to a smooth curve C_0 in two nodes p_1, p_2. Assume that $\mathrm{Aut}^\circ(C) \simeq \mathbb{G}_m$, which implies that Γ is a 2-pointed rosary of hyperelliptic A-singularities of odd length, i.e. it has an odd number of singularities (and thus an even number of components).

We can describe the tangent space

$$T_{[C]}\mathcal{M}^r_{g,n} \simeq \mathrm{Def}_{(C_0,p_1,p_2)} \oplus T^1_{p_1} \oplus T^1_{p_2} \oplus \mathrm{Def}_{(\Gamma,p_1,p_2)}$$

where $T^1_{p_1}$ (respectively $T^1_{p_2}$) is the deformation space of the node p_1 (respectively p_2). Thanks to Proposition 3.36, we have a subrepresentation

$$\mathrm{Def}^h_{(\Gamma,p_1,p_2)} \subset \mathrm{Def}_{(\Gamma,p_1,p_2)}$$

where $\mathrm{Def}^h_{(\Gamma,p_1,p_2)}$ represents the deformation space of (Γ, p_1, p_2) which coincides with the deformations preserving the property of being an hyperelliptic chain. We denote by $\mathrm{Def}^{nh}_{(\Gamma,p_1,p_2)}$ the $\mathbb{G}_m$-equivariant complement of $\mathrm{Def}^h_{(\Gamma,p_1,p_2)}$ in $\mathrm{Def}_{(\Gamma,p_1,p_2)}$. The main property of this decomposition is that $\mathbb{G}_m$ acts with strictly positive weights on Def^h and strictly negative weights on Def^{nh}, or viceversa.

Because of the condition (c_r) in Definition 4.4, we have that $\mathcal{Z}_{[C]}$ contains (in this specific example it is actually equal to) the linear subspace

$$\mathrm{Def}_{(C_0,p_1,p_2)} \oplus 0 \oplus 0 \oplus \mathrm{Def}^h_{(\Gamma,p_1,p_2)}.$$

Suppose there exists a cocharacter $\mathbb{G}_m \to \mathbb{G}_m \subset \mathrm{Aut}(C)$, i.e. an integer $a \in \mathbb{Z}$, such that $\mathcal{Z}_{[C]}$ does not contain neither $(T_{[C]}\mathcal{M}^r_{g,n})^>$ or $(T_{[C]}\mathcal{M}^r_{g,n})^\leq$. Notice that the action induced by the cocharacter will have weight a on $T^1_{p_1}$ and $T^1_{p_2}$, (strictly) positive multiples of a on $\mathrm{Def}^{nh}_{(\Gamma,p_1,p_2)}$, (stricly) negative multiples of a on $\mathrm{Def}^h_{(\Gamma,p_1,p_2)}$ and it will be trivial on $\mathrm{Def}_{(C_0,p_1,p_2)}$, again for Proposition 3.36. Exactly as in Example 5.3, the condition $(T_{[C]}\mathcal{M}^r_{g,n})^> \nsubseteq \mathcal{Z}_{[C]}$ implies that $a > 0$ while the condition $(T_{[C]}\mathcal{M}^r_{g,n})^\leq \nsubseteq \mathcal{Z}_{[C]}$ implies $a \leq 0$. Thus we have a contradiction.

We are ready for the main lemma for the hyperelliptic chains.

Lemma 5.6. *If $C : \mathrm{Spec}\, k \to \mathcal{Z} \subset \mathcal{M}^r_{g,n}$ is a curve that does not verify the hypothesis of Proposition A.7, then C verifies conditions (c_r) and (d_r) of Definition 4.4*

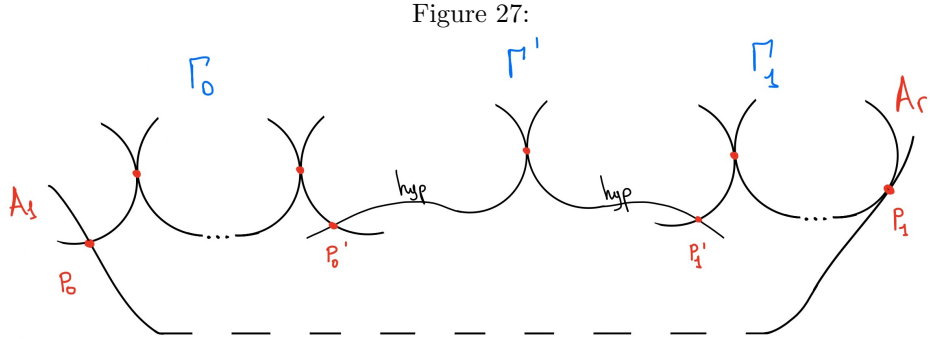
Proof. The idea of the proof is to reduce the statement to a purely combinatorial one, using the natural combinatorial structure of toric representation. We deal with the combinatorial problem in Appendix B.

By contradiction, suppose C does not verifies one of the conditions, thus there exists an A_1/A_1 (or A_1/A_r or A_r/A_r) attached non-separating hyperelliptic chain of genus bounded by $(r-1)/2$. We will prove that C verifies the hypothesis of Proposition A.7, namely that

($\star$) for every $\mathbb{G}_m \to \mathrm{Aut}^\circ(C)$ we have that $\mathcal{Z}_{[C]}$ contains either $(T_{[C]}\mathcal{M}^r_{g,n})^\leq$ or $(T_{[C]}\mathcal{M}^r_{g,n})^>$.

Let (Γ', p'_0, p'_1) be a *minimal* A_i/A_j-attached non-separating hyperelliptic chain of genus bounded by $(r-1)/2$ with $i,j \in \{1,r\}$. We define $\Gamma'_0 \subset C - \Gamma'$ (respectively $\Gamma'_1 \subset C - \Gamma'$) to be the maximal subcurve of $C - \Gamma'$ which contains p'_0 (respectively p'_1) and such that $\mathrm{Aut}^\circ(C)$ acts non-trivially on every component. Notice that Γ'_0 and/or Γ'_1 can be empty. We denote by Γ_0 (respectively Γ_1) the *minimal* subcurve of Γ'_0 which contains p'_0 (respectively p'_1) and such that $\Gamma_0 \cap (\Gamma'_0 - \Gamma_0)$ (respectively $\Gamma_1 \cap (\Gamma'_1 - \Gamma_1)$) is an A_r-singularity. If p'_0 (respectively p'_1) is an A_r-singularity, we set $\Gamma_0 = \emptyset$ (respectively $\Gamma_1 = \emptyset$). Moreover, if p'_0 (respectively p'_1) is not an A_r-singularity and Γ'_0 (respectively Γ'_1) does not have A_r-singularities, we set $\Gamma'_0 = \Gamma_0$ (respectively $\Gamma'_1 = \Gamma_1$). Finally, we define $\Gamma := \Gamma_0 \cup \Gamma' \cup \Gamma_1$ and we denote by p_0 (respectively p_1) the intersection $\Gamma_0 \cap C - \Gamma$ (respectively $\Gamma_1 \cap C - \Gamma$). If Γ'_0 (respectively Γ'_1) is empty, we set $p_0 := p'_0$ (respectively $p_1 := p'_1$). Notice that because Γ' is non-separating, p_0 and p_1 always exist. See the following picture for an instance where $\Gamma_0 = \Gamma'_0$ and Γ_1 strictly contained in Γ'_1 (thus p_1 is an A_r-singularity).

Figure 27:



By construction, we have the following properties of (Γ, p_0, p_1):

(a) Γ is a connected subcurve of C intersecting its complement $C-\Gamma$ in p_0 and p_1; notice that it is not guaranteed that the two points are distinct;

(b) there are no A_r-singularities of C in Γ except for possibly p_0 and p_1;

(c) Γ contains a non-separating hyperelliptic chain of genus bounded by $(r-1)/2$.

(d) Γ_0 (respectively Γ_1) is a (possibly empty) sequence of A_1/A_1-attached rosaries of hyperelliptic A-singularities, except for the chain containing p_0 (respectively p_1) which could also be an A_1/A_r-attached rosary of hyperelliptic A-singularities.

Notice that while A_1/A_1-attached rosaries of A-singularities have length at least two (i.e. they have at least one A-singularity) because of the stability condition, the same is not true for A_1/A_r-attached rosaries, indeed we can have a single $\mathbb{P}^1$ attached by a node and by an (hyperelliptic) A_r-singularity to the rest of the curve. We say that such chain has length 1.

Property (d) allows us to describe Γ as the sequence

$$\overset{A_1/A_r}{\underset{p_0}{\text{———}}}\ n_k\ \overset{A_1}{\text{———}}\ \dots\ \overset{A_1}{\text{———}}\ n_1\ \overset{A_1}{\text{———}}\ \Gamma'\ \overset{A_1}{\text{———}}\ m_1\ \overset{A_1}{\text{———}}\ \dots\ \overset{A_1}{\text{———}}\ m_h\ \overset{A_1/A_r}{\underset{p_1}{\text{———}}}$$

where the positive integers n_i and m_j indicate the length of the corresponding rosaries of hyperelliptic A-singularities, while the edges represent the singularities used to connect such chains. The leftmost and rightmost edges represent the points p_0 and p_1 (possibly equal). Notice that for every such rational chain there is a copy of $\mathbb{G}_m$ inside $\mathrm{Aut}^\circ(C)$ which acts as described in Proposition 3.43.

We will first deal with the case where $p_0 \neq p_1$. In this scenario, we claim the following property of Γ holds:

(•) the action of $\mathrm{Aut}^\circ(C)$ on $T^1_{p_0}$ (respectively on $T^1_{p_1}$) is completely determined by the copy of $\mathbb{G}_m$ associated to the rosary of length n_k (respectively m_h).

This follows by the construction: if p_0 is a node, then we know that the action of $\mathrm{Aut}^\circ(C)$ on $T_{p_0}(C-\Gamma)$ is trivial, for the maximality of Γ'_0, which is equal to Γ_0 if p_0 is a node. If p_0 is an A_r-singularity, then we know that the action is either trivial, or p_0 belongs to some A_1/A_1-attached rosary of hyperelliptic A-singularity, which has a unique $\mathbb{G}_m$ action and it is completely determined by the component.

The idea is to study the deformation space of Γ to see that for every cocharacter $\mathbb{G}_m \to \mathrm{Aut}^\circ(C)$ we have that $\mathcal{Z}_{[C]}$ contains either $(T_{[C]}\mathcal{M}^r_{g,n})^{\leq}$ or $(T_{[C]}\mathcal{M}^r_{g,n})^{>}$. We do a further simplification. We claim that we can assume that Γ' is a rosary of hyperelliptic A-singularities of odd length (where no singularity of type A_r appears), or equivalently that there is a copy of $\mathbb{G}_m$ inside $\mathrm{Aut}^\circ(C)$ that acts non-trivially on Γ'. Indeed, let us call C' the curve obtain from C by substituting Γ' with a rosary of hyperelliptic A-singularities $\widetilde{\Gamma}$ of odd-length (where no singularity of type A_r appears). Then if we can find a cocharacter χ such that the condition on $\mathcal{Z}_{[C]}$ is not verified, we can extend it to a cocharacter $\widetilde{\chi}$ such that the condition on $\mathcal{Z}_{[C']}$ in the deformation space $T_{[C]}\mathcal{M}^r_{g,n}$ is not verified, by setting the component of the cocharacter acting on $\widetilde{\Gamma}$ equal to zero. Thus, without loss of generality, we can assume Γ' is a rosary of hyperelliptic A-singularities of odd length.

Finally, by abuse of notation, we can describe Γ as the following sequence:

$$\overset{A_1/A_r}{\underset{p_0}{\text{———}}}\ n_1\ \overset{A_1}{\underset{p_1}{\text{———}}}\ \dots\ \overset{A_1}{\underset{p_{k-1}}{\text{———}}}\ n_k\ \overset{A_1/A_r}{\underset{p_k}{\text{———}}}$$

where n_i a is nonnegative integer representing the rosary Γ_i of hyperelliptic A_r-singularities of length n_i, for $i = 1, \dots, k$. Again by abuse of notation, we will denote by p_i the node connecting Γ_i with Γ_{i+1} for $i = 1, \dots, k-1$. We denote by p_0 the leftmost edge and by p_k the rightmost one.

To verify $(\star)$, we need to describe $\mathcal{Z}_{[C]}$ inside the deformation space of C in $\mathcal{M}^r_{g,n}$. First of all, we have the following $\mathrm{Aut}^\circ(C)$-equivariant decomposition of $T_{[C]}\mathcal{M}^r_{g,n}$

$$\mathrm{Def}_{(C-\Gamma,p_0,p_1)} \oplus \bigoplus_{i=1}^{k}(\mathrm{Def}^h_{(\Gamma_i,p_{i-1},p_i)} \oplus \mathrm{Def}^{nh}_{(\Gamma_i,p_{i-1},p_i)}) \oplus \bigoplus_{i=0}^{k}(T^1_{p_i} \oplus \mathrm{Cr}_{p_i})$$

where

- $\mathrm{Def}^h_{(\Gamma_i,p_{i-1},p_i)}$ parametrizes deformations of Γ_i which preserve the property of being a chain of hyperelliptic curves;
- $\mathrm{Def}^{nh}_{(\Gamma_i,p_{i-1},p_i)}$ is the complementary representation with the respect to the action of $\mathrm{Aut}^\circ(C)$.

See Proposition 3.43. Notice that except for $i = 0, k$, we have that Cr_{p_i} is zero and $T^1_{p_i}$ is one-dimensional. To ease the notation, we denote

- by $D(i)_h$ (respectively $D(i)_{nh}$) the subspace $\mathrm{Def}^h_{(\Gamma_i,p_{i-1},p_i)}$ (respectively $\mathrm{Def}^{nh}_{(\Gamma_i,p_{i-1},p_i)}$) for $i = 1, \dots, k$;
- by $T(i)$ the subspace $T^1_{p_i}$ and by $C(i)$ the subspace Cr_{p_i} for $i = 0, \dots, k$;
- by $D(0)$ the subspace $\mathrm{Def}_{(C-\Gamma,p_0,p_1)}$.

Moreover, given a component of the given decomposition, say V, we denote by V^c its complementary representation, i.e. the whole direct sum except for the addendum V.

By definition, $\mathcal{Z}_{[C]}$ contains the locus where the deformations preserve at least one hyperelliptic chain of genus bounded by $(r-1)/2$. Pick now a integer $1 \le h \le k$ such that n_h is odd, which we know exists by construction. Then we know that

$$T(i-1)^c \cap D(i)^c_h \cap T(i)^c \subset \mathcal{Z}_{[C]}$$

because the left-hand side represent the deformations where Γ_h is still an attached hyperelliptic chain of genus bounded by $(r-1)/2$. More generally, we say that a triplet $1 \le i \le h \le j \le k$ is *odd-minimal* if

$$n_t \text{ is odd for } i \le t \le j \iff t = h;$$

for every odd-minimal triplet $i \le h \le j$ we can define the subspace

$$V_{i,j}(h) := T(i-1)^c \cap \bigcap_{s=i}^{h} D(s)^c_h \cap \bigcap_{s=h+1}^{j} D(s)^c_{nh} \cap T(j)^c$$

which will be contained inside $\mathcal{Z}_{[C]}$, as $V_{i,j}(h)$ classifies the deformations of Γ which makes the union $\bigcup_{i\le s\le j}\Gamma_s \subset \Gamma$ an attached hyperelliptic chain of genus bounded by $(r-1)/2$ (remember that because of condition (b) above, there cannot be hyperelliptic curves in such chain of genus $(r-1)/2$).

Because of the property $(\bullet)$, we can ignore what happens to the subcurve $C - \Gamma$, i.e. it is enough to prove that for every cocharacter

$$\chi : \mathbb{G}_m \to \prod_{i=1}^{k} \mathbb{G}_m = \mathrm{Aut}^\circ(\Gamma, p_0, p_k) \subset \mathrm{Aut}^\circ(C)$$

there exists a $V_{i,j}(h)$ as above such that it contains either $(T_{[C]}\mathcal{M}^r_{g,n})^{\le}$ or $(T_{[C]}\mathcal{M}^r_{g,n})^{>}$. Equivalently, we will prove that there is no cocharacter χ such that $V_{i,j}(h)$ does not contains neither $(T_{[C]}\mathcal{M}^r_{g,n})^{\le}$ or $(T_{[C]}\mathcal{M}^r_{g,n})^{>}$.

To do so, we need to understand the signs of the weights induced by any cocharacter $\chi : \mathbb{G}_m \to \mathbb{G}^k_m$. From now on, we will identify the cocharacters $\chi \in \mathrm{Hom}(\mathbb{G}_m, \mathbb{G}^k_m)$ with vectors of integers $\mathbf{a} \in \mathbb{Z}^k$. We will use the sign convention as in the proof of Lemma 5.4. We have the following sign relations:

- $s(D(i)_h) = a_i$ while $s(D(i)_{nh}) = -a_i$ for every $i = 1, \dots, k$;
- $s(T(i)) = a_{i+1} - (-1)^{n_i} a_i$ for $i = 0, \dots, k$ (where we can set $a_0 = a_{k+1} = 0$ by property $(\bullet)$).

Therefore, for every odd-minimal triplet $i \leq h \leq j$ we have that

$$(T_{[C]}\mathcal{M}^r_{g,n})^{>} \nsubseteq V_{i,j}(h)$$

implies that one of the following is true:

- $a_i > (-1)^{n_{i-1}} a_{i-1}$;
- $a_s > 0$ for some $i \leq s \leq h$;
- $a_s < 0$ for some $h+1 \leq s \leq j$
- $a_{j+1} > (-1)^{n_j} a_j$.

and similarly for $(T_{[C]}\mathcal{M}^r_{g,n})^{\leq}$. Moreover, notice that the only information we need regarding the n_i's is the parity, both for the definition of the odd-minimal triplet and for the signs of the action. Then we can reduce the problem by assuming that the n_i's are either zeros or ones, depending on the parity. We have finally arrived to the core of the proof, which is a purely combinatorial statement. In Appendix B, we rewrite the combinatorial setup and Theorem B.8 gives us that for every vectors of integers $\mathbf{a} \in \mathbb{Z}^k$ there always exists an odd-minimal triplet $i \leq h \leq j$ such that none of the conditions above (or their respective counterpart with $\leq$) are verified, which means that either $(T_{[C]}\mathcal{M}^r_{g,n})^{>}$ or $(T_{[C]}\mathcal{M}^r_{g,n})^{\leq}$ are contained in $V_{i,j}(h)$.

To finish the proof of the theorem, we need to consider the case $p_0 = p_1$. The combinatorics of such problem is a bit more convoluted, because we cannot set as before $a_{k+1} = 0$ and $a_0 = 0$ by construction; indeed the right convention would be $a_{k+1} := a_1$ and $a_0 := a_k$, because the curve Γ actually coincide with C and it is a cycle. Notice that this can only happen if $n = 0$. We deal with this as before in Appendix B, specifically Theorem B.15. □

5.3 Condition (e_r)

Finally, we end this section by dealing with the last condition.

Lemma 5.7. *If $C \in \mathcal{Z} \subset \mathcal{M}^r_{g,n}$ is a curve that does not verify the hypothesis of Proposition A.7, then C verifies condition (e_r) of Definition 4.4*

Proof. The proof of this lemma is sensibly easier as the curves that do not verify condition (e_r) are very easy to describe. Moreover, it is enough to verify the statement for those curves C that have non-trivial positive dimensional stabilizers. Such a curve C is constructed in one of the following way (see Definition 4.21):

(1) C is obtained by pinching a $\mathbb{P}^1$ in 0 and ∞, creating an hyperelliptic A_{2k}-singularity and an hyperelliptic A_{2h}-singularity with $k + h = g$ and $h > g$;

(2) C is obtained by gluing two $\mathbb{P}^1$ in 0 and ∞ creating an A_{2k+1}-singularity and an hyperelliptic A_{2h+1}-singularity, with $k + h = g$ and $h > g$;

(3) C is obtained by pinching a $\mathbb{P}^1$ in ∞ creating a A_{2g}-singularity;

(4) C is obtained by gluing two $\mathbb{P}^1$ in ∞ creating an A_{2g+1}-singularity.

We will show how case (1) works. We leave it to the reader to verify it in the other cases. We know that the deformation space decomposes as

$$\mathrm{Def}_C = \mathrm{Def}_k \oplus \mathrm{Def}_h$$

where $\mathrm{Def}_k := T^1_k \oplus \mathrm{Cr}_k$ and the same for Def_h. By T^1_k we indicate the deformation space of the A_{2k+1}-singularity and by Cr_k its crimping space (and similarly for h). Moreover, $\mathrm{Aut}^\circ(C) \simeq \mathbb{G}_m$ where

$$s(T^1_k) = s(\mathrm{Cr}_h) = -s(\mathrm{Cr}_k) = -s(T^1_h).$$

We have that $\mathcal{Z}_{[C]}$ contains the subspace

$$V := (T^1_k \oplus 0) \oplus (0 \oplus \mathrm{Cr}_h).$$

Clearly then for every cocharacter $a : \mathbb{G}_m \to \mathbb{G}_m$ we have that V contains either $\mathrm{Def}^{>}_C$ or $\mathrm{Def}^{\leq}_C$, because otherwise we will have both $a > 0$ and $a \leq 0$. $\square$

6 Θ-completeness of $\mathcal{U}^r_{g,n}$

This section is dedicated to proof of the following theorem.

Theorem 6.1. *The algebraic stack $\mathcal{U}^r_{g,n}$ is Θ-complete.*

Proof. The proof is a direct consequence of Proposition 6.9 and Theorem 5.1. □

We start by setting the notation and make some easy reductions.

Remark 6.2. Let $\Theta_R \setminus 0 \longrightarrow \mathcal{X}$ be a morphism of algebraic stacks, where R is a DVR with fraction field Q and residue field k. Clearly, if the composition

$$\Theta_Q \hookrightarrow \Theta_R \setminus 0 \to \mathcal{X}$$

factors through the good moduli space morphism $\Theta_Q \to \operatorname{Spec} Q$, then we can consider the morphism

$$\Theta_R \to \operatorname{Spec} R \subset \Theta_R \setminus 0 \to \mathcal{X}$$

and an easy exercise shows that it extends the morphism $\Theta_R \setminus 0 \longrightarrow \mathcal{X}$. We will call such morphisms *trivial*. We only need to fill non-trivial morphisms to verify Θ-completeness.

Existence of filling

This subsection will be dedicated to prove Proposition 6.9. We start by proving some auxiliary lemmas which will help us start the induction procedure in the proof.

Lemma 6.3. *Let $C \to \operatorname{Spec} R$ a family of A_r-prestable curves over a DVR and suppose that there exists a section q which hits a lonely singularity on the generic fiber. Then q is equisingular.*

Proof. The lemma follows straightforwardly from Proposition 2.10 of [AFSvdW17]. □

Lemma 6.4. *Let R be a DVR with fraction field Q and residue field k. Let $C : \Theta_R \setminus 0 \longrightarrow \mathcal{U}^r_{g,n}$ be a non trivial morphism and suppose that the composition*

$$C_{0,1} : \mathcal{B}\mathbb{G}_{m,Q} \hookrightarrow \Theta_R \setminus 0 \longrightarrow \mathcal{U}^r_{g,n}$$

is not represented by an atom, namely one of the two (possibly pointed) curves defined in Definition 3.1. Then there exists an equisingular section of the curve represented by C, i.e. there exists a lifting

$$\begin{tikzcd} & \mathcal{A}_h \arrow[d] \\ \Theta_R \setminus 0 \arrow[ur, dashed] \arrow[r] & \mathcal{M}^r_{g,n} \end{tikzcd}$$

for some positive h.

Proof. The idea of the proof is to use the study of isotrivial degenerations of A_r-stable curves carried out in Section 3 to deal with the isotrivial degeneration over Q, whereas we use Proposition 3.13 to ensure that the limit of an outer (indeed lonely) A_h-singularities is still an A_h-singular.

We denote by $C_{1,1}$ the restriction of $C : \Theta_R \setminus 0 \to \mathcal{M}^r_{g,n}$ through the generic point

$$\operatorname{Spec} Q \hookrightarrow \Theta_Q \hookrightarrow \Theta_R \setminus 0$$

and by $C_{0,1}$ the restriction through the closed point

$$\mathcal{B}\mathbb{G}_{m,Q} \hookrightarrow \Theta_Q \hookrightarrow \Theta_R \setminus 0;$$

moreover, we denote by $C_{1,0}$ the restriction of $C : \Theta_R \setminus 0 \to \mathcal{M}^r_{g,n}$ through the closed point

$$\operatorname{Spec} k \hookrightarrow \operatorname{Spec} R \hookrightarrow \Theta_R \setminus 0.$$

Because the morphism C is not trivial, this implies that we have an induced morphism

$$\mathbb{G}_m \to \operatorname{Aut}^\circ(C_{0,1})$$

which is not trivial. Following the proof of Corollary 3.25, we can normalize along the outer nodes of $C_{0,1}$ to get a connected component of the normalization Γ such that $\mathbb{G}_m \to \operatorname{Aut}^\circ(\Gamma, I_\Gamma \cup P_\Gamma)$ is surjective. Thus $(\Gamma, I_\Gamma \cup P_\Gamma)$ is one of the possible cases described in Proposition 3.22.

In case (iii), $\Gamma = C_{0,1}$ and the curve $C_{0,1}$ is either an atom (which we are excluding), or an irreducible rational curve with two hyperelliptic even A-singularities. Condition (e_r) implies that this can only happen if $r = g$, g is even and the two singularities are of type A_g. Example 3.32 gives us that one of the two singularities will be preserved and thus it will be preserved along the isotrivial degeneration

$$C_Q : \Theta_Q \subset \Theta_R \setminus 0 \to \mathcal{U}^g_{g,n}$$

giving a lifting

$$\begin{tikzcd} & \mathcal{A}_g \arrow[d] \\ \Theta_Q \arrow[ur, dashed] \arrow[r] & \mathcal{M}^g_{g,n}. \end{tikzcd}$$

Because $r = g$, the morphism $\mathcal{A}_g \to \mathcal{M}^g_{g,n}$ is finite (the A_g-singularity cannot degenerate further), allowing us to construct the desired lifting from $\Theta_R \setminus 0$, as $\Theta_R \setminus 0 = \Theta_Q \cup_{\operatorname{Spec} Q} \operatorname{Spec} R$.

In case (ii), $\Gamma = C_{0,1}$ and the curve $C_{0,1}$ is a closed rosary of hyperelliptic A-singularities. Let $2s$ be the number of irreducible components (and A-singularities) with $s \geq 2$. Thus, Proposition 3.38 ensures us that the generic fiber $C_{1,1}$ of the isotrivial degeneration $\Theta_Q \to \mathcal{U}^r_{g,n}$ have (at least) s hyperelliptic components attached by s singularities pairwise. If $s \geq 3$, then every singularity is lonely, therefore by Lemma 6.3, we are done. If $s = 2$, then $C_{1,1}$ consists of two hyperelliptic subcurves attached by two odd A-singularities. Because of Proposition 3.13, these two outer singularities either converge to the same point in $C_{1,0}$ or they are preserved through the vertical degeneration

$$\operatorname{Spec} R \hookrightarrow \Theta_R \setminus 0 \to \mathcal{M}^r_{g,n}.$$

The first situation is excluded by condition (t_r), as it would give rise to an hyperelliptic tail of genus $< r/2$.

Finally, suppose we are in case (i). If I_Γ has exactly one point, then every odd singularity is separating, therefore it is preserved through the vertical degeneration. If the node (in I_Γ) is preserved through the isotrivial degeneration, then it is also preserved through the vertical one, as it is separating (thus lonely). See Lemma 6.3. Suppose it is not the case. If there is at least one odd singularity, then one of them has to be preserved, because of Proposition 3.43. If there are no odd singularities, then Γ has to be an even atom, attached to $C_{1,0}$ by a node. Condition (t_r) implies that r is even and it is an A_r-singularity. The even singularity has to be preserved through the isotrivial degeneration by Corollary 3.31 (because the node is not) and therefore through the vertical one because it cannot degenerate any further, since any such degeneration would worsen the singularity, i.e. the degeneration would leave $\mathcal{M}^r_{g,n}$.

If I_Γ does not contain any node of $C_{0,1}$ and Γ is not an atom, then again every odd singularity is separating and we have at least two singularities. If at least two of them are odd, one of the two has to be preserved through the isotrivial degeneration because of Example 3.32. Again, we can conclude because of Corollary 3.31 (see also Lemma 6.3 for an explicit computation). If we only have two singularities, one of them is even and the odd one is not preserved through the isotrivial degeneration (while the even one is). Condition (t_r) tells us that r is even and the even singularity is an A_r-singularity. Then it has to be preserved through the degeneration over R.

Finally, if I_Γ contains 2 points, then it has to be an A_1/A_1-attached rosary of hyperelliptic singularities. If it is separating, then Γ is attached to the rest of the curve in two separating nodes, thus either one node or one worse-than-nodal separating singularity has to be preserved. If the bridge is not separating, we can apply a similar reasoning to case (ii) to get the result. □

Remark 6.5. Since the previous proof is a bit convoluted and deals with a lot of cases, to help the reader we sum up the possible cases and how they are handled in what follows.

The proof starts by finding a subcurve $\Gamma \subset C_{0,1}$ without outer nodes where $\mathbb{G}_m$ acts non trivially. Then it proceeds to prove the statement studying all the possible Γ's that can appear, using the cases described in Proposition 3.22.

- (a) Case (iii) (not atom): follows from condition (e_r);
- (b) Case(ii), i.e. closed rosaries:
  - (b_1) number of singularities greater or equal than 3: follows from Lemma 6.3;
  - (b_2) number of singularities equal to 2: follows from condition (t_r);
- (c) Case (i), i.e. rosary of hyperelliptic A-singularities with $I_\Gamma \leq 2$:
  - (c_1) $|I_\Gamma| = 0$, thus at least two A-singularities (otherwise $C_{0,1}$ is an atom):
    - $(c_{1,1})$ at least two singularities are odd: follows from Example 3.32;
    - $(c_{1,2})$ one singularity is even: follows from condition (t_r);
  - (c_2) $|I_\Gamma| = 1$, which is a node:
    - $(c_{2,1})$ if the node is preserved: follows from Lemma 6.3;
    - $(c_{2,2})$ if the node is not preserved:
      - $(c_{2,2,1})$ if Γ has at least one odd singularity: follows from Proposition 3.43;
      - $(c_{2,2,2})$ if there are no odd singularities: follows from condition (t_r);
  - (c_3) $|I_\Gamma| = 2$, i.e. it is an A_1/A_1-attached rosary: similar to case (b) for rosaries.

Lemma 6.4 has an interesting consequence.

Corollary 6.6. *Let $g \geq 3$. Then the open embedding $\mathcal{M}^{0,nh}_{g,n} \subset \mathcal{U}^r_{g,n}$ is Θ-surjective. In particular, $\mathcal{U}^r_{g,n}$ has a closed point with finite stabilizers.*

Proof. Let $\mathcal{M}^{0,nh}_{g,n}$ be the open in $\mathcal{U}^r_{g,n}$ parametrizing n-pointed smooth non-hyperelliptic curves. The idea is to prove that the open embedding $\mathcal{M}^{0,nh}_{g,n} \subset \mathcal{U}^r_{g,n}$ is Θ-surjective (see [AHLH23, Remark 3.26]), which in particular implies that sends closed points to closed points (see [AHLH23, Lemma 3.24]).

Let $C : \Theta \to \mathcal{U}^r_{g,n}$ be any non-trivial morphism. If the generic fiber C_1 is smooth, then the special fiber C_0 has to be a pointed atom, because of Lemma 6.4. Thus, we can assume $n \leq 2$. In both the cases $n = 1, 2$, if C_0 is a n-pointed atom, then C_1 is an hyperelliptic curve, because of Proposition 3.29. The

same is true if $n = 0$ thanks to Proposition 3.28. Therefore it cannot lies in $\mathcal{M}^{0,nh}_{g,n}$, the morphism $\Theta \to \mathcal{U}^r_{g,n}$ is trivial and we won. □

Now, we would like to prove that for every morphism

$$\Theta_R \setminus 0 \longrightarrow \mathcal{A}_h \times_{\mathcal{M}^r_{g,n}} \mathcal{U}^r_{g,n}$$

the partial pointed normalization (over $\Theta_R \setminus 0$) along the equisingular section still verifies the conditions in Definition 4.4. This is immediate, except for a very specific situation that has to be handled separately. We introduce a weaker version of condition (t_r), namely

(t'_r) C does not have A_{2k+1}-attached hyperelliptic tails of genus $< r/2$.

It is slightly weaker, as we are allowing A_0-attached hyperelliptic tails of genus $< r/2$ (which appears only if $n = 1$ and the tail is the whole curve).

Lemma 6.7. *Given a geometric point $C \in \mathcal{A}'_h := \mathcal{A}_h \times_{\mathcal{M}^r_{g,n}} \mathcal{U}^r_{g,n}$, we have that the stabilization of the partial normalization of C along the equisingular section still verifies conditions (t'_r), (c_r), (d_r) and (e_r).*

Proof. Notice that a tail or a chain as the one considered in Definition 4.4 does not have markings (see Definition 3.34), except for A_0-attached hyperelliptic curves of genus $g < r/2$. Because the crimping morphism will never change subcurves that do not contain the markings given by the pointed stabilization of the normalization (either one or two points as they are A-singularities), we get the statement.

□

In the last lemma, we deal with the only case left out by Lemma 6.7, namely the case when the stabilization is an A_0-attached hyperelliptic curve of genus $< r/2$.

Lemma 6.8. *Suppose $h = 2k \leq r$ and let $(C, p) : \Theta_R \setminus 0 \to \mathcal{A}'_h$ be a non-trivial morphism. Then the stabilization of the (pointed) partial normalization*

$$(\widetilde{C}, p) : \Theta_R \setminus 0 \to \mathcal{M}^r_{g-k,n+1}$$

either

- *factors through the open substack $\mathcal{U}^r_{g-k,n+1}$ or;*
- *$h = g$, $n = 0$, $r \geq 2g$ and it factors through the substack $\mathcal{H}^{r,\circ}_{g/2,w} \subset \mathcal{M}^r_{g/2,1}$ (parametrizing honest A_r-stable hyperelliptic curves with a marked Weierstrass point).*

Proof. First of all, we claim that if the geometric point associated with the morphism

$$(\widetilde{C}_{0,1}, p_{0,1}) : \mathcal{B}\mathbb{G}_{m,Q} \subset \Theta_R \setminus 0 \longrightarrow \mathcal{M}^r_{g-k,n+1}$$

is an A_0-attached hyperelliptic curve of genus $g - k < r/2$ (namely $(\widetilde{C}_{1,0}, p_{1,0}) \in \mathcal{H}^{r,\circ}_{g-k,w}$), then $n = 0$ (clear by the definition), $h = g$ and the morphism

$$(\widetilde{C}, p) : \Theta_R \setminus 0 \longrightarrow \mathcal{M}^r_{g/2,1}$$

factors through $\mathcal{H}^{r,\circ}_{g/2,w}$. Indeed, Lemma 6.7 gives us that $(\widetilde{C}_{0,1}, p_{0,1})$ does verify conditions (t'_r), (c_r), (d_r) and (e_r). By Lemma 4.7 (see also Remark 4.8), we can conclude $(\widetilde{C}_{0,1}, p_{0,1}) \in \mathcal{H}^{r,\circ}_{g-k,w} \subset \mathcal{H}^r_{g-k,w}$, i.e. it is a $2:1$-cover of $\mathbb{P}^1$ and $p_{0,1}$ is a smooth Weierstrass point. Because the morphism

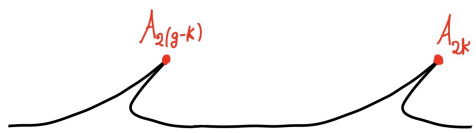
$$(C, p) : \Theta_R \setminus 0 \longrightarrow \mathcal{A}'_{2(g-k)}$$

is non-trivial, we have that $\mathrm{Aut}^\circ(C_{0,1}, p_{0,1})$ is non-trivial, thus Proposition 2.52 implies that $(\widetilde{C}_{0,1}, p_{0,1})$ is a 1-pointed even atom of genus $g-k$. Moreover, the corresponding point

$$(C_{0,1}, p_{0,1}) : \mathcal{B}\mathbb{G}_{m,Q} \longrightarrow \mathcal{A}'_{2(g-k)}$$

will be a $A_{2(g-k)}$-self-attached hyperelliptic curve of genus k and a A_{2k}-self-attached hyperelliptic curve of genus $g-k$.

Figure 28:



Because C satisfies condition (e_r) of Definition 4.4, we have that $2k = g$, thus $C_{0,1}$ can be constructed by pinching 0 and ∞ on $\mathbb{P}^1$ creating two hyperelliptic A_g-singularities. By Lemma 3.5 (see also Example 3.32), we have that the morphism

$$(C_{\Theta_Q}, p_{\Theta_Q}) : \Theta_Q \subset \Theta_R \setminus 0 \longrightarrow \mathcal{A}'_g$$

is represented by a family of A_g-self-attached hyperelliptic curves of genus $g/2$, and the same is true for (C, p) as it is a closed condition (see the proof of Lemma 4.23). This implies that the normalization $(\widetilde{C}, p)$ is an A_0-attached hyperelliptic curve of genus $g/2$, thus

$$(\widetilde{C}, p) : \Theta_R \setminus 0 \longrightarrow \mathcal{H}^r_{g/2,w} \subset \mathcal{M}^r_{g/2,1}.$$

As before, because $(\widetilde{C}, p)$ verifies conditions (t'_r), (c_r), (d_r) and (e_r), we have that the morphism factors through $\mathcal{H}^{r,\circ}_{g/2,w} \subset \mathcal{H}^r_{g/2,w}$ as claimed.

Finally, suppose that $(\widetilde{C}_{0,1}, p_{0,1})$ is not in $\mathcal{H}^r_{g-k,w}$, and thus the same is true for the composition

$$\Theta_Q \subset \Theta_R \setminus 0 \longrightarrow \mathcal{M}^r_{g-k,1}$$

as $\mathcal{H}^r_{g-k,w}$ is closed inside $\mathcal{M}^r_{g-k,1}$. We want to prove that the point

$$(\widetilde{C}_{1,0}, p_{1,0}) : \mathrm{Spec}\, k \subset \mathrm{Spec}\, R \subset \Theta_R \setminus 0 \longrightarrow \mathcal{M}^r_{g-k,1}$$

does verify condition (t_r) of Definition 4.4, i.e. it is not an A_0-attached hyperelliptic curve of genus $g-k < r/2$. By contradiction, suppose $(\widetilde{C}_{1,0}, p_{1,0}) \in \mathcal{H}^r_{g-k,w}(k)$. Notice that because $(C_{1,0}, p_{1,0})$ verifies condition (e_r) of Definition 4.4, we have that $2k \leq g$. As before, combining Lemma 6.7 and Lemma 4.7, we get that $(\widetilde{C}_{1,0}, p_{1,0}) \in \mathcal{H}^{r,\circ}_{g-k,w}(k)$. Thus $\widetilde{C}_{1,0}$ is a $2:1$-cover of $\mathbb{P}^1$ and $p_{1,0}$ is a smooth Weierstrass point. This implies $\widetilde{C}_{1,0}$ is irreducible. By the semicontinuity of the number of irreducible components for proper and flat families, we know that

$$(\widetilde{C}_{1,1}, p_{1,1}) : \mathrm{Spec}\, Q \subset \Theta_R \setminus 0 \longrightarrow \mathcal{M}^r_{g-k,1}$$

is also irreducible. Therefore we have a non-trivial morphism

$$(\widetilde{C}_{\Theta_Q}, p_{\Theta_Q}) : \Theta_Q \subset \Theta_R \setminus 0 \longrightarrow \mathcal{M}^r_{g-k,1}$$

such that

- the family of A_r-stable curves $(\widetilde{C}_{\Theta_Q}, p_{\Theta_Q})$ verifies condition (t'_r), (c_r), (d_r) and (e_r);

- the generic fiber $(\widetilde{C}_{1,1}, p_{1,1})$ is a 1-pointed A_r-stable irreducible curve of genus $g-k<r/2$.

We claim that this is impossible. Because the morphism $(\widetilde{C}, p)$ is non-trivial, we have that the group $\mathrm{Aut}^\circ(\widetilde{C}_{0,1}, p_{0,1})$ is not trivial. Therefore there exists a subcurve $\Gamma \subset \widetilde{C}_{0,1}$ where $(\Gamma, I_\Gamma \cup P_\Gamma)$ is of type (i) as in Proposition 3.22. Notice that it cannot be of type (ii) or (iii) because

$$I_\Gamma = \emptyset \implies \widetilde{C}_{0,1} = \Gamma \implies P_\Gamma = \{p_{0,1}\}.$$

Suppose that $P_\Gamma = \emptyset$, thus Γ is attached to $\widetilde{C}_{0,1} - \Gamma$ with either one or two nodes. Notice that if Γ is a rosary of *separating* A-singularities with at least one odd singularity, then every isotrivial degeneration would preserve at least one separating singularity (possibly the attaching node), thus $C_{1,1}$ would not be irreducible. Therefore, we only have two possible scenarios: either Γ is an A_1-attached hyperelliptic tail or it is an A_1/A_1-attached non-separating rosary of (odd) hyperelliptic A-singularities. The first case is not allowed because of condition (t'_r). Moreover, we have that Γ has length 1, otherwise every isotrivial degeneration would preserve at least two irreducible components. Nevertheless, if Γ has length 1, then Γ is an A_1/A_1-attached hyperelliptic chain (of length 1) of genus $< r/2$, which is impossible because of condition (c_r).

Finally, suppose $P_\Gamma = \{p_{0,1}\}$. Then the curve $C_{0,1}$ would have an A_{2l+1}-attached hyperelliptic tail of genus $k \leq g-k < r/2$ with $l \geq 1$. This is excluded because condition (t_r) of Definition 4.4. □

We end the section with the proof of one of the main results used in Theorem 6.1.

Proposition 6.9. *Let $f^\circ : \Theta_R \setminus 0 \to \mathcal{U}^r_{g,n}$ be a morphism. Then there exists a morphism $f : \Theta_R \to \mathcal{M}^r_{g,n}$ which extends f°, i.e. they fit in a commutative diagram*

$$\begin{tikzcd} \Theta_R \setminus 0 \arrow[r, "f^\circ"] \arrow[d, hook] & \mathcal{U}^r_{g,n} \arrow[d, hook] \\ \Theta_R \arrow[r, "f"] & \mathcal{M}^r_{g,n} \end{tikzcd}$$

where the vertical maps are the natural open immersions.

Proof. We will proceed by induction on the dimension of $\mathcal{M}^r_{g,n}$, namely $3g-3+n$. If $3g-3+n=1$, then $\mathcal{U}^r_{g,n}$is either $\mathcal{M}^r_{0,4}$ or $\mathcal{M}^r_{1,1}$. The first one is always a scheme while the second one admits a good moduli space for any r, and hence it is Θ-complete, proving the base case.

Turning to the induction step, we can assume f° is not trivial, because of Remark 6.2.

Let $C_{0,1}$ be the curve associated to the composition

$$\mathcal{B}\mathbb{G}_{m,Q} \subset \Theta_Q \subset \Theta_R \setminus 0 \to \mathcal{U}^r_{g,n}$$

where $R \subset Q$ is the fraction field extension of the DVR R. First of all, suppose $C_{0,1}$ is not a (possibly pointed) atom. Then we can apply Lemma 6.4 to find a lifting $\Theta_R \setminus 0 \to \mathcal{A}_h$ for some $h \geq 1$. If $h = 2k$ is even, we can use the crimping morphism

$$c_{2k} : \mathcal{A}_{2k} \longrightarrow \mathcal{M}_{g-k,n,[2k]}$$

which is an affine bundle, thanks to Proposition 2.18. Notice that if $g = k$ and $n \leq 1$, then $C_{0,1}$ will be the even atom. This case will be treated separately. Thus Proposition 2.17 gives us that identification $\mathcal{M}_{g-k,n,[2k]} \simeq \mathcal{M}^r_{g-k,n+1} \times [\mathbb{A}^1/\mathbb{G}_m]$. Moreover, Lemma 6.7 and Lemma 6.8 assure us that the morphism $\Theta_R \setminus 0 \to \mathcal{M}^r_{g-k,n+1} \times [\mathbb{A}^1/\mathbb{G}_m]$ either lands in the open $\mathcal{U}^r_{g-k,n+1} \times [\mathbb{A}^1/\mathbb{G}_m]$ or

$g = 2k$, $n = 0$ and the morphism factors through $\mathcal{H}^{r,\circ}_{g/2,w} \subset \mathcal{M}^r_{g/2,1}$ with $r \geq 2g$. In the first case, we have that $3(g-k) - 3 + (n+1) < 3g - 3 + n$, thus we can apply the induction step to get a morphism

$$\Theta_R \longrightarrow \mathcal{M}^r_{g-k,n+1} \times [\mathbb{A}^1/\mathbb{G}_m]$$

which extends

$$\Theta_R \setminus 0 \to \mathcal{U}^r_{g-k,n+1} \times [\mathbb{A}^1/\mathbb{G}_m].$$

In the second case, we can use Proposition 2.52 to get that the morphism from $\Theta_R \setminus 0$ actually extends to a morphism from Θ_R to $\mathcal{H}^{r,\circ}_{g/2,w}$, because $r \geq 2g$. The morphism c_{2k} is affine, thus Θ-complete, which allows us to construct a lifting

$$\Theta_R \to \mathcal{A}_{2k} \to \mathcal{M}^r_{g,n}$$

which is the desired morphism.

Suppose now that $h = 2k+1$ with $k \geq 1$. Firstly, suppose that the equisingular section given by Lemma 6.4 is separating. Then there exists an integer $i \leq g$ and $S \subset \{1, \dots, n\}$ such that the morphism from $\Theta_R \setminus 0$ actually lands in $\mathcal{A}^{i,S}_{2k+1}$ as it is a connected component of $\mathcal{A}_{2k+1}$. Proposition 2.25 allows us to lift the morphism

$$\Theta_R \setminus 0 \to I^{i,S}_{2k+1}$$

as $I^{i,S}_{2k+1} \to \mathcal{A}^{i,S}_{2k+1}$ is finite étale. Consider the crimping morphism

$$c^{i,S}_{2k+1} : I^{i,S}_{2k+1} \to \mathcal{M}^r_{i,s,[k]} \times \mathcal{M}^r_{g-k-i+1,n-s,[k]}$$

which by *loc.cit.* is affine. Notice that if $i = g - k - i + 1 = 0$ then $g = k+1$, thus $C_{0,1}$ is an atom. We will deal with this case later. Without loss of generality, we can assume $g + 1 > i + k$, thus

$$\mathcal{M}^r_{g-k-i+1,n-s,[k]} \simeq \mathcal{M}^r_{g-k-i+1,n-s+1} \times [\mathbb{A}^1/\mathbb{G}_m]$$

and exactly as in the even case, we can find a filling

$$\begin{tikzcd}
\Theta_R \setminus 0 \arrow[r, "f^\circ"] \arrow[d, hook] & \mathcal{U}^r_{g-k-i+1,n-s+1} \times [\mathbb{A}^1/\mathbb{G}_m] \arrow[d, hook] \\
\Theta_R \arrow[r, "f"] & \mathcal{M}^r_{g-k-i+1,n-s+1} \times [\mathbb{A}^1/\mathbb{G}_m].
\end{tikzcd}$$

Notice that f° lands in the open $\mathcal{U}^r_{g-k-i+1,n-s+1} \times [\mathbb{A}^1/\mathbb{G}_m]$ becaues of Lemma 6.7. Now we need to deal with the other factor of the product, namely $\mathcal{M}^r_{i,s,[k]}$. If $2i - 2 + s \geq 0$, then we can apply the same idea as before. If $(i,s) = (0,1)$, then $\mathcal{M}^r_{i,s,[k]} \simeq \mathcal{B}\mathbb{G}_m$ which is Θ-complete, thus everything still works. If $(i,s) = (0,0)$, then $\mathcal{M}^r_{i,s,[k]} \simeq \mathcal{B}(\mathbb{G}_m \ltimes \mathbb{G}_a)$ which is not Θ-complete. Luckily, Remark 2.41 gives us that the composition

$$I^{0,\emptyset}_{2k+1} \to \mathcal{M}^r_{g-k+1,n,[k]} \times \mathcal{B}(\mathbb{G}_m \ltimes \mathbb{G}_a) \to \mathcal{M}^r_{g-k+1,n,[k]}$$

is still affine, which allows us to conclude as before.

As for the non-separating case, if we apply Proposition 2.24, we get that it is enough to use the induction on $\mathcal{M}^r_{g-k,n,2[k]}$ and then apply the argument as above.

Finally, we deal with the case when $C_{0,1}$ is an atom. In this case, we cannot really use the induction and thus we need to do it by hand.

Suppose $C_{0,1}$ is a n-pointed odd atom of genus g with $n \leq 2$, thus $r \geq 2g+1$. Notice that if $n = 0$, then the atom is not in $\mathcal{U}^r_{g,n}$ because of condition (e_r). Consider now the curve

$$C_{1,1} : \operatorname{Spec} Q \subset \Theta_Q \subset \Theta_R \setminus 0 \to \mathcal{U}^r_{g,n};$$

using Proposition 3.29 (or Corollary 3.31) we know that $C_{1,1}$ is either

- a 1-pointed hyperelliptic curve with a non-Weierstrass marking;
- a 2-pointed hyperelliptic curve where the two markings are exchanged by the involution;
- a n-pointed $\mathbb{P}^1$ with an A_{2g+1}-singularity (possibly not hyperelliptic), with $n \leq 2$.

If $C_{1,1}$ is a 1-pointed hyperelliptic curve with a non-Weierstrass point as marking, then Lemma 4.7 together with condition (t_r) gives us that the curve $C_{1,0}$, defined as the composition

$$\operatorname{Spec} k \subset \operatorname{Spec} R \subset \Theta_R \setminus 0 \to \mathcal{U}_{g,n}^{2g+1},$$

is a $2:1$-cover of $\mathbb{P}^1$ with a marking which is not a ramification point. Thus $\Theta_R \setminus 0 \to \mathcal{U}_{g,n}^{2g+1}$ factors through $\mathcal{H}_{g,g_1^2}^{2g+1,\circ} \subset \mathcal{U}_{g,n}^{2g+1}$ (see Example 4.20). The statement of the theorem follows from the fact that $\mathcal{H}_{g,g_1^2}^{2g+1,\circ}$ admits a good moduli space (which is actually the point) by Proposition 2.54.

If $C_{1,1}$ is a 2-pointed hyperelliptic curve with the two markings exchanged by the involution, then Lemma 4.7 together with condition (t_r) gives us that the curve $C_{1,0}$, defined as the composition

$$\operatorname{Spec} k \subset \operatorname{Spec} R \subset \Theta_R \setminus 0 \to \mathcal{U}_{g,n}^{2g+1},$$

is a $2:1$-cover of $\mathbb{P}^1$ with the two markings still exchanged by the involution. Thus again $\Theta_R \setminus 0 \to \mathcal{U}_{g,n}^{2g+1}$ factors through $\mathcal{H}_{g,g_1^2}^{2g+1,\circ} \subset \mathcal{U}_{g,n}^{2g+1}$.

If $C_{1,1}$ is a n-pointed odd atom of genus g with $n \leq 2$, then such singularity has to be preserved along the degeneration $\operatorname{Spec} R \subset \Theta_R \setminus 0 \to \mathcal{U}_{g,n}^{r}$, thus allowing us the lift the morphism

$$\Theta_R \setminus 0 \to \mathcal{A}_{2g+1}.$$

In this case, the statement follows as $\mathcal{A}_{2g+1}$ is an affine bundle over $\mathcal{B}\mathbb{G}_m$.

Suppose $C_{0,1}$ is a 1-pointed even atom of genus g, thus $r \geq 2g$. Notice that if $n = 0$, then the atom is not in $\mathcal{U}_{g,n}^{r}$ because of condition (e_r). Consider now the curve

$$C_{1,1} : \operatorname{Spec} Q \subset \Theta_Q \subset \Theta_R \setminus 0 \to \mathcal{U}_{g,n}^{r};$$

using Proposition 3.29 we know that either $C_{1,1}$ is 1-pointed hyperelliptic curve with a Weiestrass point as marking, or it is a 1-pointed $\mathbb{P}^1$ with an A_{2g}-singularity (possibly not hyperelliptic). Notice that if $r = 2g+1$, than the first case does not appear because of condition (t_r) of Definition 4.4. We leave it to the reader to check that the statement still holds in these cases. $\square$

7 S-completeness of $\mathcal{U}^r_{g,n}$

This section is dedicated to prove the following theorem.

Theorem 7.1. *The moduli stack $\mathcal{U}^r_{g,n}$ is S-complete for $r \leq 5$.*

In Section 7.1, we introduce a topological version of S-completeness and use it to prove that S-completeness can be checked generically in the affine diagonal case. Moreover, we use it in Section 7.2 to reduce proving that $\mathcal{U}^r_{g,n}$ is S-complete to proving it is *weakly S-complete* (cf Definition 7.27). Section 7.3 is dedicated to explaining how the stabilization result for A-singularities can be promoted to a morphism from $\overline{\mathrm{ST}}_R$ globally, which will be main tool for proving the weak S-completeness inductively. Finally, in Section 7.4 we write down the details of the proof of Theorem 7.1 whereas in Section 7.5 we explain why the restriction on r is needed and we explicit what the obstacle is for the induction to work for every r.

In all that follows, R will be used to denote a DVR, Q will be its fraction field and k its residue field. Moreover, given a morphism $f : \operatorname{Spec} R \to \mathcal{X}$, we will denote by $f(1)$ the image of the generic point $\operatorname{Spec} Q \subset \operatorname{Spec} R$ and by $f(0)$ the image of the closed point $\operatorname{Spec} k \subset \operatorname{Spec} R$. We will use the same convention for morphisms $f : \Theta \to \mathcal{X}$.

We recall that

$$\mathrm{ST}_R = [\operatorname{Spec}(R[s,t]/(st-\pi)) \setminus 0/\mathbb{G}_m] \cong \operatorname{Spec} R \cup_{\operatorname{Spec} K} \operatorname{Spec} R$$

and

$$\overline{\mathrm{ST}}_R := [\operatorname{Spec}(R[s,t]/(st-\pi))/\mathbb{G}_m]$$

where $\mathbb{G}_m$ acts on s with weight 1 and on t with weight -1, as introduced in [Hei17, Section 2.B] and [AHLH23, Section 3.5].

A morphism of locally noetherian algebraic stacks $\mathcal{X} \to \mathcal{Y}$ is said to be S-complete if for all solid commutative diagrams

$$\begin{tikzcd} \mathrm{ST}_R \arrow[r] \arrow[d] & \mathcal{X} \arrow[d] \\ \overline{\mathrm{ST}}_R \arrow[r] \arrow[ur, dotted] & \mathcal{Y} \end{tikzcd}$$

there exists a unique dotted arrow making the diagram commutative. We say that a locally noetherian algebraic stack $\mathcal{X}$ over a base S is S-complete, if the morphism $\mathcal{X} \to S$ is S-complete. Notice that, as it happens for the valuative criteria of separatedness, the uniqueness part in the S-complete lifting criteria corresponds to the S-completeness of the diagonal.

7.1 Generic S-completeness

In this section, we are going to prove that the existence part of the valuative criterion for S-completeness can be checked generically, exactly like the valuative criterion for universal closedness of schemes. To avoid some technicality (specifically how to define the property of being *locally reductive* for morphisms of stacks), we will deal with morphisms $\mathcal{X} \to S$ where S is a algebraic space (everything works also if S is an algebraic stack with locally quasi-finite inertia). We will omit writing the base S for the fiber products if it is clear from the context.

Let us start by recalling the definition of local reductivity (see [AHLH23, Definition 2.5]).

Definition 7.2. Let $\mathcal{X}$ be a quasi-separated algebraic stack with affine stabilizers. We say that $\mathcal{X}$ is *locally reductive* if every point of $\mathcal{X}$ specializes to a closed point and every closed point admits an étale quotient presentation around that point (see [AHLH23, Definition 2.1]). Moreover, we say that

$\mathcal{X}$ is *s-locally reductive* if moreover the étale quotient presentation can be chosen representable and separated.

Notice that since $\mathcal{X}$ is quasi-separated, we have that the étale quotient presentation morphism is quasi-compact. Before going forward, we need to prove two technical facts regarding local reductivity.

Lemma 7.3. *Let $f : \mathcal{X}_1 \to \mathcal{X}_2$ a universally closed and affine morphism of quasi-separated algebraic stacks with affine stabilizers. If $\mathcal{X}_2$ is (s-)locally reductive then $\mathcal{X}_1$ is (s-)locally reductive.*

Proof. Since f is universally specializing and $\mathcal{X}_2$ is locally reductive, the closure of every point $x_1 \in \mathcal{X}_1$ intersects non-trivially the fiber of a closed point in $\mathcal{X}_2$. The intersection is a quasi-compact closed subset of $\mathcal{X}_1$, since f is universally closed, thus quasi-compact. Therefore every point $x_1 \in \mathcal{X}_1$ has a closed point in its closure.

Let now x_1 be a closed point in $\mathcal{X}_1$. We want to costruct an étale quotient presentation of x_1. Let $x_2 := f(x_1)$ and $\alpha_2 : \mathcal{U}_2 \to \mathcal{X}_2$ an étale quotient presentation where $u_2 \in \mathcal{U}$ is a closed point such that $\alpha(u_2) = x_2$. Since the stacks are quasi-separated (thus decent), we can rewrite the property of being stabilizers preserving at u_2 as the commutative diagram:

$$\begin{tikzcd} & \mathcal{U}_2 \arrow[d, "\alpha_2"] \\ G_{u_2} \simeq G_{x_2} \arrow[ur, hook] \arrow[r, hook] & \mathcal{X}_2 \end{tikzcd}$$

where the two closed immersions are (isomorphic to) the two residual gerbe embeddings. Pulling back through the morphism f, we get the diagram

$$\begin{tikzcd} & \mathcal{U}_1 \arrow[d, "\alpha_1"] \\ T \arrow[ur, hook] \arrow[r, hook] & \mathcal{X}_1 \end{tikzcd}$$

where T is a universally closed and affine stack over the residul gerbe of x_2 in $\mathcal{X}_2$. Notice that since $\mathcal{U}_1$ is affine over $\mathcal{U}_2$, it is still a quotient of an affine scheme by GL_N for some N. Since by construction $x_1 \in T$, it follows that $\alpha_1 : (\mathcal{U}_1, u_1) \to (\mathcal{X}_1, x_1)$ is an étale quotient presentation of x_1. □

Lemma 7.4. *Let $\mathcal{X}$ be a locally reductive algebraic stack with affine stabilizers of finite type over a quasi-separated algebraic space S, and let $x_1, x_2 : \operatorname{Spec} k \to \mathcal{X}$ be two geometric points (i.e. k is algebraically closed) such that $\overline{\{x_1\}} \cap \overline{\{x_2\}} \neq \emptyset$ in $\mathcal{X}_k := \mathcal{X} \times_S \operatorname{Spec} k$. Then there exists two morphisms $d_i : \Theta_k \to \mathcal{X}$ for $i = 1, 2$ such that $d_i(1) \simeq f_i(0)$ for $i = 1, 2$ and $d_1(0) \simeq d_2(0)$ is a closed point of $\mathcal{X}_k$.*

Proof. The idea is to try and reduce to the case when $\mathcal{X}_k$ is fundamental, i.e. $\mathcal{X}_k$ is the quotient stack of an affine scheme by GL_N for some N. First of all, since $\mathcal{X}$ is quasi-compact over S, we have $\mathcal{X}_k$ is quasi-compact, therefore there exists a point $x_0 \in \overline{\{x_1\}} \cap \overline{\{x_2\}}$ which is closed in $\mathcal{X}_k$. Let y_0 any closed point in the closure of x_0 in $\mathcal{X}$ and $\alpha : \mathcal{U} \to \mathcal{X}$ be an étale quotient presentation around y_0, with $\mathcal{U} \to U$ the adequate moduli space morphism over S and $\mathcal{U}$ fundamental. Consider now the base-change of α along $\operatorname{Spec} k \to S$: the morphism $\mathcal{U}_k \to \mathcal{X}_k$ is étale and surjective on x_0 (and thus on x_1 and x_2). Let $u_0 \in \mathcal{U}_k$ such that $\alpha_k(u_0) = x_0$ and consider the morphism $\mathcal{U}_k \to U_k$: it is adequately affine (since it is a property stable by base change with algebraic spaces), $\mathcal{U}_k$ is a global quotient (since $\operatorname{Spec} k \to S$ is representable) and thus by considering an étale affine chart of (the image of) u_0 in U_k, we can assume $\mathcal{U}_k$ is fundamental. The result follows by using the Hilbert-Mumford criterion on $\mathcal{U}_k$, see for instance the proof of [AHLH23, Lemma 3.24]. □

Remark 7.5. Notice that, in the notation of the proof of Lemma 7.4, $\mathcal{U}_k$ is not necessarily an étale quotient presentation for x_0, since we do not know if the morphism is stabilizer-preserving in u_0.

Remark 7.6. It is important to remark that the previous lemma is the only reason why we need the quasi-compactness of $\mathcal{X}$ over S.

From now on, since we are going to talk about valuative criteria for DVRs, we will assume to work in the category of *quasi-separated locally noetherian* algebraic stacks (or algebraic spaces). It may possible to remove the Noetherian hypothesis dealing with valuation rings.

We start by proving a simple statement, probably well-known to experts.

Lemma 7.7. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a representable, locally of finite presentation morphism. If f is S-complete, then f is separated.*

Proof. This follows easily using the valuative criterion of separatednes and [AHLH23, Lemma 3.45]. □

As a byproduct, we get the following result.

Lemma 7.8. *Let $\mathcal{X}$ be an algebraic stack which satisfies the uniqueness part of the valuative criterion of S-completeness (or equivalently whose diagonal is S-complete). Then $\mathcal{X}$ has separated diagonal.*

As a consequence, we get the following remark.

Remark 7.9. If $\mathcal{X}$ is a locally reductive algebraic stack whose diagonal is S-complete, then $\mathcal{X}$ is s-locally reductive. Indeed, Lemma 7.8 gives us that $\mathcal{X}$ has separated diagonal, therefore, up to shrinking the étale quotient presentation $\mathcal{U} \to \mathcal{X}$, we can assume it is representable (see [AHLH23, Remark 2.7]). Moreover, since $\mathcal{U}$ is S-complete and the diagonal of $\mathcal{X}$ is S-complete, the étale quotient presentation is an S-complete, representable morphism, thus it is separated thanks to Lemma 7.7.

We now introduce a more topologically flavoured notion of S-completeness. This definition will appear in a more precise form in the future work of David Rydh on topological moduli spaces. Moreover, the *almost topological S-completeness* is inspired by the concept of *weak separatedness* introduced in [ASvdW10]. In what follows, we say that a morphism $f : X \to Y$ is *generically étale* if there exists a dense open $U \subset X$ such that $f|_U$ is étale.

Definition 7.10. We say that an algebraic stack $\mathcal{X}$ over S is *almost topologically S-complete* (over S) if for any complete DVR R with fraction field Q and residue field k and commutative diagram

$$\begin{tikzcd} \operatorname{Spec} Q \arrow[r] \arrow[d, hook] & \mathcal{X} \arrow[d, "\Delta"] \\ \operatorname{Spec} R \arrow[r, "(f_1,f_2)"] & \mathcal{X} \times_S \mathcal{X} \end{tikzcd}$$

there exists an extension of fields L/k with L algebraically closed such that $\overline{\{f_1(0)\}} \cap \overline{\{f_2(0)\}} \neq \emptyset$ in $\mathcal{X}_L := \mathcal{X} \times_S \operatorname{Spec} L$. Moreover, we say that $\mathcal{X}$ is *topologically S-complete* (over S) if for any finite generically étale morphism $\mathcal{X}' \to \mathcal{X}$, the algebraic stack $\mathcal{X}'$ is almost topologically S-complete (over S).

Remark 7.11. Notice that if $\mathcal{X}$ is locally reductive and of finite type over S, in the definition of almost topologically S-complete the conclusion can be made stronger: namely there exists two morphisms $d_i : \Theta_L \to \mathcal{X}$ for $i = 1, 2$ such that $d_i(1) \simeq f_i(0)$ for $i = 1, 2$ and $d_1(0) \simeq d_2(0)$ is a closed point of $\mathcal{X}_L$. This follows from Lemma 7.4.

We are going to show that topological S-completeness is equivalent to S-completeness, in the locally reductive and affine diagonal case. Before that, we recall a simple fact to construct extensions of complete DVRs.

Remark 7.12. Given a complete DVR R with residue field k and an arbitrary extension L/k of fields, there exists a complete DVR R' and a weakly unramified extension $R \subset R'$ (see [Sta23, Tag 09E4] for

the definition of weakly unramified) such that the morphism induced between the residue fields is the extension $k \subset L$. The existence of such an extension can be proved easily by mimicking the proof of [Sta23, Tag 0328].

We are finally ready for the main proposition. It is important to us to remark that the following result was discovered during a conversation of the third author with David Rydh, who we thank immensely for all the time invested in conversation with the three authors.

Proposition 7.13. *Let $\mathcal{X}$ be a s-locally reductive algebraic stack, of finite type over S. Suppose that $\mathcal{X}$ is topologically S-complete. Then it verifies the existence part of the valuative criterion for S-completeness up to (weakly unramified) extensions of (complete) DVRs.*

Proof. Recall that it is enough to prove the existence part of S-completeness for complete DVRs (see [AHLH23, Lemma 3.41(1)]). We start with a complete DVR R with residue field k and fraction field Q and a morphism $\mathrm{ST}_R \to \mathcal{X}$, or equivalently a commutative diagram

$$\begin{tikzcd} \operatorname{Spec} Q \arrow[r] \arrow[d] & \mathcal{X} \arrow[d, "\Delta"] \\ \operatorname{Spec} R \arrow[r, "(f_1,f_2)"] & \mathcal{X} \times_S \mathcal{X}. \end{tikzcd}$$

By the topological S-completeness (and Remark 7.11), we get an extension of fields L/k and two morphisms $d_1, d_2 : \Theta_L \to \mathcal{X}$ such that $d_1(1) = f_1(0)$, $d_2(1) = f_2(0)$ and $x_0 := d_1(0) = d_2(0)$ is a closed point in $\mathcal{X}_L$. Since, we want to prove the existence part of the valuative criterion up to weakly unramified extensions of DVRs, Remark 7.12 allows us to extend the field k arbitrarily. Therefore, for simplicity, we will just avoid writing down the extensions, and from now on k will be an algebraically closed field. Let now $y_0 \in \mathcal{X}$ be a closed point in the closure of $\{x_0\}$, and let $u : \mathcal{U} \to \mathcal{X}$ be a representable separated étale quotient presentation of $\mathcal{X}$ around y_0. Since $\mathcal{X}$ is quasi-separated and $\mathcal{U}$ is quasi-compact, u is quasi-compact thus quasi-finite. We can then apply Zariski Main Theorem (see for instance Theorem 16.5 of [LMB00]) to get the commutative diagram

$$\begin{tikzcd} \mathcal{U} \arrow[r, hook] \arrow[dr, "u"'] & \mathcal{V} \arrow[d, "v"] \\ & \mathcal{X} \end{tikzcd}$$

where v is finite, and $\mathcal{U}$ is a dense open substack of $\mathcal{V}$ (we can always take the closure of $\mathcal{U}$ in $\mathcal{V}$). Since y_0 is in the image of the morphism u, the same is true for x_0. Thanks to the étaleness of u we can lift the morphism d_1 (respectively f_1) to a morphism $c_1 : \Theta_k \to \mathcal{U}$ where we define $w_0 := c_1(0)$ (respectively $h_1 : \operatorname{Spec} R \to \mathcal{U}$ such that $h_1(0) = c_1(1)$). To construct c_1, note that the pullback of $u : \mathcal{U} \to \mathcal{X}$ along $B\mathbb{G}_m \subset \Theta_k \to \mathcal{X}$ is a disjoint union of $B\mathbb{G}_m$'s, as u is étale and k is algebraically closed, so we may choose a section and then apply Lemma A.5. The existence of $h_1 : \operatorname{Spec} R \to \mathcal{U}$ follows similarly. Moreover, the finiteness of v allows us to lift the morphism f_2 to a morphism $h_2 : \operatorname{Spec} R \to \mathcal{V}$ such that $h_2(1) = h_1(1) \in \mathcal{U}$. Therefore we have a morphism $(h_1, h_2) : \mathrm{ST}_R \to \mathcal{V}$ given by gluing the two morphisms h_1 and h_2. Since $\mathcal{X}$ is topologically S-complete and v is finite and generically étale, the same is true for $\mathcal{V}$, thus we get two morphisms $b_1, b_2 : \Theta_k \to \mathcal{V}$ such that $b_i(1) = h_i(0)$ for $i = 1, 2$ and $b_1(0) = b_2(0)$ (this is again Remark 7.11 and Lemma 7.3). To help visualizing the combinatorics of

the specializations, we have the following diagram

$$\begin{tikzcd}
w_0 & h_1(0) = b_1(1) = c_1(1) \arrow[l, dashed, "c_1"'] \arrow[dl, dashed, "b_1"] & \\
b_1(0) = b_2(0) & & h_1(1) = h_2(1) \arrow[ul, "h_1"'] \arrow[dl, "h_2"] \\
& h_2(0) = b_2(1) \arrow[ul, dashed, "b_2"] &
\end{tikzcd}$$

where the solid arrows represents specializations over R while the dashed arrows are isotrivial degenerations. Notice that, if $b_1(0) = b_2(0) \in \mathcal{U} \subset \mathcal{V}$, then the morphism (h_1, h_2) factors through $\mathcal{U}$, which is S-complete since it has an affine adequate moduli space. Composing with the morphism u, we win.

Therefore, it is enough to prove that $b_1(0) = b_2(0) \in \mathcal{U}$. Notice that any morphism from Θ_k induces a morphism from $\operatorname{Spec}(k[[t]])$ by composition, therefore we can see c_1 and b_1 as two morphisms from $\operatorname{Spec}(k[[t]])$ which agree on the generic point. Topological S-completeness gives us again two morphisms $a, a' : \Theta_k \to \mathcal{V}$ such that $a(1) = w_0$, $a'(1) = b_1(0)$ and $a(0) = a'(0)$. We end up with a diagram of specialization

$$\begin{tikzcd}
& w_0 \arrow[dl, dashed, "a"'] & \\
a(0) = a'(0) & & c_1(1) = b_1(1) \arrow[ul, dashed, "c_1"'] \arrow[dl, dashed, "b_1"] \\
& b_1(0) \arrow[ul, dashed, "a'"] &
\end{tikzcd}$$

inside $\mathcal{V}_k := \mathcal{V} \times_S \operatorname{Spec} k$ which tells us it is enough to prove that $a(0) \in \mathcal{U}_k := \mathcal{U} \times_S \operatorname{Spec} k$. Since $a(1) = w_0 \in \mathcal{U}_k$, it is enough to prove that $a(0) = a(1)$.

Consider the morphism $v \circ a : \Theta_k \to \mathcal{X}_k$. By construction, $(v \circ a)(1) = x_0$, which is a closed point in $\mathcal{X}_k$, thus the morphism $v \circ a$ factors through the residual gerbe of x_0. This gives us that $a : \Theta_k \to \mathcal{V}$ factors through a finite stack over the residual gerbe of x_0. Since Θ is connected, it means that the topological image is one point, thus $a(0) = a(1)$. This concludes the proof. □

Before going forward, let us discuss the finiteness assumptions.

Remark 7.14. First of all, notice that we assumed that $\mathcal{X}$ is locally of finite type over S only to be able to apply [AHLH23, Lemma 3.24] in Lemma 7.4 and thus be able to produce isotrivial degenerations (as it happens for normal specializations). Regarding the relative quasi-compactness, we tried to avoid using it, but we did not manage to. The only obstruction is a proof of Lemma 7.4 without the quasi-compactness. We did not manage to prove that if $\mathcal{X}$ is locally reductive, then every point in the geometric fiber $\mathcal{X}_s$ over $s \in S$ has a closed point in its closure.

Nevertheless, notice that both the assumptions are used in the proof of Lemma 7.4. Indeed, it is possible to rewrite the proof of Proposition 7.13 avoiding the use of Θ as a 0-dimensional DVR and thus obtain the exact same result without the stack being of finite type. Such a proof relies on the language of *topological moduli space*, thus we decided to leave it out of this elaborate, as it will probably appear in the future work of David Rydh on topological moduli spaces.

Using Proposition 7.13, we will prove two general results, which turn to be useful in dealing with S-completeness. The first result allows us to check the existence part of the valuative criterion of S-completeness "generically" (see definition below) and we will spend the rest of this section proving it. We leave the other result for the next section. Let us now give the precise definition of generic S-completeness for an algebraic stack over a base S. We take the opportunity to remind that every stack (and algebraic space) in the coming discussion is quasi-separated and locally noetherian.

Definition 7.15. We say that an algebraic stack $\mathcal{X}$ is *generically* S-complete (over S) if the valuative criterion for S-completeness holds with the extra assumption that the morphism $\operatorname{Spec} Q \to \mathcal{X}$ lands in the set $\operatorname{Irr}(\mathcal{X})$ of generic points of irreducible components of $\mathcal{X}$.

Lemma 7.16. *The property of being generically S-complete is stable by composition with finite generically étale morphisms.*

Proof. It is enough to notice that generically étale morphisms (indeed flatness is enough) sends (the generic points of) irreducible components to irreducible components. □

Lemma 7.17. *Let $\mathcal{X}$ be an algebraic stack. If $\mathcal{X}$ is generically S-complete (over S), then $\mathcal{X}$ is topologically S-complete (over S).*

Proof. By Lemma 7.16, it is enough to show that $\mathcal{X}$ is almost topologically S-complete. We start with a morphism $\mathrm{ST}_R \to \mathcal{X}$, or equivalently a commutative diagram

$$\begin{tikzcd} \operatorname{Spec} Q \arrow[r] \arrow[d] & \mathcal{X} \arrow[d, "\Delta"] \\ \operatorname{Spec} R \arrow[r, "(f_1,f_2)"] & \mathcal{X} \times \mathcal{X}. \end{tikzcd}$$

Since $(f_1(0), f_2(0))$ is in the closure of the image of Δ, we know that there exists a DVR R' and a morphism $(g_1, g_2) : \operatorname{Spec} R' \to \mathcal{X} \times \mathcal{X}$ such that $f_i(0) = g_i(0)$ for $i = 1, 2$ and $(g_1, g_2)(1)$ is the image of a point in $\operatorname{Irr}(\mathcal{X})$ (see [LMB00, Proposition 7.2.1]). Thus, we can assume we have a diagram as above for R', and by construction $\operatorname{Spec} Q' \to \mathcal{X}$ lands in $\operatorname{Irr}(\mathcal{X})$. Therefore, by generic S-completeness there exists a morphism $G : \overline{\mathrm{ST}}_{R'} \to \mathcal{X}$ extending the morphism (g_1, g_2) and thus the closures of the two points $f_1(0)$ and $f_2(0)$ intersect non-trivially. □

Finally, we are going to prove that generic S-completeness implies S-completeness in the affine diagonal case.

Proposition 7.18. *Let $\mathcal{X}$ be a locally reductive algebraic stacks with affine diagonal which is of finite type over an algebraic space S. Then $\mathcal{X}$ is generically S-complete if and only if it is S-complete.*

Proof. The "if" statement is clear. Since affine implies S-complete, Remark 7.9 gives us that $\mathcal{X}$ is s-locally reductive and the uniqueness part of S-complete hold. Proposition 7.13 and Lemma 7.17 ensure the existence part of S-completeness up to extension of DVRs. Because $\mathcal{X}$ has affine diagonal, we have that the existence part holds without need to extend (see [AHLH23, Lemma 3.42(2)]) and we win. □

One may wonder if the result holds if $\mathcal{X}$ has S-complete diagonal, instead of affine. While the uniqueness would be verified, it is not clear the existence part of S-completeness up to extensions would be enough, since the proof of [AHLH23, Proposition 3.41(2)] relies heavily on the affine diagonal. For the rest of the section, we will try to advocate (ending with a slight generalization of Proposition 7.18) why we believe that having S-complete diagonal could be enough to prove that generic S-completeness implies S-completeness. We start by recalling an alternative description of S-completeness: it was introduced in [AHLH23, Proposition 3.42], in a slightly different form.

Remark 7.19. Everything will be over the base S. Let us start by considering the commutative diagram

$$\begin{tikzcd} \mathbb{A}^1 \arrow[r, shift left, "(1,\mathrm{Id})"] \arrow[r, shift right, "(\mathrm{Id},1)"'] & \mathbb{A}^2 \arrow[r, "m"] & \mathbb{A}^1 \end{tikzcd}$$

where m is the multiplication map. The morphisms are equivariant with respect to the following diagram of abelian groups

$$\mathbb{G}_m \underset{(\mathrm{Id},1)}{\overset{(1,\mathrm{Id})}{\rightrightarrows}} \mathbb{G}_m^2 \xrightarrow{m} \mathbb{G}_m$$

where $\mathbb{G}_m$ acts with weight 1 on $\mathbb{A}^1$ on $\mathbb{G}_m^2$ acts with weights $((1,0),(0,1))$ on $\mathbb{A}^2$. By abuse of notation, we will denote by m both the multiplication morphisms above, as well as the induced one at the level of quotient stacks.

Taking the quotient, we get the commutative diagram

$$\Theta := [\mathbb{A}^1/\mathbb{G}_m] \underset{(\mathrm{Id},1)}{\overset{(1,\mathrm{Id})}{\rightrightarrows}} \Theta^2 = [\mathbb{A}^2/\mathbb{G}_m^2] \xrightarrow{m} \Theta = [\mathbb{A}^1/\mathbb{G}_m];$$

the morphism m coincides with the good moduli space morphism $\overline{\mathrm{ST}}_R \to \operatorname{Spec} R$ when we consider R to be the 0-dimensional DVR Θ. More specifically, $\overline{\mathrm{ST}}_R \to \operatorname{Spec} R$ can be obtained pulling back m through the morphism $e : \operatorname{Spec} R \to \Theta$, which sends the variable t in $\mathbb{A}^1$ to the uniformizer π of R.

Finally, applying the Hom-stack functor $\mathrm{Hom}_\Theta(-, \mathcal{X} \times \Theta)$ (viewing Θ and Θ^2 as stacks over Θ respectively via the identity and the multiplication map) to the commutative diagram above, we get

$$\mathcal{X} \times \Theta \xrightarrow{m^*} \mathrm{Hom}_\Theta(\Theta^2, \mathcal{X} \times \Theta) \xrightarrow{s} (\mathcal{X} \times \Theta) \times_\Theta (\mathcal{X} \times \Theta);$$

where s corresponds to the pair of morphisms induced by $(1, \mathrm{Id})$ and $(\mathrm{Id}, 1)$, thus $s \circ m^*$ is the diagonal morphism. Before going forward, we should mention that it is not clear a priori that the Hom-stack is algebraic. Indeed the morphism m is a good moduli space morphism, but Theorem 10.17 in [AHR23] is only stated when the target of the morphism is an algebraic space. Nevertheless, if one consider the pullback of $S(\mathcal{X})/\Theta$ along the $\mathbb{G}_m$-torsor $\mathbb{A}^1 \to \Theta$, we get $S(\mathcal{X}) \times_\Theta \mathbb{A}^1 \simeq \mathrm{Map}_{\mathbb{A}^1}([\mathbb{A}^2/\mathbb{G}_m], \mathcal{X})$ which corresponds to the version of $S(\mathcal{X})$ in [AHLH23, Proposition 3.42], and it representable by an algebraic stack. Since $S(\mathcal{X}) \times_\Theta \mathbb{A}^1 \to S(\mathcal{X})$ is smooth, surjective and representable, we can apply [Sta23, Tag 05UL] and we get the algebraicity. Moreover, as in the proof of [AHLH23, Proposition 3.42], the morphism s is representable and locally of finite type. Notice that we are not claiming any finiteness condition on s: indeed we only know it has quasi-separated diagonal. In fact, we know it is not quasi-compact, as pointed out in *loc.cit.*

The valuative criterion for S-completeness can be rephrased as a valuative criterion for DVRs: an algebraic stack satisfies the existence part (respectively the uniqueness part) of the valuative criterion for S-completeness if and only if for every DVR R and every commutative diagram of solid arrows

$$\begin{tikzcd}
\operatorname{Spec} Q \arrow[r] \arrow[dd, hook] & \mathcal{X} \times \Theta \arrow[d, "m^*"] \\
& S(\mathcal{X}) := \mathrm{Hom}_\Theta(\Theta^2, \mathcal{X} \times \Theta) \arrow[d, "s"] \\
\operatorname{Spec} R \arrow[r] \arrow[ur, dashed] \arrow[dr, "e"'] & (\mathcal{X} \times \mathcal{X}) \times \Theta \arrow[d] \\
& \Theta
\end{tikzcd}$$

there exists (respectively there exists at most one, up to isomorphisms) a dotted arrow which makes the whole diagram commute. The morphism $e : \operatorname{Spec} R \to \Theta$ is determined by the association $t \mapsto \pi$ where t is the coordinate of $\mathbb{A}^1$ and π is a uniformizer of R. Notice that $\operatorname{Spec} Q \to \mathcal{X} \times \Theta$ factors through the open $\mathcal{X} \subset \mathcal{X} \times \Theta$ because of the morphism e. Nevertheless, it is good to remind that, as stated,

it is not a valuative criterion for the morphisms s. We are only checking the lifting for "horizontal degenerations", namely not contained in any of the two fibers over Θ. Indeed, if we suppose s is universally specializing (i.e. it verifies the existence part of the valuative criterion), then the diagonal of $\mathcal{X}$ would be universally closed, which is not implied by S-completeness.

Moreover, the morphism m^* is a monomorphism. Indeed, m is a retraction thus m^* is a section, and we believe (although we do not prove it) that it is a dense open embedding. We leave it to the interested reader.

Finally, we will explain how to potentially show that, in the case of S-complete diagonal, it is enough to check the existence part of S-completeness up to arbitrary extensions of DVRs.

Remark 7.20. Let us start with a diagram as in Remark 7.19

$$\begin{tikzcd}
\operatorname{Spec} Q \arrow[r] \arrow[dd, hook] & \mathcal{X} \times \Theta \arrow[d, "m^*"] \\
& S(\mathcal{X}) := \operatorname{Hom}_\Theta(\Theta^2, \mathcal{X} \times \Theta) \arrow[d, "s"] \\
\operatorname{Spec} R \arrow[r] \arrow[ur, dashed] \arrow[dr, "e"'] & (\mathcal{X} \times \mathcal{X}) \times \Theta \arrow[d] \\
& \Theta
\end{tikzcd}$$

and suppose we have the existence of the dotted arrow up to an arbitrary extension of DVRs $R \subset R'$. The idea is to use faithfully flat descent on the morphism $\operatorname{Spec} R' \to \operatorname{Spec} R$ to show that the morphism $\operatorname{Spec} R' \to S(\mathcal{X})$ descend to a morphism $\operatorname{Spec} R \to S(\mathcal{X})$. First of all, notice that the diagonal of $s : S(\mathcal{X}) \to (\mathcal{X} \times \mathcal{X}) \times \Theta$ is a monomorphism, which implies it is enough to check that the two morphisms

$$\operatorname{Spec}(R' \otimes_R R') \rightrightarrows \operatorname{Spec} R' \longrightarrow S(\mathcal{X})$$

are isomorphic. Now, if we assume $\mathcal{X}$ quasi-separated, the only obstruction to a proof of the statement is the *quasi-separatedness* of the morphism s (which we know is not quasi-compact). Indeed, if s is quasi-separated

- we can reduce to the case when m^* is dense, by considering the schematic closure $\mathcal{S}$ of the image of m^* in $S(\mathcal{X})$: s quasi-separated implies m^* quasi-compact (since it is a section of a quasi-separated morphism) therefore the image of m^* is dense in $\mathcal{S}$;
- by the uniqueness part of S-completeness, we know that the morphism s restricted to $\mathcal{S}$ is generically separated (since m^* is dominant and the diagonal of $\mathcal{X}$ is separated); moreover, we know that for a quasi-separated representable morphism, separatedness can be checked generically (this is [Sta23, Tag 089G]), thus we conclude that the restriction of s to $\mathcal{S}$ is separated.

We have two morphism

$$\operatorname{Spec}(R' \otimes_R R)' \rightrightarrows \mathcal{S}$$

which agree on $(\mathcal{X} \times \mathcal{X}) \times \Theta$ and on a dense open of $\operatorname{Spec}(R' \otimes_R R')$. Since $s : \mathcal{S} \to (\mathcal{X} \times \mathcal{X}) \times \Theta$ is separated, we know that the equalizer morphism

$$E := \operatorname{Spec}(R' \otimes_R R') \times_{\mathcal{S} \times_s \mathcal{S}} \mathcal{S} \subset \operatorname{Spec}(R' \otimes_R R')$$

is a closed embedding. By construction, the two functions agree on the open subset $\operatorname{Spec}(Q' \otimes_Q Q')$ of $R' \otimes_R R'$, where Q and Q' are the fraction fields of R and R'. Therefore, we have that $\operatorname{Spec}(Q' \otimes_Q Q') \subset E$. Moreover, thanks to the flatness of the extension, we have that $R' \otimes_R R' \subset Q' \otimes_Q Q'$, thus $E = R' \otimes_R R'$ and we win.

We do not know if s is quasi-separated, yet. Nevertheless, the discussion in Remark 7.20 helps us slightly generalizing Proposition 7.18 as follows.

Proposition 7.21. *Let $\mathcal{X}$ be locally reductive algebraic stack with quasi-affine and S-complete diagonal which is of finite type over an algebraic space S. Then $\mathcal{X}$ is generically S-complete if and only if it is S-complete.*

Proof. By [AHR23, Theorem 10.17], $S(\mathcal{X})$ has quasi-affine diagonal if $\mathcal{X}$ does, in particular it is quasi-separated, making the conditional discussion above unconditional. □

Notice that the quasi-affineness is only used to ensure the quasi-separatedness of $S(\mathcal{X})$ (see [AHR23, Theorem 10.17]).

If Theorem B.1 in [HR19] can be proved for flat good moduli space morphisms, then the quasi-separatedness of s would follow. More specifically, we conjecture the following statement.

Conjecture 7.22. Let $\mathcal{X}$ be a quasi-separated algebraic stack. Then the stack $S(\mathcal{X}) := \mathrm{Hom}_{\Theta}(\Theta^2, \mathcal{X} \times \Theta)$ is quasi-separated.

As corollary of Conjecture 7.22, we get the following result. Again, we remind that we are working in the category of locally noetherian quasi-separated algebraic stacks.

Corollary 7.23. *Let $\mathcal{X}$ be an algebraic stack with S-complete diagonal. Then the existence part of S-completeness holds if and only if it does after arbitrary extensions of DVR's.*

Moreover, we get the following corollary, as a consequence of Corollary 7.23, together with Remark 7.14.

Corollary 7.24. *Let $\mathcal{X}$ be a locally reductive algebraic stack with S-complete diagonal over an algebraic space S. Then $\mathcal{X}$ is S-complete (over S) if and only if it is generically S-complete (over S).*

This is the perfect mirror of what happens between separatedness and universal closedness.

7.2 Weak S-completeness

We recall that

$$\mathrm{ST}_R = [\mathrm{Spec}\,(R[s,t]/(st-\pi)) \setminus 0/\mathbb{G}_m] \cong \mathrm{Spec}\, R \cup_{\mathrm{Spec}\, K} \mathrm{Spec}\, R$$

and

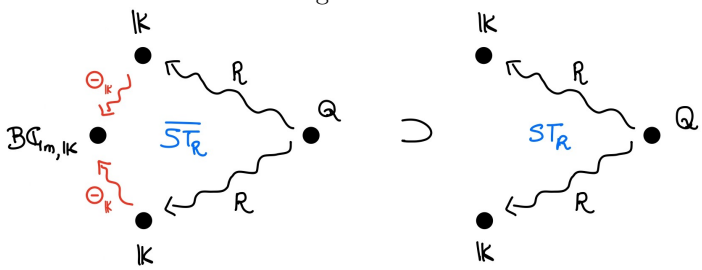
$$\overline{\mathrm{ST}}_R := [\mathrm{Spec}\,(R[s,t]/(st-\pi))\,/\mathbb{G}_m]$$

where $\mathbb{G}_m$ acts on s with weight 1 and on t with weight -1, as introduced in [Hei17, Section 2.B] and [AHLH23, Section 3.5].

As the reader probably realized already, proving S-completeness can be particularly tricky. We will make use of Proposition 7.13 one more time. Specifically, we will consider a weaker property, essentially introduced in the second part of [Mod25], which we call *weak S-completeness*, and show that it implies S-completeness for sufficiently nice stacks.

Informally, a map from $\overline{\mathrm{ST}}_R$ into an algebraic stack parametrizes a point, two different degenerations over the DVR R from that point to two other points and a specified S-equivalence between those points, given by two maps from Θ.

Figure 29:



In weak S-completeness, one allows for a zig-zag of specified isotrivial degenerations.

Definition 7.25. Let

$$\mathfrak{S}_{n,R} := \overline{\mathrm{ST}}_R \cup_{\mathrm{Spec}\, R} \dots \cup_{\mathrm{Spec}\, R} \overline{\mathrm{ST}}_R$$

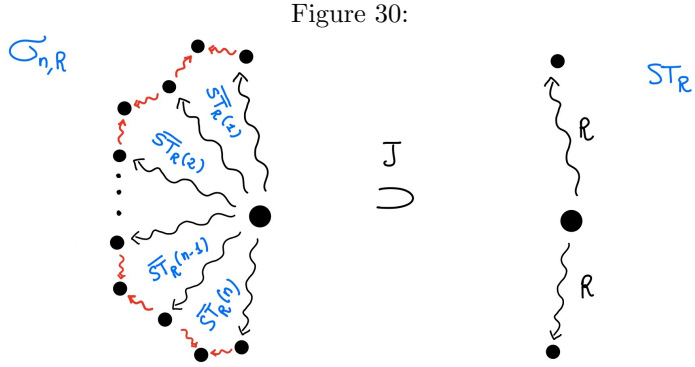
be the gluing of n copies of $\overline{\mathrm{ST}}_R$ along $\mathrm{Spec}\, R$ such that each $\mathrm{Spec}\, R \subset \mathfrak{S}_n$ is an open in either one or two copies of $\overline{\mathrm{ST}}_R$.

We denote by $j_h : \overline{\mathrm{ST}}_R(h) \subset \mathfrak{S}_{n,R}$ the open embedding associated to the h-th copy of $\overline{\mathrm{ST}}_R$. There is an unique open immersion $j : \mathrm{ST}_R \subset \mathfrak{S}_{n,R}$ such that the two copies of $\mathrm{Spec}\, R \subset \mathrm{ST}_R \subset \mathfrak{S}_{n,R}$

- factor respectively through at least one copy of $\mathrm{ST}_R \subset \mathfrak{S}_{n,R}$;
- are included respectively in exactly one copy of $\overline{\mathrm{ST}}_R \subset \mathfrak{S}_{n,R}$.

We have that the open substack ST_R associated to j intersects non-trivially only $\overline{\mathrm{ST}}(1)$ and $\overline{\mathrm{ST}}(n)$ (in a copy of $\mathrm{Spec}\, R$).

Figure 30:



We also get a structural morphism $\mathfrak{S}_{n,R} \to \mathrm{Spec}\, R$.

Finally, we denote by $\mathfrak{S}^\circ_{n,R} \subset \mathfrak{S}_{n,R}$ the open complement of the stacky points of $\mathfrak{S}_{n,R}$, which is the local model of the line with $n+1$ origins, i.e. $\mathrm{Spec}\, R \cup_{\mathrm{Spec}\, Q} \dots \cup_{\mathrm{Spec}\, Q} \mathrm{Spec}\, R$..

Remark 7.26. Let $f : \mathfrak{S}_{n,R} \to \mathcal{Z}$ be any morphism where $\mathcal{Z}$ is an algebraic stack locally of finite presentation and quasi-separated over a quasi-separated algebraic space. If $\mathcal{Z}$ has quasi-finite diagonal, then f factors through the structural morphism $\mathfrak{S}_{n,R} \to \mathrm{Spec}\, R$. This follows from the fact that the restriction $f \circ j_h : \overline{\mathrm{ST}}_R(h) \subset \mathfrak{S}_{n,R} \to \mathcal{Z}$ factors through the $\overline{\mathrm{ST}}_R \to \mathrm{Spec}\, R$, which is a consequence of Lemma 3.45 of [AHLH23].

We recall again that, since we are discussing valuative criteria, we restrict ourselves to the category of locally noetherian, quasi-separated algebraic stacks.

Definition 7.27. We say that an algebraic stack $\mathcal{X}$ over an algebraic space S is weakly S-complete if for all solid commutative diagrams

$$\begin{tikzcd} & \mathrm{ST}_R \arrow[r] \arrow[d] \arrow[dl, dotted, "j"'] & \mathcal{X} \arrow[d] \\ \mathfrak{S}_{n,R} \arrow[r, dotted] \arrow[ur, dotted] & \operatorname{Spec} R \arrow[r] & S \end{tikzcd}$$

there exists an integer n and a dotted arrow as above making the diagram commute, up to an extension $R \subset R'$ of DVRs.

Remark 7.28. Contrary to the definition of S-completeness given in [AHLH23], the definition of weak S-completeness is stated only for morphisms to algebraic spaces. The reason is that there are several possible extensions to arbitrary morphisms, depending on what is the diagram of solid arrows one would like to start with. We decide not to include a discussion regarding those and it seemed not relevant to this result. Nevertheless, all the notions we came up with agree in the case of a morphism to an algebraic space (or to an algebraic stack with quasi-finite diagonal).

Notice that S-completeness for the morphism $\mathcal{X} \to S$ implies weak S-completeness. Moreover, it is clear that weak S-completeness is stable under base change for morphisms $S' \to S$ between algebraic space.

Lemma 7.29. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a universally closed and affine morphism of locally noetherian algebraic stacks over an algebraic space S. If $\mathcal{Y}$ is weakly S-complete, then $\mathcal{X}$ is weakly S-complete.*

Proof. First of all, suppose given a commutative diagram of solid arrows

$$\begin{tikzcd} \mathrm{ST}_R \arrow[r, "j"] \arrow[d] & \mathfrak{S}_{n,R} \arrow[r] \arrow[d, dashed] & \operatorname{Spec} R \arrow[d] \\ \mathcal{X} \arrow[r, "f"] & \mathcal{Y} \arrow[r] & S; \end{tikzcd}$$

and, because $\mathcal{Y}$ is weakly S-complete, then we get the dashed arrow as above. Because f is universally closed, we have that the morphism $\mathfrak{S}^{\circ}_{n,R} \subset \mathfrak{S}_{n,R} \to \mathcal{Y}$ can be lifted to a morphism $\mathfrak{S}^{\circ}_{n,R} \to \mathcal{X}$ extending $\mathrm{ST}_R \to \mathcal{X}$. Moreover, we have the following commutative diagram

$$\begin{tikzcd} \mathrm{ST}_R(h) \arrow[r, hook] \arrow[d, hook] & \mathfrak{S}^{\circ}_{n,R} \arrow[r] \arrow[d, hook] & \mathcal{X} \arrow[d, "f"] \\ \overline{\mathrm{ST}}_R(h) \arrow[r, hook] & \mathfrak{S}_{n,R} \arrow[r] & \mathcal{Y} \end{tikzcd}$$

where the left-hand side square is cartesian and $\mathrm{ST}_R(h) \subset \overline{\mathrm{ST}}_R(h)$ is the h-th copy of $\overline{\mathrm{ST}}_R$ in $\mathfrak{S}_{n,R}$. Because f is affine, we can find a morphism $\overline{\mathrm{ST}}_R(h) \to \mathcal{X}$ extending the morphism $\mathrm{ST}_R(h) \to \mathcal{X}$. The claim follows by gluing the morphisms. $\square$

Lemma 7.30. *Let $\mathcal{X}$ be a locally reductive algebraic stack (with affine stabilizers) which is of finite type over an algebraic space S. If $\mathcal{X}$ is Θ-complete, for every geometric point $x_1 : \operatorname{Spec} k \to \mathcal{X}$ there exists a unique closed point x_0 in the closure of x_1 in $\mathcal{X}_k$ and a morphism $\Theta_k \to \mathcal{X}_k$ such that the closed point of Θ_k maps to x_0.*

Proof. This is a direct consequence of the proof of Lemma 3.24 and Lemma 3.25 of [AHLH23], imitating the strategy adopted in Lemma 7.4. Notice that the quasi-compactness matters for this proof. $\square$

Corollary 7.31. *Let $\mathcal{X}$ be a locally reductive algebraic stack, of finite type over an algebraic space S. Suppose that $\mathcal{X}$ is Θ-complete and let $f_1, f_2 : \Theta_k \to \mathcal{X}_k$ two morphisms over a geometric point $\operatorname{Spec} k \to S$. Moreover, let $x_i^1 \in \mathcal{X}(k)$ (respectively $x_i^0 \in \mathcal{X}(k)$) be the image of the generic point (respectively the closed point) of Θ_k through f_i for $i = 1, 2$. If $x_1^1 \simeq x_2^1$, then the closures of x_1^0 and x_2^0 have the same unique closed point.*

Proof. This is a simple consequence of the fact that x_1^0 and x_2^0 are contained in the closure of $x_1^1 \simeq x_2^1$. □

We are ready to prove the main result of this subsection.

Proposition 7.32. *Let $\mathcal{X}$ be a locally reductive algebraic stack of finite type over an algebraic space S. If $\mathcal{X}$ is Θ-complete and weakly S-complete, then $\mathcal{X}$ is topologically S-complete.*

Proof. First of all, notice that both weak S-completeness and Θ-completeness lift along finite morphisms (see Lemma 7.29), thus it is enough to prove almost topological S-completeness. Let R be a DVR with residue field k. By weak S-completeness, given a morphism $f^\circ = (f_1, f_2) : \mathrm{ST}_R \to \mathcal{X}$ there exists an integer $n > 0$ and a morphisms $f : \mathfrak{S}_{n,R} \to \mathcal{X}$ extending f°. If we restrict along the closed fiber of $\operatorname{Spec} R \to S$, we get a morphisms $d : D_n \to \mathcal{X}_k$ where D_n is the fiber of the structural morphism $\mathfrak{S}_{n,R} \to \operatorname{Spec} R$. Such a fiber corresponds to $2n$ morphisms $d_i : \Theta_k^{(i)} \to \mathcal{X}$ such that $d_1(1) = f_1(0)$, $d_{2n}(1) = f_2(0)$, $d_{2i-1}(0) = d_{2i}(0)$, $d_{2i}(1) = d_{2i+1}(1)$ for every $1 \leq i \leq n-1$ (this is the zig-zag we refer to before Definition 7.25). Finally, we can apply Corollary 7.31 to conclude. □

Corollary 7.33. *Let $f : \mathcal{X} \to S$ be a locally reductive algebraic stack with affine diagonal which is of finite presentation, Θ-complete and weakly S-complete over an algebraic space S. Then $\mathcal{X}$ is S-complete over S.*

Proof. This follows from Proposition 7.32 together with Proposition 7.13. □

Remark 7.34. It is again worth specifying that Proposition 7.32 also holds without the "finite type" hypothesis, using a more refined topological proof, as pointed out in Remark 7.14. Moreover, if Conjecture 7.22 holds, we have that the statement of Corollary 7.33 can be generalized as follows: let $\mathcal{X}$ be a locally reductive algebraic stack with S-complete diagonal over S; if $\mathcal{X}$ is Θ-complete and weakly S-complete, then it is S-complete.

7.3 Algebraic stabilization over $\overline{\mathrm{ST}}_R$

In this subsection, we will show how the stabilization procedure of A-singularities can be carried out over $\overline{\mathrm{ST}}_R$, which turns out to be incredibly useful in the proof of S-completeness. This will be done in two steps essentially: we first state and prove an abstract stabilization result at the level of the (formal) deformation space; afterwards we show how to pass from local to global and to provide the stabilization result at the level of the moduli stack of curves.

It is important to remark that the abstract stabilization result is not new. In fact, we are essentially applying Langton's algorithm to the case of a $\mathbb{G}_m$-representation. See [AHLH23, Theorem 6.3].

Abstract stabilization

Recall that given a morphism $f : \operatorname{Spec} R \to \mathcal{X}$ we denote by $f_1(0)$ (respectively $f_1(1)$) the image in $\mathcal{X}$ of the closed point (respectively the generic point). The same notation will be used for morphisms from Θ. Moreover, we denote by $j_1 : \operatorname{Spec} R \subset \overline{\mathrm{ST}}_R$ and $j_2 : \operatorname{Spec} R \subset \overline{\mathrm{ST}}_R$ the two open immersions corresponding to $t \neq 0$ and $s \neq 0$.

Before proving the main lemma of the section, we write down a remark which will be surprisingly useful.

Remark 7.35. Let R be a DVR with uniformizer π. Consider the scheme $C_n := \operatorname{Spec} R[s,t](st-\pi^n)$ and let $\mathbb{G}_m$ acts on C_n with weights $(1,-1)$ on (s,t) and trivially on $\operatorname{Spec} R$. Let $\mathcal{C}_n$ be the algebraic quotient stack $[C_n/\mathbb{G}_m]$ and notice that the complement of $(s,t)=(0,0)$ is a copy of ST_R. Then there exists a morphism $p_n : \overline{\mathrm{ST}}_R \to \mathcal{C}_n$ and the composition

$$\mathrm{ST}_R \subset \overline{\mathrm{ST}}_R \to \mathcal{C}_n$$

coincides with the open embedding described above. The morphism p_n can be constructed in the following way: consider the morphism

$$q_n : \operatorname{Spec} R[s,t]/(st-\pi) \longrightarrow C_n := \operatorname{Spec} R[s,t](st-\pi^n)$$

such that $s \mapsto s^n$ and $t \mapsto t^n$. If we endow source and target with the $\mathbb{G}_m$-action of weight 1, we have that q_n is equivariant with respect to the homomorphism $\mathbb{G}_m \to \mathbb{G}_m$ given by the association $\lambda \mapsto \lambda^n$. Therefore, after the quotient we obtain the desired morphism $p : \overline{\mathrm{ST}}_R \to \mathcal{C}_n$. Notice that p is not representable: indeed it is a relative coarse moduli space morphism. It gives rise to the cartesian diagram

$$\begin{tikzcd}
\big[\operatorname{Spec}\big(R[s,t]/(st-\pi)\big)/\mu_n\big] \arrow[r, "p'_n"] \arrow[d] & C_n \arrow[d] \\
\overline{\mathrm{ST}}_R \arrow[r] & \mathcal{C}_n
\end{tikzcd}$$

where μ_n acts on (s,t) with weights $(1,-1)$. Moreover, p'_n is the coarse moduli space morphism which describes C_n as a finite quotient singularity.

Again, we recall that the following statement follows directly from [AHLH23, Theorem 6.3]. Nevertheless, we decided to write down the proof of this much simpler case to give the reader an idea on how it works.

Lemma 7.36. *Let R be a DVR and V be a finite dimensional $\mathbb{G}_m$-representation with trivial isotopic component of weight 0. Suppose given a morphism $f_1 : \operatorname{Spec} R \to V$ such that $f_1(0) \in V^>$ and $f_1(1) \notin V^>$. Then, up to extension of DVRs, there exists a commutative diagram*

$$\begin{tikzcd}
\operatorname{Spec} R \arrow[r, hook, "j_1"'] \arrow[rr, bend left, "f_1"] & \overline{\mathrm{ST}}_R \arrow[r, "F"'] & {[V/\mathbb{G}_m]}
\end{tikzcd}$$

such that $f_2(0) := (F \circ j_2)(0) \in (V^< \setminus 0)$ (and thus $f_2(0) \notin V^>$).

Proof. First of all, we have that $V \simeq V^> \oplus V^<$ because the isotopic component of weight 0 is trivial. Let $\mathbf{a} := (a_1,\dots,a_k)$ (respectively $\mathbf{b} := (b_1,\dots,b_h)$) be the vector of positive (respectively negative) weights of the $\mathbb{G}_m$-representation $V^>$ (respectively $V^<$) of dimension k (respectively h). Notice that we are not assuming that they are distinct. The morphism f_1 can be described as the vector $\{p_1,\dots,p_k,q_1,\dots,q_h\}$ in R^{h+k}. Let us denote by ν the discrete valuation associated to R. By hypothesis, q_i is either 0 or $\nu(q_i) \geq 1$, for every $i = 1,\dots,h$. Moreover, there exists an index i_0 such that $\nu(q_{i_0}) \geq 1$, because $f_2(1) \notin V^>$. First of all, we have that $\operatorname{Spec} R \simeq [\operatorname{Spec} R[t]_t/\mathbb{G}_m]$ where $\mathbb{G}_m$ acts on t with weight 1 and the morphism f_1 can be described as the $\mathbb{G}_m$-equivariant morphism $\operatorname{Spec} R[t]_t \to V$ given by the vector

$$\{p_1 t^{a_1}, p_2 t^{a_2}, \dots, p_k t^{a_k}, q_1 t^{b_1}, q_2 t^{b_2}, \dots, q_h t^{b_h}\}.$$

Up to extending R by taking a big enough root of its uniformizer, we can assume that either q_i is 0 or $\nu(q_i)$ is a multiple of b_i. Therefore, without loss of generalities, we can write f_1 as the $\mathbb{G}_m$-equivariant morphism given by the vector

$$\{p_1 t^{a_1}, p_2 t^{a_2}, \dots, p_k t^{a_k}, (q_1/t)^{-b_1}, (q_2/t)^{-b_2}, \dots, (q_h/t)^{-b_h}\}$$

where again q_i is either 0 or $\nu(q_i) \geq 1$. Moreover, we still have $\nu(q_{i_0}) \geq 1$. Let $n := \min\{\nu(q_i) | q_i \neq 0\} \geq 1$; without loss of generality, we can assume $\nu(q_1) = n$. Then the morphism

$$\operatorname{Spec} R[t]_t \to V^{>} \oplus V^{<}$$

factors through the subalgebra $R[t, \pi^n/t] \subset R[t]_t$. Therefore we get a $\mathbb{G}_m$-equivariant morphism

$$\operatorname{Spec} R[t]_t \subset \operatorname{Spec} \frac{R[s,t]}{(st - \pi^n)} \to V;$$

if we quotient by $\mathbb{G}_m$, we get the morphism $\mathcal{C}_n \to [V/\mathbb{G}_m]$: see Remark 7.35. The same remark gives us the morphism $F : \overline{\mathrm{ST}}_R \to [V/\mathbb{G}_m]$ precomposing with $p_n : \overline{\mathrm{ST}}_R \to \mathcal{C}_n$. Finally, we need to check that $f_2 := F \circ j_2 : \operatorname{Spec} R \to [V/\mathbb{G}_m]$ does not send the closed point of $\operatorname{Spec} R$ to the closed subset $V^{>}$. By construction, we know that the morphism $F : \overline{\mathrm{ST}}_R \to V$ is determined by the vector

$$\{p_1 t^{na_1}, p_2 t^{na_2}, \dots, p_k t^{na_k}, s^{-nb_1}, q_2' s^{-nb_2}, \dots, q_h' s^{-nb_h}\}$$

for some $q_1', \dots, q_h' \in R$. Because j_2 corresponds to the open substack given by the equation $s \neq 0$, we can conclude. □

Stabilization of A_r-stable curves

In what follows, we are going to explain how to pass from the local result above to a global result in $\mathcal{M}^r_{g,n}$, specifically Lemma 7.38.

Before proving the main lemma, we will recall a useful result, which will help us passing from the global picture to the local picture. This follows from Lemma 3.39 and Lemma 3.41. From now on, $r \geq 2$.

Lemma 7.37. *Let r, s be positive integers such that $2 \leq s \leq r$. Let C_1 be an A_r-stable curve over an algebraically closed field k of genus g and p_1 be an A_s-singularity. Then there exists an A_r-stable curve $C \to \Theta_k$ such that*

- *its generic fiber is isomorphic to C_1;*
- *if $s = 2h+1$, its closed fiber $C_0 \to \mathcal{B}\mathbb{G}_m$ can be constructed gluing (nodally) the stabilization of the pointed normalization of C_1 at p_1 with the 2-pointed odd atom Γ_h of genus h; the $\mathbb{G}_m$ is identified with the automorphisms of Γ_h;*
- *if $s = 2h$ is even, its closed fiber $C_0 \to \mathcal{B}\mathbb{G}_m$ can be constructed gluing (nodally) the stabilization of the pointed normalization of C_1 at p_1 with the 1-pointed even atom Γ_h of genus h; the $\mathbb{G}_m$ is identified with the automorphisms of Γ_h.*

We are ready to prove the (algebraic) stabilization result for the atomic case.

Lemma 7.38. *Let R be a DVR with residue field k and $f_1 : \operatorname{Spec} R \to \mathcal{M}^r_{g,n}$ be a morphism such that $C_1(1) := f_1(1) \in \mathcal{M}^{r-1}_{g,n} \subset \mathcal{M}^r_{g,n}$. Moreover, let $C_1(0) := f_1(0)$ be an A_r-stable curve with an A_r-singularity p; then, up to an extension of DVRs, there exists a morphism $F : \overline{\mathrm{ST}}_R \to \mathcal{M}^r_{g,n}$ such that*

- *$F \circ j_1 = f_1$;*

- *if $r = 2h+1$, then $C_2(0) := (F \circ j_2)(0)$ is a curve constructed gluing (nodally) the stabilization of the pointed normalization of $C_1(0)$ at p with an hyperelliptic bridge $(\Gamma_h, q_1, q_2) \in \mathcal{H}^{r-1,\circ}_{h,g_1^2}$;*
- *if $r = 2h$, then $C_2(0) := (F \circ j_2)(0)$ is a curve constructed gluing (nodally) the stabilization of the pointed normalization of $C_1(0)$ at p with an hyperelliptic tail $(\Gamma_h, q) \in \mathcal{H}^{r-1,\circ}_{h,w}$;*
- *F sends the closed point of $\overline{\mathrm{ST}}_R$ to the curve C_0 described in Lemma 7.37 (depending on the parity of r);*

Moreover, if $C_1(0) \in \mathcal{U}^r_{g,n}$, then the same is true for C_0 (and thus for $C_2(0)$).

Proof. We will explicitly deal with the case $r = 2h+1$. We leave it to the interested reader to check that the same exact idea works for $r = 2h$.

Lemma 7.37 gives us a morphism $g : \Theta_{\overline{k}} \to \mathcal{M}^r_{g,n}$ such that $g(1) \simeq C_1(0)$ and $C_0 := g(0)$ can be constructed gluing (nodally) the stabilization of the pointed normalization of $C_1(0)$ at p with the 2-pointed odd atom Γ_h of genus h. This in particular says that C_0 is in the closure of $C_1(0)$.

Using the (formal) local structure as described in Remark A.4, we can reduce to study the same problem in the deformation space of C_0 in $\mathcal{M}^r_{g,n}$, up to possibly extending the DVR. More precisely, we have

- a linearly reductive group $G := \mathrm{Aut}(C_0)$ and a G-representation $V := T_{[C_0]}\mathcal{M}^r_{g,n}$;
- a 1-parameter subgroup $\mathbb{G}_m \subset G$ given by the inclusion $\mathbb{G}_m \simeq \mathrm{Aut}^\circ(\Gamma, p_1, p_2) \subset \mathrm{Aut}(C_0) = G$;
- a morphism $f_1 : \mathrm{Spec}\, R \to V$ which corresponds to $f_1 : \mathrm{Spec}\, R \to \mathcal{M}^r_{g,n}$;

moreover, Proposition 3.29 gives us the sign decomposition of the $\mathbb{G}_m$-action induced by the 1-parameter subgroup, therefore

- $V^>$ coincides with the deformations of C_0 which preserve the A_r-singularity of the odd atom Γ;
- $V^<$ coincides with the deformations of C_0 which preserve the two nodes p_1 and p_2 and deform Γ into an hyperelliptic bridge of genus h.

By abuse of notation, we denote by f_1 the morphism $\mathrm{Spec}\, R \to [V^> \oplus V^</\mathbb{G}_m]$. Notice that $f_1(1) \in \mathcal{M}^{r-1}_{g,n}$ implies that $f_1(1) \notin V^>$, while $f_1(0) \in V^>$. Lemma 7.36 implies that there exists a morphism $F : \overline{\mathrm{ST}}_R \to [V^> \oplus V^</\mathbb{G}_m]$ which extends f_1 and such that $(F \circ j_2)(0)$ is in $(V^< \setminus 0)$, thus the A_r-singularity p is not preserved. If we consider now the composition

$$\overline{\mathrm{ST}}_R \to [V^> \oplus V^</\mathbb{G}_m] \to [V^> \oplus V^</\mathbb{G}_m] \times V^0 \to [V/G]$$

this induces a morphism

$$F : \overline{\mathrm{ST}}_R \longrightarrow \mathcal{M}^r_{g,n}$$

such that $F \circ j_1$ coincides with f_1 and $C_2(0) := (F \circ j_2)(0)$ is constructed as stated.

Regarding the moreover part, we need to prove that substituting the singularity p with an A_1/A_1-attached odd atom Γ_h of genus $h = (r-1)/2$ does not produce a failure of the conditions in Definition 4.4. By contradiction, suppose that the curve C_0, does not satisfy condition (c_r) of Definition 4.4. Therefore, there exists a subcurve $\Gamma' \subset C_0$ which is an A_1/A_1-attached hyperelliptic chain of genus bounded by h. Because Γ_h has genus h, this implies that Γ_h is not contained in Γ'. Moreover, $\Gamma_h \cap \Gamma' \neq \emptyset$, because $C_1(0)$ satisfies (c_r) by hypothesis. But if we substitute Γ_h with the A_r-singularity p, then $\Gamma' \subset C_1(0)$ would be an A_1/A_r (or A_r/A_r)-attached hyperelliptic chain of genus bounded by h, which it is not possible since $C_1(0)$ verifies condition (d_r). The same idea gives us that C_0 verifies condition (d_r). Similarly, it is easy to see that C_0 cannot contain an $A_{2h'+1}$-attached hyperelliptic curve of genus $< h$, as otherwise the same would hold for $C_1(0)$. Moreover, C_0 cannot be an A_0-attached hyperelliptic curve of genus $< (r-1)/2 = h$, because $\Gamma_h \subset C_0$ has genus h. Therefore C_0 verifies (t_r).

Finally, suppose C_0 does not verify condition (e_r) of Definition 4.4. Then C_0 is an A_s-self-attached hyperelliptic curve of genus $g < s$, but $g \geq h+1+(s-1)/2$ because C_0 contains a A_1/A_1-attached curve of genus h and an A_s-singularity, whose contribution to the genus is always $\geq (s-1)/2$. Therefore we get

$$s > g \geq h + \frac{s+1}{2} \implies s > 2h+1 = r$$

but this is impossible because $s \leq r$. □

Remark 7.39. Lemma 7.36 is the reason why the section is called algebraic stabilization over $\overline{\mathrm{ST}}_R$. Indeed, the point $C_2(0)$ coincides with the (partial) stabilization described for instance in Section 7 of [Fed13].

Corollary 7.40. *Let R be a DVR with residue field k and $f_1 : \operatorname{Spec} R \to \mathcal{M}^r_{g,n}$ be a morphism such that $f_1(1) \in \mathcal{M}^{r-1}_{g,n} \subset \mathcal{M}^r_{g,n}$. Then, it exists an integer m and a morphism $F : \mathfrak{S}_{m,R} \to \mathcal{M}^r_{g,n}$ such that*

- *$F \circ j_1$ coincides with f_1;*
- *$(F \circ j_m)(0) \in \mathcal{M}^{r-1}_{g,n}$;*

up to an extension of the DVR. Moreover, if f_1 lands in $\mathcal{U}^r_{g,n}$, the same is true for F.

Proof. The statement follows from an inductive argument on the number of A_r-singularities in $f_1(0)$, applying iteratively Lemma 7.36. □

7.4 Proof of S-completeness ($r \leq 5$)

We are finally ready to prove Theorem 7.1.

The proof will rely on an induction procedure on r. Notice that if $r = 0$ then $\mathcal{U}^r_{g,n} = \mathcal{M}_{g,n}$ and if $r = 1$ then $\mathcal{U}^r_{g,n} = \overline{\mathcal{M}}_{g,n}$. Therefore we can assume $r \geq 2$. Moreover, thanks to Corollary 7.33, it is enough to show generic S-completeness. We can assume that the generic fiber of our families are smooth.

Let us start with a morphism $\mathrm{ST}_R \to \mathcal{U}^r_{g,n}$, where R is a DVR with residue field k and fraction field Q. We want to prove that there exists a morphism $\overline{\mathrm{ST}}_R \to \mathcal{U}^r_{g,n}$ extending $\mathrm{ST}_R \to \mathcal{U}^r_{g,n}$. To briefly summarize the idea: since C_Q does not have A_r-singularities, we can use the algebraic stabilization procedure we described in Section 7.3 to construct a filling $\mathrm{ST}_R \subset \mathfrak{S}_{m,R}$ gluing the $\overline{\mathrm{ST}}_R$-morphisms given by the induction procedure and the stabilization results (see for instance Corollary 7.40). Finally, we will use the Θ-completeness of $\mathcal{U}^r_{g,n}$ to conclude, thanks to Corollary 7.33. In what follows, we will constantly use the fact that it is enough to find the filling up to an extension of R, as $\mathcal{U}^r_{g,n}$ has affine diagonal. We will deal with the odd case, and explain why for the even case it also holds, if $r \leq 5$. Suppose $r = 2k+1$ is odd and let $f : \mathrm{ST}_R \to \mathcal{U}^r_{g,n}$ be any morphism. We will denote by f_1 and f_2 the two morphisms $\operatorname{Spec} R \to \mathcal{U}^r_{g,n}$ induced by the one from ST_R. We have that f_1 satisfies the hypothesis of Corollary 7.40, therefore, up to extending the DVR, there exists an integer m_1 and a morphism $F_1 : \mathfrak{S}_{m_1,R} \to \mathcal{U}^r_{g,n}$ such that $F_1 \circ j_1 = f_1$ and $(F_1 \circ j_{m_1})$ lands in $\mathcal{M}^{r-1}_{g,n}$. The same is true for f_2. Then, we can glue the two morphisms $(F_1 \circ j_{m_1})$ and $(F_2 \circ j_{m_2})$ in the common generic point to get a morphism

$$\mathrm{ST}_R \longrightarrow \mathcal{U}^r_{g,n} \cap \mathcal{M}^{r-1}_{g,n}$$

and, thanks to Lemma 4.24, we get that $\mathcal{U}^r_{g,n} \cap \mathcal{M}^{r-1}_{g,n} \subset \mathcal{U}^{r-1}_{g,n}$. By inductive hypothesis on r, we have that there exists a filling $G : \overline{\mathrm{ST}}_R \to \mathcal{U}^{r-1}_{g,n}$. Because $\mathcal{U}^{r-1}_{g,n} \subset \mathcal{M}^{r-1}_{g,n} \subset \mathcal{M}^r_{g,n}$, we have the following

commutative diagram

$$\begin{tikzcd} \mathrm{ST}_R \arrow[r, hook] \arrow[d] & \overline{\mathrm{ST}}_R \arrow[d, "G"] \\ \mathcal{U}^r_{g,n} \arrow[r, hook] & \mathcal{M}^r_{g,n} \end{tikzcd}$$

and Theorem 5.1 ensures us that the lifting $G : \overline{\mathrm{ST}}_R \to \mathcal{U}^{r-1}_{g,n}$ lands in $\mathcal{U}^r_{g,n}$ as well. Summing up, we have found

- a morphism $F_1 : \mathfrak{S}_{m_1,R} \to \mathcal{U}^r_{g,n}$ such that $F_1 \circ j_1 = f_1$;
- a morphism $F_2 : \mathfrak{S}_{m_2,R} \to \mathcal{U}^r_{g,n}$ such that $F_2 \circ j_1 = f_2$;
- a morphism $G : \overline{\mathrm{ST}}_R \to \mathcal{U}^r_{g,n}$ such that $G \circ j_1 = F_1 \circ j_{m_1}$ and $G \circ j_2 = F_2 \circ j_{m_2}$;

we can then glue the morphisms F_1, F_2 and G to get a morphism $\mathfrak{S}_{m_1+m_2+1,R} \to \mathcal{U}^r_{g,n}$ which extends the morphism $f : \mathrm{ST}_R \to \mathcal{U}^r_{g,n}$. This concludes the proof in the case when r is odd. The even case follows from the following remark.

Remark 7.41. The parity of r has been used in the following implication

$(\star)$ r is odd $\implies \mathcal{U}^r_{g,n} \cap \mathcal{M}^{r-1}_{g,n} \subset \mathcal{U}^{r-1}_{g,n}$.

Indeed, if a n-pointed A_r-stable curve lives in $\mathcal{U}^r_{g,n}$, then it satisfies condition (t_{r-1}), (c_{r-1}) and (e_{r-1}) because of Lemma 4.24. If r is odd, then $r-1$ is even and thus condition (d_{r-1}) is also verified, as it is empty. Clearly, the same is not true if r is even. Nevertheless, if $r \leq 5$, then condition (d_{r-1}) is always verified, as (d_3) and (d_1) are also empty, thus $(\star)$ also holds for r even if $r \leq 5$.

Thanks to Remark 7.41, we have that Theorem 7.1 holds for $r \leq 5$. This finally concludes the proof of Theorem 7.1.

7.5 What happens for $r \geq 6$?

As explained in Remark 7.41, the strategy adopted for $r \leq 5$ cannot be used for $r \geq 6$. In this subsection, we will try to explain in more detail why.

The only obstacle to a complete proof in the case $r \geq 6$ is the implication $(\star)$ in the case when r is even. More precisely, it is not true that $\mathcal{U}^r_{g,n} \cap \mathcal{M}^{r-1}_{g,n} \subset \mathcal{U}^{r-1}_{g,n}$ if r is an even integer with $r \geq 6$. Specifically, condition (d_r) of Definition 4.4 is empty if r is even: for instance, in $\mathcal{U}^6_{g,n}$ we allow curves with A_1/A_5-attached (or A_5/A_5-attached) non-separating hyperelliptic chains, that are not allowed in $\mathcal{U}^5_{g,n}$, thus after the (algebraic) stabilization procedure, we still cannot use the inductive procedure.

Nevertheless, we state the following conjecture.

Conjecture 7.42. The algebraic stack $\mathcal{U}^r_{g,n}$ is S-complete, and therefore it admits a separated good moduli space, for any $r \geq 0$.

In the remaining part of the section, we will explain what are the obstacles in making the argument works and some possible tentative solutions. We will do so in an informal way, as we believe it could be easier to understand these issues without the full load of details one would need to prove every claim stated in what follows.

A_1/A_{r-1}-attached chains are not problematic

To ease the discussion, we can suppose that $r = 6$. Everything we say actually applies that any even integer r.

Recall that the issue comes from condition $(d_{5,1})$ of Definition 4.4: namely, after the algebraic stabilization procedure, we could end with curves in $\mathcal{U}^6_{g,n} \cap \mathcal{M}^5_{g,n}$ that have A_1/A_5-attached non-separating hyperelliptic chains.

Consider a morphism $C : \operatorname{Spec} R \to \mathcal{U}^6_{g,n}$ such that the special fiber C_0 has an A_1/A_5-attached non-separating elliptic chain Γ.

Since we want to prove generic S-completeness we can assume the generic fiber does not have a A_1/A_5-attached non-separating hyperelliptic chain Γ of genus 1 (i.e. elliptic chain). Proposition 3.43 tells us that there exists a family of A_6-stable curves over Θ such that the generic fiber coincide with C_0 and the special fiber C_0' can be constructed by attaching nodally to $C_0 - \Gamma$ a 2-pointed rosary Γ' of hyperelliptic A-singularities. The chain Γ' is uniquely determined by Γ.

If Γ is A_1/A_5-attached, it is easy to see that $C_0' \in \mathcal{U}^6_{g,n}$. Therefore, in a similar fashion to Lemma 7.38, we can apply Lemma 7.36 to get a morphism $G : \overline{\mathrm{ST}}_R \to \mathcal{M}^6_{g,n}$ such that $G \circ j_1$ coincides with C while the closed point of $G \circ j_2$ does not have the A_1/A_5-attached non-separating hyperelliptic chain anymore, i.e. $G \circ j_2$ lands in $\mathcal{U}^5_{g,n}$. Indeed, the closed point of $G \circ j_2$ is of the form described in Proposition 3.43 which is not the A_1/A_5-attached non-separating hyperelliptic chain. Notice that G still factors through $\mathcal{U}^6_{g,n}$.

A_{r-1}/A_{r-1}-attached chains are problematic

We have already explained why A_1/A_{r-1}-attached non-separating hyperelliptic chains are not problematic. The problem occurs when there exists a subcurve Γ which is an A_5/A_5-attached hyperelliptic chain, i.e. condition $(d_{5,2})$ is not satisfied. In this case, Proposition 3.43 gives us a curve C_0' and a subcurve Γ' of C_0' which is an A_1/A_1-attached non-separating hyperelliptic chain of genus bounded by $(6-1)/2$, thus C_0' is not in $\mathcal{U}^6_{g,n}$.

One may hope that there exists another family of A_6-stable curves over Θ such that the generic fiber is C_0, the special fiber is in $\mathcal{U}^6_{g,n}$ and Lemma 7.36 applies to get us another family of curves over $\operatorname{Spec} R$ which lands in $\mathcal{U}^5_{g,n}$. This is not the case, as the following example shows.

Consider an A_6-stable curve C_0 such that

- it contains an A_5/A_5-attached honestly hyperelliptic curve Γ of genus 1 (for instance a smooth genus 1 curve);
- $C_0 - \Gamma$ is smooth and not hyperelliptic;

Then a simple argument shows that if C is a family of A_6-stable curves over Θ with C_0 as the generic fiber and C_0' as the closed fiber, then $C_0' \in \mathcal{U}^6_{g,n}$ implies $C_0' = C_0$ (and thus the morphism $C : \Theta \to \mathcal{U}^6_{g,n}$ is trivial). Informally, the curve $C_0 - \Gamma$ does not really play any role, whereas the special fiber of a Θ-degeneration of C_0 corresponds to (a degeneration of) a curve C_0' with an A_1/A_1-attached rosary Γ' of hyperelliptic A-singularities of length 3 with singularities $A_5 - A_3 - A_5$.

In conclusion, degenerations over DVRs in $\mathcal{U}^6_{g,n}$ whose closed point has an A_5/A_5-attached (non-separating) chain cannot be "stabilized" by means of a morphism from $\overline{\mathrm{ST}}_R$ to $\mathcal{U}^6_{g,n}$.

First tentative solution

We recall the setting for the reader convenience. Let us start with a morphism $F : \mathrm{ST}_R \to \mathcal{U}^6_{g,n}$ which corresponds to two morphisms $F_1 : \operatorname{Spec} R \to \mathcal{U}^6_{g,n}$ and $F_2 : \operatorname{Spec} R \to \mathcal{U}^6_{g,n}$ which agree on the generic point of $\operatorname{Spec} R$, i.e. $F_1(1) \simeq F_2(1)$. Thanks to Lemma 7.38 and Theorem 6.1, we can assume without loss of generality that F lands in $\mathcal{U}^6_{g,n} \cap \mathcal{M}^5_{g,n}$.

As we discussed above, given a morphism $F : \operatorname{Spec} R \to \mathcal{U}^6_{g,n} \cap \mathcal{M}^5_{g,n}$, we would like to construct a morphism $G : \overline{\mathrm{ST}}_R \to \mathcal{U}^6_{g,n}$ such that $G \circ j_1 = F$ and $G \circ j_2$ lands in $\mathcal{U}^5_{g,n}$, so that we can use the

inductive hypothesis. As we explained in Section 7.5, we can find a morphism $G : \overline{\mathrm{ST}}_R \to \mathcal{M}^6_{g,n}$ with the necessary properties, but it will never land in $\mathcal{U}^6_{g,n} \subset \mathcal{M}^6_{g,n}$. This stops the inductive procedure. In fact, let G_1 (respectively G_2) a morphism as above associated to F_1 (respectively F_2). The morphism $F' : \mathrm{ST}_R \to \mathcal{U}^5_{g,n}$ induced by the pair $G_1 \circ j_2$ and $G_2 \circ j_2$ can be completed to a morphism $\overline{F'} : \overline{\mathrm{ST}}_R \to \mathcal{U}^5_{g,n}$, by inductive hypothesis. Gluing G_1, G_2 and $\overline{F}'$, we get a morphism $\mathfrak{S}_3 \to \mathcal{M}^6_{g,n}$ whose extremal rays are F_1 and F_2.

However, $\mathcal{M}^6_{g,n}$ is not Θ-complete (in fact not even N-separated), thus we cannot use Corollary 7.33 to get the result. One could hope that there exists an open substack $\mathcal{X}^5_{g,n} \subset \mathcal{M}^5_{g,n}$ which is Θ-complete (or even N-separated) and contains both $\mathcal{U}^6_{g,n} \cap \mathcal{M}^5_{g,n}$ and $\mathcal{U}^5_{g,n}$. Nevertheless, the natural choice does not work either.

Namely, let $\mathcal{V}^5_{g,n} := \mathcal{U}^6_{g,n} \cap \mathcal{M}^5_{g,n}$; alternatively, we can define $\mathcal{V}^5_{g,n}$ as the open substack of $\mathcal{M}^5_{g,n}$ described by the conditions (t_5), (c_6) and (e_5) of Definition 4.4. Recall that $\mathcal{U}^5_{g,n}$ is the open substack of $\mathcal{M}^5_{g,n}$ defined by the conditions (t_5), (c_5), (d_5) and (e_5). There is natural candidate for $\mathcal{X}^5_{g,n}$, namely the open substack of $\mathcal{M}^5_{g,n}$ defined by conditions (t_5), (c_5) and (e_5). One can show that $\mathcal{X}^5_{g,n}$ is not Θ-complete as well.

Argument in favor of the conjecture

Finally, we explain why we believe Conjecture 7.42 to be true. The main idea is that one can prove that $\mathcal{U}^6_{g,n}$ is almost topologically S-complete (or weakly separated as defined in [ASvdW10]).

Let us explain how to prove that $\mathcal{U}^6_{g,n}$ is almost topologically S-complete. Let $F : \mathrm{ST}_R \to \mathcal{U}^6_{g,n}$ be a morphism as in Section 7.5; we denote by f_1 and f_2 the two morphisms from $\operatorname{Spec} R$ induced by F. Using the stabilization technique explained in *loc.cit.*, we can find a morphism $F' : \mathrm{ST}_R \to \mathcal{U}^5_{g,n}$ such that the generic point of ST_R is sent to the same point by F and F'. The induction step provides us with a morphism $\overline{F'} : \overline{\mathrm{ST}}_R \to \mathcal{U}^5_{g,n}$ which extends F'. Up to extending the DVR, we can find a morphism $f : \operatorname{Spec} R \to \mathcal{U}^5_{g,n}$ which sends the generic point (respectively the closed point) of R to the image of the generic point (respectively the closed point) of $\overline{\mathrm{ST}}_R$ through $\overline{F'}$. If we apply the stabilization technique again (with opposite cocharacter), it is not difficult to show that we get a morphism $f : \operatorname{Spec} R \to \mathcal{U}^6_{g,n}$ such that $f(0)$ is in the closure of $f_1(0)$ and $f_2(0)$. Together with the fact that $\mathcal{U}^6_{g,n}$ is Θ-complete, we get that $\mathcal{U}^6_{g,n}$ is almost topologically S-complete.

The fact that $\mathcal{U}^6_{g,n}$ is almost topologically S-complete is not enough to prove the S-completeness. The difficulty of passing from almost topological S-complete to topological S-complete (and thus to S-complete by Proposition 7.13) lies in the difficulty of using only topology to discern separatedness for non-locally separated algebraic spaces (see [ASvdW10] for a more detailed discussion about it). More specifically, checking topological S-completeness requires dealing with abstract finite generically étale covers, which are hard to control for most moduli problems, ours in particular. It is not clear what could be the right generalization to algebraic stacks of being locally separated. One possible solution is to consider stacks that are Zariski-locally S-complete, namely that there exists a covering by open substacks that are S-complete. Notice that an algebraic space which is Zariski-locally separated is also locally separated, but the converse is not true. Nevertheless, one can easily prove the following lemma.

Lemma 7.43. *Let $\mathcal{X}$ be a Noetherian algebraic stack. Then $\mathcal{X}$ is S-complete if and only if $\mathcal{X}$ is Zariski-locally S-complete and almost topologically S-complete.*

Notice that the étale local structure theorem (see [AHR23]) gives us that every algebraic stack with linearly reductive stabilizers is étale-locally S-complete. Nevertheless, being Zariski-locally S-complete is a hard-to-establish property for moduli problems, exactly as being a scheme (or a Zarisky- locally separated algebraic space) is for a fine moduli problem, in particular when there are no line bundles to check quasi-projectivity (see Section 8).

8 Properness and non-projectivity of $U^5_{g,n}$

This section is dedicated to proving the following theorem.

Theorem 8.1. *The moduli space $U^r_{g,n}$ is a proper algebraic space for $r \leq 5$. If $r = 5$ and $g \geq 5$, it is not projective. Finally, it is not a scheme for $r = 5$, $n = 0$ or $g = 7$ and $g \geq 9$.*

In Section 8.1, we show the universal closedness of $\mathcal{U}^r_{g,n}$. Moreover, in Section 8.2 we compute the Picard group of $U^r_{g,n}$ for $r \geq 5$ and $g \geq \lfloor (r+1)/2 \rfloor$. The computation gives us that for $r = 5$ and $g \geq 5$, the (rational) Picard group is trivial. Finally, in Section 8.3 we show that it is not a scheme when $n = 0$, $g = 7$ and $g \geq 9$. Thus Theorem 8.1 follows.

8.1 Properness

The goal of this subsection is to prove the following proposition.

Proposition 8.2. *The moduli stack $\mathcal{U}^r_{g,n}$ is universally closed for $r \leq 5$.*

To do that, we will show that the moduli stack $\mathcal{U}^r_{g,n}$ is *generically* universally closed, namely that every family of smooth curves over the fraction field Q of a DVR R extends to a family in $\mathcal{U}^r_{g,n}$ over R. As always, we can do this up to taking a (finite) extension of DVRs. Together with the following lemma, we will get the universally closedness of $\mathcal{U}^5_{g,n}$ for $r \leq 5$.

Lemma 8.3. *Let $\mathcal{X}$ be an algebraic stack of finite type over a quasi-compact quasi-separated noetherian algebraic space S and assume that it has a good moduli space $\pi : \mathcal{X} \to X$ such that X is separated over S. Moreover, suppose that there exists a dense open $\mathcal{U} \subset \mathcal{X}$ with the following property:*

($\star$) *for every DVR R with fraction field Q and for every commutative diagram*

$$\begin{tikzcd} \operatorname{Spec} Q \arrow[r, "f_1"] \arrow[d, hook] & \mathcal{X} \arrow[d] \\ \operatorname{Spec} R \arrow[r] & S \end{tikzcd}$$

such that f_1 factors through $\mathcal{U}$, there exists a morphism $f : \operatorname{Spec} R \to \mathcal{X}$ extending f_1, up to extensions of DVRs.

Then $\mathcal{X}$ is universally closed over S.

Proof. Since $\mathcal{U}$ is dense, we can reduce to the case when $\mathcal{X}$ (and thus X) is irreducible. Since π is universally closed, it is enough to prove that X is universally closed. Because X is separated and of finite type, we can apply Nagata compactification (relative to S) for algebraic spaces (see [CLO12, Theorem 1.2.1]) to get an open embedding $X \subset \overline{X}$ where $\overline{X}$ is proper over S. Up to taking the closure of X in $\overline{X}$, we can assume X to be dense in $\overline{X}$. Finally, it is enough to prove that $X = \overline{X}$. By contradiction, let $x_0 \in \overline{X} \setminus X$; then there exists a DVR R with fraction field Q and a morphism $f : \operatorname{Spec} R \to \overline{X}$ such that $f(0) = x_0$ and $f(1)$ coincides with the generic point of X. By surjectivity, up to extending the DVR, we can lift the morphism $f_1 := f|_{\operatorname{Spec} Q} : \operatorname{Spec} Q \to X$ to a morphism $f'_1 : \operatorname{Spec} Q \to \mathcal{U} \subset \mathcal{X}$. By assumption, we can extend it to a morphism $f' : \operatorname{Spec} R \to \mathcal{X}$. Then we have two morphisms $f' \circ \pi$ and f who agree on the generic point of $\operatorname{Spec} R$, therefore they are equal by the separatedness of $\overline{X}$. Therefore $x_0 \in X$, which is absurd. $\square$

Remark 8.4. Lemma 8.3 is the perfect generalization of the same statement for separated algebraic spaces (or DM stacks). Moreover, Lemma 8.3 can be formulated more generally for morphisms of algebraic stacks, but we do not need it, so we avoid dealing with the technicalities.

Remark 8.5. It is not too hard to prove that $\mathcal{U}^r_{g,n}$ is universally closed for every r (not only when it is S-complete as well). Since it would not improve the reading and it is not necessary for Theorem 8.1, we decided to not include it in the paper.

Proof. This follows combining the fact that $\mathcal{U}^r_{g,n}$ has a separated good moduli space for $r \leq 5$ together with Lemma 8.3 and Lemma 8.6. □

In the rest of the section, we are going to prove the following lemma. We will do it using an inductive argument exactly as we did in the proof of Theorem 7.1. The idea is the same, but it is somehow reversed.

Lemma 8.6. *The open $\mathcal{M}^0_{g,n} \subset \mathcal{U}^r_{g,n}$ (i.e. the moduli stack of n-pointed smooth curves of genus g) verifies the property $(\star)$ in Lemma 8.3.*

Proof. Notice that $\mathcal{M}^0_{g,n}$ is contained in $\mathcal{U}^r_{g,n}$ for every r. We proceed by induction on r. If $r = 1$, we have that $\mathcal{U}^1_{g,n}$ is the Deligne-Mumford compactification, therefore there is nothing to prove. Suppose now that $\mathcal{M}^0_{g,n}$ in $\mathcal{U}^{r-1}_{g,n}$ satisfies $(\star)$. Given a morphism $g_0 : \operatorname{Spec} Q \to \mathcal{M}^0_{g,n}$, i.e. a family of n-pointed smooth curves over $\operatorname{Spec} Q$, we want to find a morphism $g_r : \operatorname{Spec} R \to \mathcal{U}^r_{g,n}$ which extends g_0. By inductive hypothesis, we know that there is a morphism $g_{r-1} : \operatorname{Spec} R \to \mathcal{U}^{r-1}_{g,n}$ which extends g_0. Notice that if $g_{r-1}(0) \in \mathcal{U}^r_{g,n}$, then we are done. Thus we can assume $g_{r-1}(0) \notin \mathcal{U}^r_{g,n}$. We will show the result analyzing the two cases depending on the parity.

r is odd

Regarding the odd case, it is clear that $(c_{r-1}) \implies (c_r)$ if r is odd and condition (e_r) does not really depend on r. Moreover, since $g_{r-1}(0) \in \mathcal{U}^{r-1}_{g,n}$, it clearly verifies condition (d_r). Thus it remains to deal with the case when $g_{r-1}(0)$ satisfies (t_{r-1}) but not (t_r), namely it admits either an A_0-attached or A_{2k+1}-attached hyperelliptic tail of genus $(r-1)/2$. We start by dealing with the case of A_{2k+1}-attached hyperelliptic tails.

Suppose there is exactly one A_{2k+1}-attached hyperelliptic tail of genus $(r-1)/2$. Lemma 3.42 gives a morphism $h : \Theta \to \mathcal{M}^r_{g,n}$ such that $h(1) = g_{r-1}(0)$. Exactly as in the proof of Lemma 7.38 (we will also use the same notation), we can reduce to study the same problem in the deformation space of $C_0 := h(0)$ in $\mathcal{M}^r_{g,n}$, up to possibly extending the DVR. More precisely, we have

- a linearly reductive group $G := \operatorname{Aut}(C_0)$ and a G_0-representation $V := T_{[C_0]}\mathcal{M}^r_{g,n}$;
- a 1-parameter subgroup $\mathbb{G}_m \subset G_0$ given by the inclusion $\mathbb{G}_m \simeq \operatorname{Aut}^\circ(D, \widetilde{q}) \subset \operatorname{Aut}(C_0) = G_0$ (see Lemma 3.42 for the description of $(D, \widetilde{q})$);
- a morphism $g'_{r-1} : \operatorname{Spec} R \to V := T_{[C_0]}\mathcal{M}^r_{g,n}$ which corresponds to $g_{r-1} : \operatorname{Spec} R \to \mathcal{M}^r_{g,n}$;

moreover, Proposition 3.29 gives us the sign decomposition of the $\mathbb{G}_m$-action induced by the 1-parameter subgroup, therefore

- $V^<$ coincides with the deformations of C_0 which preserve the A_{r-1}-singularity of D and the node $\widetilde{q}$;
- $V^>$ coincides with the deformations of C_0 which preserve the A_{2k+1}-singularity of D.

By abuse of notation, we denote by g'_{r-1} the morphism $\operatorname{Spec} R \to [V^> \oplus V^< / \mathbb{G}_m]$. Notice that $g_{r-1}(1) \in \mathcal{M}^0_{g,n}$ implies that $g'_{r-1}(1) \notin V^>$, while $g'_{r-1}(0) \in V^>$. Lemma 7.36 implies that there exists a morphism $G' : \overline{\mathrm{ST}}_R \to [V^> \oplus V^< / \mathbb{G}_m]$ which extends g'_{r-1} and such that $(G' \circ j_2)(0)$ is in $(V^< \setminus 0)$, thus the A_{2k+1}-attached hyperelliptic tails does not appear anymore. Beware that if the

A_{2k+1}-singularity deforms, an A_1-attached hyperelliptic tails could appear, but it would have genus $(r-1)/2+k > r/2$ as $k \geq 1$. If we consider now the composition

$$\overline{\mathrm{ST}}_R \to [V^> \oplus V^</\mathbb{G}_m] \to [V^> \oplus V^</\mathbb{G}_m] \times V^0 \to [V/G_0]$$

this induces a morphism

$$G : \overline{\mathrm{ST}}_R \longrightarrow \mathcal{M}^r_{g,n}$$

such that $G \circ j_1$ coincides with g_{r-1} and $g_r(0) := (G \circ j_2)(0)$ does not have the hyperelliptic tail anymore. Finally, we get that $g_r \in \mathcal{U}^r_{g,n}$ and we are done. If there is more than one hyperelliptic tail, then we can just apply the same idea iteratively. Notice that exchanging a hyperelliptic tail of genus $(r-1)/2$ with an A_{r-1}-singularity never affects non-separating chains, thus the curve $g_r(0)$ will still verify condition (d_r) and (c_r).

The case of A_0-attached hyperelliptic curves is identical.

r is even

By construction, condition (d_r) is empty if r is even, while (e_r) does not really depend on r. Moreover, it is easy to see that $(t_{r-1}) \implies (t_r)$: indeed, the numerical conditions $< (r-1)/2$ and $< r/2$ (for integers) coincide if r is even. Therefore, it remains to deal with the case when $g_{r-1}(0)$ verifies (c_{r-1}), but not (c_r), or equivalently there is an A_1/A_1-attached non-separating hyperelliptic chain of genus bounded by $(r-1)/2$ (but none of genus bounded by $(r-2)/2$).

Consider a morphism $g_{r-1} : \operatorname{Spec} R \to \mathcal{U}^{r-1}_{g,n}$ such that the special fiber $g_{r-1}(0)$ has one A_1/A_1-attached non-separating hyperelliptic chain Γ of genus bounded by $(r-1)/2$.

Proposition 3.43 tells us that there exists a family of A_r-stable curves over Θ such that the generic fiber coincide with $g_{r-1}(0)$ and the special fiber C_0 can be constructed by attaching nodally to $C_0 - \Gamma$ a 2-pointed rational chain Γ' of hyperelliptic A-singularities. The chain Γ' is uniquely determined by Γ.

Therefore, in a similar fashion to the odd case, we can apply Lemma 7.36 to get a morphism $G : \overline{\mathrm{ST}}_R \to \mathcal{M}^r_{g,n}$ such that $G \circ j_1$ coincides with g_{r-1} while the closed point of $G \circ j_2$ does not have the A_1/A_1-attached non-separating hyperelliptic chain anymore, i.e. $G \circ j_2$ lands in $\mathcal{U}^r_{g,n}$. Indeed, the closed point of $G \circ j_2$ is of the form described in Proposition 3.43 which is not the A_1/A_1-attached non-separating hyperelliptic chain.

If there are more than one A_1/A_1-attached non-separating hyperelliptic chain, we can apply the procedure iteratively to get the result. □

8.2 The Picard groups of $U^r_{g,n}$

First, we describe the line bundles that descend to the good moduli space $\mathcal{U}^r_{g,n} \to U^r_{g,n}$. The computations are carried out in Theorem 8.9. Namely, we compute the rational Picard group of $U^r_{g,n}$ (when it exists) and we show that if $\mathcal{U}^r_{g,n}$ admits a good moduli space, then the Picard group of the latter is torsion for $g, r \geq 5$ thus implying that $U^5_{g,n}$ is not projective for $g \geq 5$.

Proposition 8.7. *Let (g,n) be a pair of positive integers with $g > 2$ and define*

$$T_{g,n} = \Big\{(i,I) \;\Big|\; 0 \leq i \leq g,\ I \subset [n],\ \textit{with } (i,I) \notin \{(0,\emptyset),(g,[n])\}\Big\}/\sim,$$

where $(i,I) \sim (j,J)$ if and only if $(i,I) = (j,J)$ or $(i,I) = (g-j,[n]\setminus J)$. Then every line bundle $L \in \operatorname{Pic}(\mathcal{M}^r_{g,n})$ for $r \geq 3$, can be written uniquely as an integral combination

$$L = a\lambda + b_{\mathrm{irr}}\delta_{\mathrm{irr}} + \sum_{[i,I]\in T_{g,n}} b_{i,I}\delta_{i,I},$$

where we use the notation $\delta_{0,\{i\}} = -\psi_i$.

Proof. Let $\mathcal{M}^{\mathrm{lci}}_{g,n}$ be the moduli stack parametrizing curves with local complete intersection singularities and ample log-dualizing sheaf. Then $\overline{\mathcal{M}}_{g,n} \subset \mathcal{U}^{\mathrm{lci}}_{g,n}$ is a big open inclusion of smooth stacks, hence $\mathrm{Pic}(\mathcal{U}^{\mathrm{lci}}_{g,n}) = \mathrm{Pic}(\overline{\mathcal{M}}_{g,n})$. The latter group is described in [FV20]. □

For a noetherian algebraic stack with a good moduli space $\mathcal{X} \to X$ we can compute the subspace $\mathrm{Pic}_{\mathbb{Q}}(X) \subset \mathrm{Pic}_{\mathbb{Q}}(\mathcal{X})$ using [Alp13, Theorem 10.3]. Since we are interested in the rational Picard group, it is enough to compute the weights of the one-dimensional $\mathbb{G}_m$-representations corresponding to the pullback of line bundles $L \in \mathrm{Pic}(\mathcal{X})$ via maps $B\mathbb{G}_m \to \mathcal{X}$ induced by one-parameter subgroups of the stabilizer of a closed point. For $\mathcal{X} = \mathcal{U}^r_{g,n}$, we can apply the computations in [AFS11], which hinges on the modularity of the problem.

Fix a point $x \in |\mathcal{U}^r_{g,n}|$. Given a one-parameter subgroup $\rho\colon \mathbb{G}_m \to \mathrm{Stab}_x \mathcal{U}^r_{g,n}$ and an element $L \in \mathrm{Pic}_{\mathbb{Q}}(U^r_{g,n})$ we write $\langle L, \rho\rangle$ for the weight described above.

Proposition 8.8. *Let* $(C, \{p_i\}) \in \mathcal{M}^r_{g,n}(k)$ *be a closed point, where* k *is an algebraically closed field and suppose* $\Gamma \subset C$ *is a subcurve such that* $P_\Gamma = \emptyset$, I_Γ *only contains nodes of* C *and* (Γ, I_Γ) *is either a* n*-pointed atom with* $n = 1, 2$. *Let* $\rho\colon \mathbb{G}_m \to \mathrm{Stab}_{(C,\{p_i\})} \mathcal{M}^r_{g,n}$ *be the one-parameter subgroup which acts non-trivially only on* Γ *and whose action on the tangent space(s) in* Γ *of the point(s) in* I_Γ *has weight* 1. *Let* L *be one of the generators of* $\mathrm{Pic}(\mathcal{M}^r_{g,n})$ *as in Proposition 8.7, then the following holds:*

1) if Γ *is a* 1*-pointed even atom of genus* j*, then:*

$$\langle L, \rho\rangle = \begin{cases} 4j(2j+1) & \text{if } L = \delta_{\mathrm{irr}}, \\ -1 & \text{if } L = \delta_{j,\emptyset}, \\ j^2 & \text{if } L = \lambda, \\ 0 & \text{otherwise.} \end{cases}$$

2) if Γ *is a* non-separating 2*-pointed odd atom of genus* k*, then:*

$$\langle L, \rho\rangle = \begin{cases} 2k(2k+3) & \text{if } L = \delta_{\mathrm{irr}}, \\ \dfrac{k^2+k}{2} & \text{if } L = \lambda, \\ 0 & \text{otherwise.} \end{cases}$$

3) if $\Gamma_{i,I}$ *is a* 2*-pointed odd atom of genus* k*, such that* $C - \Gamma$ *has two connected components (resp. one connected component) with genus and marked points* (i, I) *and* $(g-k-i, I^c)$ *(resp. with genus and marked point* $(g-k, I)$ *with* $|I| = n-1$*). Then we have:*

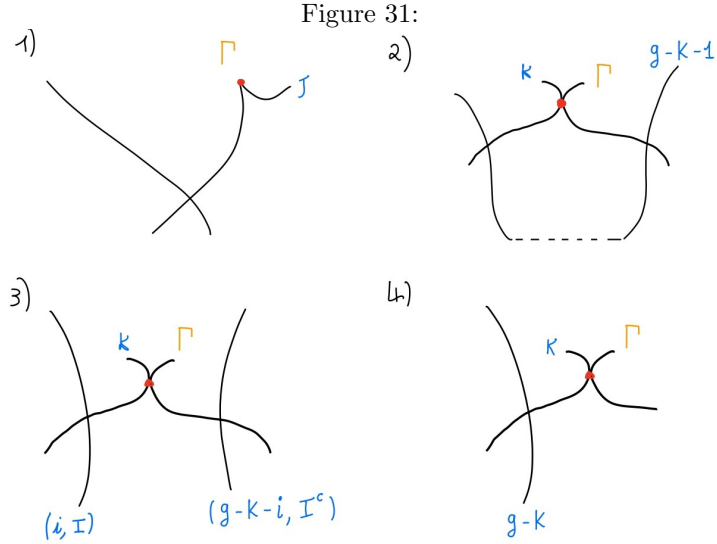
$$\langle L, \rho\rangle = \begin{cases} (2k+1)(2k+2) & \text{if } L = \delta_{\mathrm{irr}}, \\ -1 & \text{if } L = \delta_{i,I}, \\ -1 & \text{if } L = \delta_{g-k-i,I^c}, \\ \dfrac{k^2+k}{2} & \text{if } L = \lambda, \\ 0 & \text{otherwise.} \end{cases}$$

4) if Γ *is a* 1*-pointed odd atom of genus* $k \geq 2$*, such that* $C - \Gamma$ *is connected with genus* $g - k$ *and*

contains all the marked points. Then:

$$\langle L, \rho \rangle = \begin{cases} (2k+1)(2k+2) & \text{if } L = \delta_{\mathrm{irr}}, \\ -1 & \text{if } L = \delta_{g-k,[n]}, \\ \dfrac{k^2+k}{2} & \text{if } L = \lambda, \\ 0 & \text{otherwise.} \end{cases}$$

Figure 31:



Proof. When L is a component of the boundary δ, the computations are carried out as in [AFS11, Examples 4.8, 4.9, 4.16] using Proposition 4.7 *loc. cit.*. The computation for $L = \lambda$ can be found in [AFS11, Table 2]. □

We can now compute the rational line bundles that descend.

Theorem 8.9. *Assume $g \geq \max\{5, \lfloor (r+1)/2 \rfloor\}$ and that the stack $\mathcal{U}^r_{g,n}$ admits a good moduli space $\mathcal{U}^r_{g,n} \to U^r_{g,n}$. Then*

$$\mathrm{Pic}_{\mathbb{Q}}(U^r_{g,n}) = 0 \qquad \text{for } r \geq 5.$$

Proof. Let $L \in \mathrm{Pic}_{\mathbb{Q}}(\mathcal{U}^r_{g,n})$ be a $\mathbb{Q}$-line bundle that descend to the good moduli space, then for any morphism

$$f \colon B\mathbb{G}_m \longrightarrow \mathcal{U}^r_{g,n}$$

we have that f^*L descends to the good moduli space $B\mathbb{G}_m \to \operatorname{Spec} k$. Equivalently, f^*L corresponds to the one-dimensional trivial representation of $\mathbb{G}_m$. Thus we consider all morphisms $B\mathbb{G}_m \to \mathcal{U}^r_{g,n}$ arising from a curve $(C, \{p_i\}) \in \mathcal{U}^r_{g,n}(k)$ together with a cocharacter $\rho \colon \mathbb{G}_m \to \mathrm{Aut}(C, \{p_i\})$ as in Proposition 8.8. Therefore for a line bundle that descends to $U^r_{g,n}$ the following holds:

$$\begin{cases} \dfrac{k^2+k}{2} a + 2k(2k+3) b_{\mathrm{irr}} = 0 & \text{if } r = 2k+1, \\ j^2 a + 4j(2j+1) b_{\mathrm{irr}} - b_{j,\emptyset} = 0 & \text{if } r = 2j, \\ \dfrac{k^2+k}{2} a + (2k+1)(2k+2) b_{\mathrm{irr}} - b_{i,I} - b_{g-k-i,I^c} = 0 & \text{for } 1 \leq k < r/2, \\ \dfrac{k^2+k}{2} a + (2k+1)(2k+2) b_{\mathrm{irr}} - b_{g-k,[n]} = 0 & \text{for } 2 \leq k < r/2. \end{cases} \tag{1}$$

Notice that Cases 1 and 2 in *loc. cit.* occur respectively for r odd and even.

Denote for simplicity $x_k(a, b_{\mathrm{irr}}) := \frac{k^2+k}{2}a + (2k+1)(2k+2)b_{\mathrm{irr}}$. From the third equation used for the k specified, we have:

$$b_{i,I} = \frac{1-(-1)^i}{2}x_1(a, b_{\mathrm{irr}}) + (-1)^i b_{2,I} \qquad \text{for } k = 1, \tag{2}$$

$$b_{2,I} = \frac{1}{2}x_2(a, b_{\mathrm{irr}}) \qquad \text{for } k = 1, 2, \tag{3}$$

$$x_3(a, b_{\mathrm{irr}}) - x_1(a, b_{\mathrm{irr}}) = 5a + 42b_{\mathrm{irr}} = 0 \qquad \text{for } k = 1, 3. \tag{4}$$

Indeed, the equation for $k = 1$ together with the relation $\delta_{i,I} = \delta_{g-i,I^c}$ gives $b_{i,I} + b_{i+1,I} = x_1(a, b_{\mathrm{irr}}) = b_{i+1,I} + b_{i+2,I}$, implying equation (2). Equations (3) and (4) are deduced similarly. The atoms needed to obtain the above equations from Proposition 8.8 impose the following constraints on the genus and the marked points: Equation (3) holds when $g \geq 6$ or when $g \geq 4$ and $|[n] \setminus I| \geq 1$, while (4) holds when either $(g \geq 3, n \geq 2)$ or $(g \geq 5, n = 1)$ or $g \geq 7$.

Case 1: r odd and $(g, n) \neq (5, 0)$. The fourth equation of (1) together with (3) gives $x_2(a, b_{\mathrm{irr}}) = 0$ (this again uses the relation $\delta_{i,I} = \delta_{g-i,I^c}$). The first equation of (1) (for $r = 2k+1$) gives $a = -8\frac{r+2}{r+1}b_{\mathrm{irr}}$. If $r \neq 19$, these two relations force $a = b_{\mathrm{irr}} = 0$. If $r = 19$, then $g \geq 10$, and combining (4) with $x_2(a, b_{\mathrm{irr}}) = 0$ again implies $a = b_{\mathrm{irr}} = 0$. Then (2) and (3) complete the claim.

Case 2: $r = 2j$ even and $(g, n) \neq (5, 0)$. As before, $x_2(a, b_{\mathrm{irr}}) = 0$. Using the second equation of (1) together with (2) and (3), we obtain

$$j^2 a + 4j(2j+1)b_{\mathrm{irr}} - \frac{1-(-1)^j}{2}x_1(a, b_{\mathrm{irr}}) = 0.$$

The latter equation together with $x_2(a, b_{\mathrm{irr}}) = 0$ has an unique solution $a = b_{\mathrm{irr}} = 0$ for all $j > 0$. The claim now follows as in the odd case.

Case 3: $(g, n) = (5, 0)$. In this case, the claim follows because system (1) has three independent linear equations in three variables for any r. We omit the direct computation. □

For the sake of completeness, we also compute the Picard group for small values of (g, n).

Proposition 8.10. *Let (g, n) satisfy $5 > g \geq \lfloor (r+1)/2 \rfloor$ and assume that the stack $\mathcal{U}^r_{g,n}$ admits a good moduli space $\mathcal{U}^r_{g,n} \to U^r_{g,n}$. Then for $(g, n) = (3, n)$ with $n \geq 2$ or $(g, n) = (4, n)$ with $n \geq 1$, we have $\mathrm{Pic}_{\mathbb{Q}}(U^r_{g,n}) = 0$ for all $r \geq 5$. Moreover we have:*

$$\mathrm{Pic}_{\mathbb{Q}}(U^r_{4,0}) = \begin{cases} \langle \delta_{\mathrm{irr}} + 30\delta_2, \lambda + 3\delta_2 \rangle_{\mathbb{Q}}, & \text{if } r = 5, 6, \\ \langle \delta_{\mathrm{irr}} - 9\lambda + 3\delta_2 \rangle_{\mathbb{Q}}, & \text{if } r = 7. \end{cases}$$

$$\mathrm{Pic}_{\mathbb{Q}}(U^r_{3,0}) = \langle 3\delta_{\mathrm{irr}} - 28\lambda \rangle_{\mathbb{Q}} \qquad \text{for } 5 \leq r \leq 6.$$

$$\mathrm{Pic}_{\mathbb{Q}}(U^r_{3,1}) = \begin{cases} \langle 3\delta_{\mathrm{irr}} - 28\lambda + 6\delta_{2,\emptyset} + 2\delta_{0,\{1\}} \rangle_{\mathbb{Q}}, & \text{if } r = 5, \\ \langle 11\delta_{\mathrm{irr}} - 102\lambda + 24\delta_{2,\emptyset} + 6\delta_{0,\{1\}} \rangle_{\mathbb{Q}}, & \text{if } r = 6. \end{cases}$$

The cases above are an exhaustive list of rational Picard groups for $r \geq 5$.

Proof. When $\mathrm{Pic}_{\mathbb{Q}}(U^r_{g,n}) = 0$, the statement follows from a direct analysis of system (1), in the same spirit as the proof of Theorem 8.9. We omit the details.

For the remaining cases $(g, n) \in \{(4, 0), (3, 0), (3, 1)\}$, one can explicitly classify all closed points for any r such that $g \geq \lfloor (r+1)/2 \rfloor$. All possible cocharacters of closed points are products of the types appearing in Proposition 8.8. Solving the resulting linear systems case by case yields the expressions listed above. We omit the computations. □

For completeness, we compute the rational Picard for $r = 4$. Notice that for $r \leq 3$, the admissible open coincides with the α_c-stable one from the Hassett-Keel program. In that cases the Picard is already computed in [CTV23, Example 6.5], an explicit description with generators can be found in [Gor26, Sec. 4.1.1].

Proposition 8.11. *Let (g, n) be a hyperbolic pair with $g \geq 4$ or $g \geq 2$ and $n \geq 1$. Then $\mathrm{Pic}_{\mathbb{Q}}(U^4_{g,n})$ is generated by the elements above when the conditions on g and n are satisfied:*

$$\begin{cases} \delta_{\mathrm{irr}} - 10\lambda + 2\sum_{i=0}^{g} \delta_{i,I} & \text{if } n \geq 2 \text{ and } I \neq \emptyset, [n], \\ \delta_{\mathrm{irr}} - 10\lambda + 2\sum_{i=1}^{\lfloor (g-1)/2 \rfloor} \delta_{2i+1,\emptyset} & \text{if } n \geq 1, \\ 3\delta_{\mathrm{irr}} - 28\lambda + 8\sum_{i=1}^{\lfloor g/2 \rfloor} \delta_{2i,\emptyset} & \text{if } n \geq 1, \\ \delta_{\mathrm{irr}} - 10\lambda + 2\sum_{i=1}^{\lfloor (g-2)/4 \rfloor} \delta_{2i+1} & \text{if } n = 0 \text{ and } 2 \mid g, \\ 3\delta_{\mathrm{irr}} - 28\lambda + 8\sum_{i=1}^{\lfloor g/4 \rfloor} \delta_{2i} & \text{if } n = 0 \text{ and } 2 \mid g, \\ 7\delta_{\mathrm{irr}} - 68\lambda + 8\sum_{i=1}^{\lfloor g/2 \rfloor} \delta_{i} & \text{if } n = 0 \text{ and } 2 \nmid g. \end{cases}$$

Proof. The computation follows similarly to Theorem 8.9. In particular the following subset of equations in Equation (1) holds:

$$\begin{cases} 4a + 40b_{\mathrm{irr}} - b_{2,\emptyset} = 0 \\ a + 12b_{\mathrm{irr}} - b_{i,I} - b_{g-k-i,I^c} = 0 \end{cases}$$

Finally, the only non-trivial cocharacters that are not products of the types appearing in Proposition 8.8 act non-trivially on non-separating attached $\mathbb{G}_m$-rosaries, which have weight 0 with respect to any line bundle (because they have odd length). Therefore, the equations above are the only constraints for descending line bundles. The claim then follows from a computation that we omit. □

8.3 The moduli space U^5_g is not a scheme

We end the section by showing that U^5_g is not a scheme (if the genus is big enough). To do that, we first need to prove some simple facts regarding Alper's $\mathcal{O}$-semistability (see [Alp13, Definition 11.1]). The third named author would like to thank Andres Fernandez Herrero for explaining the main idea to him. Indeed, for a more thorough investigation of the difference between algebraic spaces and schemes, the reader is invited to look at [FHWZ24]. In fact what we do is to find a point which can be shown to be schematically trivial in the sense of [FHWZ24, Definition 2.9], see Remark 8.18.

For the rest of the section, if Y is an algebraic stack and $t \in \Gamma(Y, \mathcal{L})$ is a global section for some line bundle, we denote by Y_t the open subset defined as the complement of the support of t.

Lemma 8.12. *Let $\mathcal{X}$ be a quasi-compact quasi-separated algebraic stack over a quasi-compact quasi-separated algebraic space S and suppose that $\mathcal{X}$ admits a good moduli space $\mathcal{X} \to X$ which is proper and Stein over S (i.e. the map $p : X \to S$ is proper and $p_*\mathcal{O}_X \simeq \mathcal{O}_S$). Then the $\mathcal{O}_{\mathcal{X}}$-semistable locus in $\mathcal{X}$ is saturated with respect to S, i.e. there exists an open subset $S_0 \subset S$ (possibly empty) such that $\mathcal{X}^{ss}_{\mathcal{O}} = \mathcal{X}_0 := \mathcal{X} \times_S S_0$. Moreover, S_0 coincides with the maximal open in S such that the morphism $X_0 := X \times_S S_0 \to S_0$ is finite.*

Proof. Let $S_0 \subset S$ be the subset of points in S such that the fiber of $p : X \to S$ in s is 0-dimensional. Since the morphism is proper, we know that S_0 is an open subset. First of all, we can show that every point in $\mathcal{X}_0$ is $\mathcal{O}_{\mathcal{X}}$-semistable, thus $\mathcal{X}_0 \subset \mathcal{X}^{ss}_{\mathcal{O}}$. Indeed, by definition, if $x \in \mathcal{X}_0$, we can choose the open subset $S_0 \subset S$ and the section $1_{S_0} \neq 0 \in \Gamma(X_0, \mathcal{O}_{X_0})$ such that $(\mathcal{X}_0)_1 = \mathcal{X}_0 \to S_0$ is cohomologically

affine (since it is a composition of a good moduli space morphism with a finite morphism). Notice that this is always true, without the hypothesis on p.

Let $x \in \mathcal{X}^{ss}_{\mathcal{O}}$ be a $\mathcal{O}_{\mathcal{X}}$-semistable point over S (as in [Alp13, Definition 11.1]), then there exists an open $S' \subset S$ and a section $t \in \Gamma(S', p_*\mathcal{O}_X)$ such that $(\mathcal{X} \times_S S')_t \to S'$ is cohomologically affine. Since p is proper and Stein, we have that $p_*\mathcal{O}_X \simeq \mathcal{O}_S$, thus $t \in \Gamma(S', \mathcal{O}_{S'})$ and $(\mathcal{X} \times_S S')_t = \mathcal{X} \times_S S'_t$. Thus we get that $\mathcal{X} \times_S S'_t \to S'_t$ is cohomologically affine which implies that $X \times_S S'_t \to S'_t$ is affine (see [Alp13, Remark 4.3]) and proper, i.e. finite. This implies $S'_t \subset S_0$ and thus $x \in p^{-1}S_0 = \mathcal{X}_0$ and we win. □

Lemma 8.13. *Let $\mathcal{X}$ be a noetherian algebraic stacks with affine diagonal over a quasi-compact quasi-separated algebraic space S and suppose that $\mathcal{X}$ admits a good moduli space $\pi : \mathcal{X} \to X$. If $x_0 \in \mathcal{X}^{ss}_{\mathcal{O}}$, then X is relatively quasi-affine around $\pi(x_0)$, i.e. there exists a Zariski open neighbourhood $\pi(x_0) \in U \subset X$ such that $U \to S$ is quasi-affine. Moreover, suppose that $\mathcal{X}$ is regular and $x_0 \in \mathcal{X}$ is a point such that every non-torsion line bundle $\mathcal{L} \in Pic(\mathcal{X})$ induces a non-trivial character of the (connected component of the identity of) the stabilizer group of x_0. If X is relatively quasi-affine around $\pi(x_0)$, then $x_0 \in \mathcal{X}^{ss}_{\mathcal{O}}$.*

Proof. If $x_0 \in \mathcal{X}^{ss}_{\mathcal{O}}$, then by definition $\pi(x_0)$ is contained in an open of the form $(X \times_S S')_t \subset X$, for some open $S' \subset S$ and section $t \in \Gamma(X \times_S S', \mathcal{O})$, which is quasi-affine over S (since $(X \times_S S')_t$ is affine over the open S' inside the quasi-separated algebraic space S).

Suppose now that $\mathcal{X}$ is regular and $x_0 \in \mathcal{X}$ is a point such that every non-torsion line bundle $\mathcal{L} \in Pic(\mathcal{X})$ induces a non-trivial character of (the connected component of the identity of) the stabilizer group of x_0. If X is relatively quasi-affine around x_0, then it exists an open neighbourhood $U \subset X$ of x_0 such that $U \to S$ is quasi-affine. Therefore we can apply [Alp13, Theorem 11.14(ii)] to get a line bundle $\mathcal{L}$ on $\mathcal{X}$ and a commutative diagram

$$\begin{tikzcd}
\mathcal{U} \arrow[r, hook] \arrow[d] & \mathcal{X}^{ss}_{\mathcal{L}} \arrow[r, hook] \arrow[d] & \mathcal{X} \arrow[d] \arrow[dr] & \\
U \arrow[r, hook] & X^{ss}_{\mathcal{L}} \arrow[r] & X \arrow[r] & S
\end{tikzcd}$$

where the squares are Cartesian and the vertical arrows are moduli space morphisms. Moreover, the upper horizontal arrows are open immersions. [Alp13, Theorem 11.5(ii)] gives us that the line bundle $\mathcal{L}$ actually descends to the good moduli space $X^{ss}_{\mathcal{L}}$ as an ample line bundle over S, which implies that the action of the stabilizer group of $x_0 \in \mathcal{U} \subset \mathcal{X}^{ss}_{\mathcal{L}}$ on $\mathcal{L}$ is trivial, thus by hypothesis $\mathcal{L}$ is a torsion line bundle. Thus we can assume $\mathcal{L} \simeq \mathcal{O}_{\mathcal{X}}$, which implies $x_0 \in \mathcal{X}^{ss}_{\mathcal{O}}$. □

Proposition 8.14. *Let $\mathcal{X}$ be a noetherian regular algebraic stacks with affine diagonal over a quasi-compact quasi-separated connected algebraic space S and suppose that $\mathcal{X}$ admits a good moduli space $\pi : \mathcal{X} \to X$ which is proper, flat and Stein over S. Suppose there exists a point $x_0 \in \mathcal{X}$ such that every non-torsion line bundle $\mathcal{L} \in Pic(\mathcal{X})$ induces a non-trivial character of the (connected component of the identity of) the stabilizer group of x_0. If X is relatively quasi-affine around x_0, then X is finite over S.*

Proof. Thanks to Lemma 8.13, we have that $x_0 \in \mathcal{X}^{ss}_{\mathcal{O}}$. Moreover, Lemma 8.12, together with the flatness of $X \to S$ and the connectedness of S, gives that $S_0 = S$, $\mathcal{X}^{ss}_{\mathcal{O}} = \mathcal{X}$ and thus $X \to S$ is finite. □

We use now Proposition 8.14 to show that if $n = 0$, we can prove that U^5_g is not a scheme.

Corollary 8.15. *The moduli space U^5_g is not a scheme for $g = 7$ and $g \geq 9$.*

Proof. Recall that our base scheme is $\operatorname{Spec}\kappa$, where κ is a field of characteristic zero. Moreover, $\mathcal{U}^r_{g,n}$ has affine diagonal and it is smooth and geometrically connected over κ (since the smooth locus is geometrically connected and dense in $\mathcal{U}^r_{g,n}$). Moreover, we proved that $U^5_{g,n}$ is proper over κ, and $\mathcal{U}^5_{g,n}$ contains a positive dimensional separated DM stack as a saturated open (thanks to Corollary 6.6), thus it is not finite. Therefore, thanks to Proposition 8.14, if we find a curve x_0 such that every (non-torsion) line bundle on $\mathcal{U}^5_g$ induces a non-trivial character of the (connected component of the identity of the) stabilizer group of x_0, we win.

Let $h := \lfloor g/2 \rfloor$. Consider the curve C whose combinatorial structure is described by the following graph

$$\Gamma_{2,1} \text{ —— } \Gamma^2_1 \text{ —— } \Gamma^3_1 \text{ —— } \dots \text{ —— } \Gamma^h_1 \text{ —— } C_0 \text{ ═══ } \Gamma_{2,2}$$

where

- the lines represents attachment nodes,
- Γ^i_1 is the 2-pointed odd atom of genus 1 for $i = 2, \dots, h$;
- $\Gamma_{2,1}$ is the 1-pointed odd atom of genus 2,
- $\Gamma_{2,2}$ is the 2-pointed odd atom of genus 2, and
- C_0 is a 4-pointed curve of genus $g - 2 - (h-1) - (2+1) = \lceil g/2 \rceil - 4$.

Since $g \geq 7$, then the construction makes sense. Let us show that the action of $\operatorname{Aut}^\circ(C)$ is always non-trivial on every non-torsion line bundle $\operatorname{Pic}(\mathcal{M}^5_g)$. By construction, the curve C contains a genus 1 component Γ^i_1 of type (3) of Proposition 8.8 for every $i = 2, \dots, h$. Moreover, it contains a genus 2 component $\Gamma_{2,1}$ of type (4) and a genus 2 component $\Gamma_{2,2}$ of type (2) of *loc.cit.*

Let us deal with the odd case first. Therefore a line bundle $L := a\lambda + b_{\text{irr}}\delta_{\text{irr}} + \sum_{i=2,\dots,h} b_i\delta_i$ (notice that δ_1 is contained in the complement of $\mathcal{U}^5_g$) induces the trivial character of $\operatorname{Aut}^\circ(C)$ if and only if the coefficients solve the following linear system

$$\begin{cases} 3a + 28b_{\text{irr}} = 0 \\ a + 12b_{\text{irr}} - b_i - b_{i+1} = 0 \qquad \text{for} \quad i = 2, \dots, h-1 \\ a + 12b_{\text{irr}} - 2b_h = 0 \\ 3a + 30b_{\text{irr}} - b_2 = 0 \end{cases}$$

Using the family of equations in i and the third equation, one gets $b_i = (a + 12b_{\text{irr}})/2$ for every $i = 1, \dots, h$, and thus the first and last equation gives us $a = b_{\text{irr}} = 0$, giving us the statement.

Regarding the even case, one can modify the curve by replacing Γ^h_1 with an odd atom Γ_2 of genus 2 and prove the same result. This new curve makes sense only if $g \geq 10$, thus we get the statement. □

Remark 8.16. For $g = 8$, the above technique does not work. Specifically, through an extensive analysis of the combinatorics of the curves in genus 8, there is no point in $\mathcal{U}^5_8$ such that any non-torsion line bundle induces a non-trivial character.

Remark 8.17. The same idea works for U^r_g if $r \geq 5$, if they exist (see Conjecture 1.3). Clearly, the lower bound on the genus will dependent (linearly) on r.

Remark 8.18. Using the methods of [FHWZ24], it is possible to show more, namely that the point constructed in the proof of Corollary 8.15, x_0, is schematically trivial as in [FHWZ24, Definition 2.9]. That is, for any neighborhood $x_0 \in V \subset \mathcal{U}^5_g$, any map to a scheme $V \to X$ factors through a point, in particular this holds when $V = \mathcal{U}^5_g$, giving an alternate proof that U^5_g isn't a scheme. The second named author would like to thank Dario Weißman for explaining the following argument to him. To work in the setting of [FHWZ24], we assume here that the ground field κ is algebraically closed. The schematic

fiber Fib_{x_0} of x_0 (see [FHWZ24, Definition 4.8]) can be computed as follows by [FHWZ24, Proposition 4.19]: it is the irreducible component containing x_0 of the intersection of all prime divisors of $\mathcal{U}_g^5$ passing through x_0, such that Zariski locally around x_0 the divisors are described by sections of the structure sheaf. Since $\mathcal{U}_g^5$ is a smooth stack, all Weil divisors are Cartier, so they are globally described by a line bundle with a cosection $(\mathcal{L}, s)$. Since the structure sheaf of $\mathcal{U}_g^5$ descends to the good moduli space and hence it induces the trivial character at x_0, it follows by construction that for any $(\mathcal{L}, s)$ appearing in the intersection defining Fib_{x_0}, $\mathcal{L}$ is a torsion line bundle. This implies that after taking a power of $(\mathcal{L}, s)$, the divisor is given by a section of the structure sheaf of $\mathcal{U}_g^5$ globally, i.e the divisor is the vanishing of some morphism $f : \mathcal{U}_g^5 \to \mathbb{A}^1$, which by universality of good moduli spaces is induced by a morphism $U_g^5 \to \mathbb{A}^1$. Since U_g^5 is proper, connected and reduced (by [Alp13, Theorem 4.16 (viii)]), any map $U_g^5 \to \mathbb{A}^1$ factors through a point, and the vanishing locus of $f : \mathcal{U}_g^5 \to \mathbb{A}^1$ is either the full stack or the empty set. Thus there are no prime divisors going through x_0 that Zariski locally around x_0 are described by the vanishing of a section of the structure sheaf. Hence Fib_{x_0} is an intersection in $\mathcal{U}_g^5$ indexed by the empty set, i.e. $\mathrm{Fib}_{x_0} = \mathcal{U}_g^5$. Finally, by [FHWZ24, Remark 4.15], this implies that x_0 is a schematically trivial point.

One may wonder if the same technique can be used for $n \geq 1$. This does not seem to be the case, as we now explain. The problem is that the Picard group gets too big when we increase n.

Lemma 8.19. *Let $\mathcal{X}$ be a quasi-separated algebraic stack with affine stabilizers and $x_0 \in \mathcal{X}$ is a point such that every non-torsion line bundle $\mathcal{L} \in \mathrm{Pic}(\mathcal{X})$ induces a non-trivial character of the (connected component of the identity of) the stabilizer group of x_0. Then $\dim_{\mathbb{Q}} \mathrm{Hom}(\mathrm{Aut}(x_0), \mathbb{G}_m)_{\mathbb{Q}} \geq \dim_{\mathbb{Q}} \mathrm{Pic}(\mathcal{X})_{\mathbb{Q}}$.*

Proof. Firstly, notice that $\mathrm{Hom}(\mathrm{Aut}(x_0), \mathbb{G}_m)$ is an abelian group with the multiplication induced by $\mathbb{G}_m$, thus tensoring with $\mathbb{Q}$ makes sense. Let k be any field in the equivalence class of x_0. Then we have a morphism of abelian groups

$$\mu : \mathrm{Pic}(\mathcal{X}) \longrightarrow \mathrm{Hom}(\mathrm{Aut}_k(x_0), \mathbb{G}_{m,k})$$

which becomes a morphism of $\mathbb{Q}$-vector spaces tensoring with $\mathbb{Q}$. The statement follows noticing that the line bundles inducing a trivial character corresponds to elements in the kernel of μ. $\square$

We claim that if there exists a n-pointed A_r-stable curve $x_0 \in \mathcal{M}_{g,n}^r$ such that every non-torsion line bundle $L \in \mathrm{Pic}(\mathcal{M}_{g,n}^r)$ induces a non-trivial character of $\mathrm{Aut}^\circ(x_0)$, then $n = 0$. Indeed, if such curve exists, Lemma 8.19 and Lemma 3.26 give us that

$$\dim_{\mathbb{Q}} \mathrm{Pic}(\mathcal{M}_{g,n}^r)_{\mathbb{Q}} \leq \dim \mathrm{Aut}^\circ(x_0) \leq g$$

which implies $n = 0$ (thanks to the explicit description of the generators of the Picard group given in Proposition 8.7). Therefore, we end the discussion with the following question.

Question 8.20. Is $U_{g,n}^5$ a scheme when $n \geq 1$ (for sufficiently large genera)?

A Linearization

In this section, we state some results which will allow us to study isotrivial degenerations through their linearizations, i.e studying them in the deformation space. All the results in this section are probably well-known to experts, but we collected them for the sake of readability.

We begin by recalling some easy results regarding affine linear GIT.

A.1 Affine linear GIT

In this short subsection, we analyze the problem of finding G-equivariant closed subset $Z \subset V$ of a G-representation containing 0 and with the property that the complement is Θ-complete (respectively is S-complete, it has a good moduli space).

Remark A.1. We recall the following notation: if V is a G-representation for some algebraic group G and $\mathbb{G}_m \to G$ is a cocharacter of G, we denote by $V^{>0}$ (respectively $V^{\geq 0}$, V^0, $V^{<0}$, $V^{\leq 0}$) the direct sum of the eigenspaces of V with positive (respectively nonnegative, zero, negative, non positive) weights with respect to the action of $\mathbb{G}_m$.

The following lemma is a special case of [HL14, Thm. 1.4.8].

Lemma A.2. *Let k be an algebraically closed field, G be an algebraic group over k and V be a G-representation. Suppose we are given a commutative diagram*

$$\begin{tikzcd} \mathcal{B}\mathbb{G}_{m,k} \arrow[r] \arrow[d, hook] & \mathcal{B}G \arrow[d, hook] \\ \Theta_k \arrow[r] & {[V/G]}. \end{tikzcd}$$

Then the morphism $\Theta_k \to [V/G]$ factors through the morphism $[V^{>0}/\mathbb{G}_m] \to [V/G]$. Moreover, let R be a DVR with residue field k and suppose we have a commutative diagram

$$\begin{tikzcd} \mathcal{B}\mathbb{G}_{m,k} \arrow[r] \arrow[d, hook] & \mathcal{B}G \arrow[d, hook] \\ \mathcal{B}\mathbb{G}_{m,R} \arrow[r] & {[V/G]} \end{tikzcd}$$

Then the morphism $\mathcal{B}\mathbb{G}_{m,R} \to [V/G]$ factors through the morphism $[V^0/\mathbb{G}_m] \to [V/G]$.

Proof. Firstly, we notice that we have a commutative diagram

$$\begin{tikzcd} \Theta_k \arrow[r] \arrow[dr] & {[V/\mathbb{G}_m]} \arrow[r] \arrow[d] & {[V/G]} \arrow[d] \\ & \mathcal{B}\mathbb{G}_{m,k} \arrow[r] & \mathcal{B}G \end{tikzcd}$$

where the right-hand square is cartesian. Thus without loss of generality, we can assume $G = \mathbb{G}_m$. Because Θ is coherently complete along its closed point, it is easy to prove that the morphism $\Theta_k \to [V/\mathbb{G}_m]$ is induced by a $\mathbb{G}_m$-equivariant morphism $\mathbb{A}^1 \to V$. The result follows from a straightforward computation. The proof can be adapted easily to the other case. $\square$

Proposition A.3. *Let V be a G-representation and let Z be a G-equivariant closed subset of V which contains 0 and denote by U the open complement. The following are equivalent:*

(i) there exists a cocharacter $\mathbb{G}_m \to G$ such that Z does not contain neither $V^{>0}$ nor V^0;

(ii) there exists a commutative diagram

$$\begin{tikzcd} \Theta_R \setminus 0 \arrow[r, "f^0"] \arrow[d, hook] & {[U/G]} \arrow[d, hook] \\ \Theta_R \arrow[r, "f"] & {[V/G]}. \end{tikzcd}$$

where f sends the closed point of Θ_R to the 0 of the G-representation V.

Moreover, the following are equivalent:

(i) there exists a cocharacter $\mathbb{G}_m \to G$ such that Z does not contain neither $V^{>0}$ nor $V^{<0}$;

(ii) there exists a commutative diagram

$$\begin{tikzcd} \mathrm{ST}_R \arrow[r, "f^0"] \arrow[d, hook] & {[U/G]} \arrow[d, hook] \\ \overline{\mathrm{ST}}_R \arrow[r, "f"] & {[V/G]}. \end{tikzcd}$$

where f sends the closed point of $\overline{\mathrm{ST}}_R$ to the 0 of the G-representation V.

Proof. We will deal with the Θ_R case. First of all, (ii) implies (i) follows straightforwardly from Lemma A.2. Let us assume (i). First of all, by pulling back through the cocharacter $\chi : \mathbb{G}_m \to G$, it is enough to consider the case $G = \mathbb{G}_m$, with the identity cocharacter. We have that $Z \subset V$ is a $\mathbb{G}_m$-equivariant subset of V which does not contain $V^{>0}$ nor V^0. Firstly, notice that because Z does not contain V^0, this implies we can find a DVR R and a morphism $f_0 : \operatorname{Spec} R \to V^0$ such that f sends the closed point of $\operatorname{Spec} R$ in the origin of the subrepresentation V^0 while the open point is mapped to the generic point of V^0, which is not in Z. Similarly, because Z does not contain $V^{>0}$ we can find a closed rational point $x_1 \in V^{>0}(k)$ which is not in Z. Because the weights of $\mathbb{G}_m$ on $V^{>0}$ are strictly positive, we know that $0 = \lim_{t\to 0} t.x_1$ in the quotient stack $[V^{>0}/\mathbb{G}_m]$. Therefore we get a morphism $f_+ : \Theta_k \to [V^{>0}/\mathbb{G}_m]$ such that the generic point $\operatorname{Spec} k \subset \Theta_k$ is mapped to $x_1 \in [V^{>0}/\mathbb{G}_m]$. We constructed a morphism

$$(f_+, f_0) : \Theta_R = \Theta_k \times_k \operatorname{Spec} R \longrightarrow [V^{>0} \oplus V^0/\mathbb{G}_m] \subset [V/\mathbb{G}_m];$$

we leave it to the reader to check that it verifies the desired properties.

The $\overline{\mathrm{ST}}_R$ case is identical. Indeed, (ii) implies (i) follows from Lemma A.2. We assume (i). Repeating the argument above we can construct the following diagram

$$\begin{tikzcd} {[\mathbb{A}^1/\mathbb{G}_m]} \arrow[rr, "f_+"] \arrow[dr] & & {[V^{>0}/\mathbb{G}_m]} \arrow[dr] & \\ & \mathcal{B}\mathbb{G}_m \arrow[rr, equal] & & \mathcal{B}\mathbb{G}_m \\ {[\mathbb{A}^1/\mathbb{G}_m]} \arrow[rr, "f_-"] \arrow[ur] & & {[V^{<0}/\mathbb{G}_m]} \arrow[ur] & \end{tikzcd}$$

where the source of f_+ (respectively f_-) is the quotient of $\mathbb{A}^1$ by the $\mathbb{G}_m$-action of weight 1 (respectively of weight -1). By taking the fiber product, we get a morphism

$$(f_+, f_-) : [\mathbb{A}^2/\mathbb{G}_m] \longrightarrow [V^{>0} \oplus V^{<0}/\mathbb{G}_m] \subset [V/\mathbb{G}_m]$$

where the $\mathbb{G}_m$-action of $\mathbb{A}^2$ has weights $(1,-1)$. Finally, the result follows because $\overline{\mathrm{ST}}_R$ is the local version of $[\mathbb{A}^2/\mathbb{G}_m]$, namely

$$\overline{\mathrm{ST}}_R \simeq [\mathbb{A}^2/\mathbb{G}_m] \times_{\mathbb{A}^1} \operatorname{Spec} R \longrightarrow [\mathbb{A}^2/\mathbb{G}_m]$$

where $[\mathbb{A}^2/\mathbb{G}_m] \to \mathbb{A}^1$ is the good moduli space morphism and $\operatorname{Spec} R \to \mathbb{A}^1$ is a morphism sending the closed point of $\operatorname{Spec} R$ to $0 \in \mathbb{A}^1$. In our case, we can just choose $R = k[[t]]$. □

A.2 Reduction to the deformation space

Let $\mathcal{X}$ be an algebraic stack with affine stabilizers, of finite type over a noetherian base scheme S, and $x : \operatorname{Spec} k \to \mathcal{X}$ be a point. We denote by G_x the stabilizer group and by $T_x\mathcal{X}$ the G_x-representation given by the tangent vector space of $x \in X$.

Remark A.4. Let $\mathcal{X}$ be an algebraic stack with affine stabilizers of finite type over a noetherian base scheme S and x be a point in $\mathcal{X}$ with linearly reductive stabilizer group. We know the completion exists

$$\eta_x : \widehat{\mathcal{X}}_x \longrightarrow \mathcal{X}$$

at the point $x \in \mathcal{X}$ (see Theorem 1.12 of [AHR23]). Suppose now that we are given a morphism $x : \operatorname{Spec} k \to \mathcal{X}$ such that the stabilizer group G_x is linearly reductive and x is a smooth point of $\mathcal{X}$. Then we can identify $\widehat{\mathcal{X}}_x$ with the completion $[T_x\mathcal{X}/G_x]^\wedge$ of the linearly fundamental stack $[T_x\mathcal{X}/G_x]$ at the closed point $\mathcal{B}G_x$. The identification follows from the fact that the first thickening $\mathcal{X}_x^{[0]} := \mathcal{B}G_x \subset \mathcal{X}_x^{[1]}$ of $\mathcal{B}G_x$ in $\widehat{\mathcal{X}}_x$ is actually trivial, i.e. it admits a section, because G_x is linearly reductive. Recall that the completion $[T_x\mathcal{X}/G_x]^\wedge$ coincides with the fiber product

$$\begin{tikzcd}
{[T_x\mathcal{X}/G_x]^\wedge} \arrow[r] \arrow[d] & {[T_x\mathcal{X}/G_x]} \arrow[d, "p"] \\
(T_x\mathcal{X}//G_x)^\wedge \arrow[r] & T_x\mathcal{X}//G_x
\end{tikzcd}$$

where p is the good moduli space morphism and the bottom horizontal morphism is the completion of the good moduli space in 0.

Lemma A.5. *Let $\mathcal{X}$ be an algebraic stack with affine stabilizers of finite type over a field k and $(\mathcal{Z}, \mathcal{Z}_0)$ be a coherently complete pair of algebraic stacks. Suppose given a diagram*

$$\begin{tikzcd}
\mathcal{Z}_0 \arrow[r, hook] \arrow[d] & \mathcal{Z} \arrow[d, "f"] \\
\mathcal{B}G_x \arrow[r] & \mathcal{X}
\end{tikzcd}$$

where G_x is the stabilizer group of a rational smooth point $x \in \mathcal{X}(k)$. Assume moreover that G_x is linearly reductive, then there exists a unique lifting

$$\begin{tikzcd}
(\mathcal{Z}, \mathcal{Z}_0) \arrow[r, dashed, "\exists! \hat{f}"] \arrow[dr, "f"'] & ([T_x\mathcal{X}/G_x]^\wedge, \mathcal{B}G_x) \arrow[r] \arrow[d] & {[T_x\mathcal{X}/G_x]} \\
& \mathcal{X} &
\end{tikzcd}$$

where the right side of the diagram is described in Remark A.4.

Proof. It follows by the definition of coherently complete pairs and [HR19, Theorem 1.1] as in [AHR23, Section 1.7]. □

Remark A.6. Let $\mathcal{X}$ be an algebraic stack of finite type over a noetherian base scheme S and x be a point in $\mathcal{X}$. Given $\mathcal{Z}$ be a closed subset of $\mathcal{X}$, we denote by $\widehat{\mathcal{Z}}_x$ the completion of $\mathcal{Z}$ in x, namely the preimage $\eta_x^{-1}(\mathcal{Z})$. If G_x is linearly reductive, this coincides with the preimage of a closed G_x-invariant subset $\mathcal{Z}_x \subset T_x X$ through the (completion) morphism

$$\widehat{\mathcal{X}}_x \simeq [T_x\mathcal{X}/G_x]^\wedge \longrightarrow [T_x\mathcal{X}/G_x].$$

Proposition A.7. *Let $\mathcal{X}$ be an algebraic stack smooth over a field with affine diagonal and linearly reductive stabilizers (at every point). Let $\mathcal{Z}$ be a closed subset of $\mathcal{X}$. The following are equivalent:*

(i) for every geometric point $x : \operatorname{Spec} k \to \mathcal{Z}$ and every cocharacter $\mathbb{G}_{m,k} \to G_x$, the (linearized) G_x-equivariant closed subset $Z_x \subset T_x\mathcal{X}$ contains either $T_x\mathcal{X}^{>0}$ or $T_x\mathcal{X}^0$ (respectively $T_x\mathcal{X}^{>0}$ or $T_x\mathcal{X}^{<0}$);

(ii) the open embedding $\mathcal{U} := \mathcal{X} \setminus \mathcal{Z} \subset \mathcal{X}$ is Θ-complete (respectively S-complete).

Proof. We will prove the case for Θ-completeness. The same idea works for S-completeness. We start with the implication $(i) \implies (ii)$. Let R be a complete DVR, and suppose given a commutative diagram

$$\begin{tikzcd} \Theta_R \setminus 0 \arrow[r] \arrow[d, hook] & \mathcal{U} \arrow[d, hook] \\ \Theta_R \arrow[r] & \mathcal{X}. \end{tikzcd}$$

By contradiction, we assume that the closed point of Θ_R maps to a point x in $\mathcal{Z}$. Because $\mathcal{X}$ is defined over a field, Cohen's structure theorem implies that R is a k-algebra where k is the residue field of R. Thus, we can base change the whole problem to $\operatorname{Spec} k$, and, without loss of generality, assume that x is a rational point. Using Lemma A.5 for the pair $(\Theta_R, 0)$, we can reduce to the study of the induced morphism

$$\Theta_R \longrightarrow [T_x\mathcal{X}/G_x]$$

which sends the closed point of Θ_R to the zero of the tangent space. The statement follows from Proposition A.3. For the converse, it is enough to notice that Θ-completeness and S-completeness are stable by base change, so we can reduce the statement to the local picture and use Proposition A.3. $\square$

Remark A.8. The equivalent condition of Θ-completeness in the previous proposition can be equivalently stated as follows (even if apparently stronger):

(i) for every $x : \operatorname{Spec} k \to \mathcal{Z}$ geometric point and every cocharacter $\mathbb{G}_{m,k} \to G_x$, the (linearized) G_x-equivariant closed subset $Z_x \subset T_x X$ contains either $T_x X^0$ or $T_x X^{>0} \cup T_x X^{<0}$;

B Combinatorics of problematic sequences

In this appendix, we deal with the combinatorial problem arisen from the local geometry of the open embedding $\mathcal{U}^r_{g,n} \subset \mathcal{M}^r_{g,n}$. See the proof of Theorem 5.1.

We will first set the problem. Fix a positive integer k, and suppose we are given a binary sequence $\mathbf{b} \in \{0,1\}^k$ of length k. We will draw such sequence as

$$\text{——} \; b_1 \; \text{——} \; b_2 \; \text{——} \; \dots \; \text{——} \; b_k \; \text{——}$$

where $b_i \in \{0,1\}$ for every $i = 1, \dots, k$. We will denote as e_i the edge in the previous picture between b_i and b_{i+1}. The left-most edge will be denoted by e_0 while the right-most edge will be denoted by e_k. For every $i \leq j \in 1, \dots, k$, we will denote by $\mathbf{b}(i,j)$ the subsequence in $\{0,1\}^{i-j+1}$ which start with the edge e_{i-1} and ends with the edge e_j.

Definition B.1. Fix k a positive integer and a binary sequence $\mathbf{b}$ of length k. A subsequence $\mathbf{b}(i,j)$ (or equivalently the pair $i \leq j$) will be called 1-*minimal* if there exists a unique $h \in \{i, \ldots, j\}$ such that b_h is 1.

We will now introduce all the terminology to state the problem.

Definition B.2. A vector of integers $\mathbf{a} \in \mathbb{Z}^k$ will be called $>$-*problematic* with respect to a 1-minimal subsequence $\mathbf{b}(i,j)$ of $\mathbf{b}$ with $b_h = 1$ $(i \leq h \leq j)$ if one of the following conditions hold:

(1) $a_i > (-1)^{b_{i-1}} a_{i-1}$;

(2) $a_s > 0$ for some $i \leq s \leq h$;

(3) $a_s < 0$ for some $h+1 \leq s \leq j$;

(4) $a_{j+1} > (-1)^{b_j} a_j$;

We will denote by $A_{i,j}(>)$ the set of $>$-problematic vectors for the subsequence $\mathbf{b}(i,j)$. A vector of integers $\mathbf{a} \in \mathbb{Z}^k$ will be called $>$-*problematic* with respect to $\mathbf{b}$ if it is $>$-problematic for every 1-minimal subsequence. We will denote by $A(>)$ the set of $>$-problematic vectors for $\mathbf{b}$.

Remark B.3. We set a_0 and a_{k+1} to be zero in the previous definition. For instance, if $i = 1$ then condition (1) becomes $a_i > 0$ which is already included in condition (2). Moreover, if $j = k$, then condition (4) becomes $(-1)^{b_k} a_k < 0$. If $b_k = 1$, then $h = k$ and thus condition (3) is empty while condition (4) coincides with $a_k > 0$ which is already contained in (2). If $b_k = 0$, then $h < k$ and thus condition (4) is included in condition (3).

Definition B.4. A vector of integers $\mathbf{a} \in \mathbb{Z}^k$ will be called $\leq$-*problematic* with respect to a minimal subsequence $\mathbf{b}(i,j)$ of $\mathbf{b}$ with $b_h = 1$ $(i \leq h \leq j)$ if one of the following conditions hold:

(1) $a_i \leq (-1)^{b_{i-1}} a_{i-1}$;

(2) $a_s \leq 0$ for some $i \leq s \leq h$;

(3) $a_s \geq 0$ for some $h+1 \leq s \leq j$;

(4) $a_{j+1} \leq (-1)^{b_j} a_j$;

We will denote by $A_{i,j}(\leq)$ the set of $\leq$-problematic vectors for the subsequence $\mathbf{b}(i,j)$. A vector of integers $\mathbf{a} \in \mathbb{Z}^k$ will be called $>$-*problematic* with respect to $\mathbf{b}$ if it is $\leq$-problematic for every 1-minimal subsequence. We will denote by $A(\leq)$ the set of $\leq$-problematic vectors for $\mathbf{b}$.

Remark B.5. As before, we set a_0 and a_{k+1} to be zero in the previous definition. For instance, if $i = 1$ then condition (1) becomes $a_i \leq 0$ which is already included in condition (2). Moreover, if $j = k$, then condition (4) becomes $(-1)^{b_k} a_k \leq 0$. If $b_k = 1$, then $h = k$ and thus condition (3) is empty while condition (4) coincides with $a_k \leq 0$ which is already contained in (2). If $b_k = 0$, then $h < k$ and thus condition (4) is included in condition (3).

Definition B.6. Let k be a positive integer and $\mathbf{b}$ a binary sequence of length k. Then a vector $\mathbf{a} \in \mathbb{Z}^k$ is called *problematic* (with respect to $\mathbf{b}$) if it is $>$-problematic and $\leq$-problematic. We will denote by $A \subset \mathbb{Z}^k$ the subset of problematic vectors.

Remark B.7. We have the following straightforward equalities

- $A(>) = \bigcap_{1-\min \mathbf{b}(i,j)} A_{i,j}(>)$;
- $A(\leq) = \bigcap_{1-\min \mathbf{b}(i,j)} A_{i,j}(\leq)$;
- $A = A(>) \cap A(\leq)$.

where the intersection is taken on the 1-minimal subsequences of $\mathbf{b}$.

We are ready for the main theorem of the section.

Theorem B.8. *Let k be a positive integer and $\mathbf{b}$ a binary sequence of length k. If there exists an index $1 \leq h \leq k$ such that $b_h = 1$, then there are no problematic vectors of integers.*

The proof of the Theorem follows from an easy iterative argument, which consists of two lemmas. In the first one, we prove the starting point of the iterative argument. In the second one, we describe how the iteration goes.

Lemma B.9. *Suppose we are in the hypotheses of Theorem B.8 and let $\mathbf{a} \in \mathbb{Z}^k$ be a problematic vector of integers. Let $h_1 < h_2 < \cdots < h_r$ the indexes such that $b_{h_t} = 1$ for $t = 1, \ldots, r$. Then*

$$a_{h_1} > 0 \iff a_{h_2} \leq a_{h_2-1} \leq a_{h_2-2} \leq \cdots \leq a_{h_1+1} \leq -a_{h_1} < 0.$$

The same is true for the opposite sign case, namely

$$a_{h_1} \leq 0 \iff a_{h_2} > a_{h_2-1} > a_{h_2-2} > \cdots > a_{h_1+1} > -a_{h_1} \geq 0.$$

Proof. The "only if" statement is clear. Assume then $a_{h_1} > 0$ by hypothesis and suppose $a_{h_1+1} + a_{h_1} > 0$. If we consider the 1-minimal subsequence $\mathbf{b}(h_1, h_1)$, condition (1) of Definition B.4 has to hold (because the others are not met) and then $a_{h_1} \leq a_{h_1-1}$, thus $a_{h_1-1} > 0$. Thus, we can apply condition (1) again of Definition B.4 for the subsequence $\mathbf{b}(h_1 - 1, h_1)$ and this gives us $a_{h_1-1} \leq a_{h_1-2}$, which again implies $a_{h_1-2} > 0$. We can apply this idea iteratively to get that

$$a_{h_1+1} + a_{h_1} > 0 \implies a_1 \geq a_2 \geq \cdots \geq a_{h_1-1} \geq a_{h_1} > 0.$$

This is impossible, as in this case the vector $\mathbf{a}$ would not verify any of the conditions of Definition B.4 for the subsequence $\mathbf{b}(1, h_1)$. Thus, we have that $a_{h_1+1} \leq -a_{h_1} < 0$.

We will now repeat the same idea to get $a_{h_1+2} \leq a_{h_1+1}$. Suppose $a_{h_1+2} > a_{h_1+1}$, then condition (1) of Definition B.4 has to hold for the subsequence $\mathbf{b}(h_1, h_1 + 1)$ giving us again $a_{h_1} \leq a_{h_1-1}$, thus $a_{h_1-1} > 0$. Iterating the same idea as before, we get a contradiction. Thus $a_{h_1+2} \leq a_{h_1+1}$. We can iterate this process until we meet the index h_2, because of the 1-minimality. We leave to the reader the check the opposite sign case. □

Lemma B.10. *Suppose we are in the hypotheses of Theorem B.8 and let $\mathbf{a} \in \mathbb{Z}^k$ be a problematic vector of integers. Let $h_1 < h_2 < \cdots < h_r$ the indexes such that $b_{h_t} = 1$ for $t = 1, \ldots, r$. Then for every $s = 2, \ldots, k-1$ we get that*

$$a_{h_s} > a_{h_s-1} > a_{h_s-2} > \cdots > a_{h_{s-1}+1} > -a_{h_{s-1}} \geq 0$$

implies (it is in fact equivalent to)

$$a_{h_{s+1}} \leq a_{h_{s+1}-1} \leq a_{h_{s+1}-2} \leq \cdots \leq a_{h_s+1} \leq -a_{h_s} < 0.$$

The same is true for the opposite sign case, namely

$$a_{h_s} \leq a_{h_s-1} \leq a_{h_s-2} \leq \cdots \leq a_{h_{s-1}+1} \leq -a_{h_{s-1}} < 0$$

implies (it is in fact equivalent to)

$$a_{h_{s+1}} > a_{h_{s+1}-1} > a_{h_{s+1}-2} > \cdots > a_{h_s+1} > -a_{h_s} \geq 0.$$

Proof. The proof works exactly as in the previous case. Indeed, it is enough to check the conditions of Definition B.4 for the subsequences $\mathbf{b}(h_s, j)$ for $j = h_s, \ldots, h_{s+1} - 1$. We leave it to the reader to check the details. □

Remark B.11. Notice that the proof only uses the inequalities $a_{h_s} > a_{h_s-1}$ and $a_{h_s} > 0$ to deduce the statement.

Proof of Theorem B.8. Suppose there exists $\mathbf{a} \in \mathbb{Z}^k$ a problematic vector of integers. Let $h_1 < h_2 < \cdots < h_r$ all the indexes such that $b_{h_t} = 1$ (for $t = 1, \ldots, r$). Suppose that $a_{h_1} > 0$. By Lemma B.9, this implies

$$a_{h_2} \leq a_{h_2-1} \leq a_{h_2-2} \leq \cdots \leq a_{h_1+1} \leq -a_{h_1} < 0.$$

By applying Lemma B.10 iteratively, we get that either

$$a_{h_r} \leq a_{h_r-1} \leq a_{h_r-2} \leq \cdots \leq a_{h_{r-1}+1} \leq -a_{h_{r-1}} < 0$$

or

$$a_{h_r} > a_{h_r-1} > a_{h_r-2} > \cdots > a_{h_{r-1}+1} > -a_{h_{r-1}} \geq 0.$$

In both cases we reach a contradiction, verifying that $\mathbf{a}$ does not verify the conditions in Definition B.2 (or in Definition B.4) for the subsequence $\mathbf{b}(h_r, k)$. □

Cyclic case

We will generalize the statement of Theorem B.8 to the case when the sequence is actually a circuit, i.e. in the picture

$$\text{——} \; b_1 \; \text{——} \; b_2 \; \text{——} \; \ldots \; \text{——} \; b_k \; \text{——}$$

the leftmost and rightmost edge coincide. To do this formally, we extend the binary sequence $\mathbf{b} \in \{0,1\}^k$ to an infinite binary sequence $\mathbf{b}^\circ \in \{0,1\}^{\mathbb{Z}}$ which is k-periodic, i.e. $\mathbf{b}^\circ(n) := \mathbf{b}([n])$ where $[n]$ is the class of the integer n modulo k. The definition of problematic vectors remains the same, but Remark B.5 is not true anymore. Notice that Lemma B.10 is also still valid, as the proof does not rely on the finiteness of k. Given a vector of integers $\mathbf{a} \in \mathbb{Z}^k$ of length k, we denote by $\mathbf{a}^\circ$ its periodic extension:

$$\begin{tikzcd} \mathbb{Z} \arrow[r, two heads, "\pi_k"'] \arrow[rr, bend left, "\mathbf{a}^\circ"] & \mathbb{Z}/(k) \arrow[r, "\mathbf{a}"'] & \{0,1\} \end{tikzcd}$$

where π_k is the usual quotient map. We will denote by a_i° the integer $\mathbf{a}^\circ(i)$ for some index $i \in \mathbb{Z}$.

Definition B.12. We say that $\mathbf{a} \in \mathbb{Z}^k$ is problematic for $\mathbf{b}^\circ$ if $\mathbf{a}^\circ$ satisfies the conditions of Definition B.6 for the periodic sequence $\mathbf{b}^\circ$.

Lemma B.13. *Let $\mathbf{a} \in \mathbb{Z}^k$ a vector of integers such that it exists an index $1 \leq i \leq k$ with the following properties:*

- $b_i = 1$;
- $a_i^\circ > 0$;
- $a_i^\circ > (-1)^{b_{i-1}^\circ} a_{i-1}^\circ$ *and* $a_i^\circ > -a_{i+1}^\circ$.

Then $\mathbf{a}$ is not problematic for $\mathbf{b}^\circ$. The same statement is true for the $\leq$.

Proof. Suppose that $\mathbf{a}$ is problematic. By the periodicity of $\mathbf{b}^\circ$ and $\mathbf{a}^\circ$, we can assume $i = 1$ without loss of generality. If $1 = h_1 < h_2 < \cdots < h_r \leq k$ are the indexes where $b_{h_t} = 1$, then we can iteratively apply Lemma B.10 (see Remark B.11) to get that

- if r is even, then
$$a_{h_1} \leq -a_{h_2} < a_{h_3} \leq \cdots \leq -a_{h_r} < a_{h_1+k}^\circ = a_{h_1};$$

- if r is odd, then
$$a_{h_1} \leq -a_{h_2} < a_{h_3} \leq \cdots < a_{h_r} \leq -a^\circ_{h_1+k} = -a_{h_1}$$

In both cases, we reach a contradiction. □

Lemma B.14. *Let $\mathbf{a} \in \mathbb{Z}^k$ a vector of integers such that it exists an index $1 \leq i \leq k$ with the following properties:*

- $b_i = 1$;
- $a^\circ_i > 0$;
- $a^\circ_i \leq (-1)^{b^\circ_{i-1}} a^\circ_{i-1}$ *and* $a^\circ_i \leq -a^\circ_{i+1}$.

Then $\mathbf{a}$ is not problematic for $\mathbf{b}^\circ$. The same statement is true for the $\leq$.

Proof. Suppose $\mathbf{a}$ is problematic and consider the maximal 1-minimal sequence $\mathbf{b}(j_1, j_2)$ containing i. By periodicity, we can assume $j_1 = 1$ and we set $j := j_2$. Because of the hypotheses, we know that $a_{i-1} > 0$ and $a_{i+1} < 0$. Because $\mathbf{a}$ is problematic, the condition on the subsequence $\mathbf{b}(i-1, i+1)$ gives us that either $a_{i+2} \leq a_{i+1} < 0$ or $a_{i-2} \geq a_{i-1} > 0$. Now take the largest interval $2 \leq i_1 \leq i \leq i_2 \leq j-1$ such that for every $i_1 \leq t \leq i$ we have

$$a_t \leq (-1)^{b_{t-1}} a_{t-1}$$

and for every $i \leq t \leq i_2$ we have
$$a_t \geq a_{t+1}.$$

Then we claim that either $i_1 = 2$ or $i_2 = j - 1$. In fact we get that $a_t > 0$ for every $i_1 - 1 \leq t \leq i$ and $a_t < 0$ for every $i \leq t \leq i_2 + 1$. If we consider the sequence $\mathbf{b}(i_1 - 1, i_2 + 1)$, than one of the two relations above has to be true for either $t = i_1 - 1$ or $t = i_2 + 1$. This contradicts the maximality of the interval $[i_1, i_2]$. Without loss of generality, we can assume $i_2 = j$, which implies $a_j \leq a_{j-1} < 0$. But now we have

- $b_j = 1$, because of the maximality of the 1-minimal sequence $\mathbf{b}(1, j)$;
- $a_j \leq 0$;
- $a_j \leq a_{j-1}$.

Therefore we can apply Lemma B.13 to get the result. □

Theorem B.15. *Let $\mathbf{b} \in \{0,1\}^k$ a binary sequence of length k. If there exists an index $1 \leq i \leq k$ such that $b_i = 1$, then there are no problematic vectors of integers $\mathbf{a} \in \mathbb{Z}^k$ for the periodic infinite binary sequence $\mathbf{b}^\circ$.*

Proof. The theorem follows from combining Lemma B.13 and Lemma B.14 □

Remark B.16. Notice that Theorem B.15 actually implies Theorem B.8. This can be seen by enlarging the binary sequence $\mathbf{b} \in \{0,1\}^k$ adding one element to it and setting it to 0. We denote as $\mathbf{b}' \in \{0,1\}^{k+1}$ the new binary sequence of length $k+1$. Now the problematic vectors of integers with respect to $\mathbf{b}'^\circ$ which have $a_{k+1} = 0$ coincide exactly with the problematic vectors with respect to $\mathbf{b}$.

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